

	8	7	6	5	4	3	2	1	CHINAFOX																																																															
D	BOM Groups																																																																							
	<table><tr><th>BOM GROUP</th><th colspan="5">BOM OPTIONS</th></tr><tr><td>X362_COMMON</td><td colspan="5">ALTERNATE,COMMON,X362_COMMON1,X362_COMMON2,X362_COMMON3,X362_PROGPARTS</td></tr><tr><td>X362_COMMON1</td><td colspan="5">SOC:SAMSUNG,SE:PROD,BOARD_ID:6,T208_PROG:REV5,CPUHYS,AP_TEMP,TBTTHRM_SNS,S3XCLK:1NT</td></tr><tr><td>X362_COMMON2</td><td colspan="5">EDP_ENABLE,XDP:YES,PCH_CLK:GRMCLK,TBT_DBG,SAMCONN,SKIP_S3V3:AUDIBLE,SOC_BOOT:SPI</td></tr><tr><td>X362_COMMON3</td><td colspan="5">CPUTHRM:ALRT,TBTTHRM:ALRT,LOADRC:YES,CUMULUS_IPD,S3_STATE:YES,WIFI_SAK:NO,VCCPLLLOC:S3</td></tr><tr><td>X362_PROGPARTS</td><td colspan="5">BOOTROM_PROG,SMC_PROG,AR_LT_PROG,AR_RT_PROG,WIFI_PROG,BTROM_PROG</td></tr><tr><td>X362_DEVEL:ENG</td><td colspan="5">ALTERNATE,DBGLED,USBC_DBG,XDP_CONN:YES,WIFI_DBG,S3X_DBG,DEBUG_BUTTON,LOADISNS</td></tr><tr><td>X362_DEVEL:DVT</td><td colspan="5">ALTERNATE</td></tr><tr><td>X362_DEVEL:PVT</td><td colspan="5">ALTERNATE</td></tr></table>								BOM GROUP	BOM OPTIONS					X362_COMMON	ALTERNATE,COMMON,X362_COMMON1,X362_COMMON2,X362_COMMON3,X362_PROGPARTS					X362_COMMON1	SOC:SAMSUNG,SE:PROD,BOARD_ID:6,T208_PROG:REV5,CPUHYS,AP_TEMP,TBTTHRM_SNS,S3XCLK:1NT					X362_COMMON2	EDP_ENABLE,XDP:YES,PCH_CLK:GRMCLK,TBT_DBG,SAMCONN,SKIP_S3V3:AUDIBLE,SOC_BOOT:SPI					X362_COMMON3	CPUTHRM:ALRT,TBTTHRM:ALRT,LOADRC:YES,CUMULUS_IPD,S3_STATE:YES,WIFI_SAK:NO,VCCPLLLOC:S3					X362_PROGPARTS	BOOTROM_PROG,SMC_PROG,AR_LT_PROG,AR_RT_PROG,WIFI_PROG,BTROM_PROG					X362_DEVEL:ENG	ALTERNATE,DBGLED,USBC_DBG,XDP_CONN:YES,WIFI_DBG,S3X_DBG,DEBUG_BUTTON,LOADISNS					X362_DEVEL:DVT	ALTERNATE					X362_DEVEL:PVT	ALTERNATE														
	BOM GROUP	BOM OPTIONS																																																																						
	X362_COMMON	ALTERNATE,COMMON,X362_COMMON1,X362_COMMON2,X362_COMMON3,X362_PROGPARTS																																																																						
	X362_COMMON1	SOC:SAMSUNG,SE:PROD,BOARD_ID:6,T208_PROG:REV5,CPUHYS,AP_TEMP,TBTTHRM_SNS,S3XCLK:1NT																																																																						
	X362_COMMON2	EDP_ENABLE,XDP:YES,PCH_CLK:GRMCLK,TBT_DBG,SAMCONN,SKIP_S3V3:AUDIBLE,SOC_BOOT:SPI																																																																						
	X362_COMMON3	CPUTHRM:ALRT,TBTTHRM:ALRT,LOADRC:YES,CUMULUS_IPD,S3_STATE:YES,WIFI_SAK:NO,VCCPLLLOC:S3																																																																						
	X362_PROGPARTS	BOOTROM_PROG,SMC_PROG,AR_LT_PROG,AR_RT_PROG,WIFI_PROG,BTROM_PROG																																																																						
	X362_DEVEL:ENG	ALTERNATE,DBGLED,USBC_DBG,XDP_CONN:YES,WIFI_DBG,S3X_DBG,DEBUG_BUTTON,LOADISNS																																																																						
	X362_DEVEL:DVT	ALTERNATE																																																																						
X362_DEVEL:PVT	ALTERNATE																																																																							
Module Parts																																																																								
<table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>337S00266</td><td>1</td><td>CPU,SKIU,SR2JK,PRQ,2.9,28W,1.05,B1356</td><td>U0500</td><td>CRITICAL</td><td>CPU_SKL23:2.9G</td></tr><tr><td>337S00267</td><td>1</td><td>CPU,SKIU,SR2JJ,PRQ,3.1,28W,1.1,B1356</td><td>U0500</td><td>CRITICAL</td><td>CPU_SKL23:3.1G</td></tr><tr><td>337S00268</td><td>1</td><td>CPU,SKIU,SR2JH,PRQ,3.3,28W,1.1,B1356</td><td>U0500</td><td>CRITICAL</td><td>CPU_SKL23:3.3G</td></tr><tr><td>998-04195</td><td>1</td><td>INTERPOSER,VTT ADAPTER,SKL-U,BGA1356</td><td>U0500</td><td>CRITICAL</td><td>CPU_SOCKET</td></tr></table>								PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	337S00266	1	CPU,SKIU,SR2JK,PRQ,2.9,28W,1.05,B1356	U0500	CRITICAL	CPU_SKL23:2.9G	337S00267	1	CPU,SKIU,SR2JJ,PRQ,3.1,28W,1.1,B1356	U0500	CRITICAL	CPU_SKL23:3.1G	337S00268	1	CPU,SKIU,SR2JH,PRQ,3.3,28W,1.1,B1356	U0500	CRITICAL	CPU_SKL23:3.3G	998-04195	1	INTERPOSER,VTT ADAPTER,SKL-U,BGA1356	U0500	CRITICAL	CPU_SOCKET																																			
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																			
337S00266	1	CPU,SKIU,SR2JK,PRQ,2.9,28W,1.05,B1356	U0500	CRITICAL	CPU_SKL23:2.9G																																																																			
337S00267	1	CPU,SKIU,SR2JJ,PRQ,3.1,28W,1.1,B1356	U0500	CRITICAL	CPU_SKL23:3.1G																																																																			
337S00268	1	CPU,SKIU,SR2JH,PRQ,3.3,28W,1.1,B1356	U0500	CRITICAL	CPU_SKL23:3.3G																																																																			
998-04195	1	INTERPOSER,VTT ADAPTER,SKL-U,BGA1356	U0500	CRITICAL	CPU_SOCKET																																																																			
<table><tr><td>338S00254</td><td>2</td><td>IC,TBT,ALPINE RIDGE,QSTY,PRQ,C1,CSP337</td><td>U2800,UB000</td><td>CRITICAL</td><td></td></tr><tr><td>353S00961</td><td>4</td><td>IC,CD3215,ACE,C00,USB_PWR_SW,BLANK,BGA96</td><td>U3100,U3200,UB300,UB400</td><td>CRITICAL</td><td></td></tr></table>								338S00254	2	IC,TBT,ALPINE RIDGE,QSTY,PRQ,C1,CSP337	U2800,UB000	CRITICAL		353S00961	4	IC,CD3215,ACE,C00,USB_PWR_SW,BLANK,BGA96	U3100,U3200,UB300,UB400	CRITICAL																																																						
338S00254	2	IC,TBT,ALPINE RIDGE,QSTY,PRQ,C1,CSP337	U2800,UB000	CRITICAL																																																																				
353S00961	4	IC,CD3215,ACE,C00,USB_PWR_SW,BLANK,BGA96	U3100,U3200,UB300,UB400	CRITICAL																																																																				
<table><tr><td>998-06736</td><td>1</td><td>IC,S3X CNTLR,B1,DEV,CSP900</td><td>U8600</td><td>CRITICAL</td><td></td></tr><tr><td>333S00055</td><td>1</td><td>IC,LPODR3-1600,4GBIT,25NM,A,276B</td><td>POP8600</td><td>CRITICAL</td><td>POP_4GBIT</td></tr><tr><td>333S00056</td><td>1</td><td>IC,LPODR3-1600,8GBIT,25NM,A,276B</td><td>POP8600</td><td>CRITICAL</td><td>POP_8GBIT</td></tr></table>								998-06736	1	IC,S3X CNTLR,B1,DEV,CSP900	U8600	CRITICAL		333S00055	1	IC,LPODR3-1600,4GBIT,25NM,A,276B	POP8600	CRITICAL	POP_4GBIT	333S00056	1	IC,LPODR3-1600,8GBIT,25NM,A,276B	POP8600	CRITICAL	POP_8GBIT																																															
998-06736	1	IC,S3X CNTLR,B1,DEV,CSP900	U8600	CRITICAL																																																																				
333S00055	1	IC,LPODR3-1600,4GBIT,25NM,A,276B	POP8600	CRITICAL	POP_4GBIT																																																																			
333S00056	1	IC,LPODR3-1600,8GBIT,25NM,A,276B	POP8600	CRITICAL	POP_8GBIT																																																																			
<table><tr><td>343S00147</td><td>1</td><td>IC,SL64AP41172,PAK3,STQFN20</td><td>U3620</td><td>CRITICAL</td><td></td></tr><tr><td>338S00221</td><td>1</td><td>IC,PMU,SN650839,7X7MM,BGA168</td><td>U7800</td><td>CRITICAL</td><td></td></tr></table>								343S00147	1	IC,SL64AP41172,PAK3,STQFN20	U3620	CRITICAL		338S00221	1	IC,PMU,SN650839,7X7MM,BGA168	U7800	CRITICAL																																																						
343S00147	1	IC,SL64AP41172,PAK3,STQFN20	U3620	CRITICAL																																																																				
338S00221	1	IC,PMU,SN650839,7X7MM,BGA168	U7800	CRITICAL																																																																				
<table><tr><td>353S01016</td><td>1</td><td>IC,ISL9239HIZ,PMU,TUBA,WCSF40,2.1X3.3MM</td><td>U7000</td><td>CRITICAL</td><td></td></tr><tr><td>338S00227</td><td>1</td><td>IC,PMU,PICCOLO,D2231A0,OTF-AK,WLCSP96</td><td>U9300</td><td>CRITICAL</td><td></td></tr></table>								353S01016	1	IC,ISL9239HIZ,PMU,TUBA,WCSF40,2.1X3.3MM	U7000	CRITICAL		338S00227	1	IC,PMU,PICCOLO,D2231A0,OTF-AK,WLCSP96	U9300	CRITICAL																																																						
353S01016	1	IC,ISL9239HIZ,PMU,TUBA,WCSF40,2.1X3.3MM	U7000	CRITICAL																																																																				
338S00227	1	IC,PMU,PICCOLO,D2231A0,OTF-AK,WLCSP96	U9300	CRITICAL																																																																				
C	Programmables (All Builds)																																																																							
	EFI ROM																																																																							
	<table><tr><td>341S00698</td><td>1</td><td>IC,EFI ROM (V0193) DVT,X362</td><td>U6100</td><td>CRITICAL</td><td>BOOTROM_PROG</td></tr></table>								341S00698	1	IC,EFI ROM (V0193) DVT,X362	U6100	CRITICAL	BOOTROM_PROG																																																										
	341S00698	1	IC,EFI ROM (V0193) DVT,X362	U6100	CRITICAL	BOOTROM_PROG																																																																		
	SMC ROM																																																																							
	<table><tr><td>341S00697</td><td>1</td><td>IC,SMC-B1,EXT (VXXXX) DVT,X362</td><td>U5000</td><td>CRITICAL</td><td>SMC_PROG</td></tr></table>								341S00697	1	IC,SMC-B1,EXT (VXXXX) DVT,X362	U5000	CRITICAL	SMC_PROG																																																										
	341S00697	1	IC,SMC-B1,EXT (VXXXX) DVT,X362	U5000	CRITICAL	SMC_PROG																																																																		
	TBT ROMs																																																																							
	<table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>341S00692</td><td>1</td><td>T29,AR1 (VXXX) DVT,X362</td><td>U2890</td><td>CRITICAL</td><td>AR_LT_PROG</td></tr><tr><td>341S00693</td><td>1</td><td>T29,AR2 (VXXX) DVT,X362</td><td>UB090</td><td>CRITICAL</td><td>AR_RT_PROG</td></tr></table>								PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	341S00692	1	T29,AR1 (VXXX) DVT,X362	U2890	CRITICAL	AR_LT_PROG	341S00693	1	T29,AR2 (VXXX) DVT,X362	UB090	CRITICAL	AR_RT_PROG																																														
	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																		
341S00692	1	T29,AR1 (VXXX) DVT,X362	U2890	CRITICAL	AR_LT_PROG																																																																			
341S00693	1	T29,AR2 (VXXX) DVT,X362	UB090	CRITICAL	AR_RT_PROG																																																																			
WIFI/BT ROM																																																																								
<table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>341S00695</td><td>1</td><td>IC,BT ROM (VXXXXXX) DVT,X362/X363</td><td>U3750</td><td>CRITICAL</td><td>BTROM_PROG</td></tr><tr><td>341S00696</td><td>1</td><td>WIFI ROM (PXXX) DVT,WW1,X362/X363</td><td>U3710</td><td>CRITICAL</td><td>WIFI_PROG</td></tr></table>								PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	341S00695	1	IC,BT ROM (VXXXXXX) DVT,X362/X363	U3750	CRITICAL	BTROM_PROG	341S00696	1	WIFI ROM (PXXX) DVT,WW1,X362/X363	U3710	CRITICAL	WIFI_PROG																																															
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																			
341S00695	1	IC,BT ROM (VXXXXXX) DVT,X362/X363	U3750	CRITICAL	BTROM_PROG																																																																			
341S00696	1	WIFI ROM (PXXX) DVT,WW1,X362/X363	U3710	CRITICAL	WIFI_PROG																																																																			
B	Variable BOM Groups Development/Base BOMs																																																																							
	<table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>685-00055</td><td>1</td><td>COMMON PARTS,MLB,X362</td><td>BASE</td><td>CRITICAL</td><td>BASE_BOM</td></tr><tr><td>985-00070</td><td>1</td><td>DEV,MLB,X362</td><td>DEVEL</td><td>CRITICAL</td><td>DEVEL_BOM</td></tr></table>								PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	685-00055	1	COMMON PARTS,MLB,X362	BASE	CRITICAL	BASE_BOM	985-00070	1	DEV,MLB,X362	DEVEL	CRITICAL	DEVEL_BOM																																														
	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																		
	685-00055	1	COMMON PARTS,MLB,X362	BASE	CRITICAL	BASE_BOM																																																																		
	985-00070	1	DEV,MLB,X362	DEVEL	CRITICAL	DEVEL_BOM																																																																		
	Main DRAM Parts																																																																							
	<table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>333S00069</td><td>4</td><td>IC,SDRAM,LPODR3-2133,16GBIT,20NM,BGA178</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>8G_MICRON_2133</td></tr><tr><td>333S00070</td><td>4</td><td>IC,SDRAM,LPODR3-2133,32GBIT,20NM,BGA178</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>16G_MICRON_2133</td></tr><tr><td>333S00068</td><td>4</td><td>IC,SDRAM,LPODR3-2133,16GBIT,20NM,BGA178</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>8G_SAMSUNG_2133</td></tr><tr><td>333S00050</td><td>4</td><td>IC,SDRAM,LPODR3-2133,32GBIT,20NM,BGA178</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>16G_SAMSUNG_2133</td></tr></table>								PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	333S00069	4	IC,SDRAM,LPODR3-2133,16GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	8G_MICRON_2133	333S00070	4	IC,SDRAM,LPODR3-2133,32GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	16G_MICRON_2133	333S00068	4	IC,SDRAM,LPODR3-2133,16GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	8G_SAMSUNG_2133	333S00050	4	IC,SDRAM,LPODR3-2133,32GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	16G_SAMSUNG_2133																																		
	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																		
	333S00069	4	IC,SDRAM,LPODR3-2133,16GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	8G_MICRON_2133																																																																		
	333S00070	4	IC,SDRAM,LPODR3-2133,32GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	16G_MICRON_2133																																																																		
333S00068	4	IC,SDRAM,LPODR3-2133,16GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	8G_SAMSUNG_2133																																																																			
333S00050	4	IC,SDRAM,LPODR3-2133,32GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	16G_SAMSUNG_2133																																																																			
Main DRAM SPD Straps																																																																								
<table><tr><th>BOM GROUP</th><th>BOM OPTIONS</th></tr><tr><td>RAM_8G_MICRON_2133</td><td>8G_MICRON_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG2_L,RAMCFG1_L</td></tr><tr><td>RAM_16G_MICRON_2133</td><td>16G_MICRON_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG1_L</td></tr><tr><td>RAM_8G_SAMSUNG_2133</td><td>8G_SAMSUNG_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG2_L,RAMCFG0_L</td></tr><tr><td>RAM_16G_SAMSUNG_2133</td><td>16G_SAMSUNG_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG0_L</td></tr></table>								BOM GROUP	BOM OPTIONS	RAM_8G_MICRON_2133	8G_MICRON_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG2_L,RAMCFG1_L	RAM_16G_MICRON_2133	16G_MICRON_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG1_L	RAM_8G_SAMSUNG_2133	8G_SAMSUNG_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG2_L,RAMCFG0_L	RAM_16G_SAMSUNG_2133	16G_SAMSUNG_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG0_L																																																							
BOM GROUP	BOM OPTIONS																																																																							
RAM_8G_MICRON_2133	8G_MICRON_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG2_L,RAMCFG1_L																																																																							
RAM_16G_MICRON_2133	16G_MICRON_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG1_L																																																																							
RAM_8G_SAMSUNG_2133	8G_SAMSUNG_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG2_L,RAMCFG0_L																																																																							
RAM_16G_SAMSUNG_2133	16G_SAMSUNG_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG0_L																																																																							
NAND Parts																																																																								
<table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>335S00124</td><td>4</td><td>NAND,12NM,64GB,T0GG,HPN,128G,LGA60</td><td>U9100,U9120,U9200,U9220</td><td>CRITICAL</td><td>NAND_SDISK_256GB</td></tr><tr><td>335S00125</td><td>4</td><td>NAND,12NM,128GB,T0GG,HPN,128G,LGA60</td><td>U9100,U9120,U9200,U9220</td><td>CRITICAL</td><td>NAND_SDISK_512GB</td></tr><tr><td>335S00126</td><td>4</td><td>NAND,12NM,256GB,T0GG,HPN,128G,LGA60</td><td>U9100,U9120,U9200,U9220</td><td>CRITICAL</td><td>NAND_SDISK_1TB</td></tr><tr><td>335S00135</td><td>4</td><td>NAND,TGDDR2,128GX4,15NM,HP,USHD,T2,LGA60</td><td>U9100,U9120,U9200,U9220</td><td>CRITICAL</td><td>NAND_TSHBA_256GB</td></tr><tr><td>335S00136</td><td>4</td><td>NAND,TGDDR2,128GX8,15NM,HP,USHD,T2,LGA60</td><td>U9100,U9120,U9200,U9220</td><td>CRITICAL</td><td>NAND_TSHBA_512GB</td></tr><tr><td>335S00137</td><td>4</td><td>NAND,TGDDR2,128GX16,15NM,HP,UHD,T2,LGA60</td><td>U9100,U9120,U9200,U9220</td><td>CRITICAL</td><td>NAND_TSHBA_1TB</td></tr></table>								PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	335S00124	4	NAND,12NM,64GB,T0GG,HPN,128G,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_SDISK_256GB	335S00125	4	NAND,12NM,128GB,T0GG,HPN,128G,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_SDISK_512GB	335S00126	4	NAND,12NM,256GB,T0GG,HPN,128G,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_SDISK_1TB	335S00135	4	NAND,TGDDR2,128GX4,15NM,HP,USHD,T2,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_TSHBA_256GB	335S00136	4	NAND,TGDDR2,128GX8,15NM,HP,USHD,T2,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_TSHBA_512GB	335S00137	4	NAND,TGDDR2,128GX16,15NM,HP,UHD,T2,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_TSHBA_1TB																							
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																			
335S00124	4	NAND,12NM,64GB,T0GG,HPN,128G,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_SDISK_256GB																																																																			
335S00125	4	NAND,12NM,128GB,T0GG,HPN,128G,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_SDISK_512GB																																																																			
335S00126	4	NAND,12NM,256GB,T0GG,HPN,128G,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_SDISK_1TB																																																																			
335S00135	4	NAND,TGDDR2,128GX4,15NM,HP,USHD,T2,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_TSHBA_256GB																																																																			
335S00136	4	NAND,TGDDR2,128GX8,15NM,HP,USHD,T2,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_TSHBA_512GB																																																																			
335S00137	4	NAND,TGDDR2,128GX16,15NM,HP,UHD,T2,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_TSHBA_1TB																																																																			
NAND Straps																																																																								
<table><tr><th>BOM GROUP</th><th>BOM OPTIONS</th></tr><tr><td>SAND_256G</td><td>ALTERNATE,NAND_SDISK_256GB,POP_4GBIT,CAPACITY1,CAPACITY0</td></tr><tr><td>SAND_512G</td><td>ALTERNATE,NAND_SDISK_512GB,POP_4GBIT,CAPACITY2</td></tr><tr><td>SAND_1T</td><td>ALTERNATE,NAND_SDISK_1TB,POP_8GBIT,CAPACITY2,CAPACITY0</td></tr><tr><td>TOSH_256G</td><td>ALTERNATE,NAND_TSHBA_256GB,POP_4GBIT,CAPACITY1,CAPACITY0</td></tr><tr><td>TOSH_512G</td><td>ALTERNATE,NAND_TSHBA_512GB,POP_4GBIT,CAPACITY2</td></tr><tr><td>TOSH_1T</td><td>ALTERNATE,NAND_TSHBA_1TB,POP_8GBIT,CAPACITY2,CAPACITY0</td></tr></table>								BOM GROUP	BOM OPTIONS	SAND_256G	ALTERNATE,NAND_SDISK_256GB,POP_4GBIT,CAPACITY1,CAPACITY0	SAND_512G	ALTERNATE,NAND_SDISK_512GB,POP_4GBIT,CAPACITY2	SAND_1T	ALTERNATE,NAND_SDISK_1TB,POP_8GBIT,CAPACITY2,CAPACITY0	TOSH_256G	ALTERNATE,NAND_TSHBA_256GB,POP_4GBIT,CAPACITY1,CAPACITY0	TOSH_512G	ALTERNATE,NAND_TSHBA_512GB,POP_4GBIT,CAPACITY2	TOSH_1T	ALTERNATE,NAND_TSHBA_1TB,POP_8GBIT,CAPACITY2,CAPACITY0																																																			
BOM GROUP	BOM OPTIONS																																																																							
SAND_256G	ALTERNATE,NAND_SDISK_256GB,POP_4GBIT,CAPACITY1,CAPACITY0																																																																							
SAND_512G	ALTERNATE,NAND_SDISK_512GB,POP_4GBIT,CAPACITY2																																																																							
SAND_1T	ALTERNATE,NAND_SDISK_1TB,POP_8GBIT,CAPACITY2,CAPACITY0																																																																							
TOSH_256G	ALTERNATE,NAND_TSHBA_256GB,POP_4GBIT,CAPACITY1,CAPACITY0																																																																							
TOSH_512G	ALTERNATE,NAND_TSHBA_512GB,POP_4GBIT,CAPACITY2																																																																							
TOSH_1T	ALTERNATE,NAND_TSHBA_1TB,POP_8GBIT,CAPACITY2,CAPACITY0																																																																							
A	Strategic Silicon																																																																							
	<table><tr><th>PART#</th><th>STRATEGIC VALUE</th><th>COMMENT</th></tr><tr><td>337S00266</td><td>08</td><td>CPU</td></tr><tr><td>337S00267</td><td>08</td><td>CPU</td></tr><tr><td>337S00268</td><td>08</td><td>CPU</td></tr><tr><td>333S00050</td><td>07</td><td>MEMORY</td></tr><tr><td>333S00068</td><td>07</td><td>MEMORY</td></tr><tr><td>333S00069</td><td>07</td><td>MEMORY</td></tr><tr><td>333S00070</td><td>07</td><td>MEMORY</td></tr><tr><td>335S00124</td><td>02</td><td>NAND</td></tr><tr><td>335S00125</td><td>02</td><td>NAND</td></tr><tr><td>335S00126</td><td>02</td><td>NAND</td></tr><tr><td>335S00135</td><td>02</td><td>NAND</td></tr><tr><td>335S00136</td><td>02</td><td>NAND</td></tr><tr><td>335S00137</td><td>02</td><td>NAND</td></tr><tr><td>333S00025</td><td>02</td><td>S3X DRAM</td></tr><tr><td>333S00026</td><td>02</td><td>S3X DRAM</td></tr><tr><td>333S00055</td><td>02</td><td>S3X DRAM</td></tr><tr><td>333S00056</td><td>02</td><td>S3X DRAM</td></tr><tr><td>333S00107</td><td>02</td><td>S3X DRAM</td></tr><tr><td>333S00108</td><td>02</td><td>S3X DRAM</td></tr><tr><td>998-06736</td><td>02</td><td>S3X CONTROLLER</td></tr></table>								PART#	STRATEGIC VALUE	COMMENT	337S00266	08	CPU	337S00267	08	CPU	337S00268	08	CPU	333S00050	07	MEMORY	333S00068	07	MEMORY	333S00069	07	MEMORY	333S00070	07	MEMORY	335S00124	02	NAND	335S00125	02	NAND	335S00126	02	NAND	335S00135	02	NAND	335S00136	02	NAND	335S00137	02	NAND	333S00025	02	S3X DRAM	333S00026	02	S3X DRAM	333S00055	02	S3X DRAM	333S00056	02	S3X DRAM	333S00107	02	S3X DRAM	333S00108	02	S3X DRAM	998-06736	02	S3X CONTROLLER	
	PART#	STRATEGIC VALUE	COMMENT																																																																					
	337S00266	08	CPU																																																																					
	337S00267	08	CPU																																																																					
	337S00268	08	CPU																																																																					
	333S00050	07	MEMORY																																																																					
	333S00068	07	MEMORY																																																																					
	333S00069	07	MEMORY																																																																					
	333S00070	07	MEMORY																																																																					
335S00124	02	NAND																																																																						
335S00125	02	NAND																																																																						
335S00126	02	NAND																																																																						
335S00135	02	NAND																																																																						
335S00136	02	NAND																																																																						
335S00137	02	NAND																																																																						
333S00025	02	S3X DRAM																																																																						
333S00026	02	S3X DRAM																																																																						
333S00055	02	S3X DRAM																																																																						
333S00056	02	S3X DRAM																																																																						
333S00107	02	S3X DRAM																																																																						
333S00108	02	S3X DRAM																																																																						
998-06736	02	S3X CONTROLLER																																																																						
<table><tr><th>PART#</th><th>STRATEGIC VALUE</th><th>COMMENT</th></tr><tr><td>338S00227</td><td>02</td><td>PICCOLO</td></tr><tr><td>343S00095</td><td>10</td><td>T208</td></tr><tr><td>343S00096</td><td>10</td><td>T208</td></tr><tr><td>343S00097</td><td>10</td><td>T208</td></tr><tr><td>338S00193</td><td>09</td><td>BERKELIUM</td></tr><tr><td>353S3978</td><td>02</td><td>MOJAVE</td></tr><tr><td>338S00147</td><td>02</td><td>SECURE ELEMENT</td></tr><tr><td>338S00254</td><td>08</td><td>ALPINE RIDGE</td></tr><tr><td>353S00961</td><td>09</td><td>ACE</td></tr><tr><td>338S00142</td><td>09</td><td>CLIFDEN</td></tr><tr><td>353S00685</td><td>07</td><td>AUDIO AMP</td></tr><tr><td>353S4316</td><td>08</td><td>BAYSIDE</td></tr><tr><td>338S00221</td><td>08</td><td>BANJO</td></tr><tr><td>353S01016</td><td>09</td><td>TUBA</td></tr><tr><td>339S00056</td><td>05</td><td>ICEBOCK</td></tr><tr><td>343S00147</td><td>08</td><td>PAK</td></tr><tr><td>359S00006</td><td>08</td><td>GREEN CLOCK</td></tr><tr><td>353S00795</td><td>09</td><td>DEBUG MUX</td></tr></table>								PART#	STRATEGIC VALUE	COMMENT	338S00227	02	PICCOLO	343S00095	10	T208	343S00096	10	T208	343S00097	10	T208	338S00193	09	BERKELIUM	353S3978	02	MOJAVE	338S00147	02	SECURE ELEMENT	338S00254	08	ALPINE RIDGE	353S00961	09	ACE	338S00142	09	CLIFDEN	353S00685	07	AUDIO AMP	353S4316	08	BAYSIDE	338S00221	08	BANJO	353S01016	09	TUBA	339S00056	05	ICEBOCK	343S00147	08	PAK	359S00006	08	GREEN CLOCK	353S00795	09	DEBUG MUX								
PART#	STRATEGIC VALUE	COMMENT																																																																						
338S00227	02	PICCOLO																																																																						
343S00095	10	T208																																																																						
343S00096	10	T208																																																																						
343S00097	10	T208																																																																						
338S00193	09	BERKELIUM																																																																						
353S3978	02	MOJAVE																																																																						
338S00147	02	SECURE ELEMENT																																																																						
338S00254	08	ALPINE RIDGE																																																																						
353S00961	09	ACE																																																																						
338S00142	09	CLIFDEN																																																																						
353S00685	07	AUDIO AMP																																																																						
353S4316	08	BAYSIDE																																																																						
338S00221	08	BANJO																																																																						
353S01016	09	TUBA																																																																						
339S00056	05	ICEBOCK																																																																						
343S00147	08	PAK																																																																						
359S00006	08	GREEN CLOCK																																																																						
353S00795	09	DEBUG MUX																																																																						
CPU DRAM CFG Chart																																																																								
<table><tr><th>DIE REV</th><th>CFG 4</th></tr><tr><td>A</td><td>0</td></tr><tr><td>B</td><td>1</td></tr></table>								DIE REV	CFG 4	A	0	B	1																																																											
DIE REV	CFG 4																																																																							
A	0																																																																							
B	1																																																																							
<table><tr><th>SPEED</th><th>CFG 3</th></tr><tr><td>2133</td><td>0</td></tr><tr><td>1866</td><td>1</td></tr></table>								SPEED	CFG 3	2133	0	1866	1																																																											
SPEED	CFG 3																																																																							
2133	0																																																																							
1866	1																																																																							
<table><tr><th>CAPACITY</th><th>CFG 2</th></tr><tr><td>8GB</td><td>0</td></tr><tr><td>16GB</td><td>1</td></tr></table>								CAPACITY	CFG 2	8GB	0	16GB	1																																																											
CAPACITY	CFG 2																																																																							
8GB	0																																																																							
16GB	1																																																																							
<table><tr><th>VENDOR</th><th>CFG 1</th><th>CFG 0</th></tr><tr><td>HYNIX</td><td>0</td><td>0</td></tr><tr><td>MICRON</td><td>0</td><td>1</td></tr><tr><td>SAMSUNG</td><td>1</td><td>0</td></tr><tr><td>N/A</td><td>1</td><td>1</td></tr></table>								VENDOR	CFG 1	CFG 0	HYNIX	0	0	MICRON	0	1	SAMSUNG	1	0	N/A	1	1																																																		
VENDOR	CFG 1	CFG 0																																																																						
HYNIX	0	0																																																																						
MICRON	0	1																																																																						
SAMSUNG	1	0																																																																						
N/A	1	1																																																																						
BOM Configuration																																																																								
<table><tr><td colspan="2">SYNC_MASTER=SHART_344</td><td colspan="2">SYNC_DATE=11/27/2012</td></tr><tr><td colspan="4">PAGE TITLE</td><td colspan="2">DRAWING NUMBER</td><td colspan="2">STEP</td></tr><tr><td colspan="4" rowspan="2"> Apple Inc. </td><td colspan="2">051-00515</td><td colspan="2">D</td></tr><tr><td colspan="2">REVISION</td><td colspan="2">7.0.0</td></tr><tr><td colspan="4" rowspan="4">NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I1 NOT TO REPRODUCE OR COPY IT I11 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I1V ALL RIGHTS RESERVED</td><td colspan="2">BRANCH</td><td colspan="2">dvt1</td></tr><tr><td colspan="2">PAGE</td><td colspan="2">2 OF 145</td></tr><tr><td colspan="2">SHEET</td><td colspan="2">2 OF 119</td></tr><tr><td colspan="2"></td><td colspan="2"></td></tr></table>								SYNC_MASTER=SHART_344		SYNC_DATE=11/27/2012		PAGE TITLE				DRAWING NUMBER		STEP		Apple Inc.				051-00515		D		REVISION		7.0.0		NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I1 NOT TO REPRODUCE OR COPY IT I11 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I1V ALL RIGHTS RESERVED				BRANCH		dvt1		PAGE		2 OF 145		SHEET		2 OF 119																										
SYNC_MASTER=SHART_344		SYNC_DATE=11/27/2012																																																																						
PAGE TITLE				DRAWING NUMBER		STEP																																																																		
Apple Inc.				051-00515		D																																																																		
				REVISION		7.0.0																																																																		
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I1 NOT TO REPRODUCE OR COPY IT I11 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I1V ALL RIGHTS RESERVED				BRANCH		dvt1																																																																		
				PAGE		2 OF 145																																																																		
				SHEET		2 OF 119																																																																		

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00367	152S00266		ALL	
152S00368	152S00269		ALL	
152S00370	152S00270		ALL	
353S4068	353S4070		ALL	
353S00772	353S4070		ALL	
138S00086	138S0884		ALL	
107S0248	107S0250		ALL	
152S00434	152S1829		ALL	
371S00019	371S0463		ALL	
353S00107	353S3239	ANY	ALL	
353S00231	353S3987		ALL	
333S00025	333S00055	POP_4GBIT	ALL	
333S00026	333S00056	POP_8GBIT	ALL	
333S00107	333S00055	POP_4GBIT	ALL	
333S00108	333S00056	POP_8GBIT	ALL	

138S0701	138S0689		ALL	
138S00012	138S0771		ALL	
138S0772	138S00013		ALL	
152S00401	152S0529		ALL	
152S00344	152S1683		ALL	
152S00331	152S00283		ALL	
152S2034	152S00190		ALL	
197S0613	197S0612		ALL	
311S00121	311S0398		ALL	
371S0713	371S0558		ALL	
371S00074	371S0602		ALL	

BOM NUMBER	BOM NAME	BOM OPTIONS
639-01988	MLB,NO CPU,X362	BASE_BOM,DEVEL_BOM,RAM_16G_SAMSUNG_2133,SAND_512G
639-01989	MLB,CPU SOCKET,X362	BASE_BOM,DEVEL_BOM,CPU_SOCKET,RAM_16G_SAMSUNG_2133,SAND_512G
639-02521	MLB,2.9G,SAM-8G,TOSH-256G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:2.9G,RAM_8G_SAMSUNG_2133,TOSH_256G
639-02522	MLB,2.9G,SAM-16G,TOSH-256G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:2.9G,RAM_16G_SAMSUNG_2133,TOSH_256G
639-02523	MLB,2.9G,MIC-8G,TOSH-256G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:2.9G,RAM_8G_MICRON_2133,TOSH_256G
639-02524	MLB,2.9G,MIC-16G,TOSH-256G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:2.9G,RAM_16G_MICRON_2133,TOSH_256G
639-02525	MLB,2.9G,SAM-8G,TOSH-512G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:2.9G,RAM_8G_SAMSUNG_2133,TOSH_512G
639-02526	MLB,2.9G,SAM-16G,TOSH-512G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:2.9G,RAM_16G_SAMSUNG_2133,TOSH_512G
639-02527	MLB,2.9G,MIC-8G,TOSH-512G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:2.9G,RAM_8G_MICRON_2133,TOSH_512G
639-02528	MLB,2.9G,MIC-16G,TOSH-512G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:2.9G,RAM_16G_MICRON_2133,TOSH_512G
639-02529	MLB,2.9G,SAM-8G,TOSH-1T,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:2.9G,RAM_8G_SAMSUNG_2133,TOSH_1T
639-02530	MLB,2.9G,SAM-16G,TOSH-1T,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:2.9G,RAM_16G_SAMSUNG_2133,TOSH_1T
639-02531	MLB,2.9G,MIC-8G,TOSH-1T,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:2.9G,RAM_8G_MICRON_2133,TOSH_1T
639-02532	MLB,2.9G,MIC-16G,TOSH-1T,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:2.9G,RAM_16G_MICRON_2133,TOSH_1T
639-02533	MLB,3.1G,SAM-8G,TOSH-256G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.1G,RAM_8G_SAMSUNG_2133,TOSH_256G
639-02534	MLB,3.1G,SAM-16G,TOSH-256G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.1G,RAM_16G_SAMSUNG_2133,TOSH_256G
639-02535	MLB,3.1G,MIC-8G,TOSH-256G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.1G,RAM_8G_MICRON_2133,TOSH_256G
639-02536	MLB,3.1G,MIC-16G,TOSH-256G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.1G,RAM_16G_MICRON_2133,TOSH_256G
639-02537	MLB,3.1G,SAM-8G,TOSH-512G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.1G,RAM_8G_SAMSUNG_2133,TOSH_512G
639-02538	MLB,3.1G,SAM-16G,TOSH-512G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.1G,RAM_16G_SAMSUNG_2133,TOSH_512G
639-02539	MLB,3.1G,MIC-8G,TOSH-512G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.1G,RAM_8G_MICRON_2133,TOSH_512G
639-02540	MLB,3.1G,MIC-16G,TOSH-512G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.1G,RAM_16G_MICRON_2133,TOSH_512G
639-02541	MLB,3.1G,SAM-8G,TOSH-1T,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.1G,RAM_8G_SAMSUNG_2133,TOSH_1T
639-02542	MLB,3.1G,SAM-16G,TOSH-1T,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.1G,RAM_16G_SAMSUNG_2133,TOSH_1T
639-02543	MLB,3.1G,MIC-8G,TOSH-1T,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.1G,RAM_8G_MICRON_2133,TOSH_1T
639-02544	MLB,3.1G,MIC-16G,TOSH-1T,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.1G,RAM_16G_MICRON_2133,TOSH_1T
639-02545	MLB,3.3G,SAM-8G,TOSH-256G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.3G,RAM_8G_SAMSUNG_2133,TOSH_256G
639-02546	MLB,3.3G,SAM-16G,TOSH-256G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.3G,RAM_16G_SAMSUNG_2133,TOSH_256G
639-02547	MLB,3.3G,MIC-8G,TOSH-256G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.3G,RAM_8G_MICRON_2133,TOSH_256G
639-02548	MLB,3.3G,MIC-16G,TOSH-256G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.3G,RAM_16G_MICRON_2133,TOSH_256G
639-02549	MLB,3.3G,SAM-8G,TOSH-512G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.3G,RAM_8G_SAMSUNG_2133,TOSH_512G
639-02550	MLB,3.3G,SAM-16G,TOSH-512G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.3G,RAM_16G_SAMSUNG_2133,TOSH_512G
639-02551	MLB,3.3G,MIC-8G,TOSH-512G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.3G,RAM_8G_MICRON_2133,TOSH_512G
639-02552	MLB,3.3G,MIC-16G,TOSH-512G,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.3G,RAM_16G_MICRON_2133,TOSH_512G
639-02553	MLB,3.3G,SAM-8G,TOSH-1T,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.3G,RAM_8G_SAMSUNG_2133,TOSH_1T
639-02554	MLB,3.3G,SAM-16G,TOSH-1T,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.3G,RAM_16G_SAMSUNG_2133,TOSH_1T
639-02555	MLB,3.3G,MIC-8G,TOSH-1T,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.3G,RAM_8G_MICRON_2133,TOSH_1T
639-02556	MLB,3.3G,MIC-16G,TOSH-1T,X362	BASE_BOM,DEVEL_BOM,CPU_SKL23:3.3G,RAM_16G_MICRON_2133,TOSH_1T

FORM REVISION 1713 2/83		FORM REVISION 1713 2/83	
PAGE TITLE			
 Apple Inc.		DRAWING NUMBER 051-000515	SIZE D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I! NOT TO REPRODUCE OR COPY IT I! NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I! ALL RIGHTS RESERVED		REVISION 7.0.0	
		BRANCH dvt1	
		PAGE 3 OF 145	
		SHEET 3 OF 119	

D

C

B

A

D

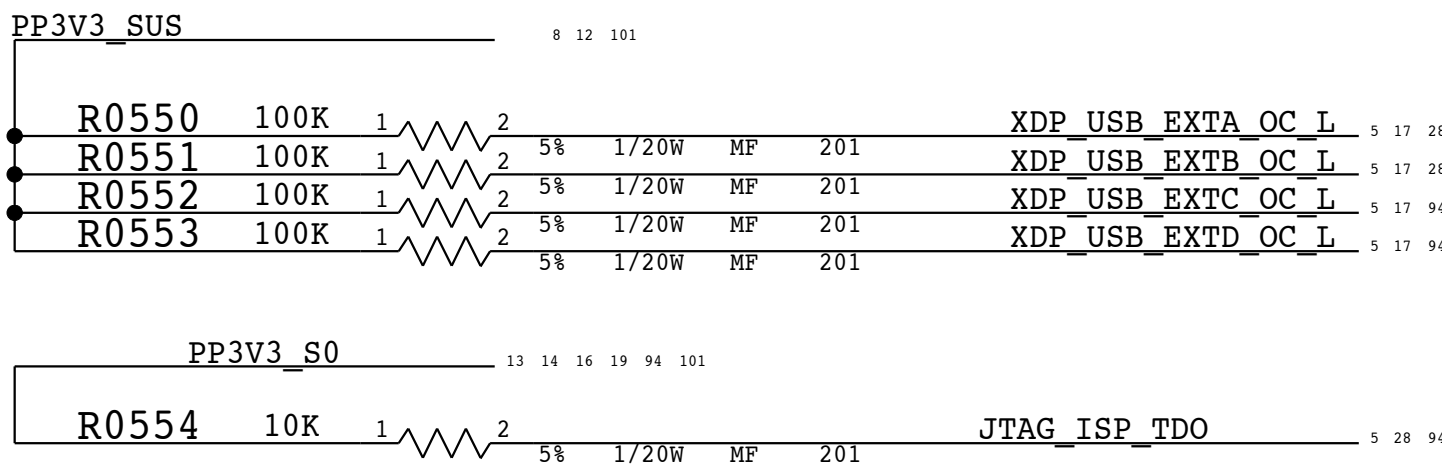
C

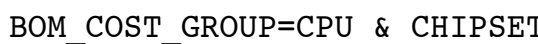
B

A

DESIGN: X502/DEV MLB U		
LAST CHANGE: Tue Apr 28 20:32:21 2015		
PAGE TITLE		
CPU GFX		
 Apple Inc.	DRAWING NUMBER	051-00515
	REVISION	7.0.0
	BRANCH	dvt1
	PAGE	5 OF 145
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I ALL RIGHTS RESERVED		SHEET 5 OF 119

BOM_COST_GROUP=CPU & CHIPSET





DESIGN: X502/DEV MLB U			
LAST CHANGE: Mon Apr 27 22:56:39 2015			
PAGE TITLE			
CPU MISC/JTAG/CFG/RSVD			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00515		D
	REVISION		
	7.0.0		
NOTICE OF PROPRIETARY PROPERTY:	BRANCH		
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	dvt1		
	PAGE	6 OF 145	
	SHEET	6 OF 119	

D

C

B

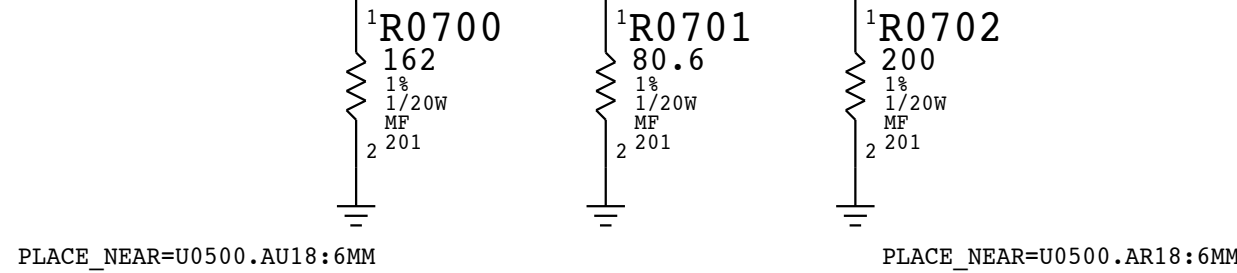
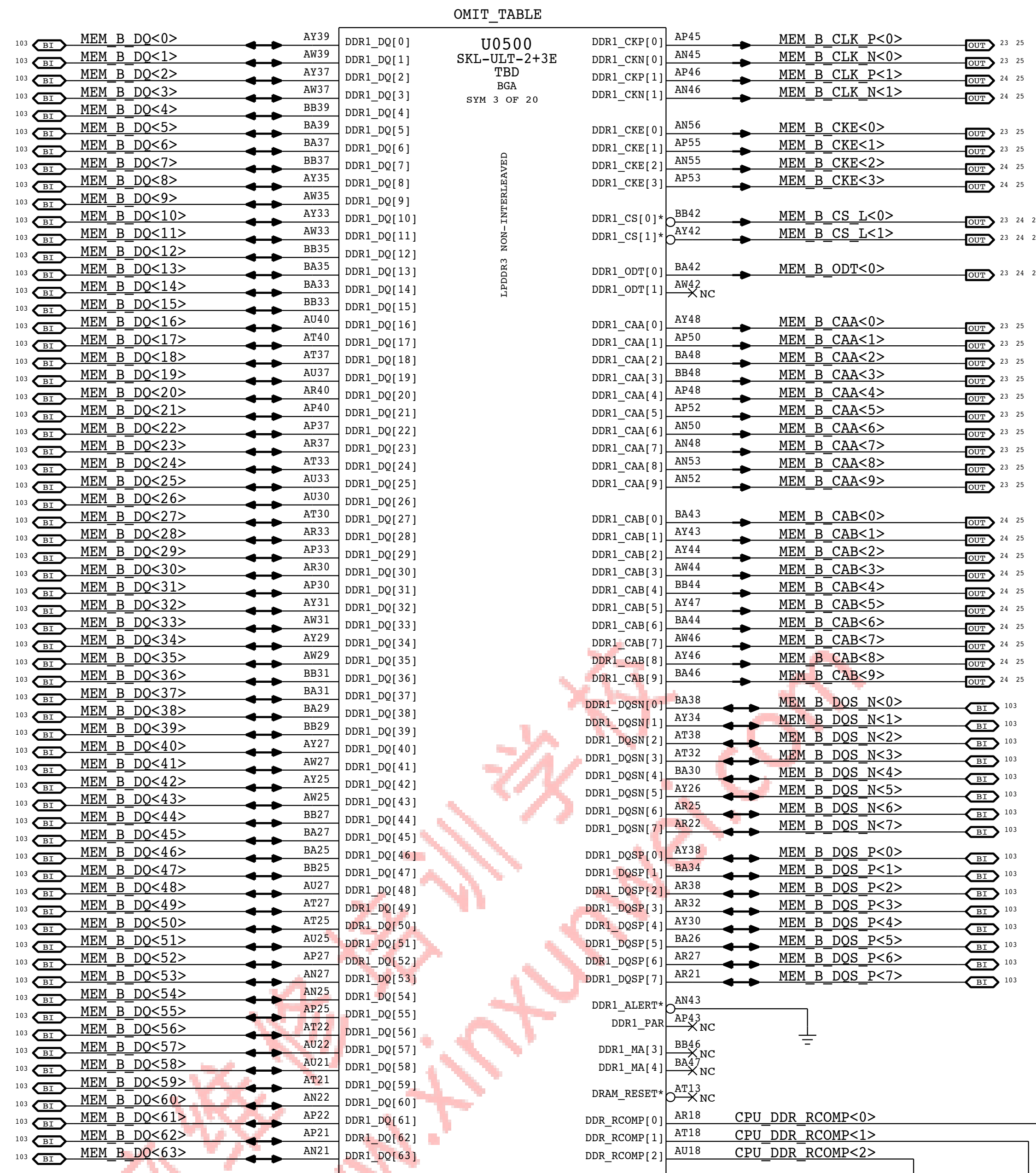
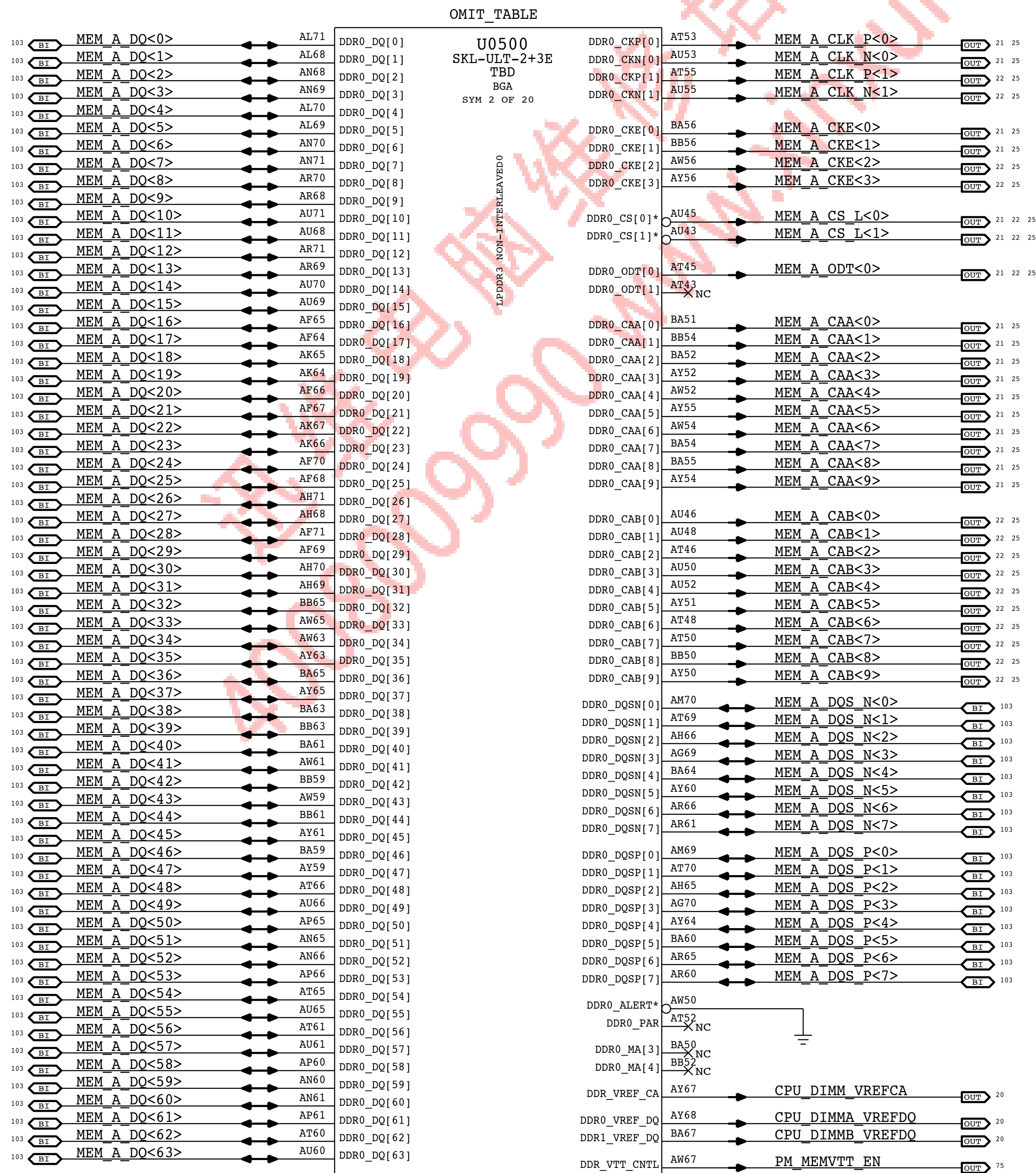
A

D

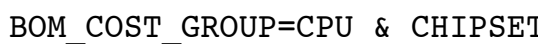
C

B

A



SYNC_MASTER=J52_MLB			SYNC_DATE=05/12/2015		
PAGE TITLE					
CPU LPDDR3 Interface					
 Apple Inc.			DRAWING NUMBER		SIZE
			051-00515		D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			REVISION		
			7.0.0		
			BRANCH		dvt1
			PAGE		7 OF 145
			SHEET		
			7 OF 119		



D

C

B

A

1

2

3

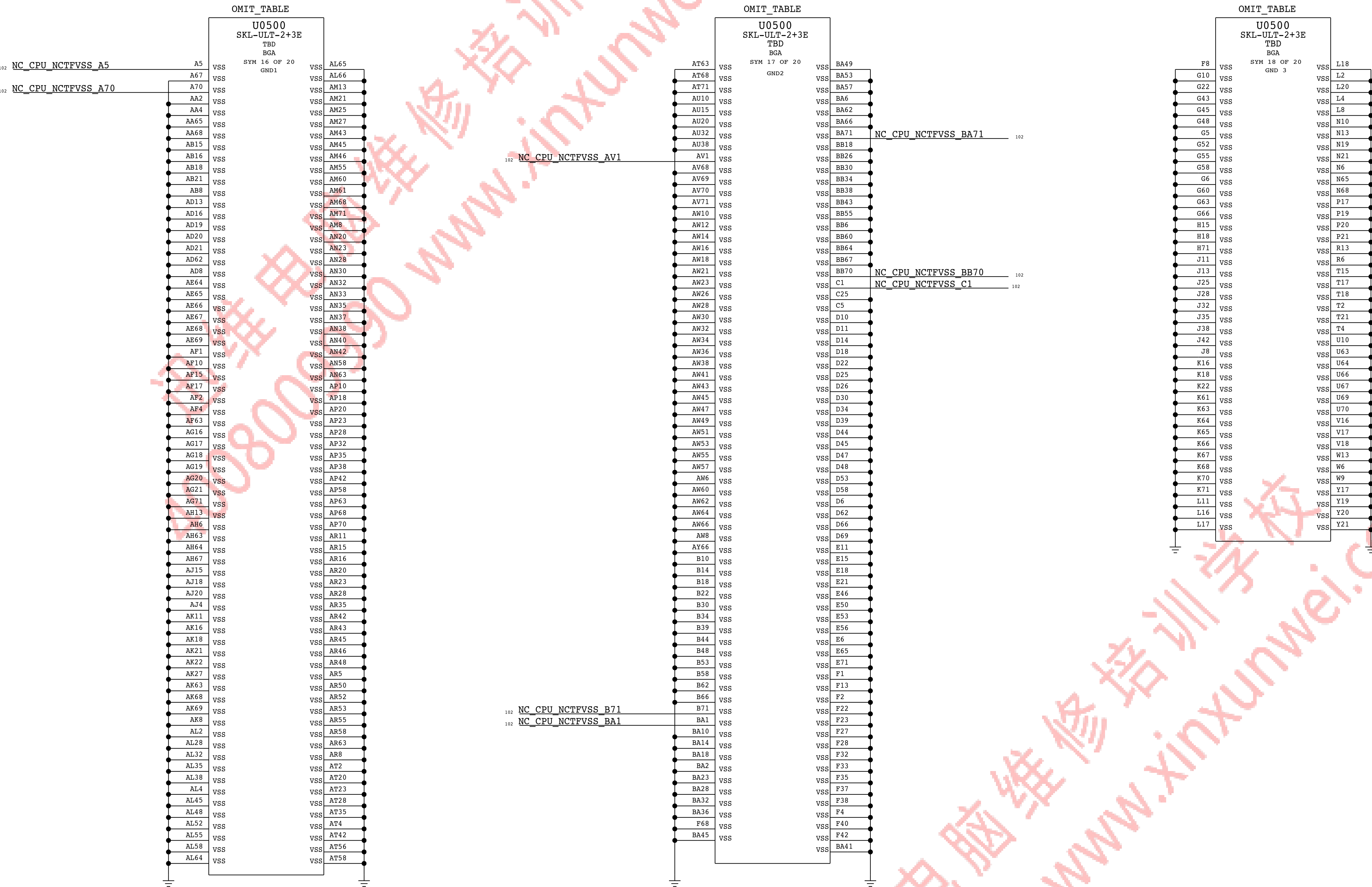
4

5

6

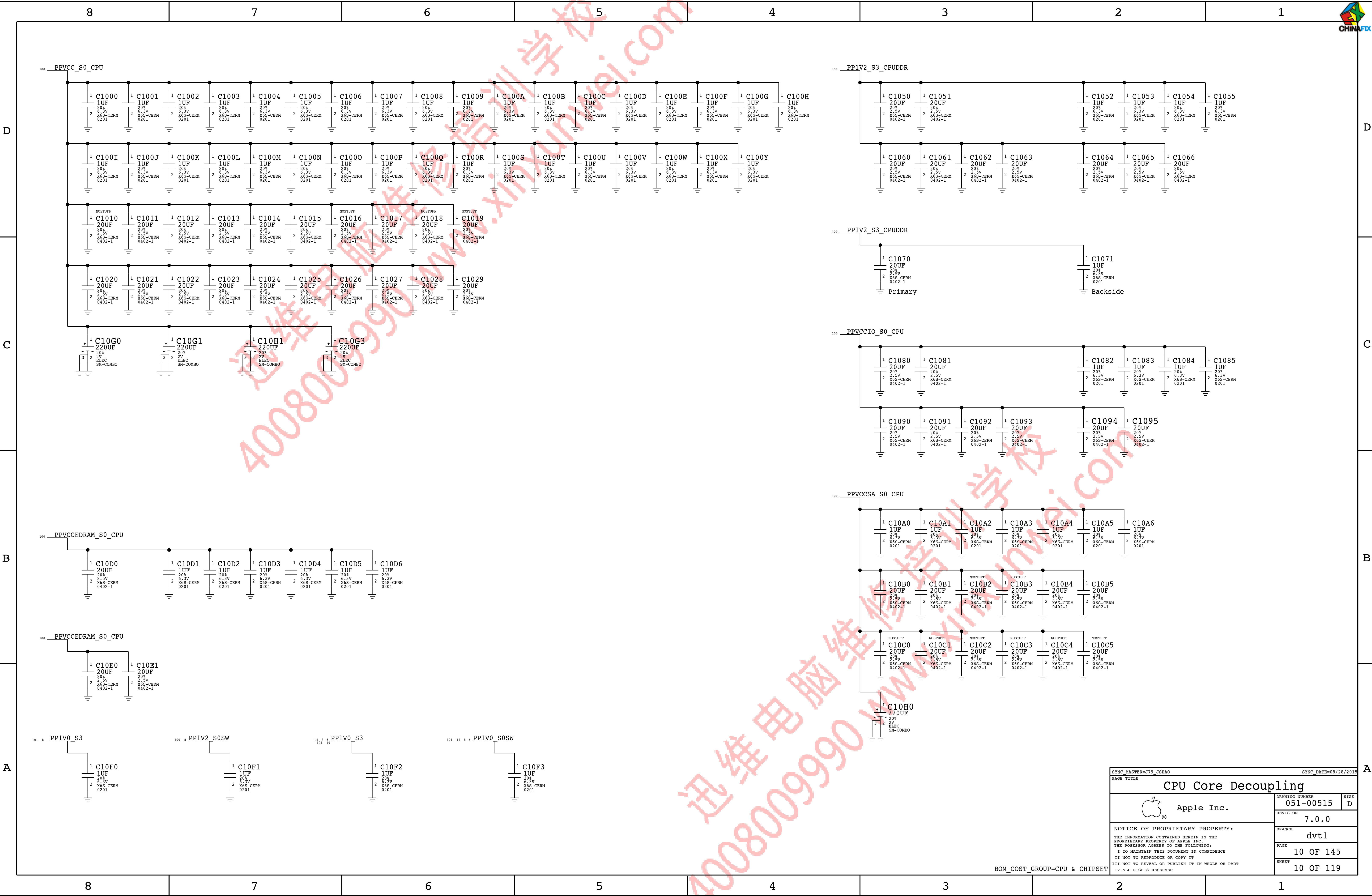
7

8



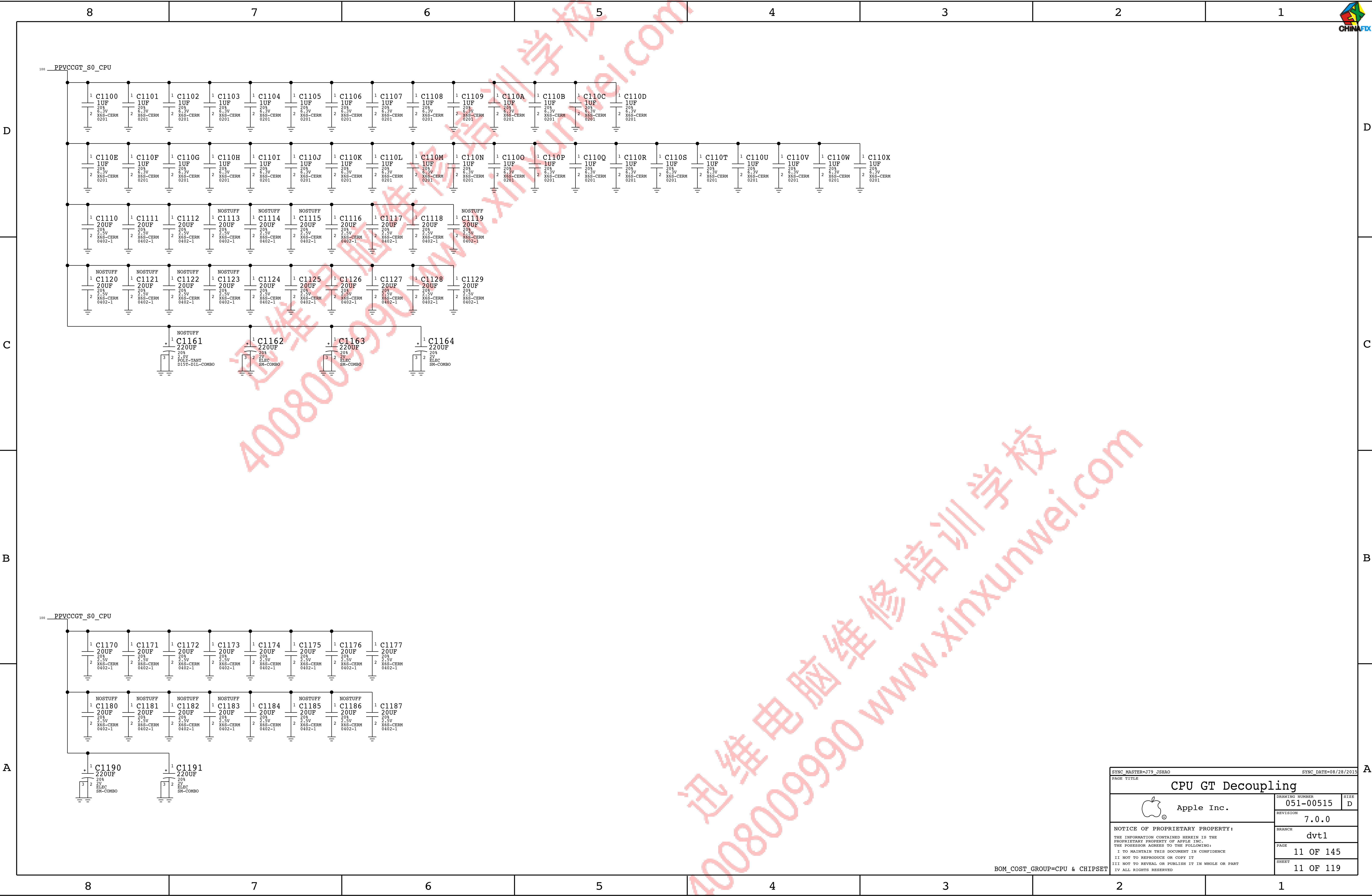
SYNC MASTER=J79 ALFRED		SYNC DATE=05/12/2015	
PAGE TITLE			
CPU & PCH Grounds			
	Apple Inc.	DRAWING NUMBER	051-00515
		REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	dvt1
		PAGE	9 OF 145
		SHEET	9 OF 119

BOM_COST_GROUP=CPU & CHIPSET



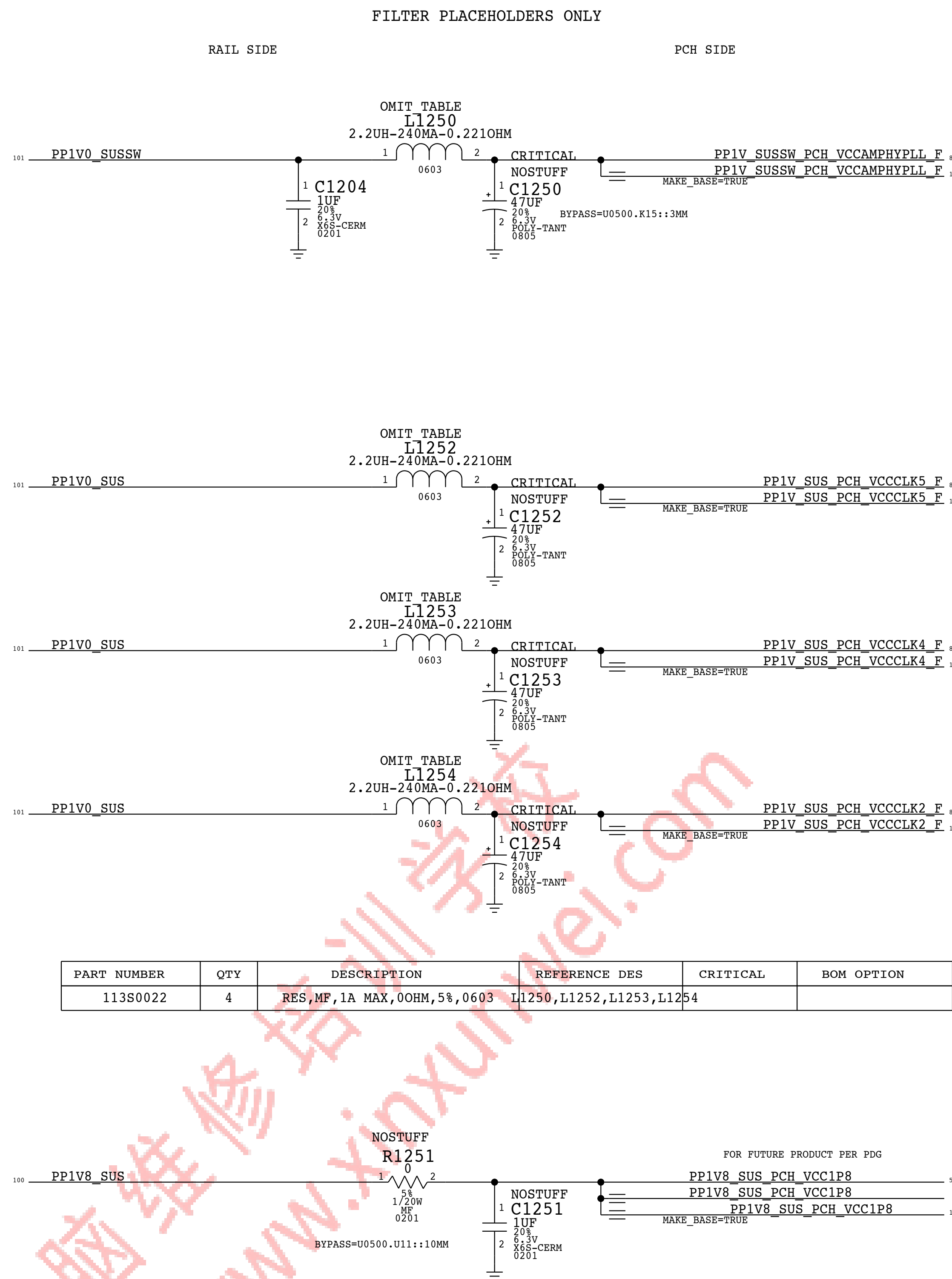
SYNC_MASTER=J79_JSHAO		SYNC_DATE=08/28/2015	
PAGE TITLE			
CPU Core Decoupling			
 Apple Inc.	DRAWING NUMBER	051-00515	SHEET
	REVISION	7.0.0	D
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I1 NOT TO REPRODUCE OR COPY IT I11 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I1V ALL RIGHTS RESERVED		dvt1	
		PAGE	10 OF 145
		SHEET	10 OF 119

BOM_COST_GROUP=CPU & CHIPSET



SYNC_MASTER=J79_JSHAO		SYNC_DATE=08/28/2015	
PAGE TITLE			
CPU GT Decoupling			
 Apple Inc.	DRAWING NUMBER	051-00515	SIZE
	REVISION	7.0.0	D
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	dvt1
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	11 OF 145
I I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	11 OF 119
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

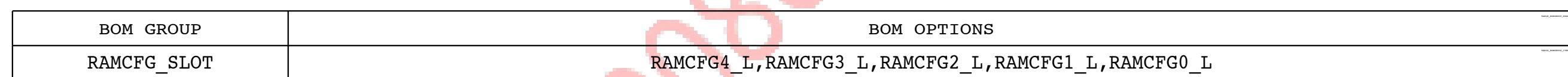
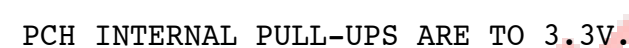
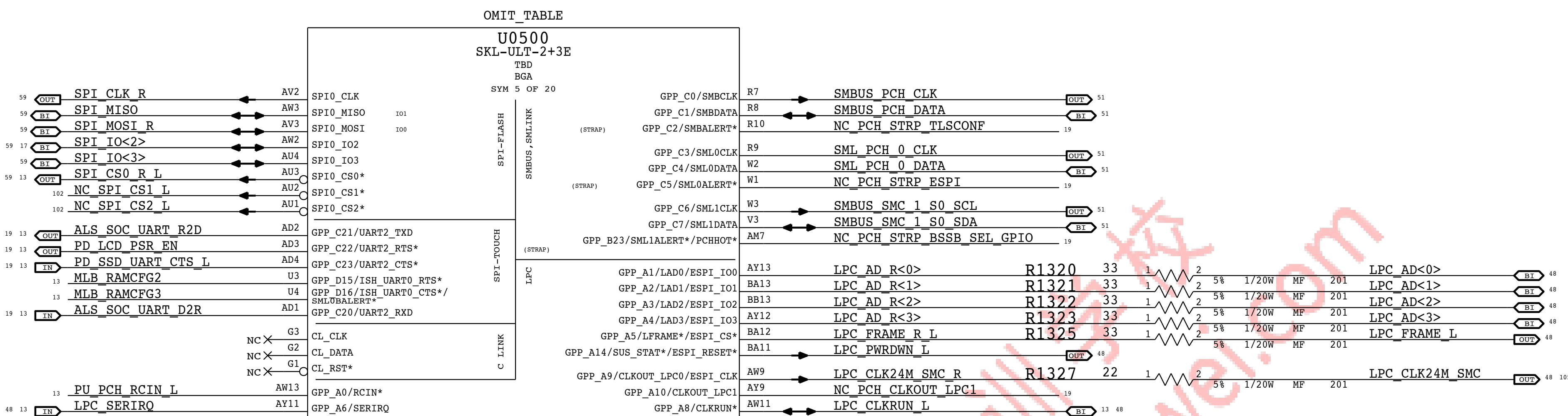
BOM_COST_GROUP=CPU & CHIPSET



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
113S0022	4	RES,MF,1A MAX,00HM,5%,0603	L1250,L1252,L1253,L1254		

PAGE TITLE		
PCH Decoupling		
 Apple Inc.	DRAWING NUMBER	SIZE
	051-00515	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	REVISION	
	7.0.0	
	BRANCH	dvt1
	PAGE	12 OF 145
	SHEET	12 OF 119

BOM_COST_GROUP=CPU & CHIPSET

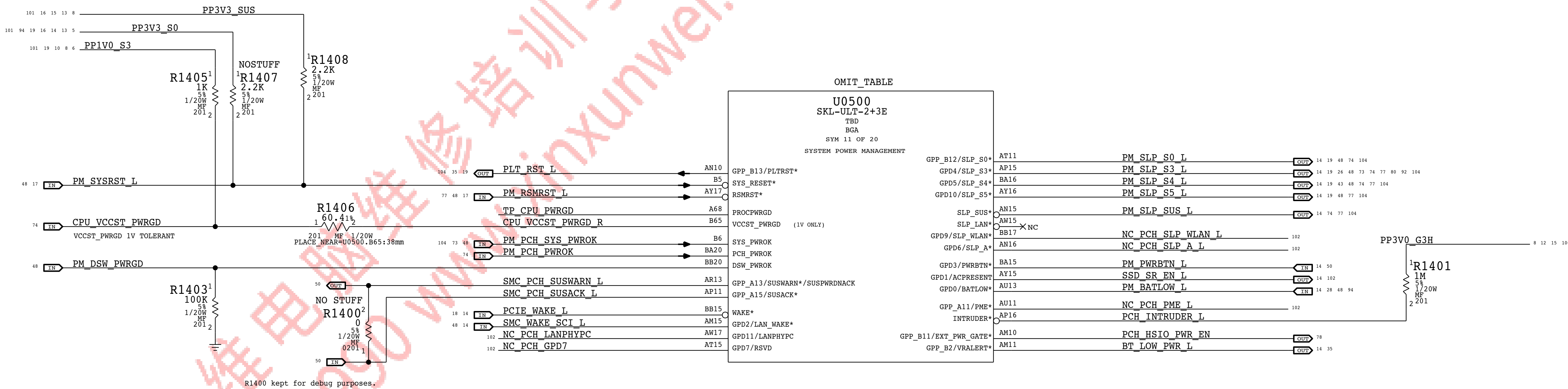


BOM COST GROUP=CPU & CHIPSET

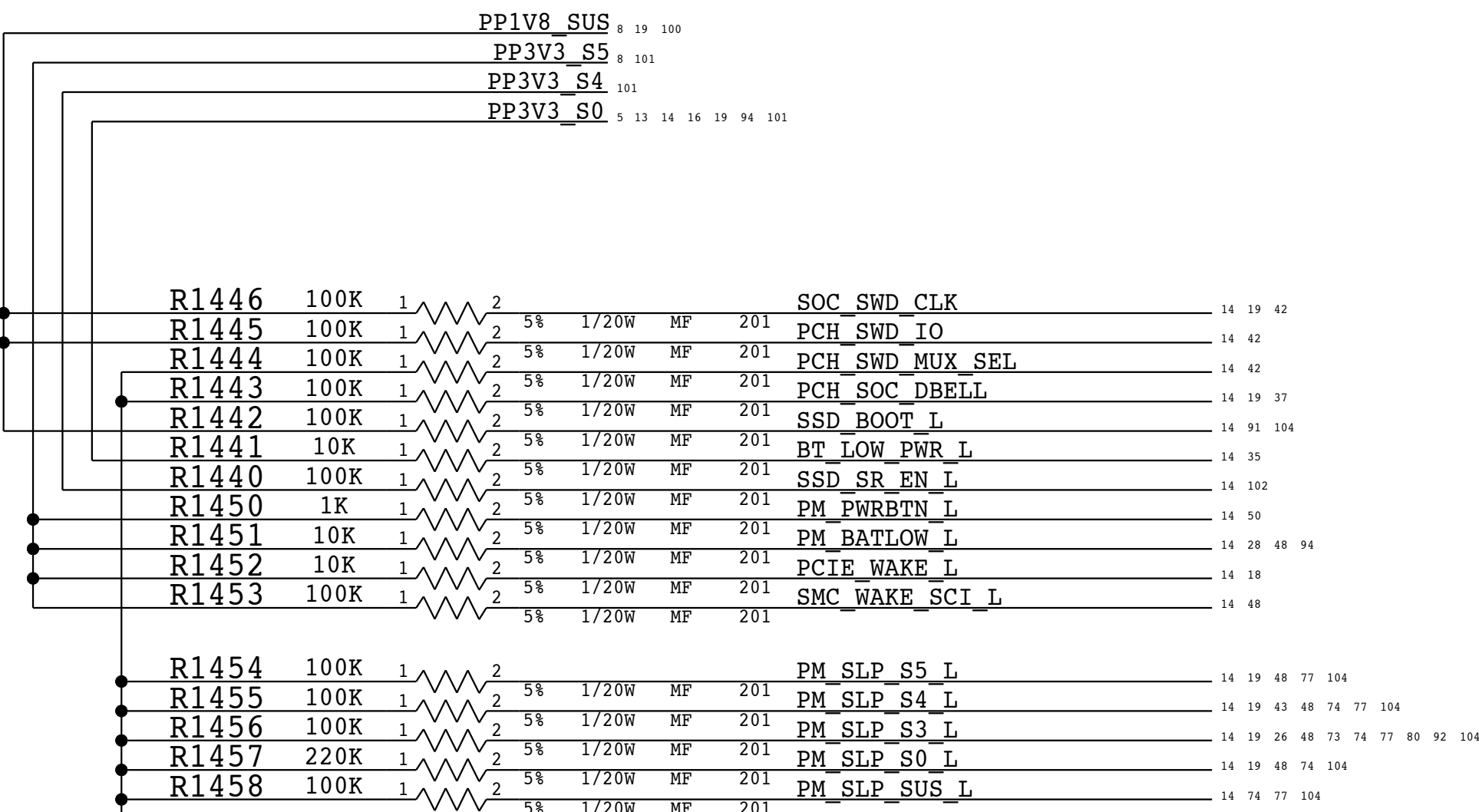
DESIGN: X502/MLB			
LAST CHANGE: Tue Feb 2 13:18:21 2016			
PAGE TITLE			
PCH Audio/LPC/SPI/SMBus			
 Apple Inc.		DRAWING NUMBER 051-00515	
		REVISION 7.0.0	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I II NOT TO REPRODUCE OR COPY IT I III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I IV ALL RIGHTS RESERVED		BRANCH dvt1	
		PAGE 13 OF 145	
		SHEET 13 OF 119	



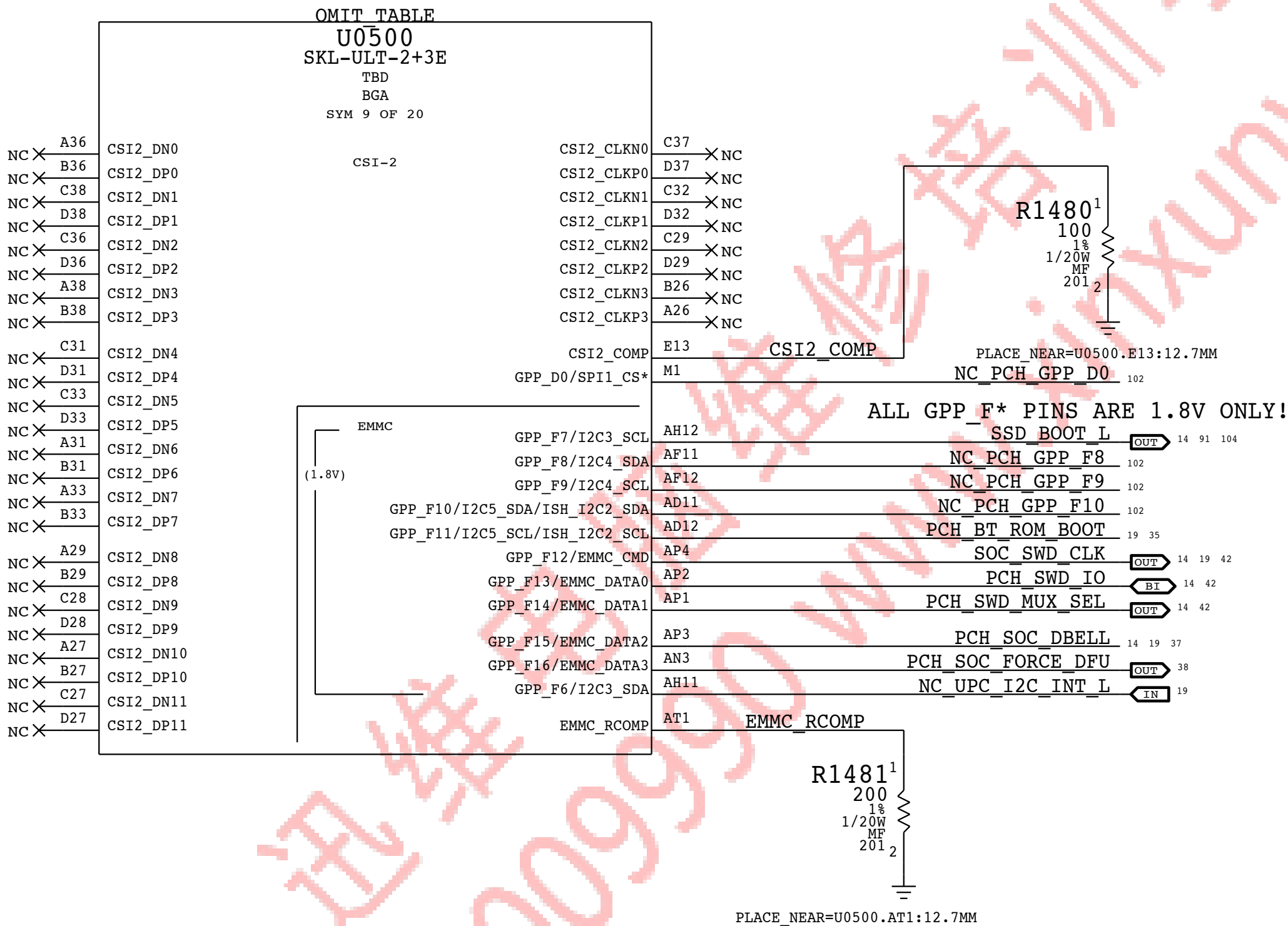
PCH Reset Button



R1400 kept for debug purposes.



NOTE: PM_SLP_S0_L HAS INTERNAL PULL-UP BEFORE RSMRST_L IS RELEASED.
THIS CAUSES A VOLTAGE DIVIDER WITH THE PULL-DOWN HERE.
THE SIGNAL IS DRIVEN HI AFTER RSMRST_L IS RELEASED.



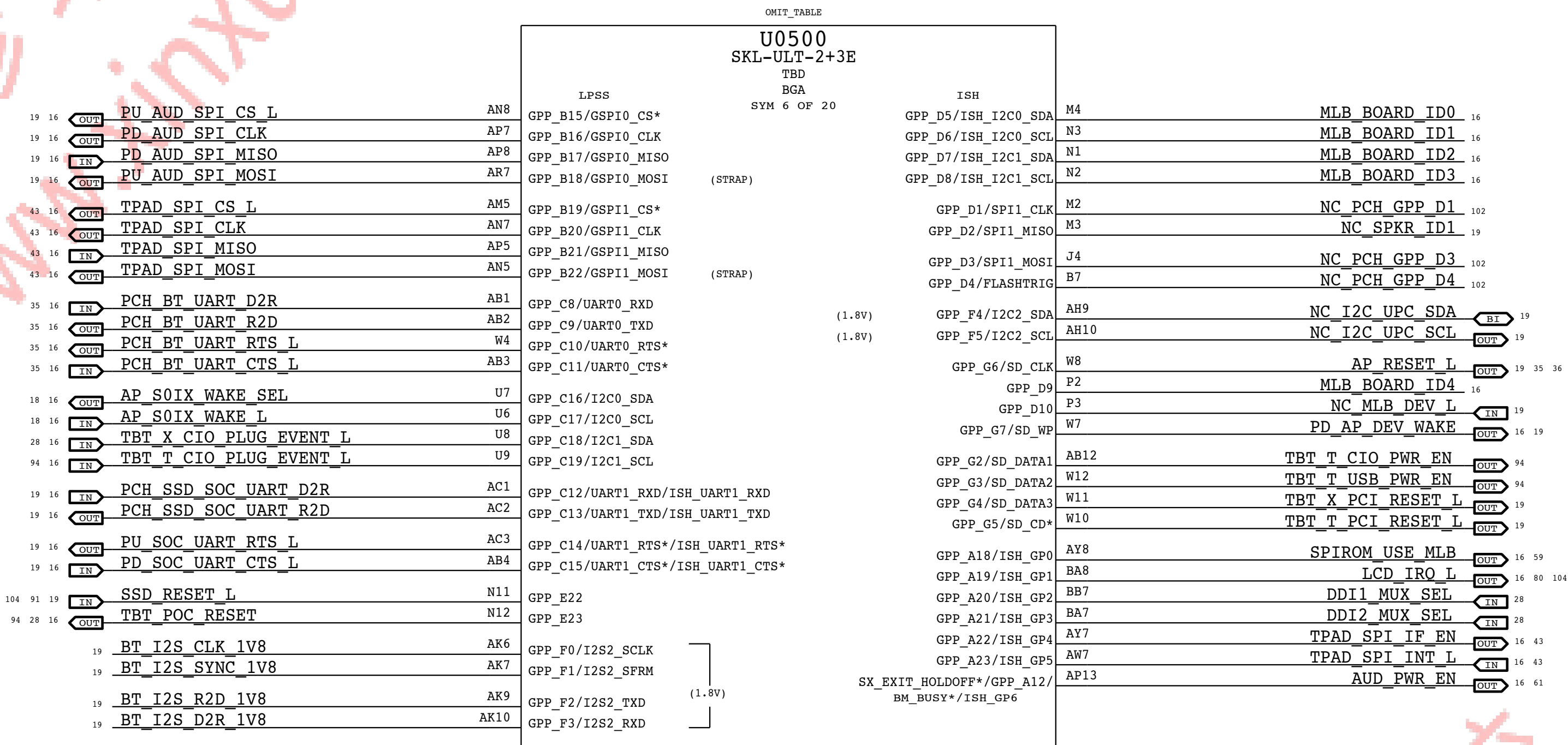
ALL GPP_F* PINS ARE 1.8V ONLY!

DESIGN: X502/MLB		
LAST CHANGE: Tue May 3 17:45:28 2016		
PAGE TITLE		
PCH Power Management		
	DRAWING NUMBER	051-00515
	REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I ALL RIGHTS RESERVED	BRANCH	dvt1
	PAGE	14 OF 145
	SHEET	14 OF 119

BOM_COST_GROUP=CPU & CHIPSET

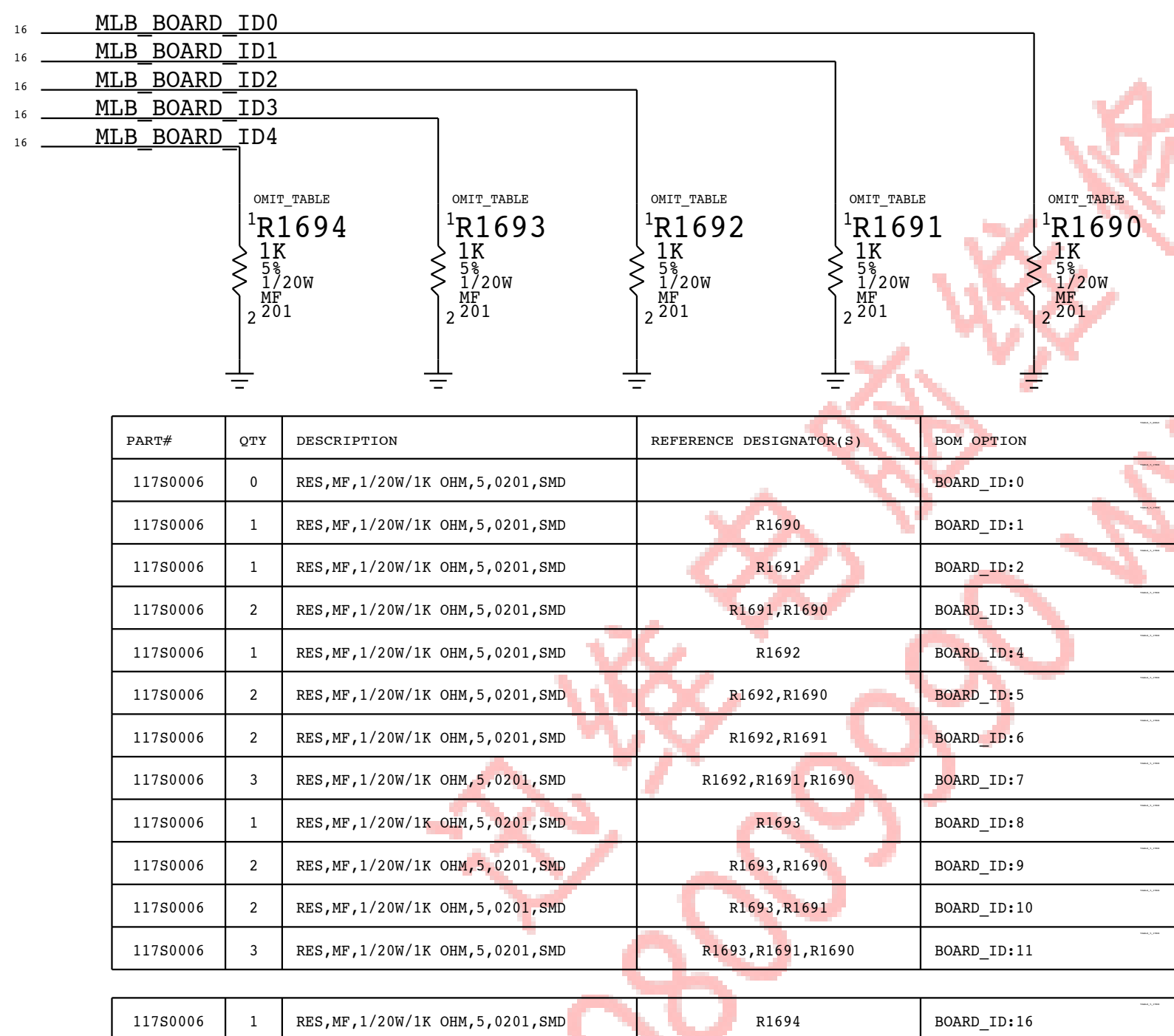


ALL GPP_F* PINS ARE 1.8V ONLY!



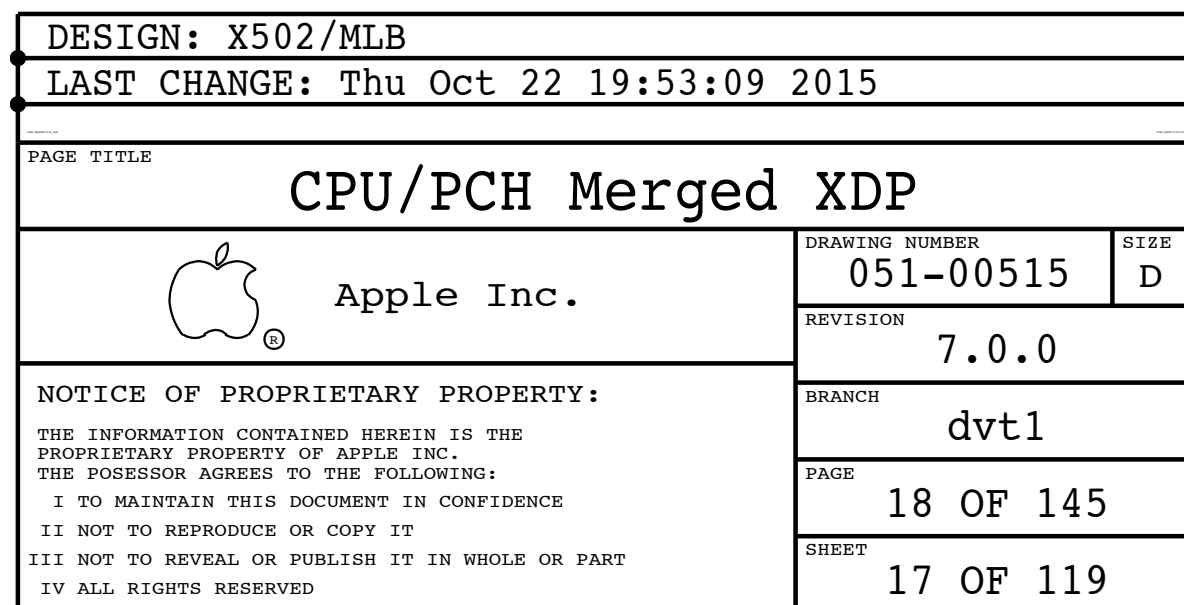
MLB ID STRAPS.

PCH INTERNAL PULL-UPS ARE TO VCCGPPD = 3.3V.



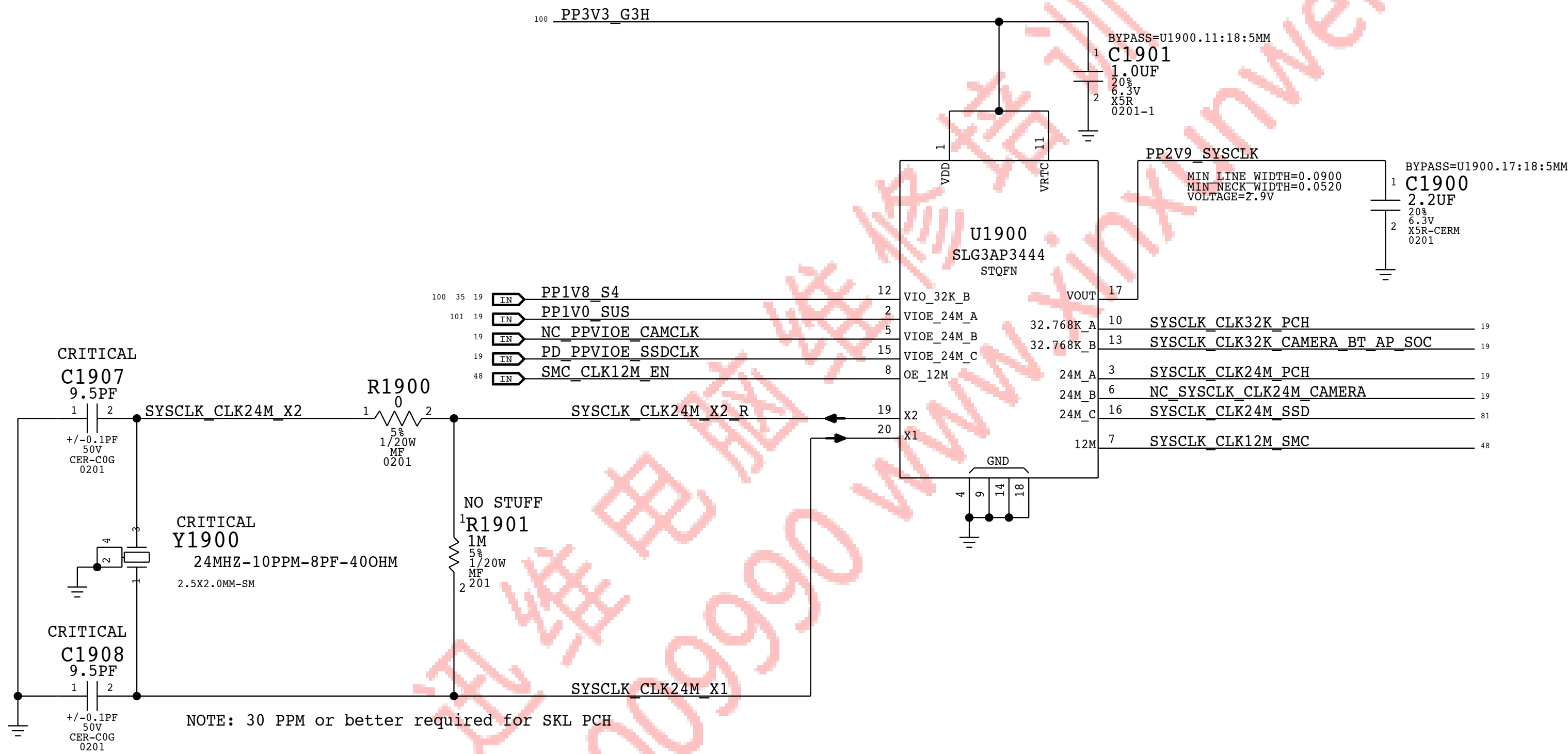
BOM_COST_GROUP=CPU & CHIPSET

DESIGN: X502/MLB		
LAST CHANGE: Wed Oct 28 12:50:22 2015		
PAGE TITLE		
PCH SPI/UART/GPIO		
	Apple Inc.	DRAWING NUMBER 051-00515
		STEP D
		REVISION 7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I ALL RIGHTS RESERVED		BRANCH dvt1
		PAGE 16 OF 145
		SHEET 16 OF 119

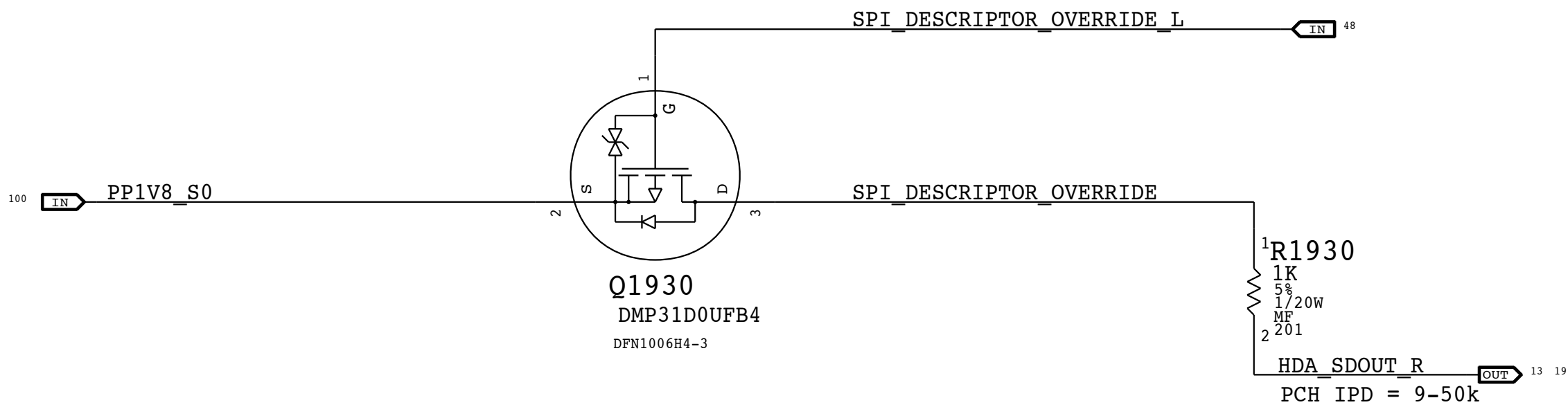


BOM COST GROUP=DEBUG

System 32kHz / 12MHz / 24MHz Clock Generator

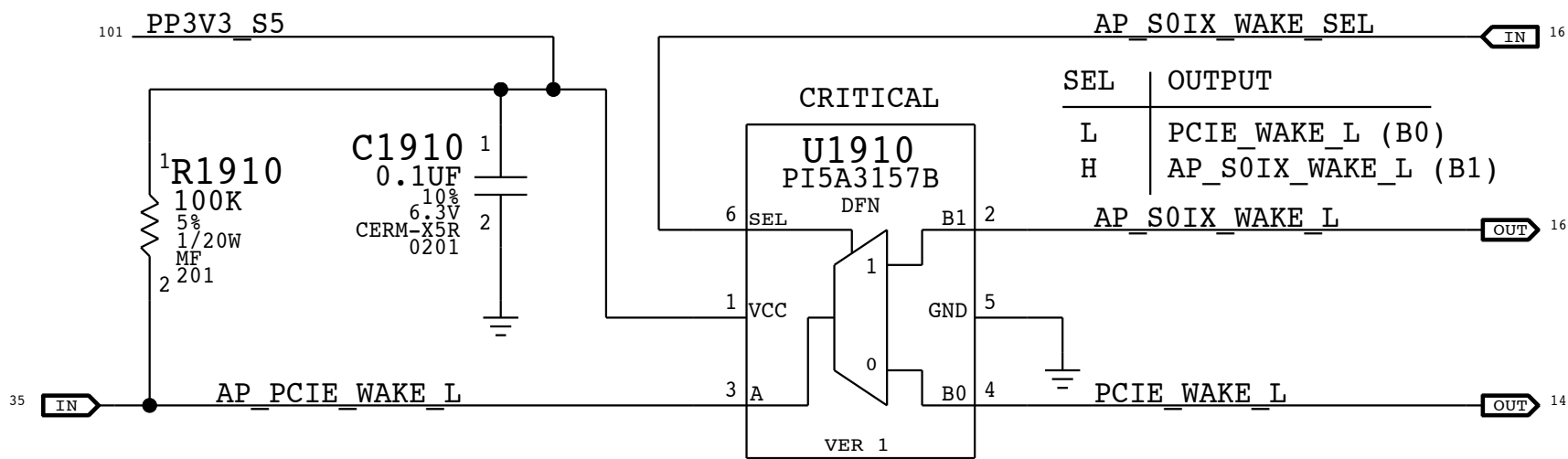


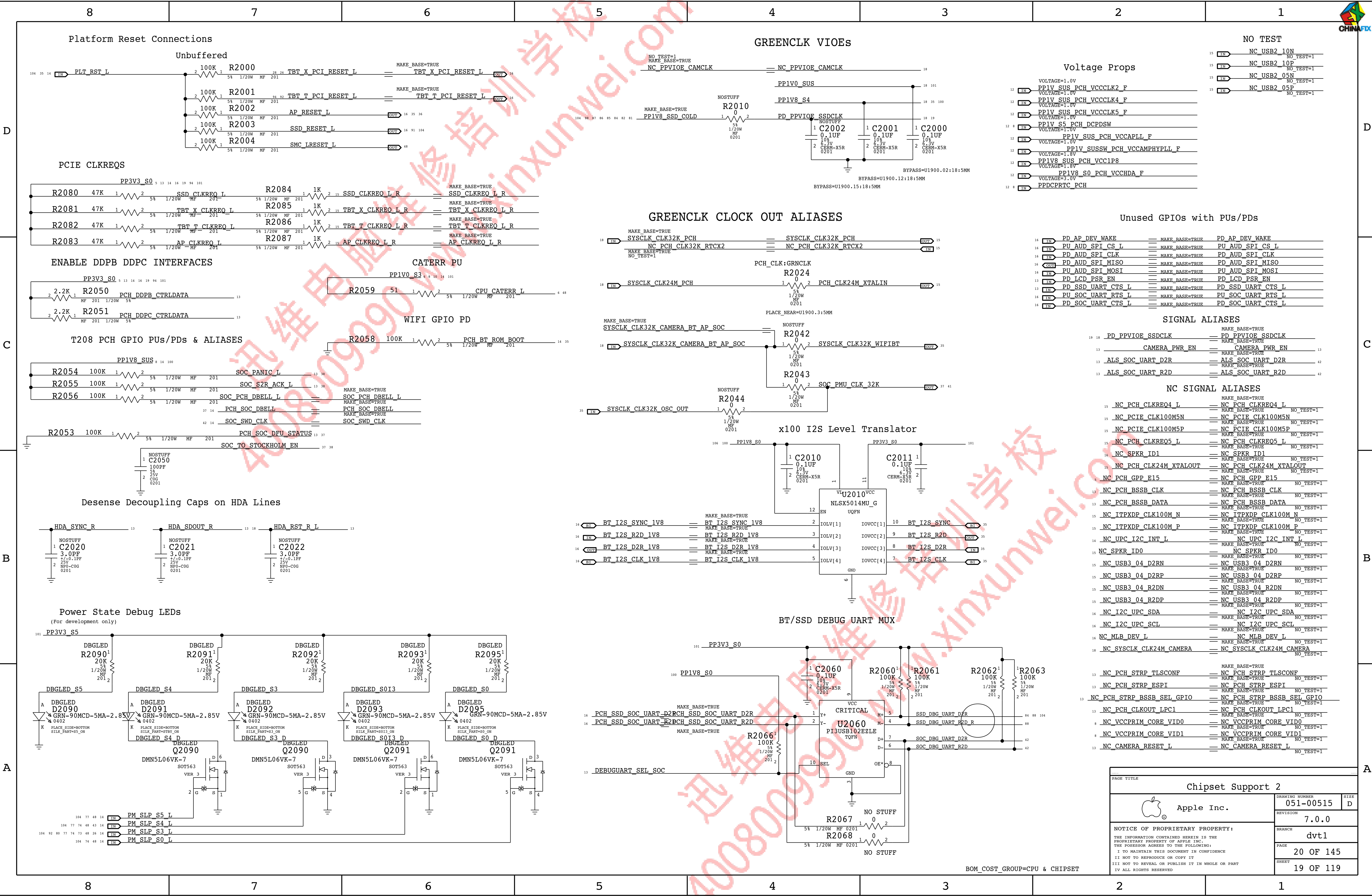
PCH ME Disable Strap



PCH uses HDA SDO as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting. ***** Circuit does not support HDA voltage >3.3V.

PCIe Wake Muxing

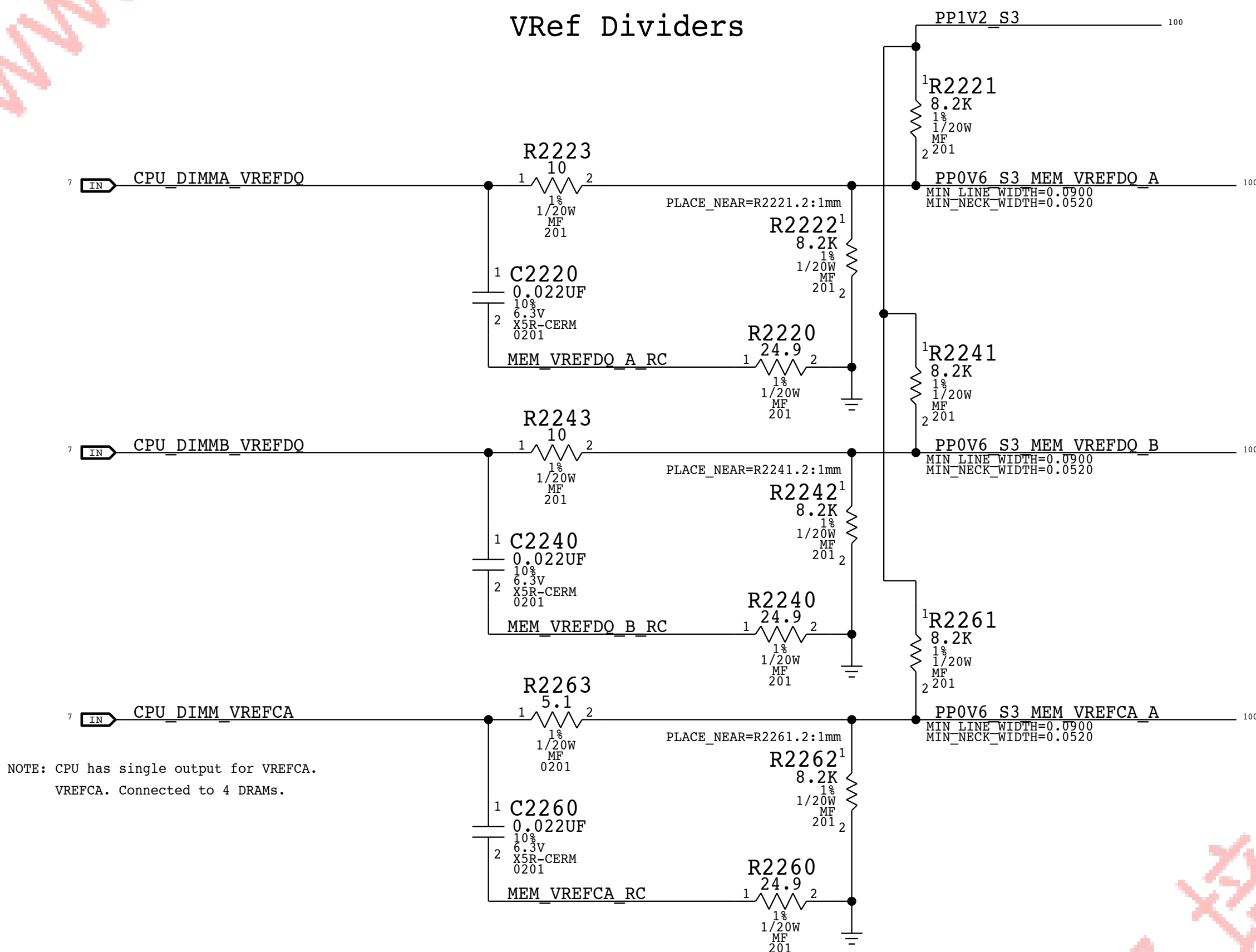




PAGE TITLE			Chipset Support 2	
		DRAWING NUMBER	051-00515	STEP
		REVISION	7.0.0	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: 1 TO MAINTAIN THIS DOCUMENT IN CONFIDENCE 11 NOT TO REPRODUCE OR COPY IT 111 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART 1111 ALL RIGHTS RESERVED		BRANCH	dvt1	
		PAGE	20 OF 145	
		SHEET	19 OF 119	

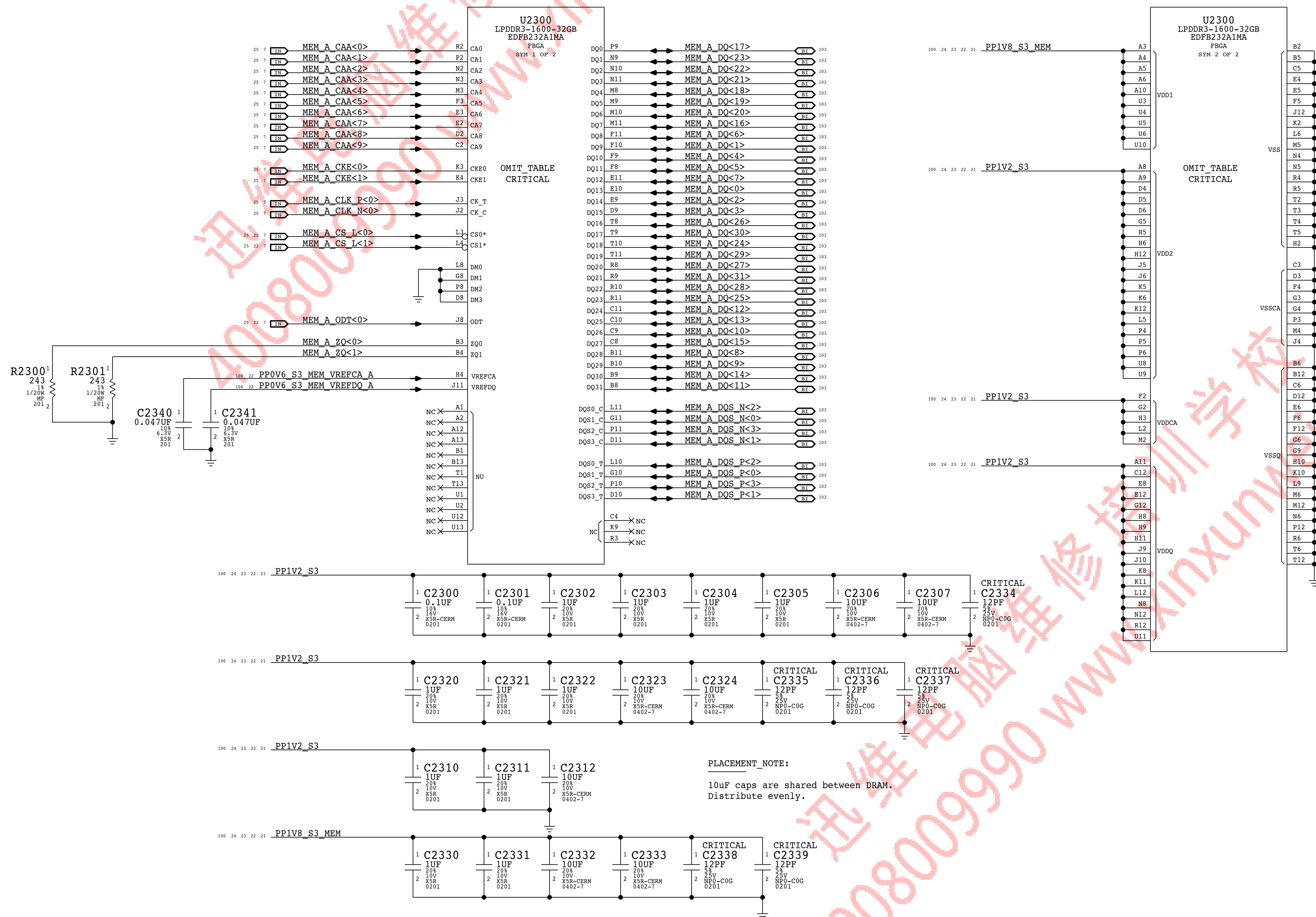
BOM_COST_GROUP=CPU & CHIPSET

CPU-Based Margining



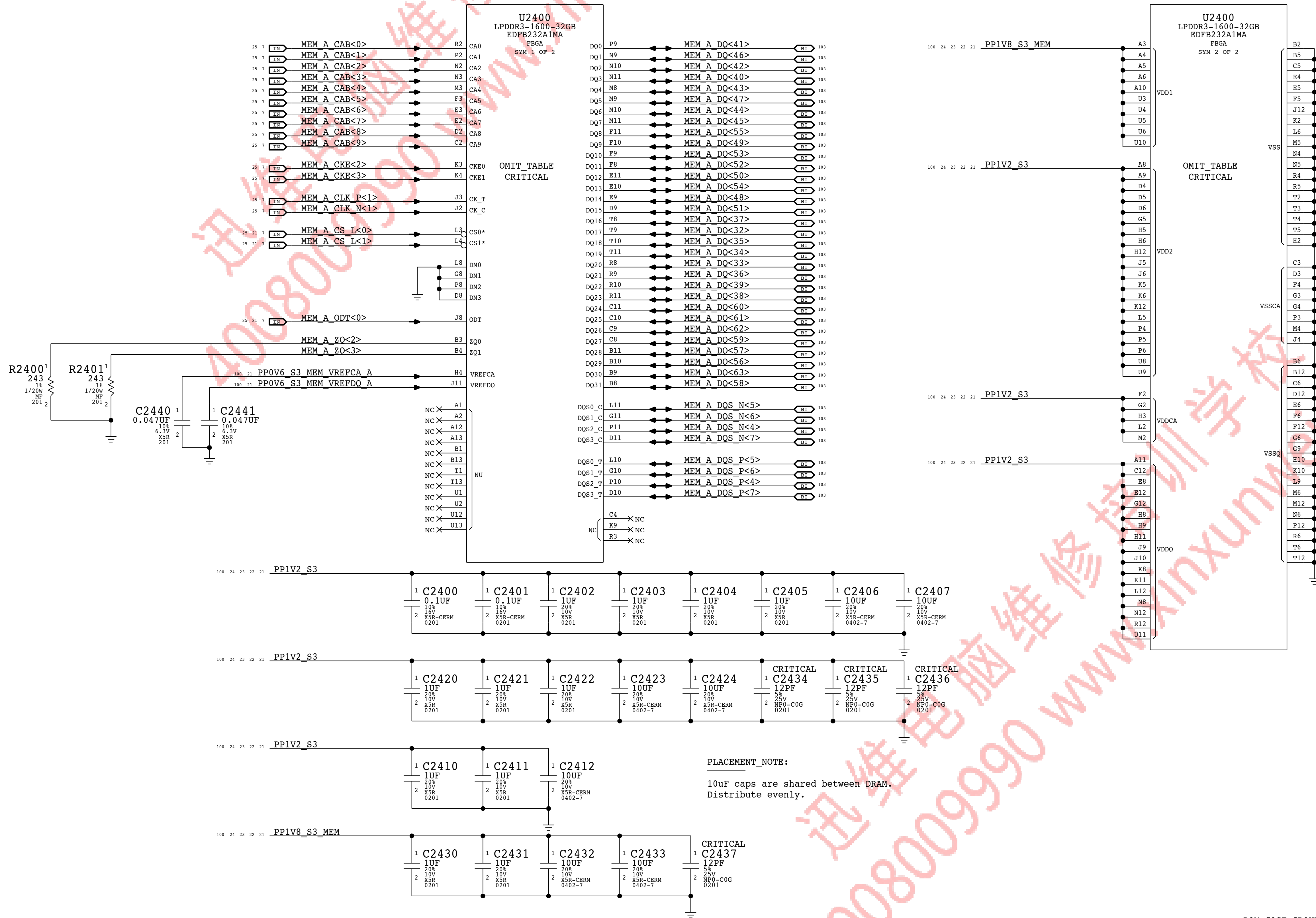
SYNC_MASTER=J52_MLB		SYNC_DATE=05/12/2015	
PAGE TITLE			
LPDDR3 VREF Margining			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00515		D
	REVISION		
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	7.0.0		
	BRANCH		
	dvt1		
	PAGE		
	22 OF 145		
SHEET			
20 OF 119			

LPDDR3 CHANNEL A (0-31)



SYNC MASTER=J52_MLB		SYNC_DATE=05/12/2015	
PAGE TITLE			
LPDDR3 DRAM Channel A (00-31)			
 Apple Inc.		DRAWING NUMBER	SIDE
		051-00515	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		7.0.0	
		BRANCH	dvt1
		PAGE	23 OF 145
		SHEET	21 OF 119

LPDDR3 CHANNEL A (32-63)

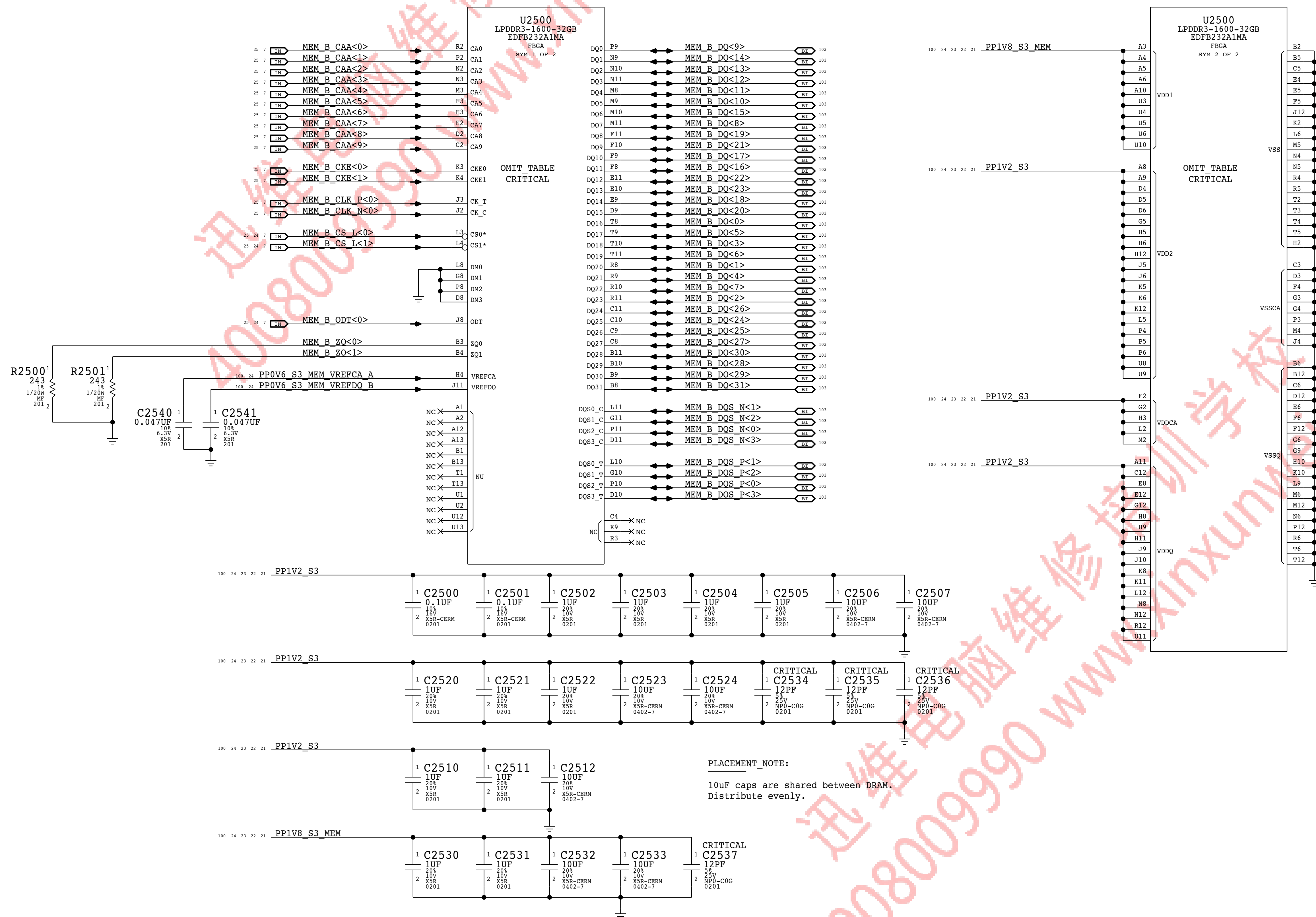


PLACEMENT NOTE:
10uF caps are shared between DRAM.
Distribute evenly.

SYNC_MASTER=J52_MLB		SYNC_DATE=05/12/2015	
PAGE TITLE			
LPDDR3 DRAM Channel A		(32-63)	
 Apple Inc.	DRAWING NUMBER	051-00515	STEP D
	REVISION	7.0.0	
	BRANCH	dvt1	
	PAGE	24 OF 145	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		SHEET	22 OF 119

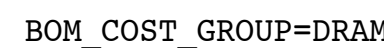
BOM_COST_GROUP=DRAM

LPDDR3 CHANNEL B (0-31)

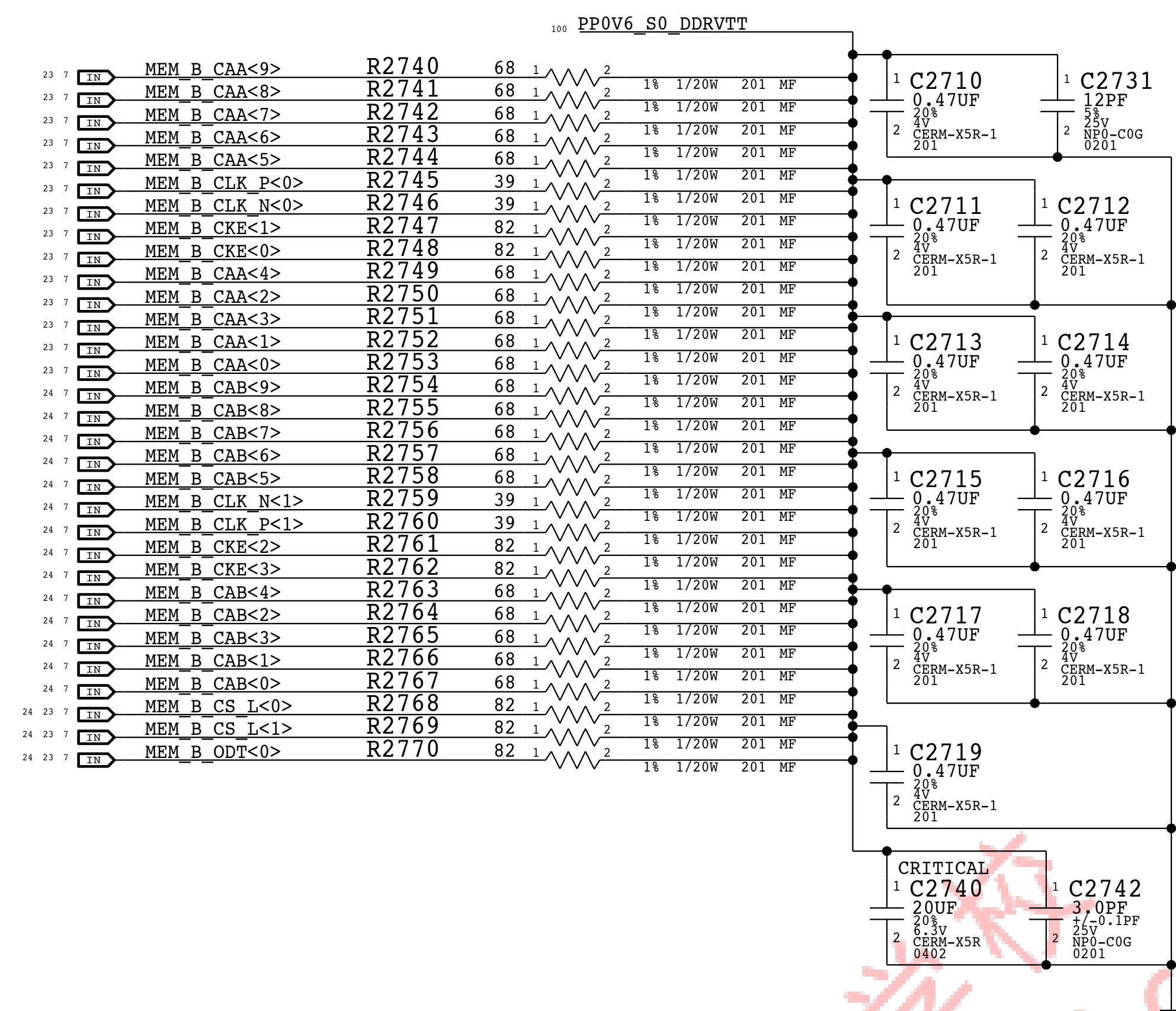
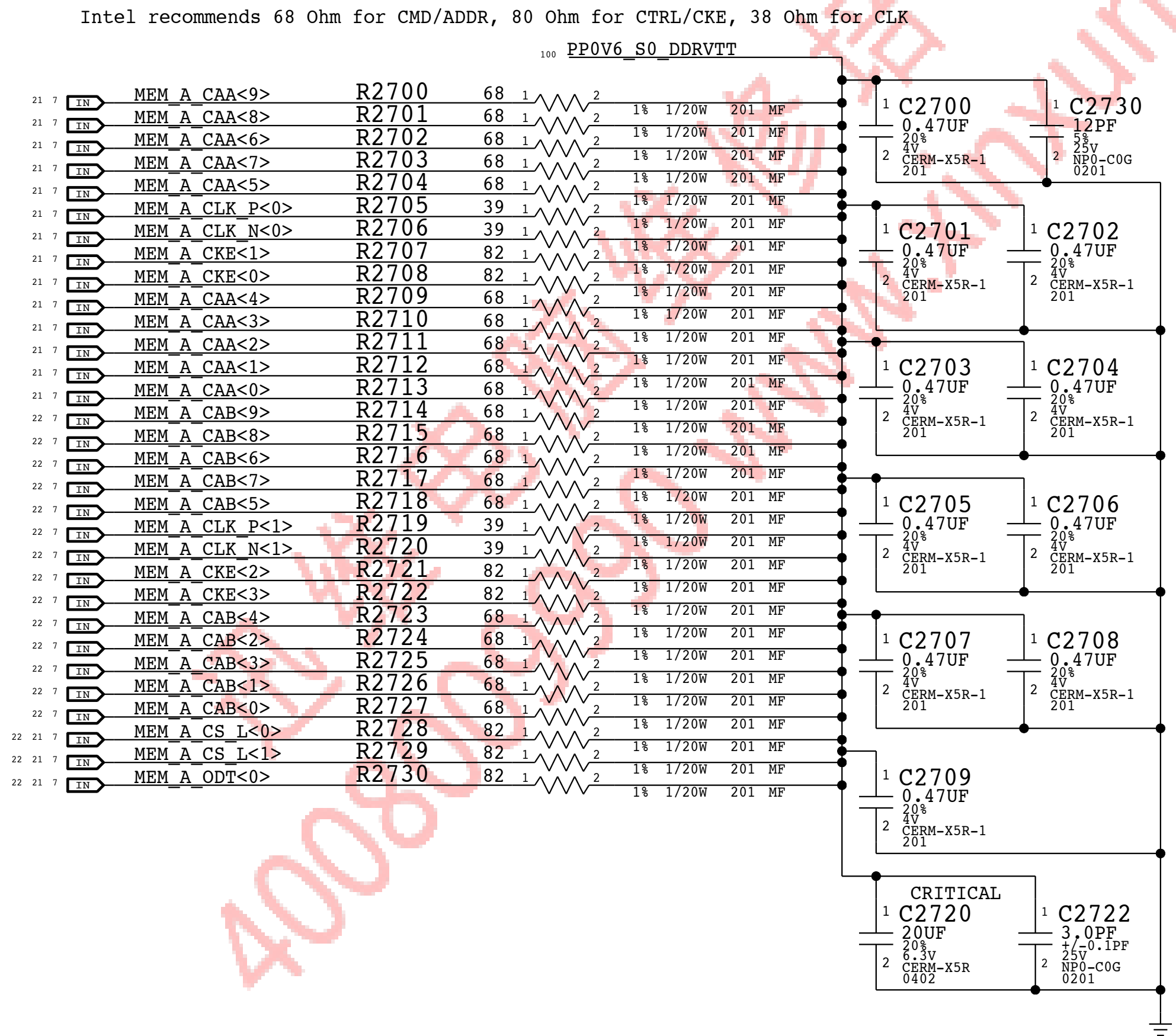


SYNC MASTER=J52 MLB		SYNC DATE=05/12/2015	
PAGE TITLE			
LPDDR3 DRAM Channel B (00-31)			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-00515	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		REVISION	
		7.0.0	
		BRANCH	
		dvt1	
		PAGE	
		25 OF 145	
		SHEET	
		23 OF 119	

10uF caps are shared between DRAM
Distribute evenly.

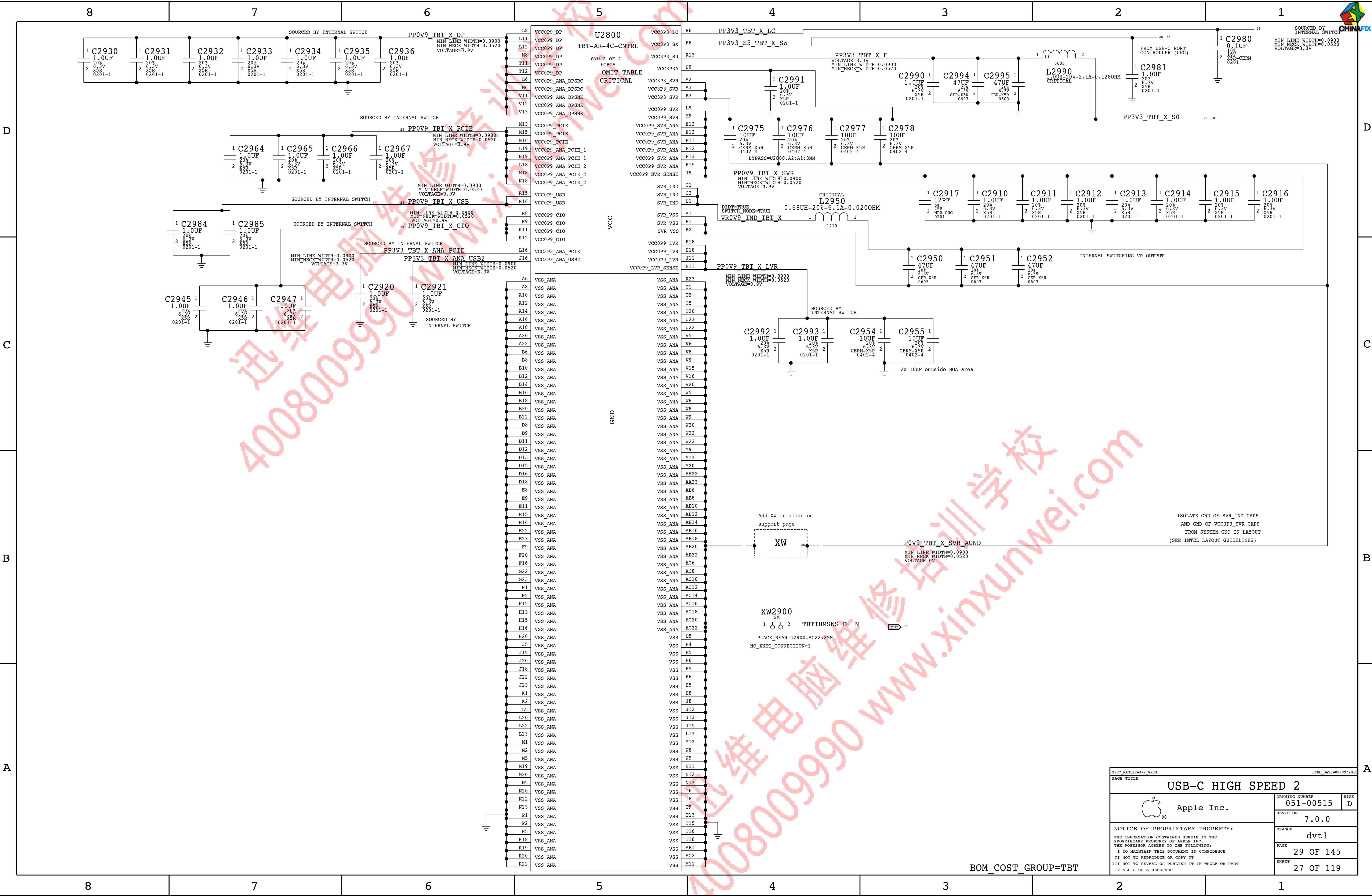


SYNC MASTER=J52 MLB		SYNC DATE=05/12/2001	
PAGE TITLE			
LPDDR3 DRAM Channel B (32-63)			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-00515	D
		REVISION	
		7.0.0	
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			
BRANCH		dvt1	
PAGE		26 OF 145	
SHEET		24 OF 119	



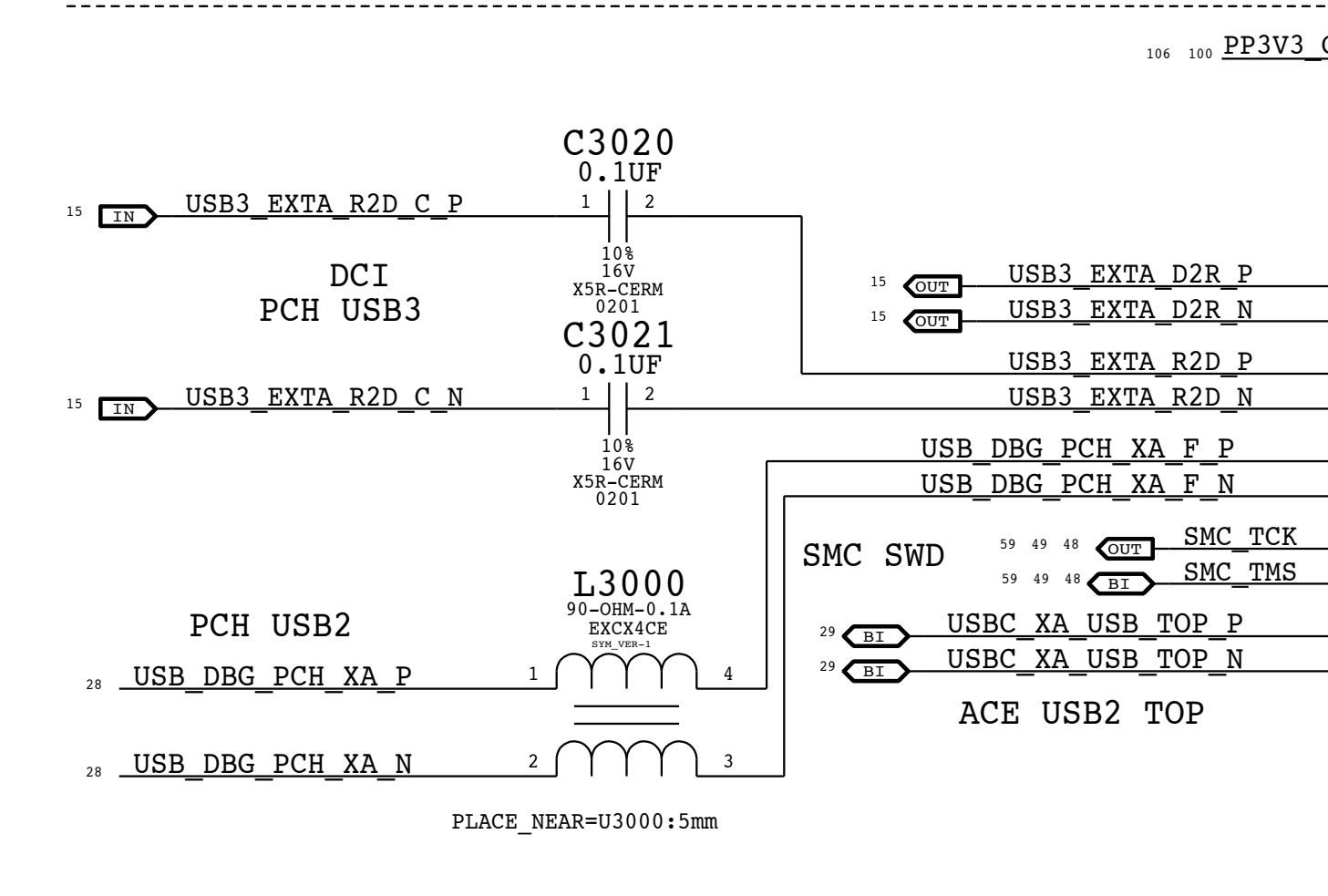
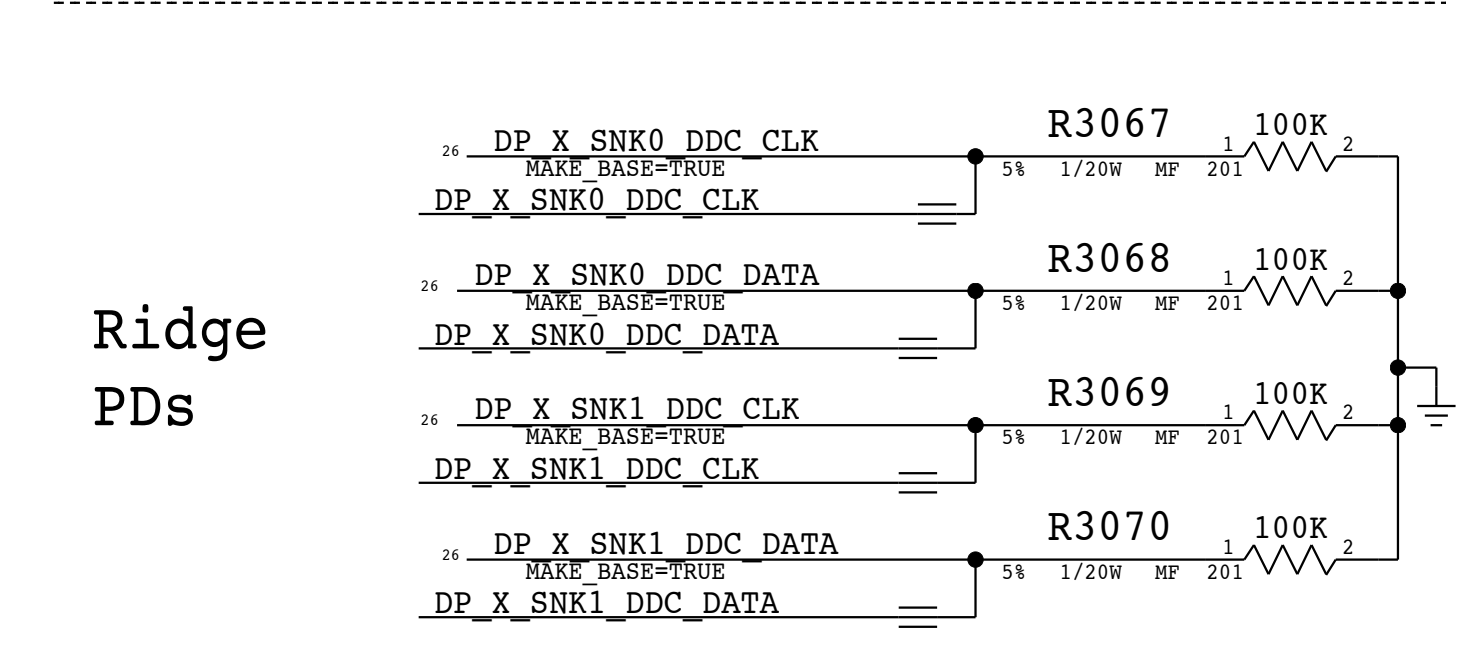
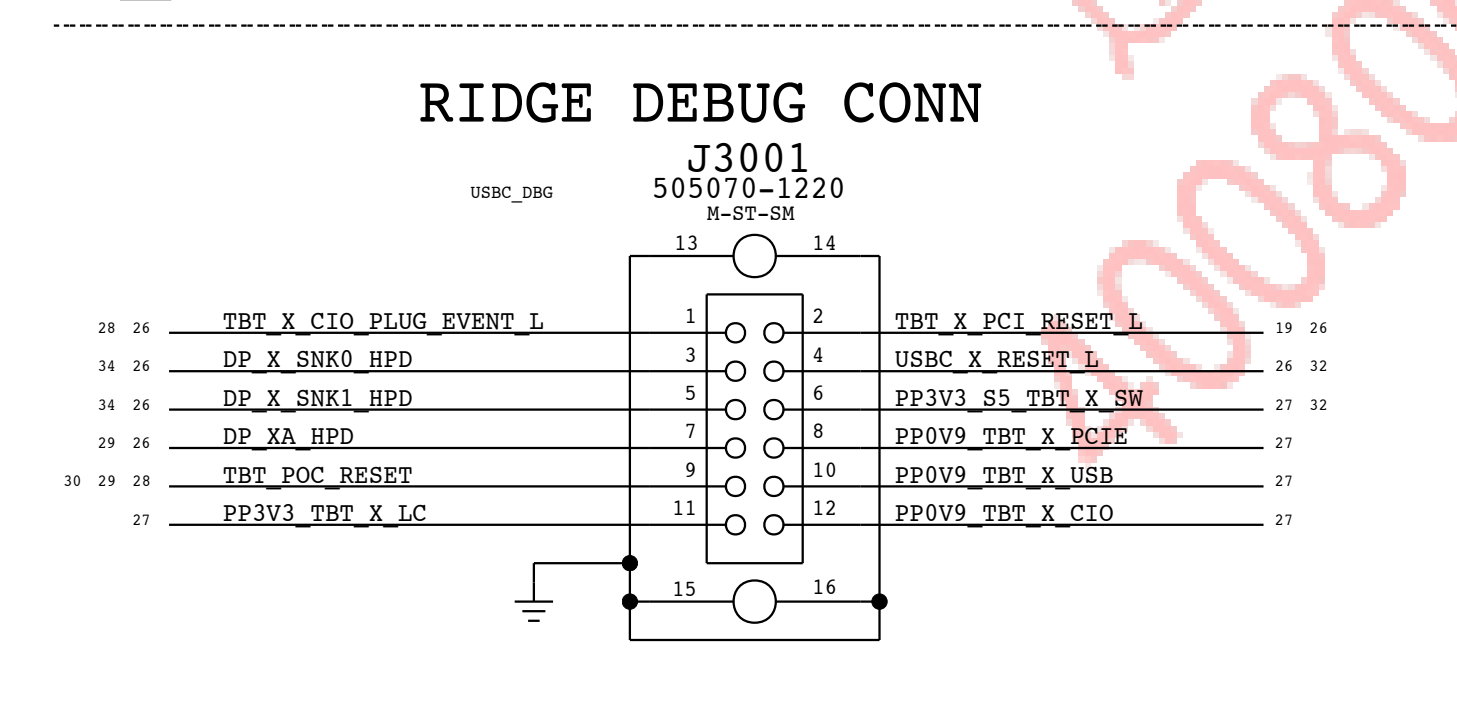
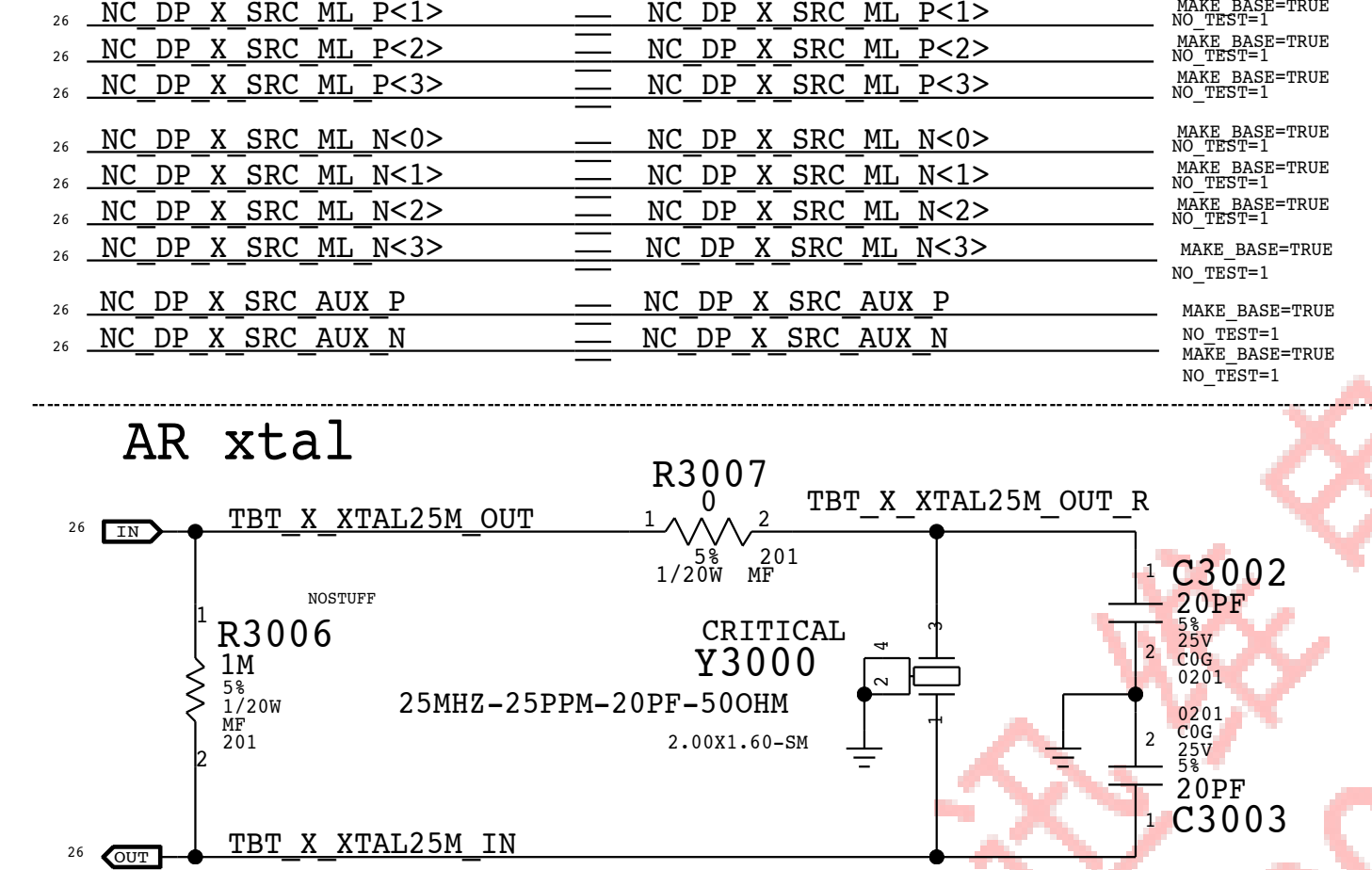
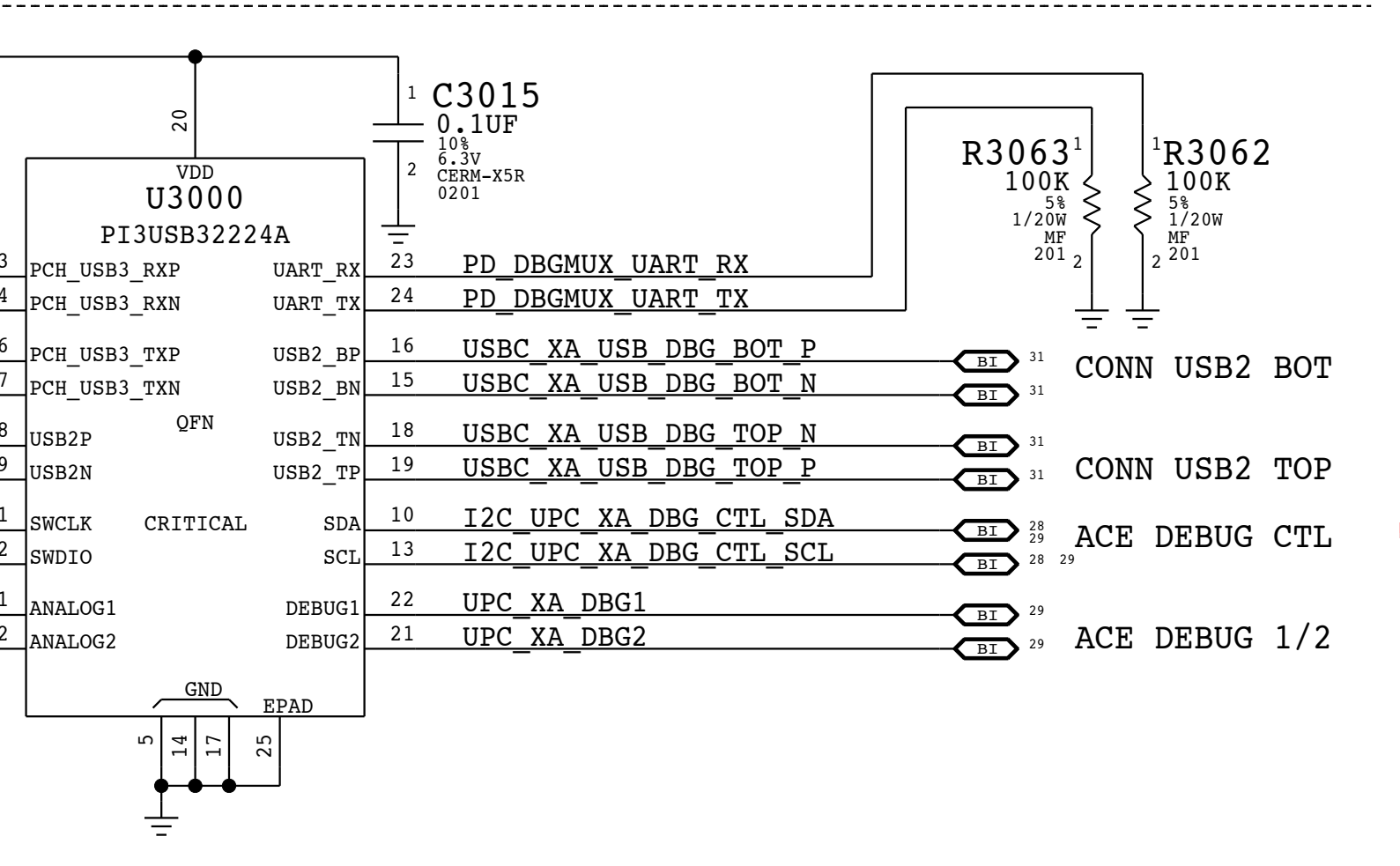
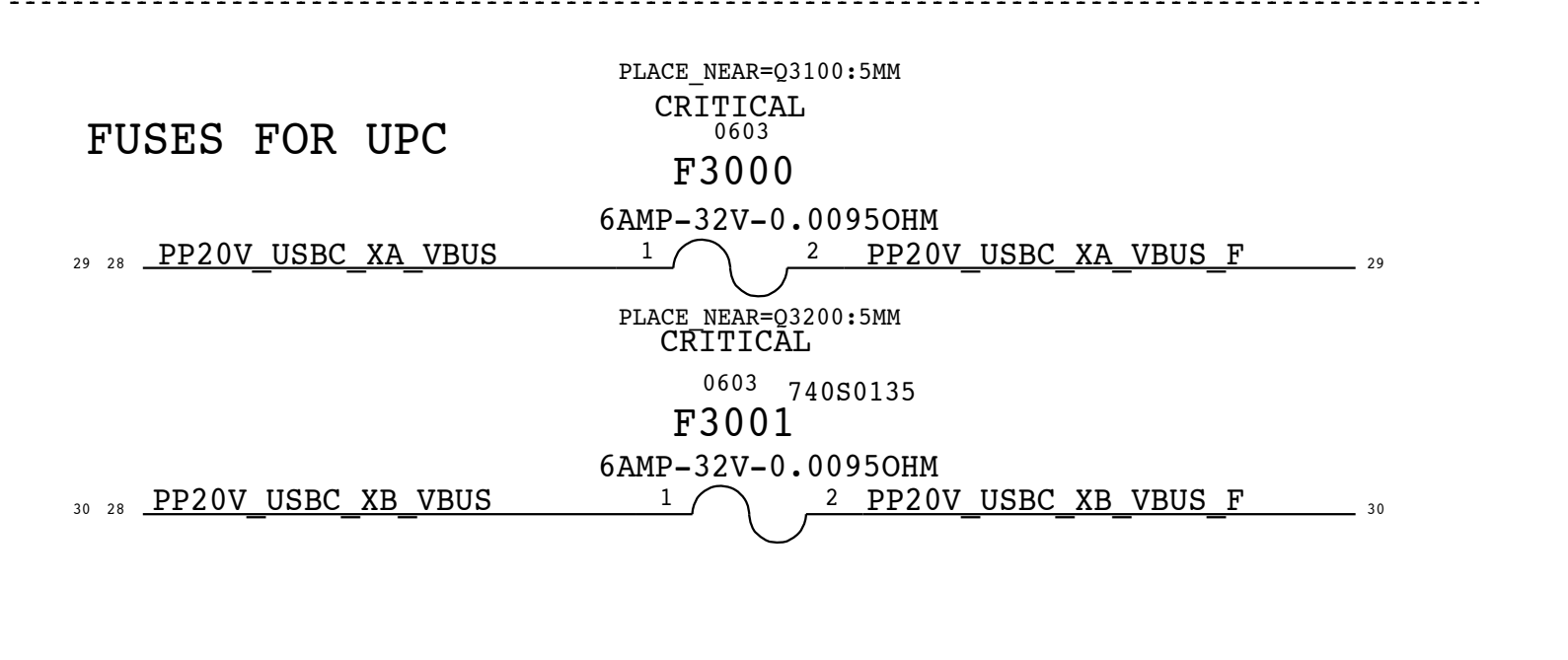
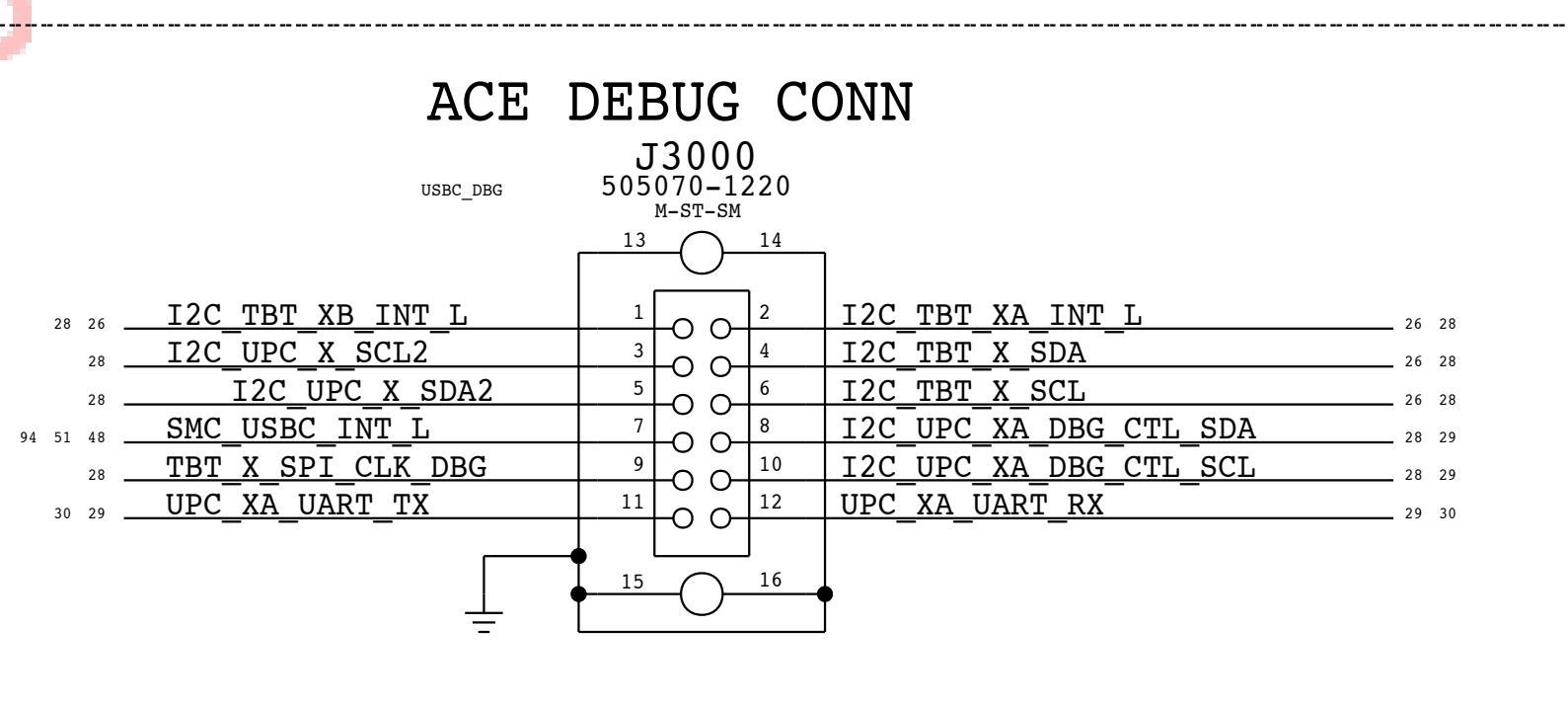
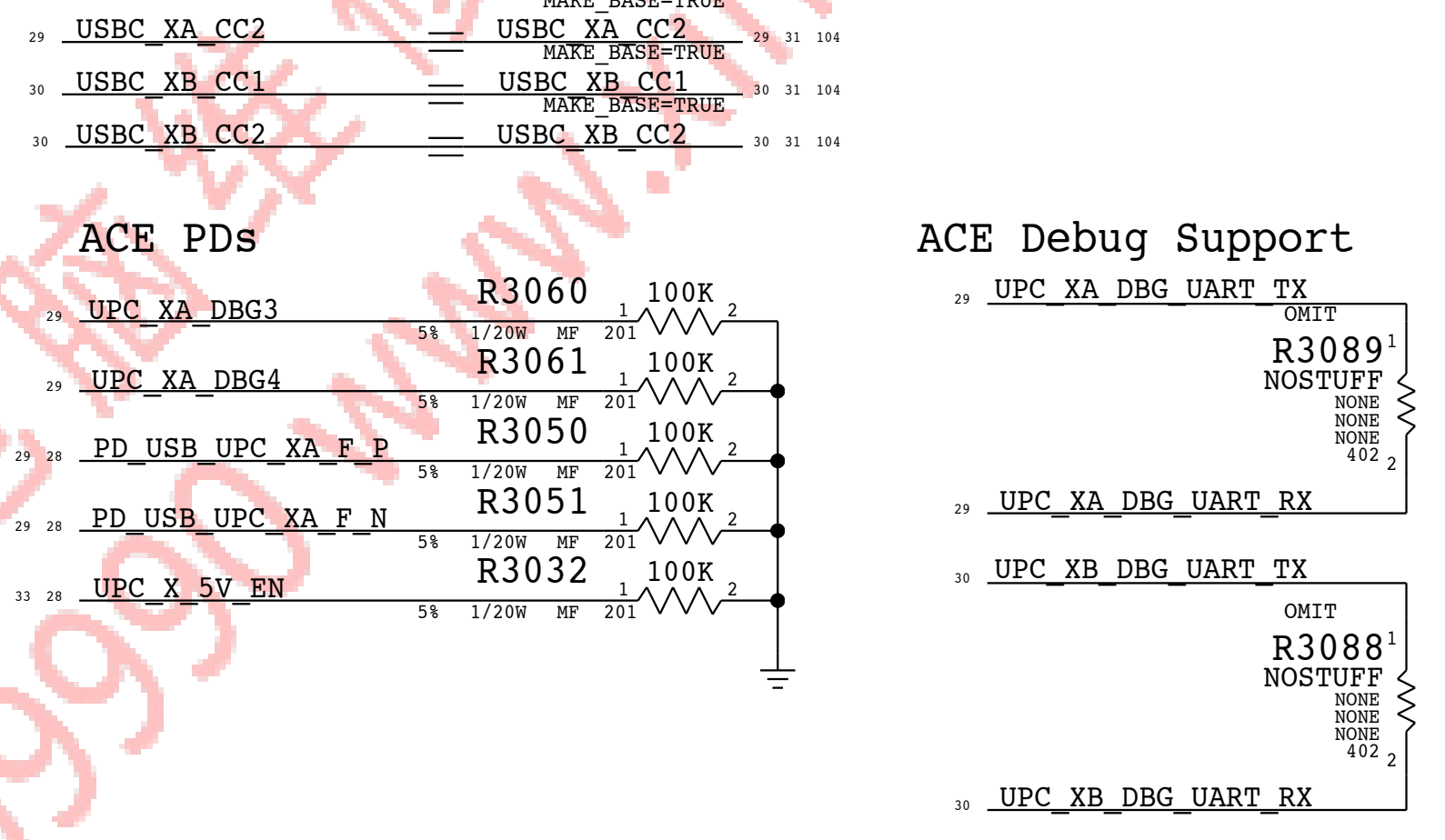
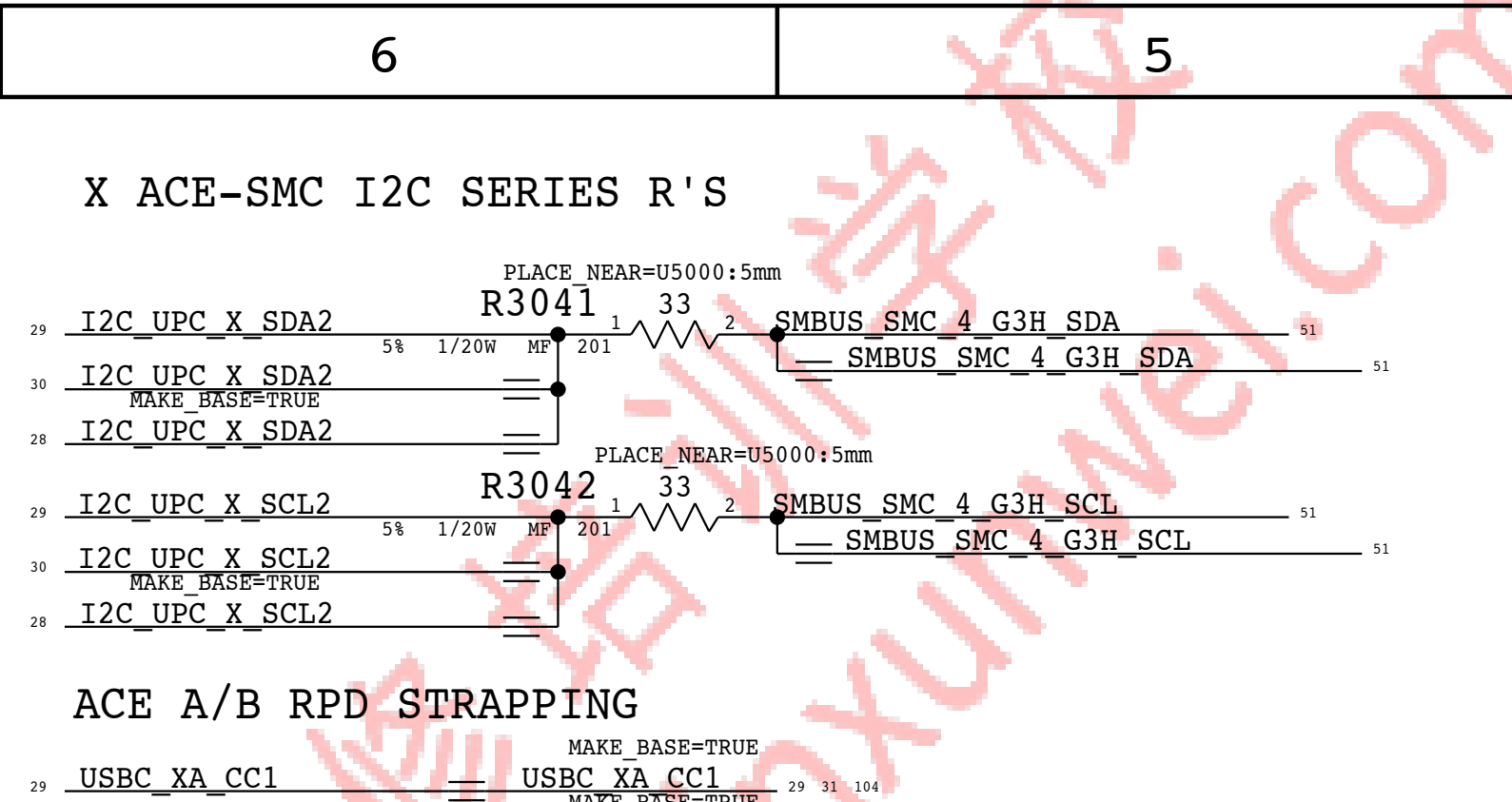
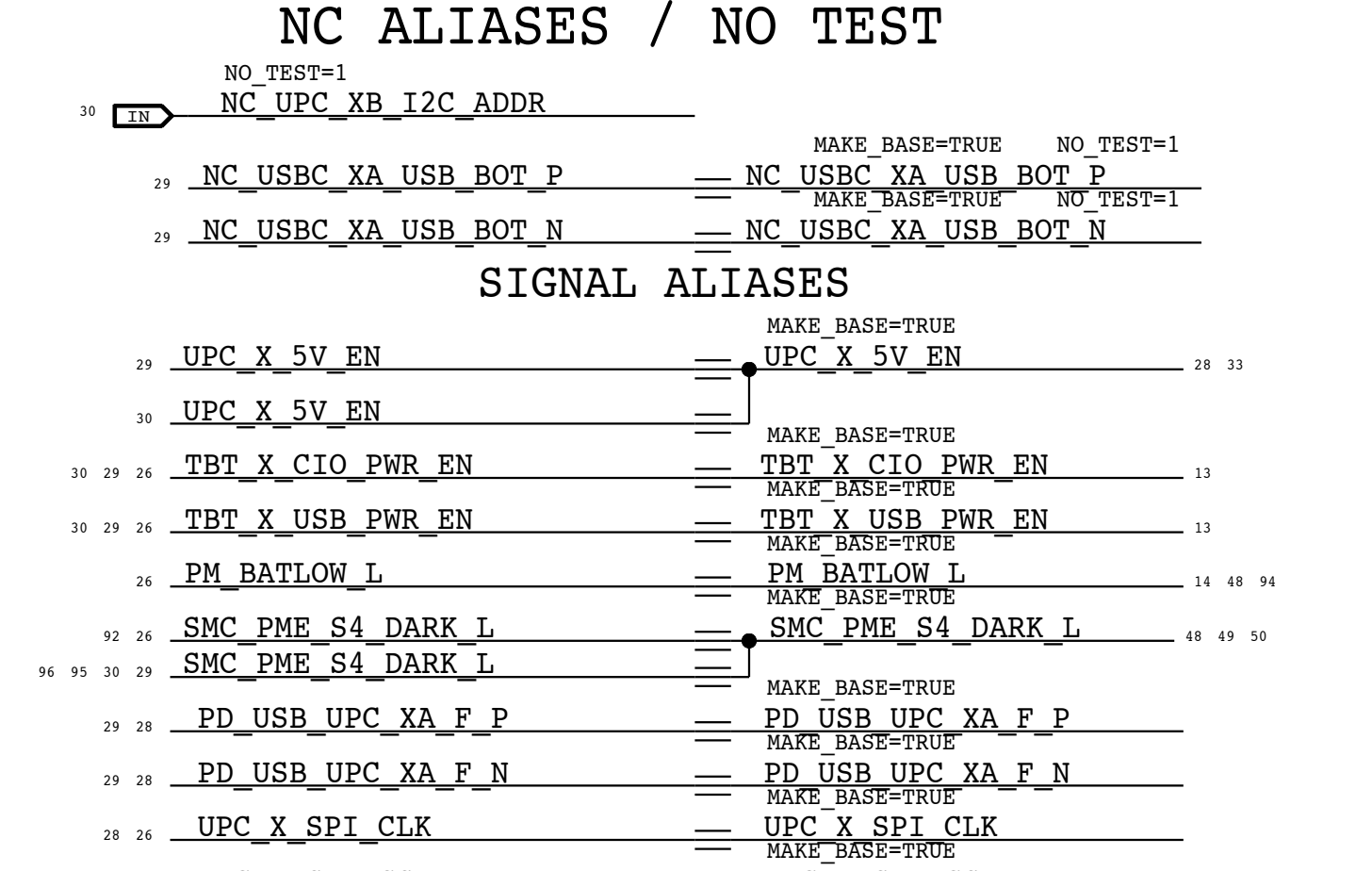
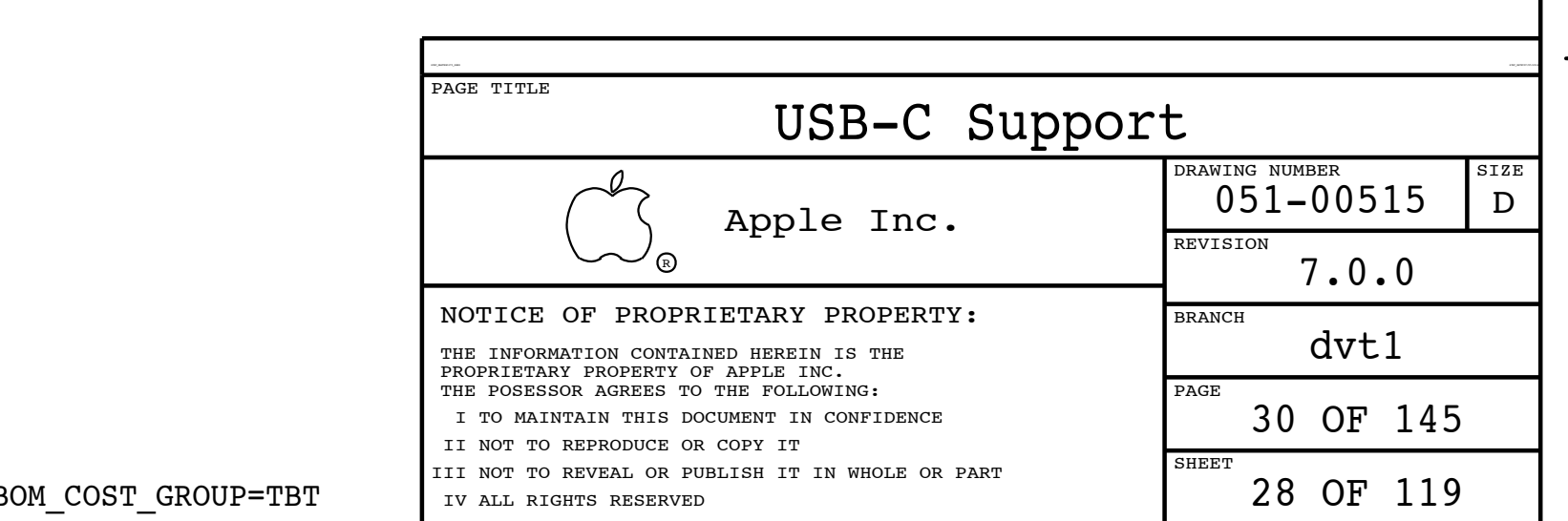
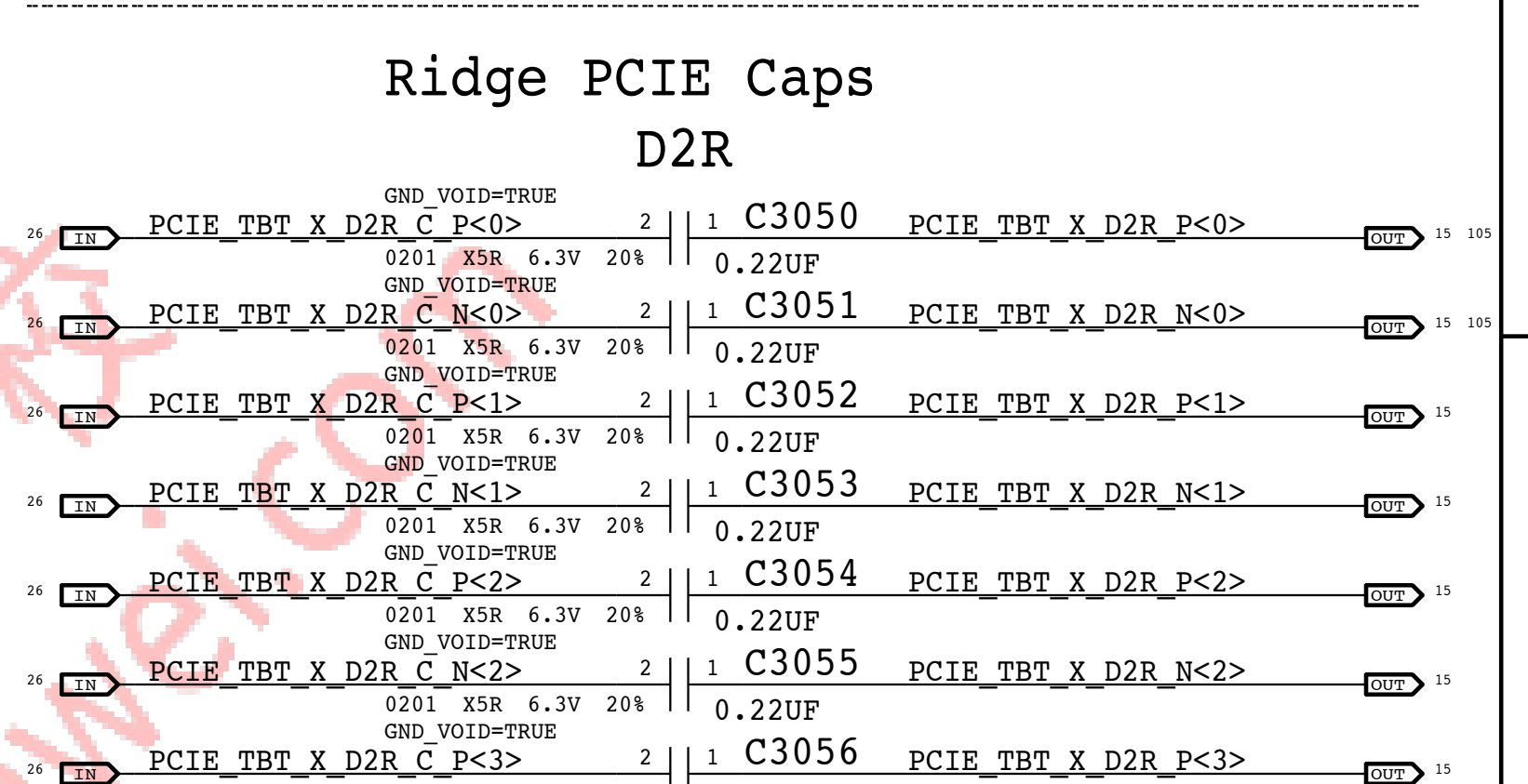
SYNC_MASTER=J52_MLB		SYNC_DATE=05/12/2015	
PAGE TITLE			
LPDDR3 DRAM Termination			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00515		D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	REVISION		
	7.0.0		
	BRANCH		
	dvt1		
	PAGE		
27 OF 145			
SHEET			
25 OF 119			

BOM_COST_GROUP=DRAM

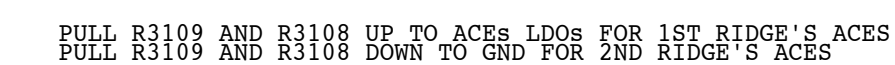


SYNC_MASTER=J79_GREG		SYNC_DATE=09/09/2015	
PAGE TITLE			
USB-C HIGH SPEED 2			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00515		D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I ALL RIGHTS RESERVED	REVISION		
	7.0.0		
	BRANCH		
	dvt1		
	PAGE		
	29 OF 145		
	SHEET		
	27 OF 119		

BOM_COST_GROUP=TBT

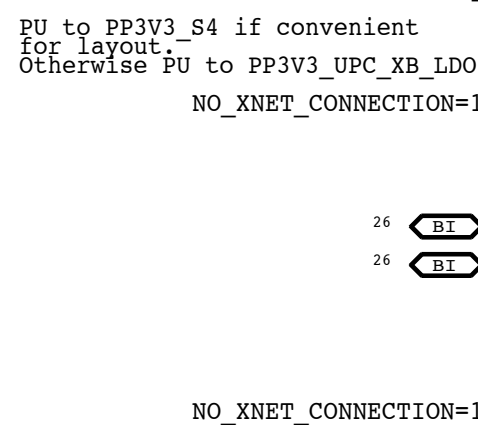


8	7	6	5
---	---	---	---



BOM_COST_GROUP=USB-C

BOM COST GROUP=USB-C



D

C

B

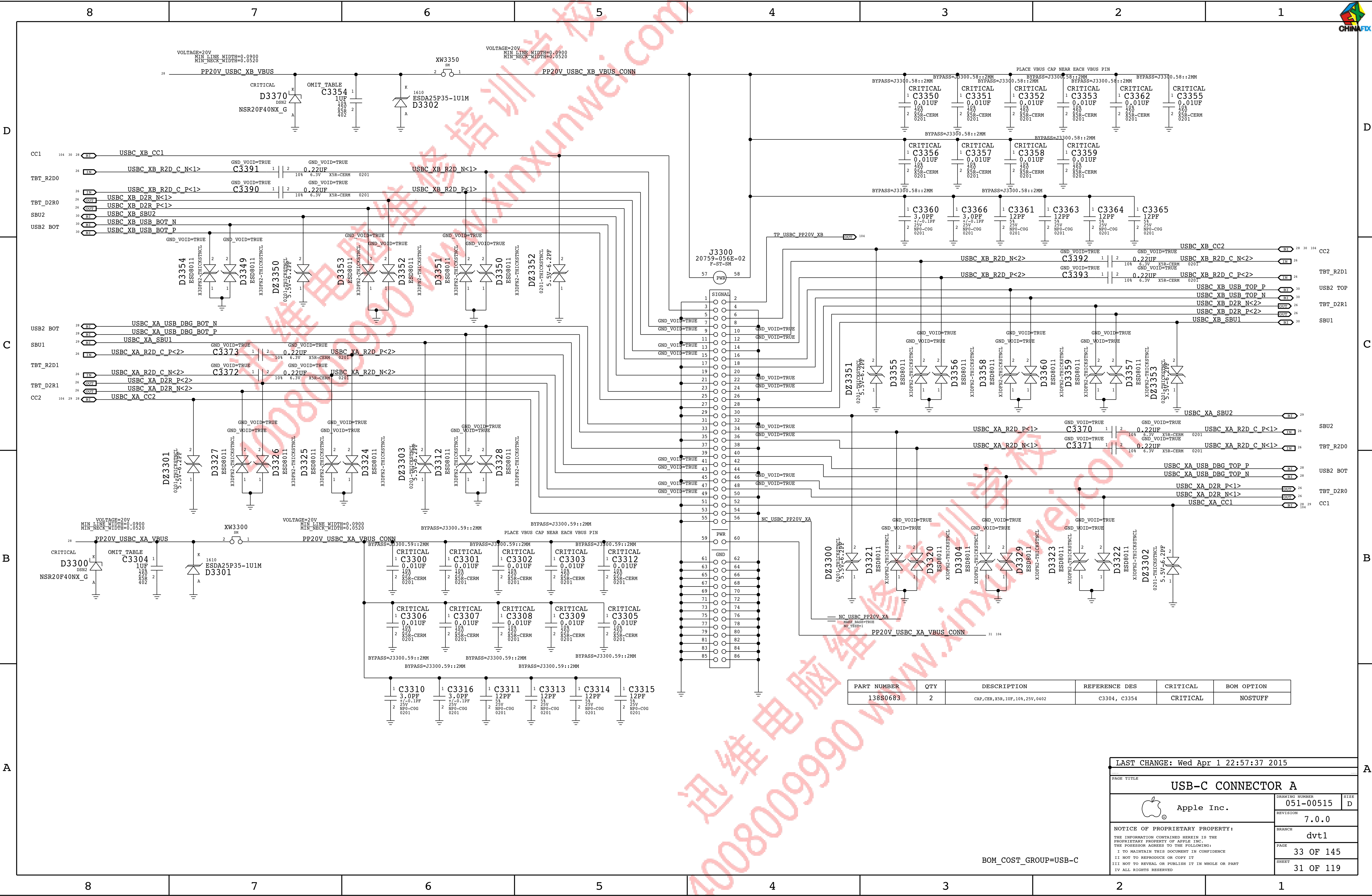
A

D

C

B

A



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
13880683	2	CAP,CER,X5R,1UF,10%,25V,0402	C3304, C3354	CRITICAL	NOSTUFF

LAST CHANGE: Wed Apr 1 22:57:37 2015

PAGE TITLE

USB-C CONNECTOR A



Apple Inc.

DRAWING NUMBER
051-00515

STEP
D

REVISION
7.0.0

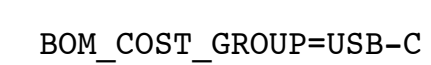
BRANCH
dvt1

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
1 TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
11 NOT TO REPRODUCE OR COPY IT
111 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
1V ALL RIGHTS RESERVED

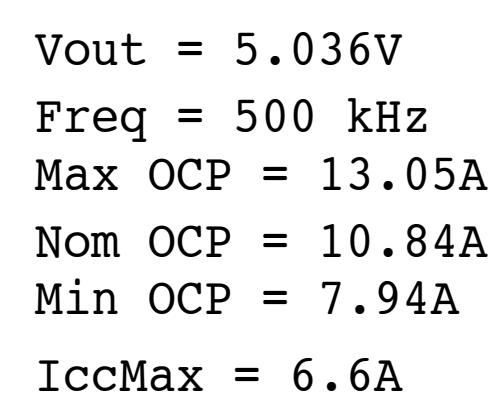
PAGE
33 OF 145

SHEET
31 OF 119

BOM_COST_GROUP=USB-C



DESIGN: X502/DEV MLB U			
LAST CHANGE: Wed Feb 18 17:12:24 2015			
SRC: MUFTR=719.0000		SRC: MUFTR=01/14/2016	
PAGE TITLE			
 USB-C CONNECTOR B Apple Inc.			
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		DRAWING NUMBER 051-00515	SIZE D
		REVISION 7.0.0	
		BRANCH dvt1	
		PAGE 34 OF 145	
		SHEET 32 OF 119	



PAGE TITLE		DRAWING NUMBER		SITE	
TBT 5V REGULATOR		051-00515		D	
 Apple Inc.		REVISION		7.0.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH		dvt1	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		PAGE		35 OF 145	
		SHEET		33 OF 119	

D

C

B

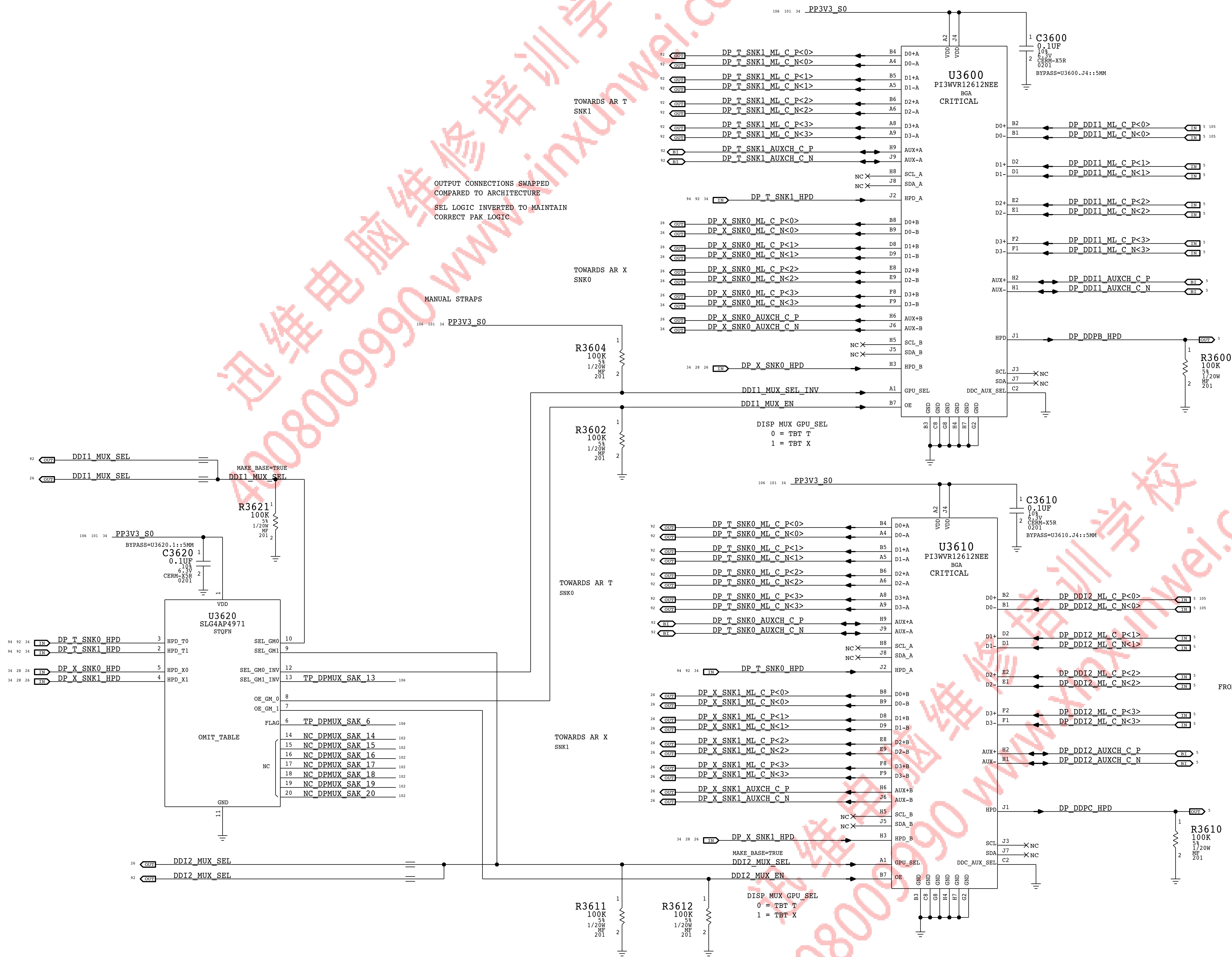
A

D

C

B

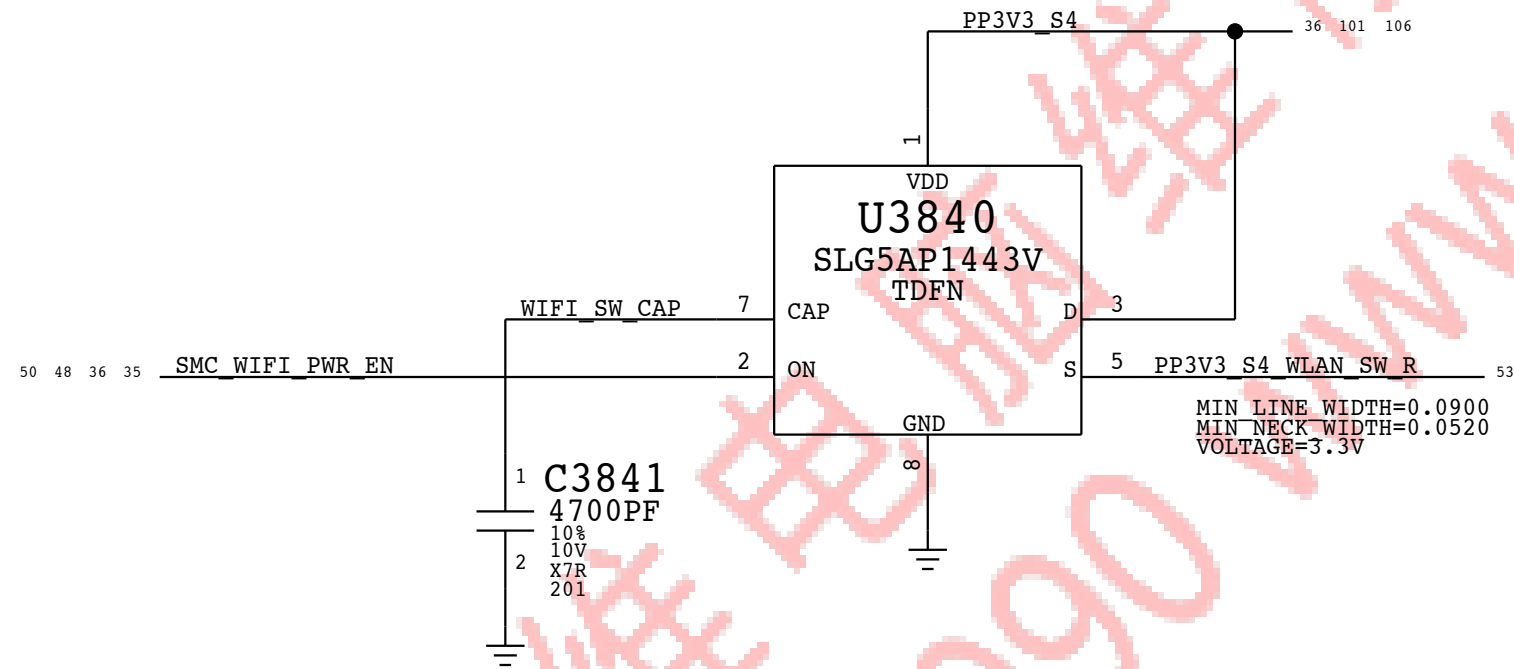
A



PAGE TITLE		
Display Mux		
	DRAWING NUMBER	051-00515
	REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: 1 TO MAINTAIN THIS DOCUMENT IN CONFIDENCE 11 NOT TO REPRODUCE OR COPY IT 111 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART 1V ALL RIGHTS RESERVED	BRANCH	dvt1
	PAGE	36 OF 145
	SHEET	34 OF 119

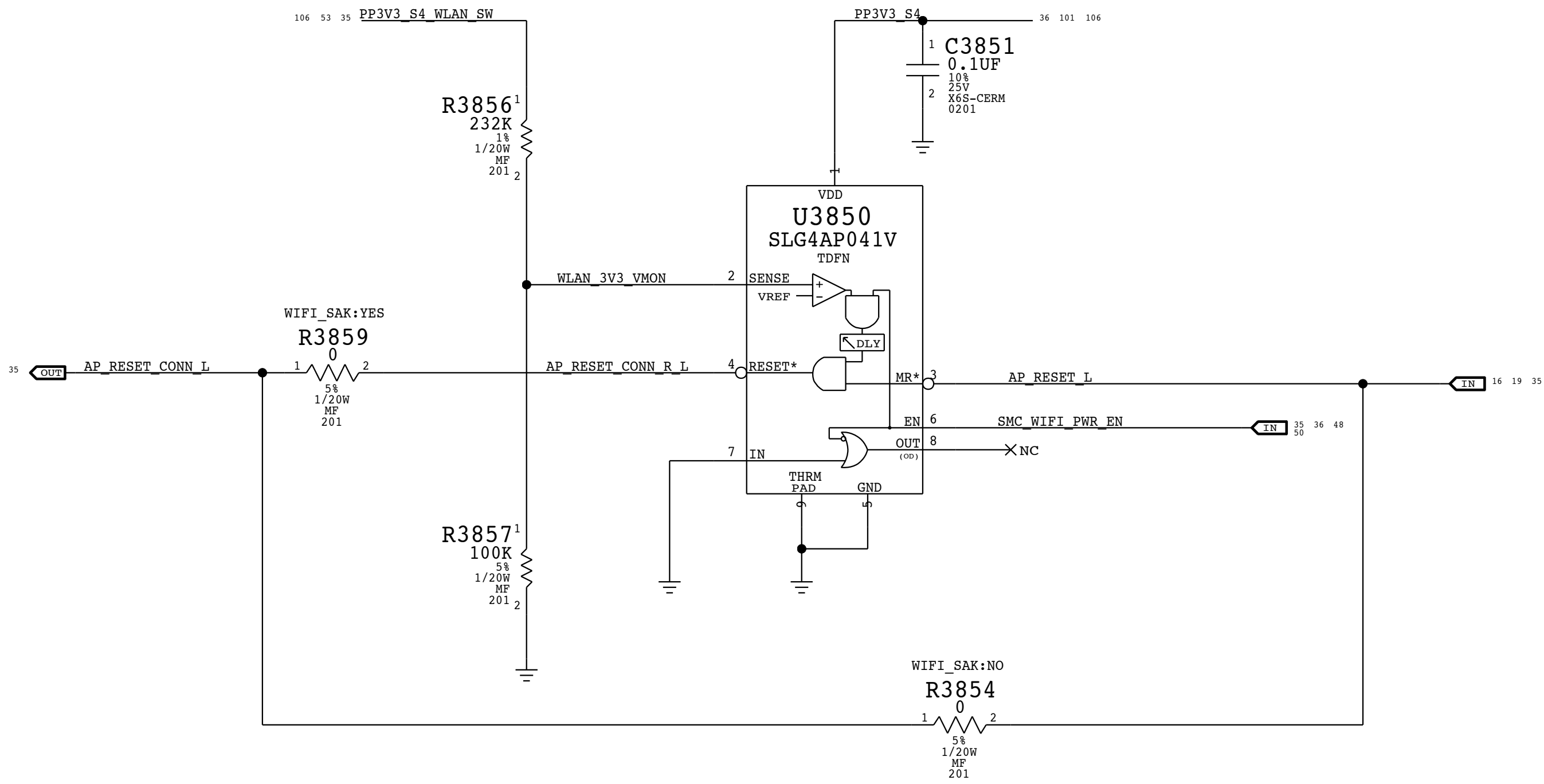
BOM_COST_GROUP=GRAPHICS

WLAN Power Switch

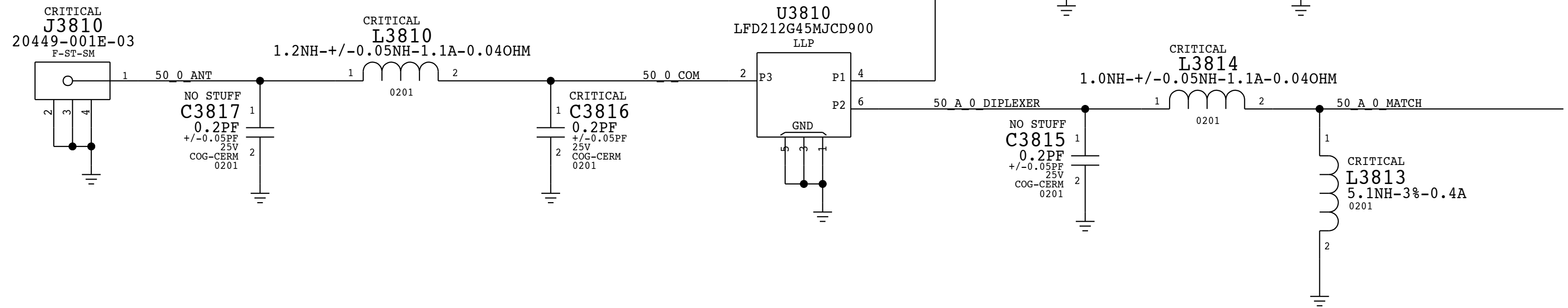


Supervisor & CLKREQ# Isolation

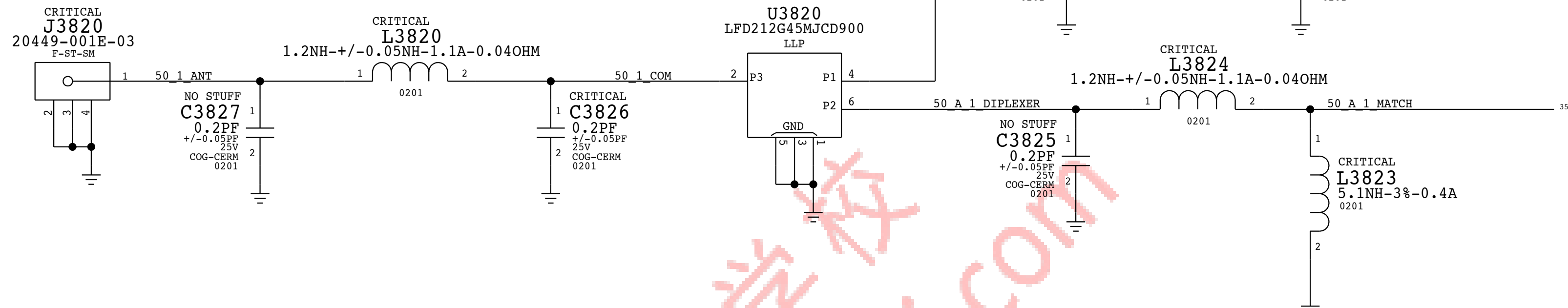
Delay = 130ms +/- 20%



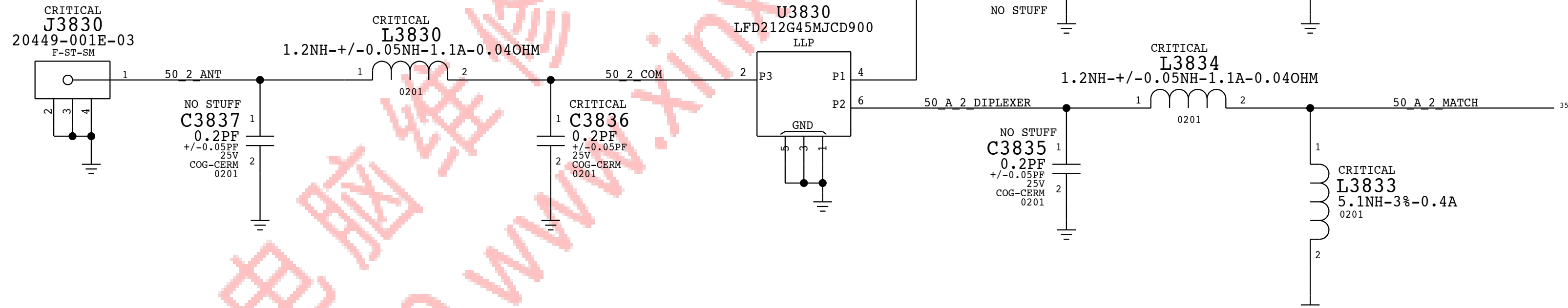
CORE0 DIPLEXER AND MATCHING



CORE1 DIPLEXER AND MATCHING



CORE2 DIPLEXER AND MATCHING



BOM_COST_GROUP=WIRELESS

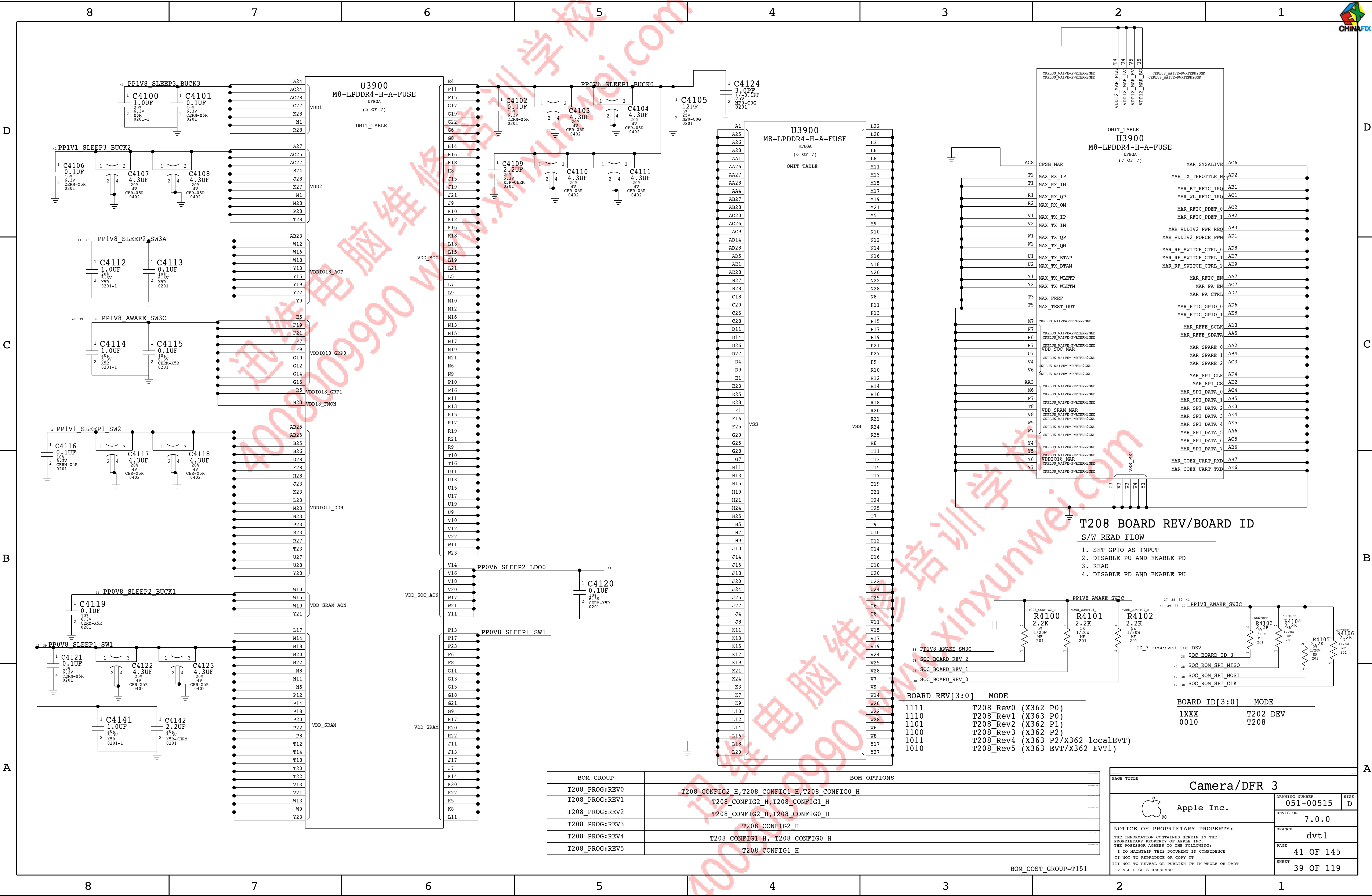
EVRD_MATER=J79_NETE		EVRD_DATE=02/02/2016	
PAGE TITLE			
WIFI/BT: MODULE 2			
	Apple Inc.	DRAWING NUMBER	051-00515
		REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	dvt1
		PAGE	38 OF 145
		SHEET	36 OF 119

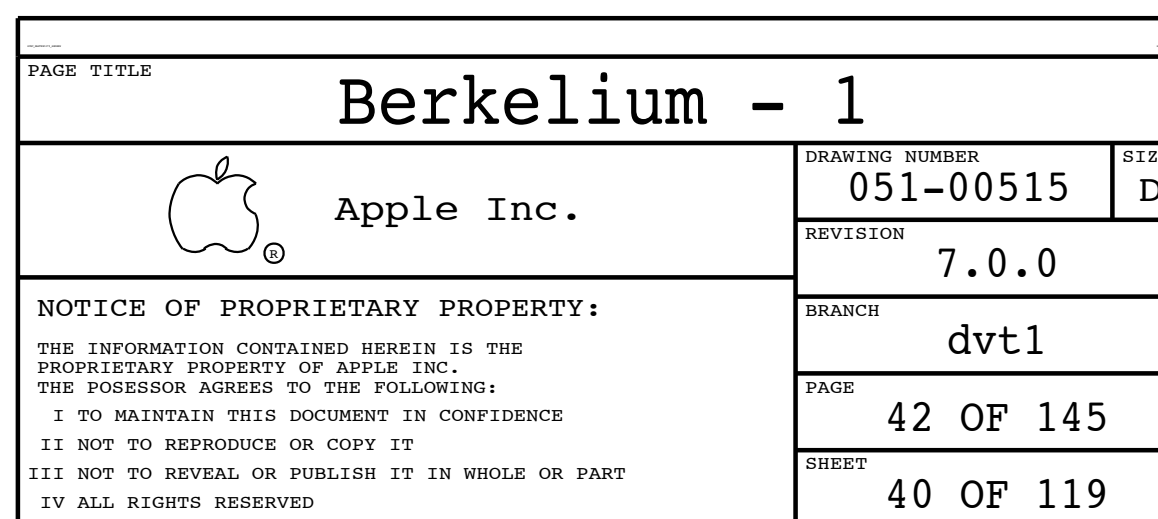
[illegible]

8 SEP ROM

[illegible][illegible]







D

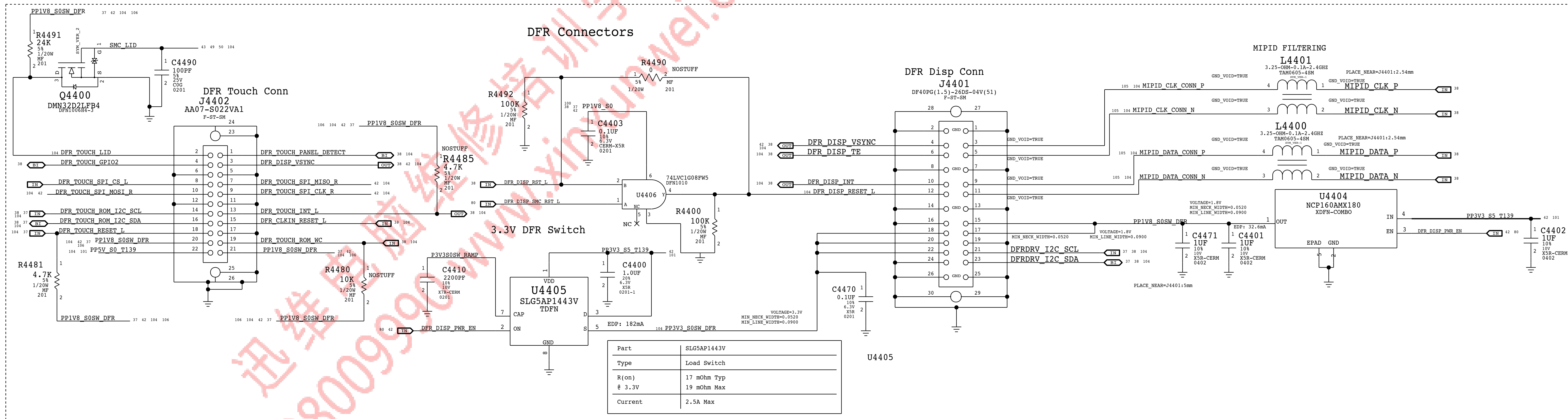


B

A

BOM_COST_GROUP=T151

T208 Support





D

C

B

A

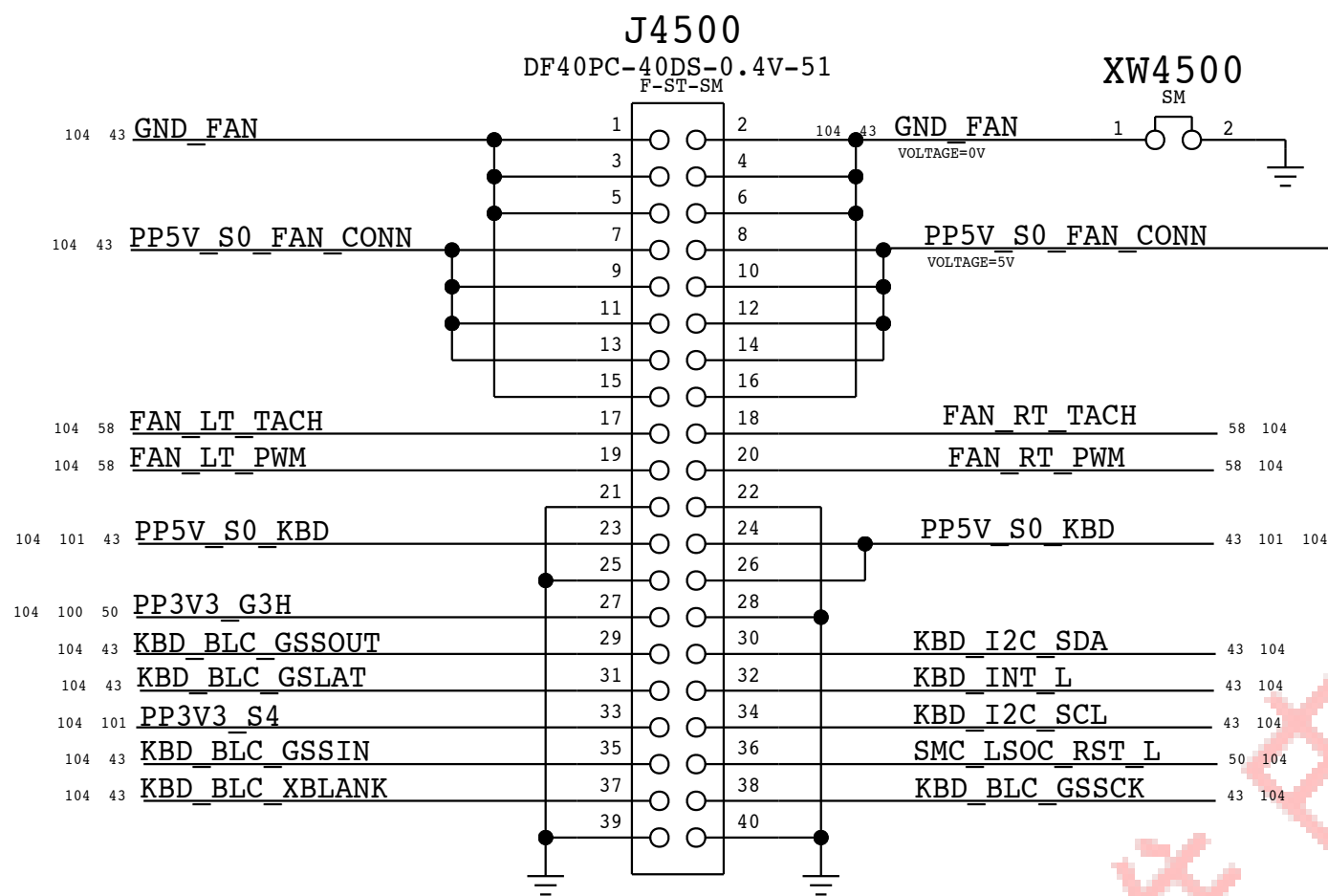
D

C

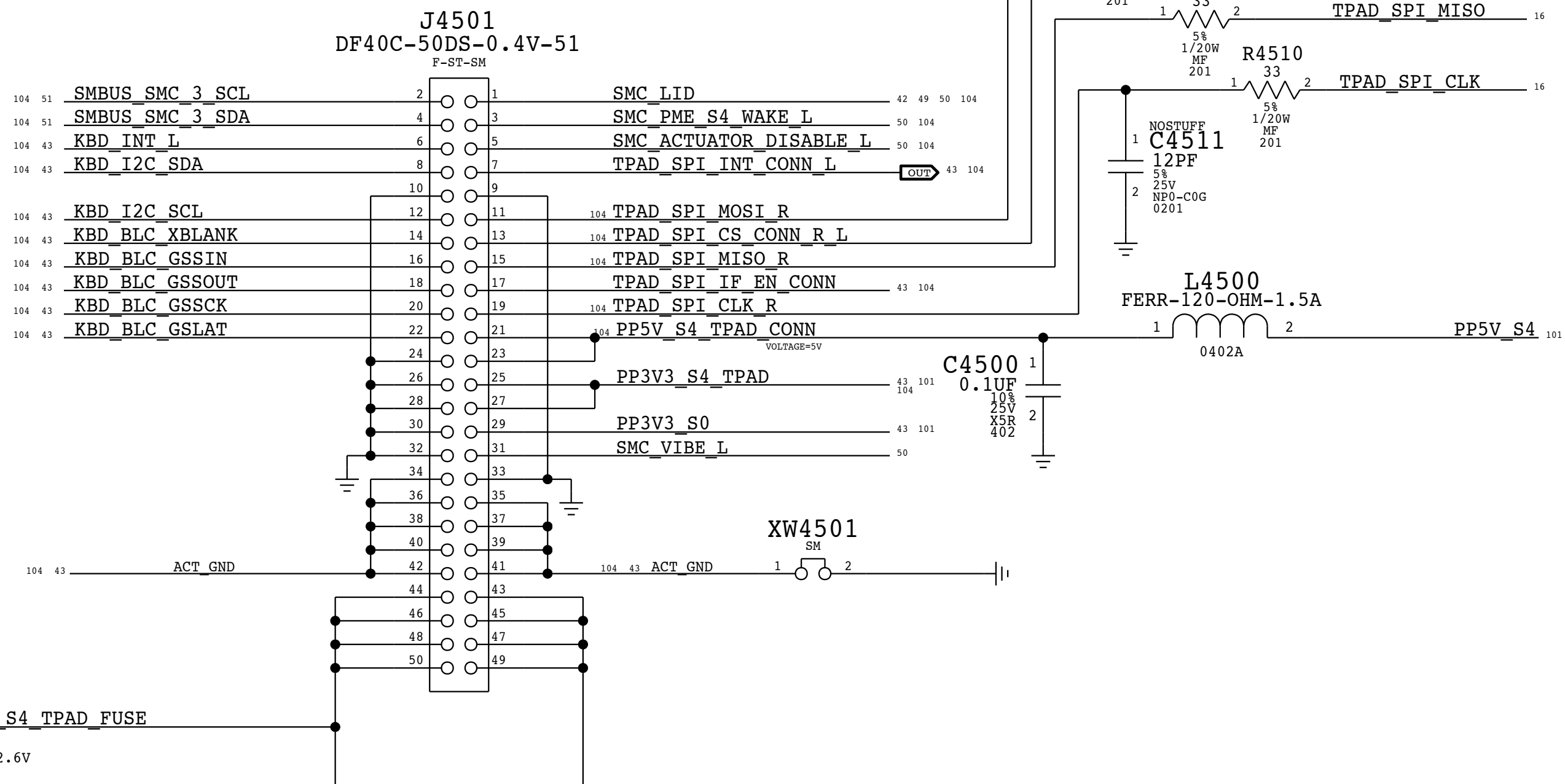
B

A

KBD CONNECTOR

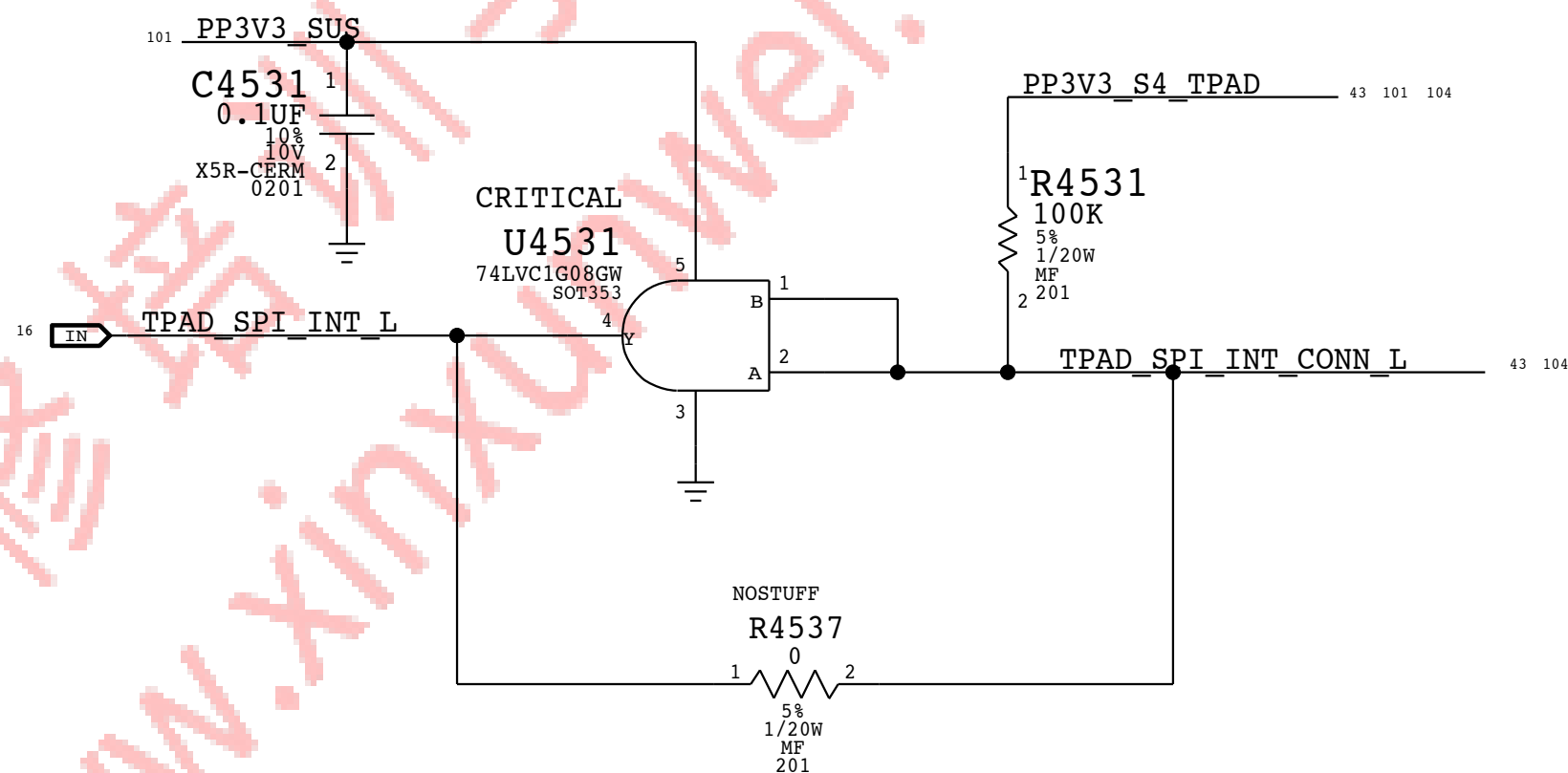
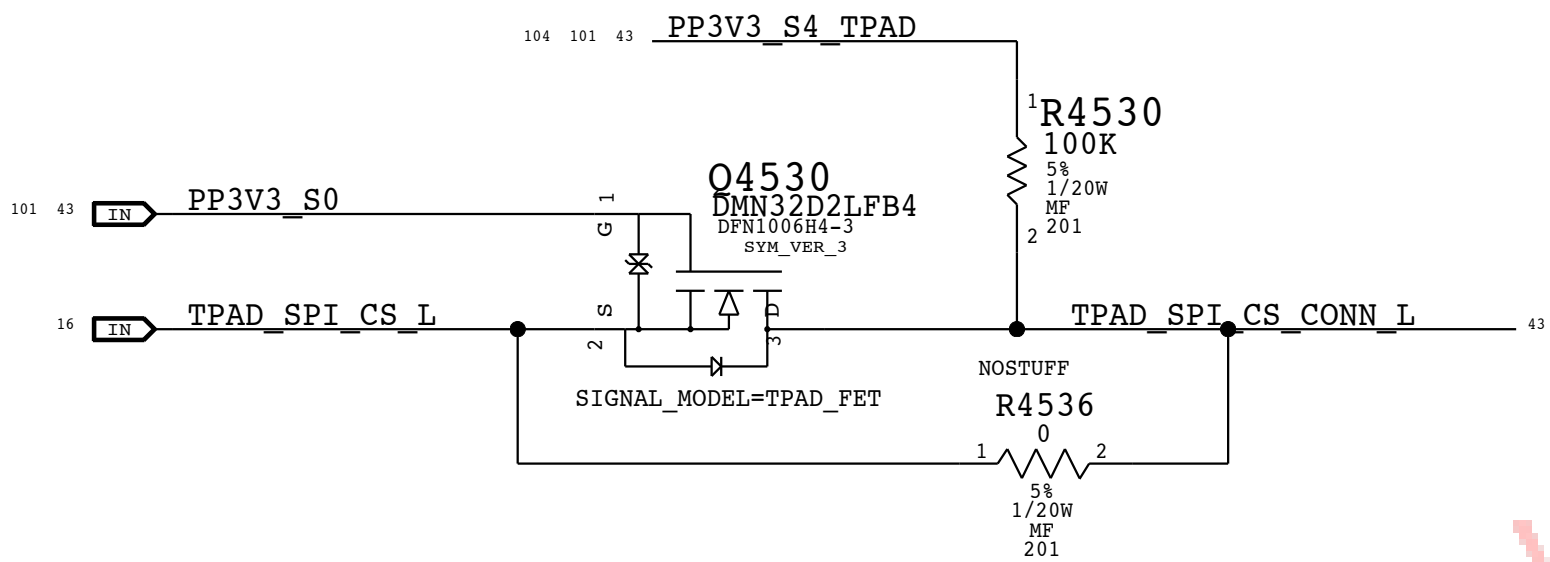
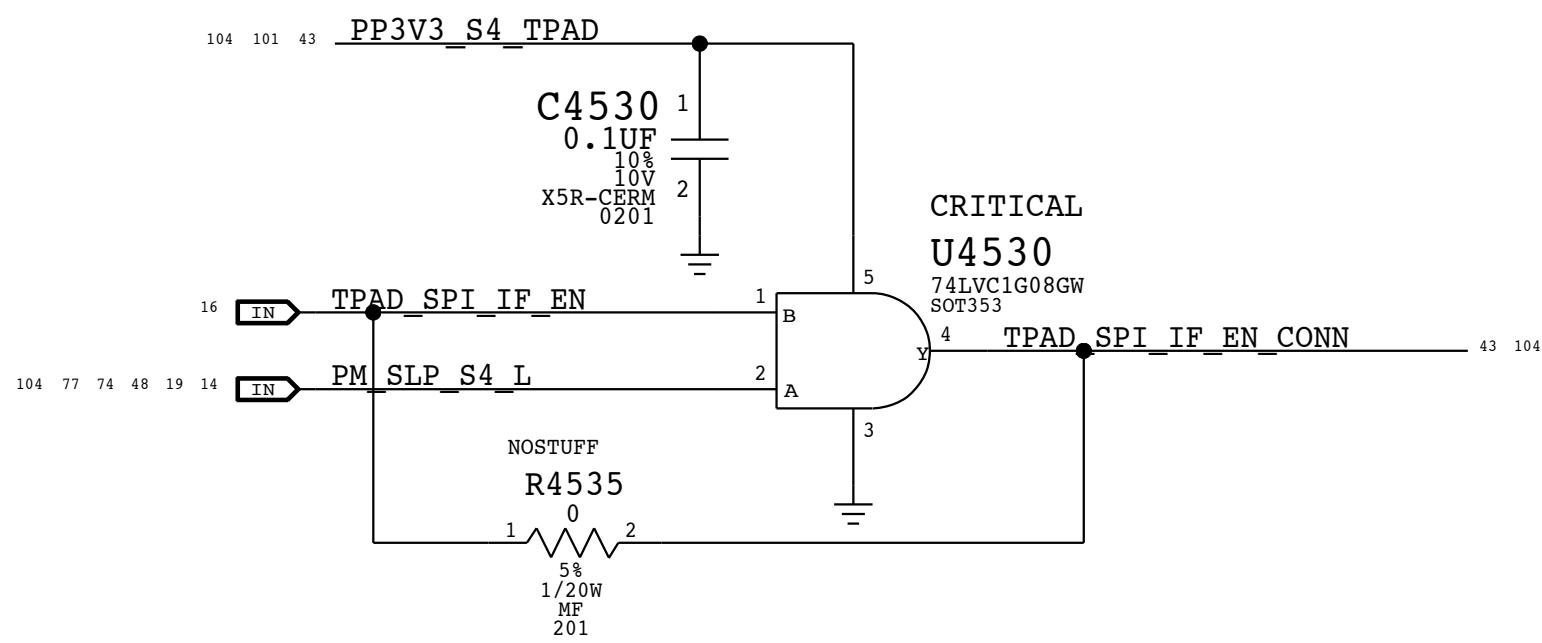


TPAD CONNECTOR



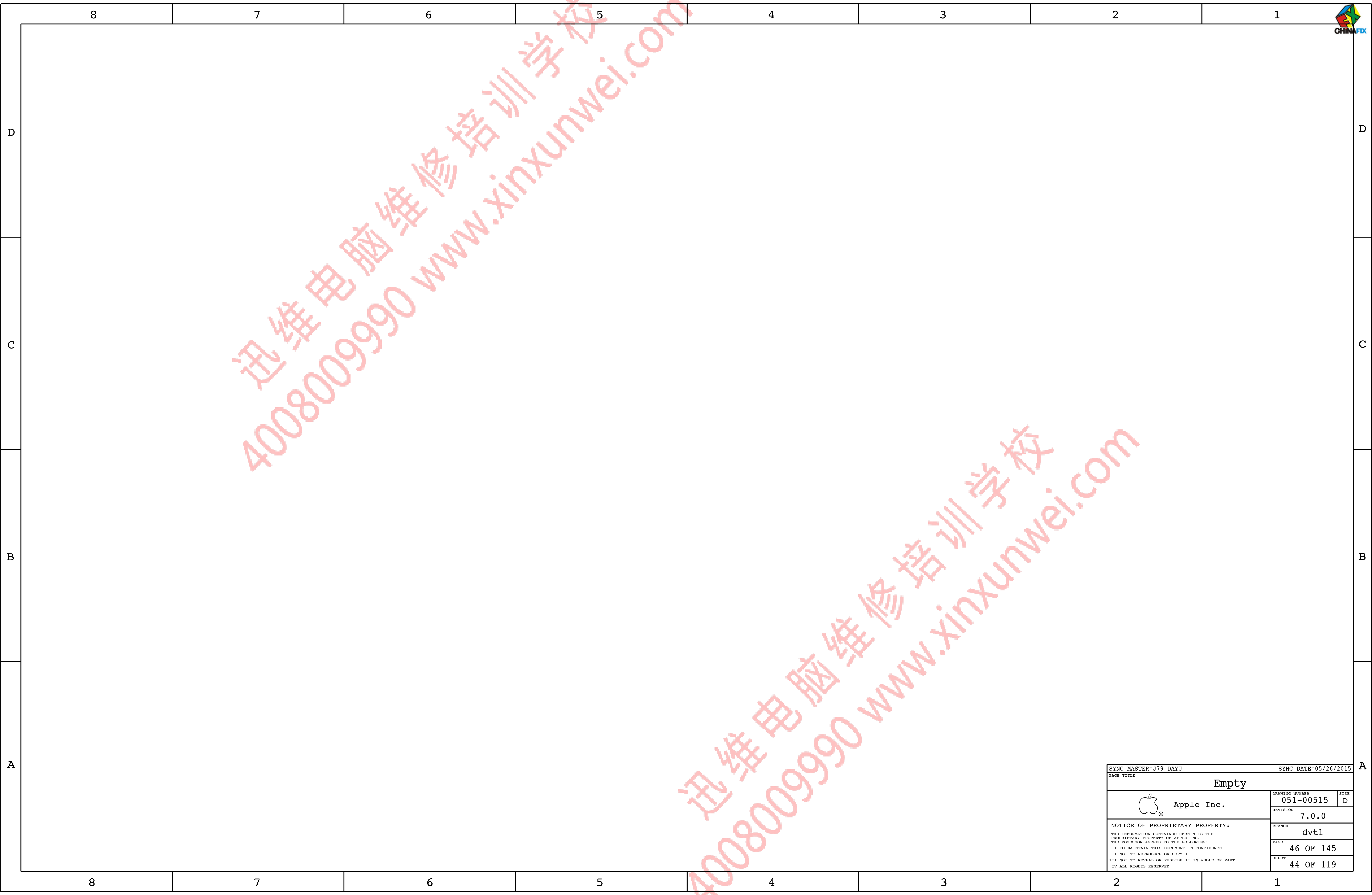
516S00187, MATE WITH 516S00188

TRACKPAD ISOLATION

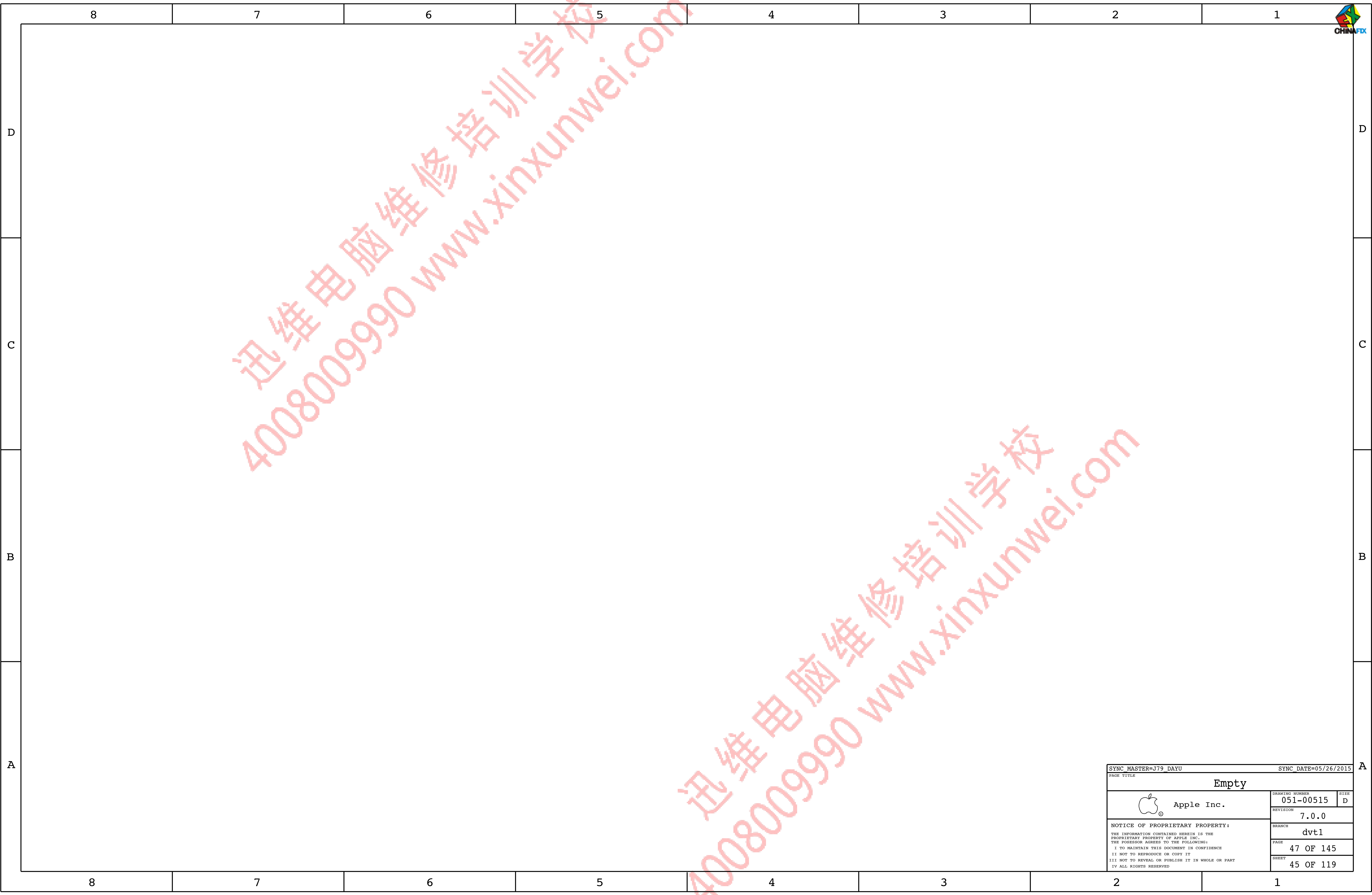


BOM_COST_GROUP=KEYBOARD

SYNC_MASTER=J79_GAREN		SYNC_DATE=11/21/2015	
PAGE TITLE			
Connectors&ESD			
	DRAWING NUMBER		SIE
	051-00515		D
Apple Inc.		REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	dvt1
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	45 OF 145
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	43 OF 119
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			



SYNC_MASTER=J79 DAYU		SYNC_DATE=05/26/2015	
PAGE TITLE			
Empty			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00515		D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	REVISION		7.0.0
	BRANCH		dvt1
	PAGE		46 OF 145
	SHEET		44 OF 119

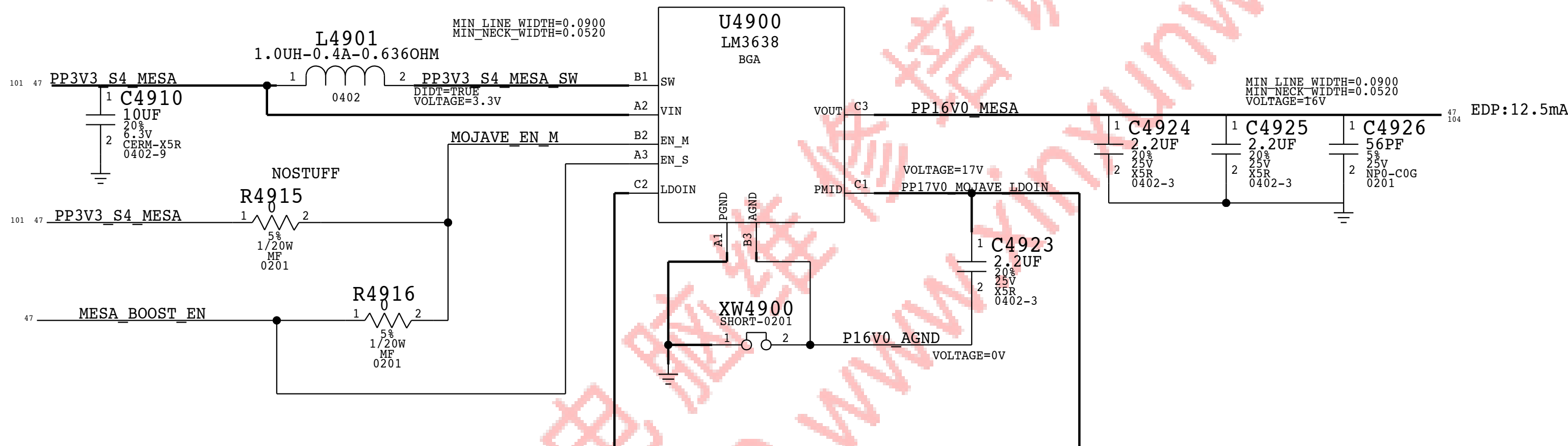


SYNC_MASTER=J79 DAYU		SYNC_DATE=05/26/2015	
PAGE TITLE			
Empty			
 Apple Inc.		DRAWING NUMBER	051-00515
		REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	dvt1
		PAGE	47 OF 145
		SHEET	45 OF 119



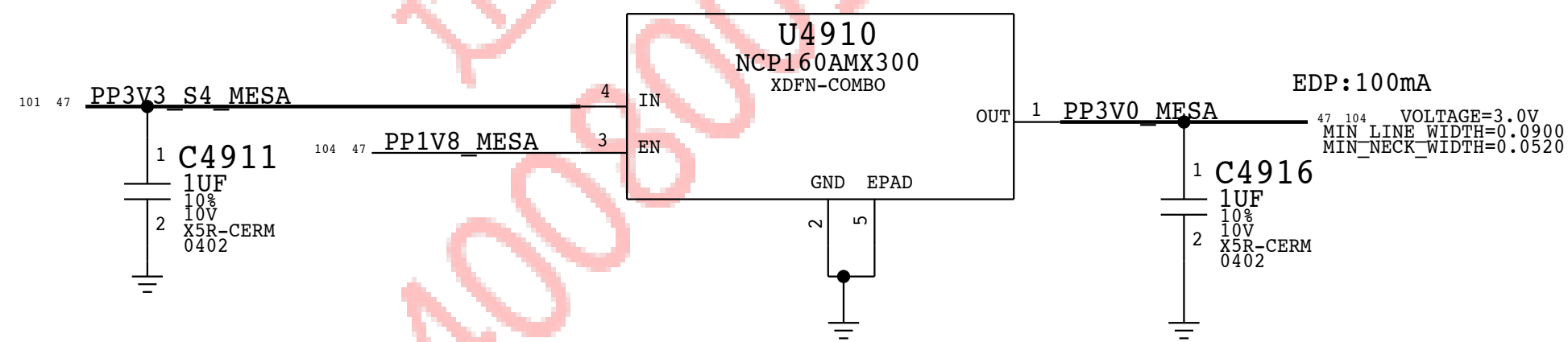
ISOLATE FROM OTHER COMPONENTS/NETS AS MUCH AS POSSIBLE

MOJAVE 16V BOOST

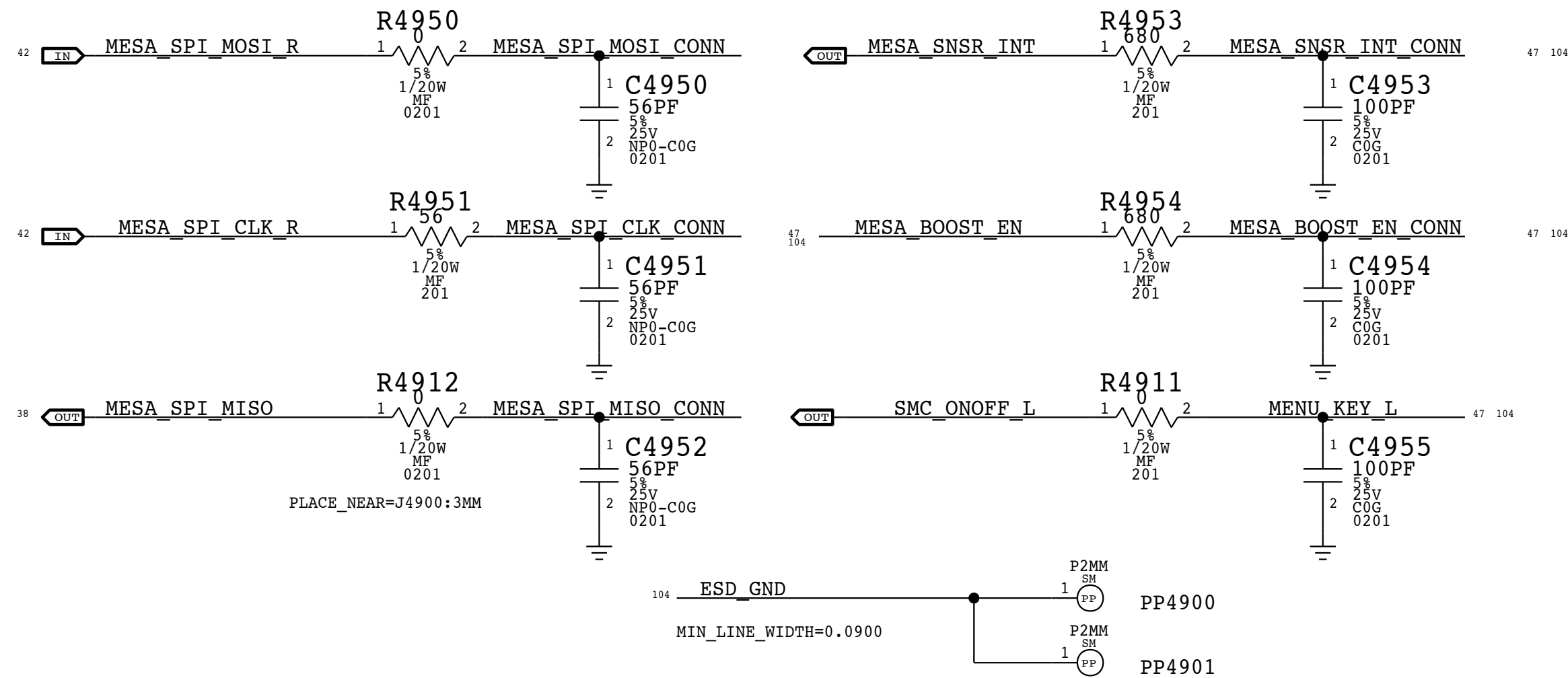
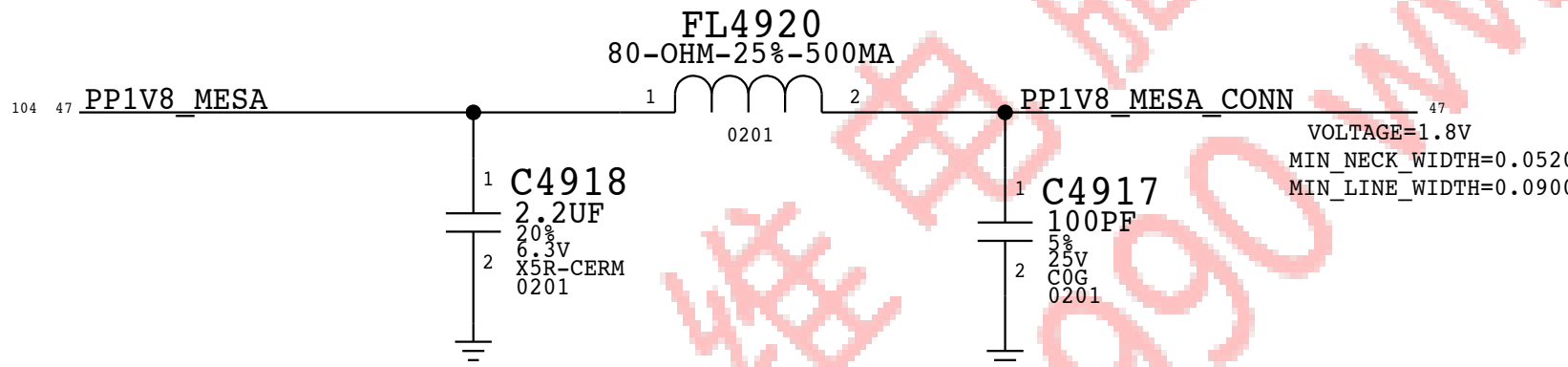
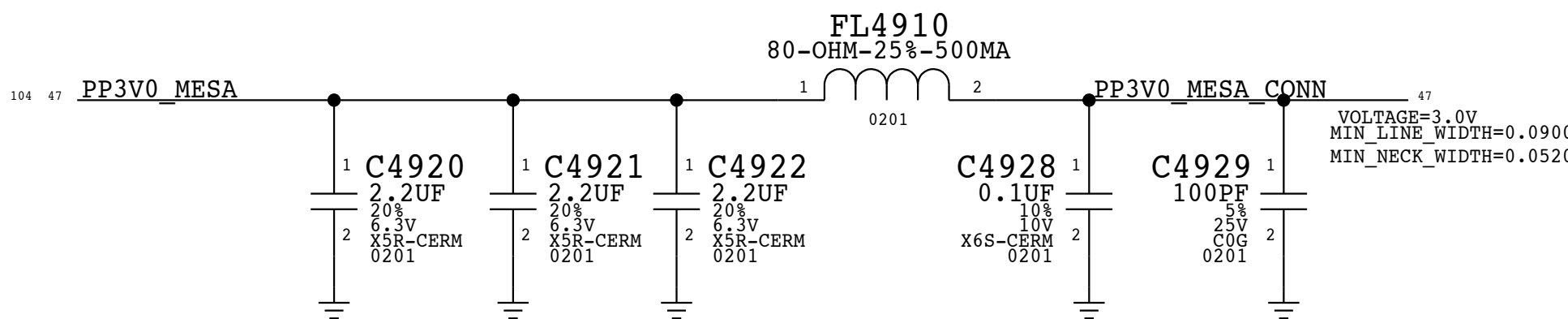
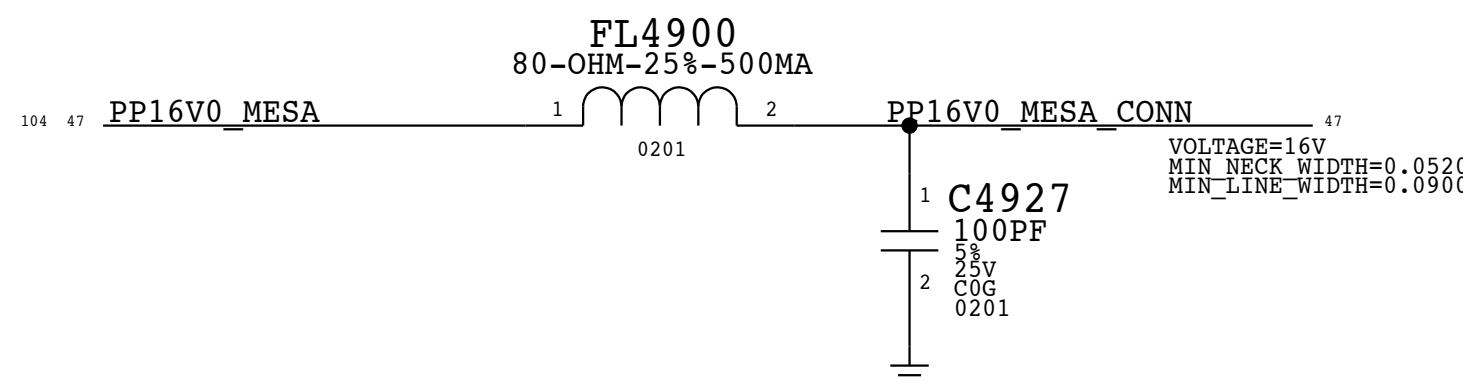
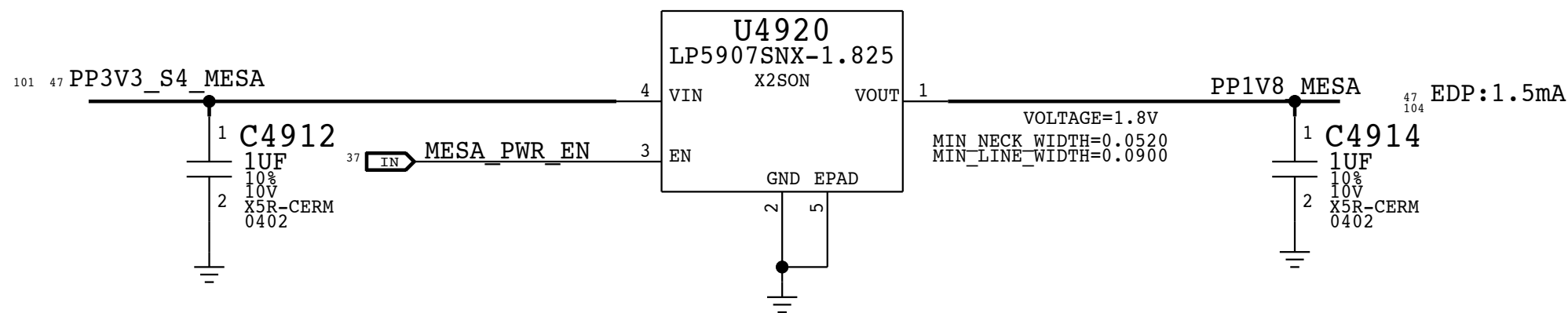


3.0V MESA

Option to feed LDO from 5V in case of dropout issue

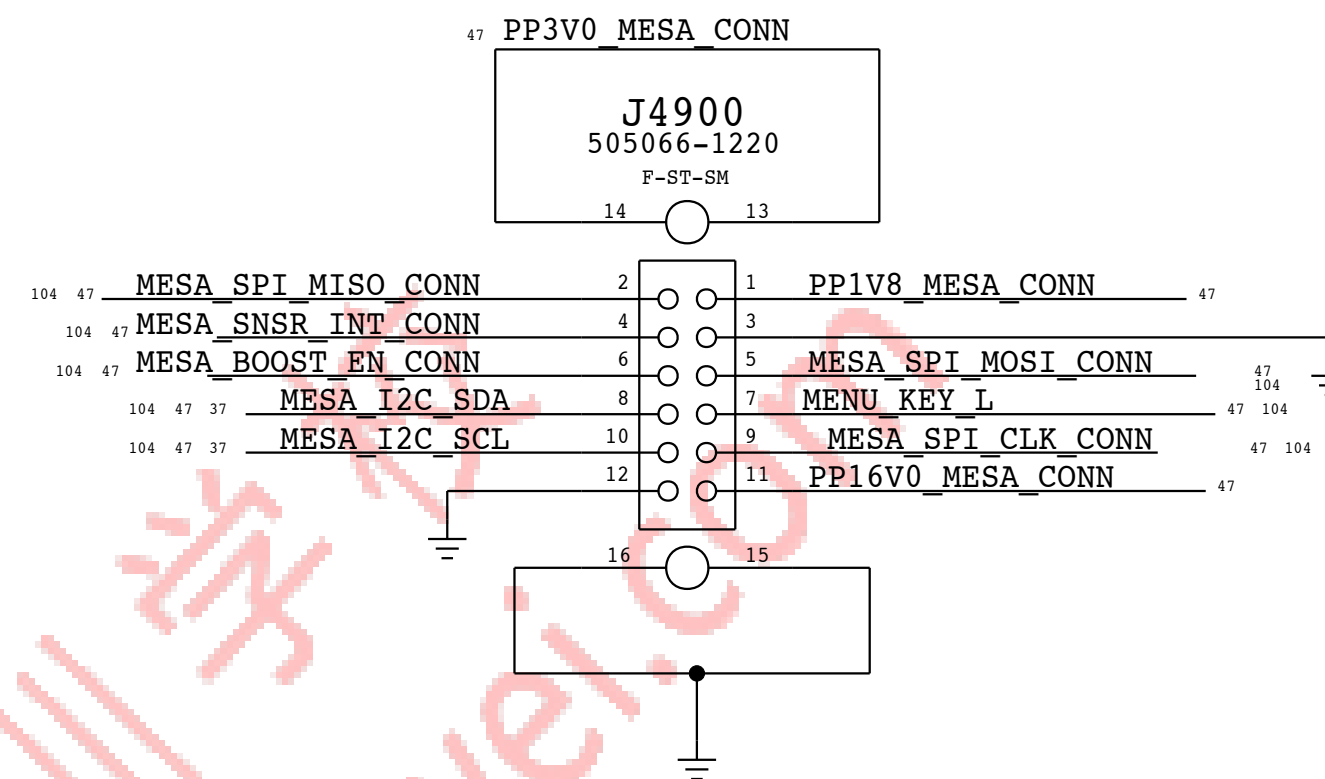


1.8V MESA



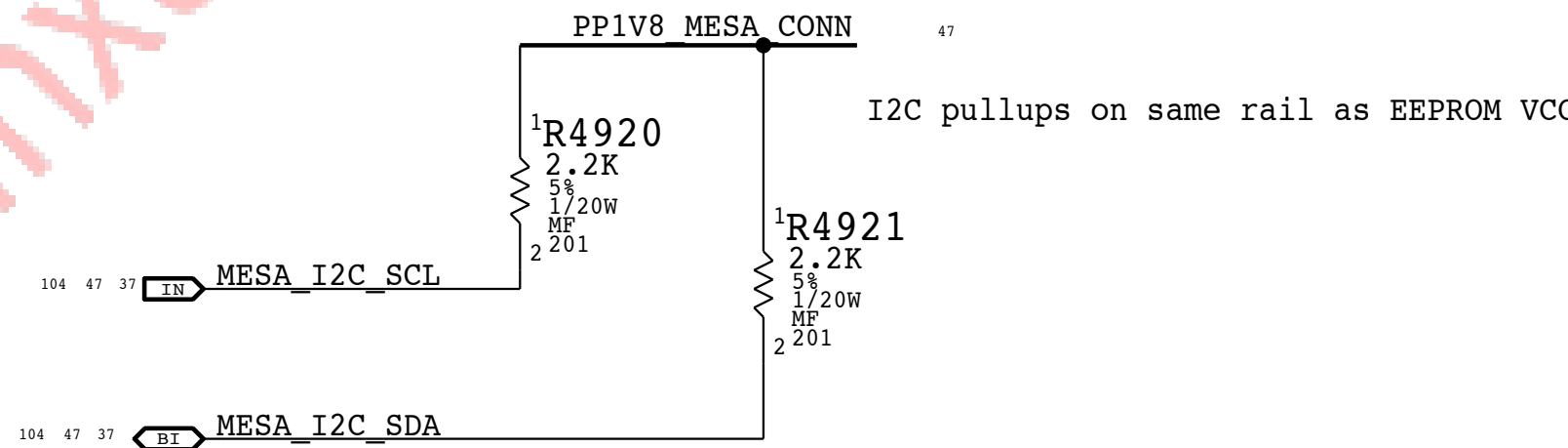
MESA FLEX CONNECTOR

Protol Connector for X434/X435 Support
PLUG (516S00115) - X434/ X435 Jumper
Recptacle (516S00203) - X362/X363 MLB



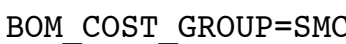
Mesa Power Sequencing Requirements

Power On: 1V8 -> 3V3 -> 16V0



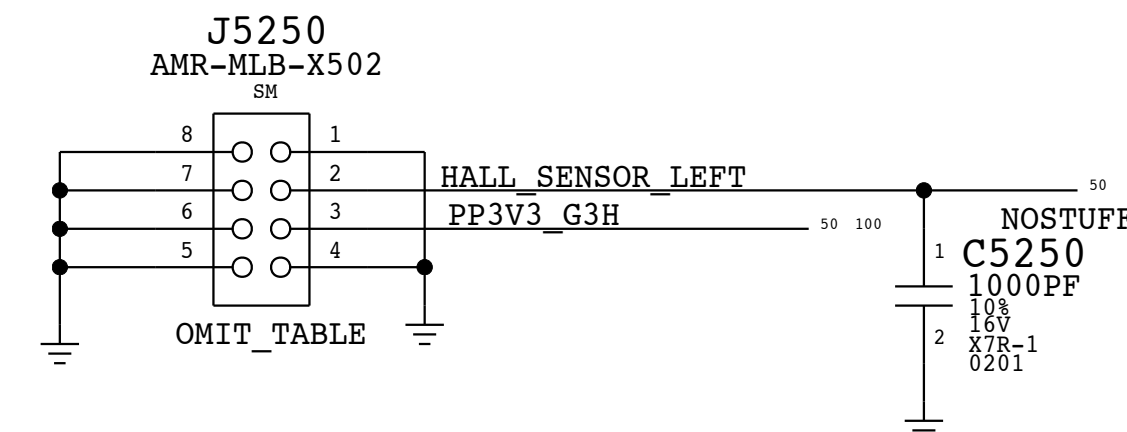
PAGE TITLE			
MESA			
	DRAWING NUMBER	051-00515	STEP
	REVISION	7.0.0	D
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	dv1
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE	49 OF 145
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET	47 OF 119
II NOT TO REPRODUCE OR COPY IT			
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
IV ALL RIGHTS RESERVED			

BOM_COST_GROUP=T151

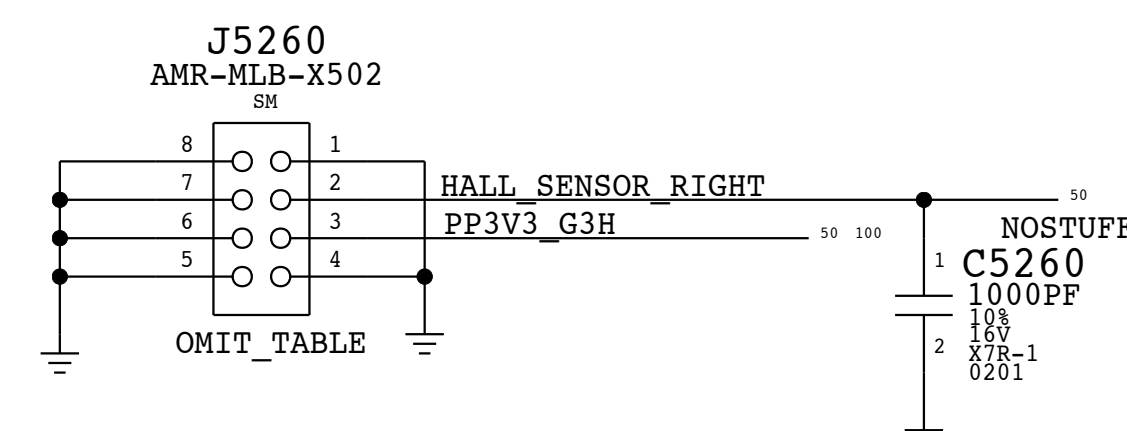




Hall Effect Pads - Left



Hall Effect Pads - Right



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
677-04255	2	SUBASSY (T&R) PCBA,HES INTERPOSER,X502	J5250, J5260	CRITICAL	

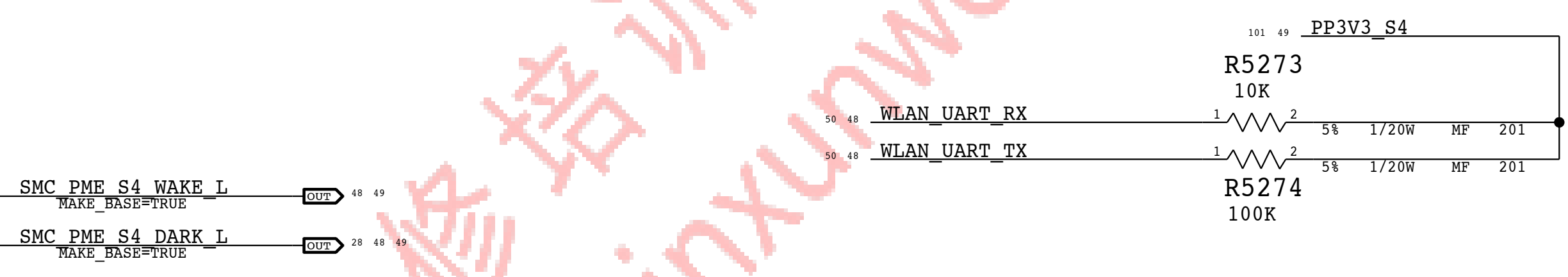
Specify one of these BOM GROUPS.

BOM GROUP	BOM OPTIONS
CPUTHRM:BOTH	CPUTHRM_THRM:SMC,CPUTHRM_ALRT:SMC
CPUTHRM:THRM	CPUTHRM_THRM:SMC,CPUTHRM_ALRT:PU
CPUTHRM:ALRT	CPUTHRM_ALRT:SMC
CPUTHRM:NONE	CPUTHRM_ALRT:PU

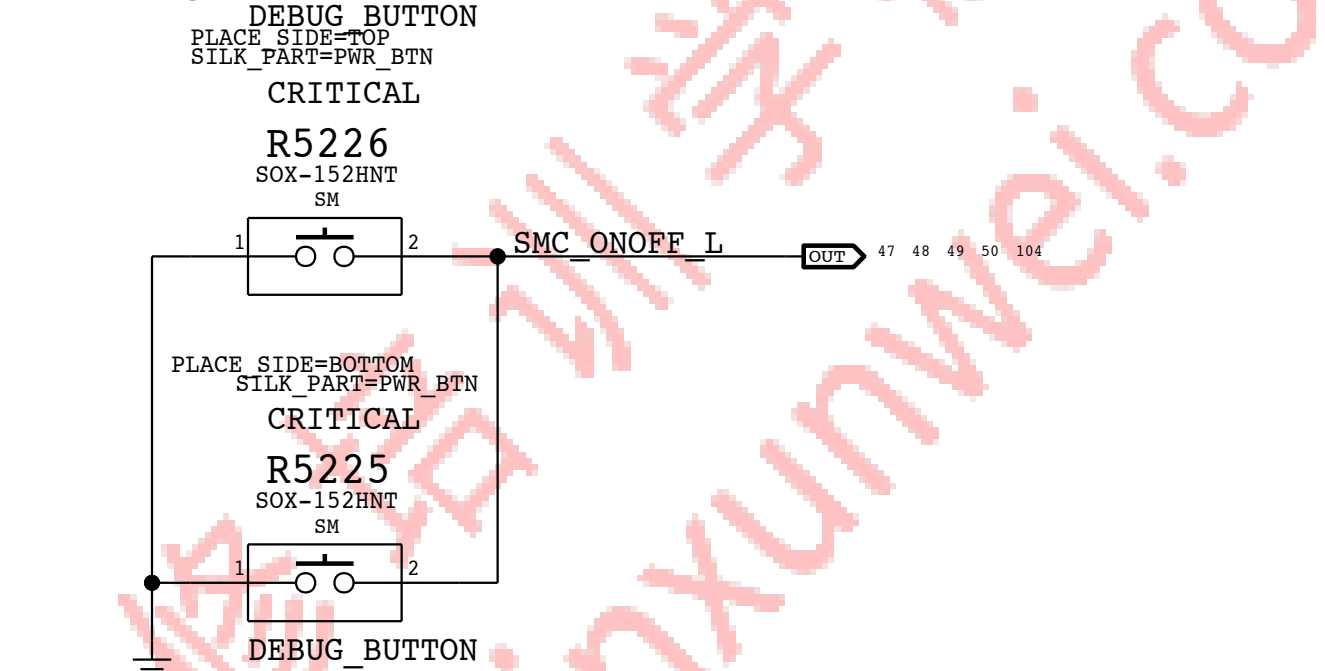
Specify one of these BOM GROUPS.

BOM GROUP	BOM OPTIONS
TBTTHRM:BOTH	TBTTHRM_THRM:SMC,TBTTHRM_ALRT:SMC
TBTTHRM:THRM	TBTTHRM_THRM:SMC,TBTTHRM_ALRT:PU
TBTTHRM:ALRT	TBTTHRM_THRM:PU,TBTTHRM_ALRT:SMC
TBTTHRM:NONE	TBTTHRM_THRM:PU,TBTTHRM_ALRT:PU

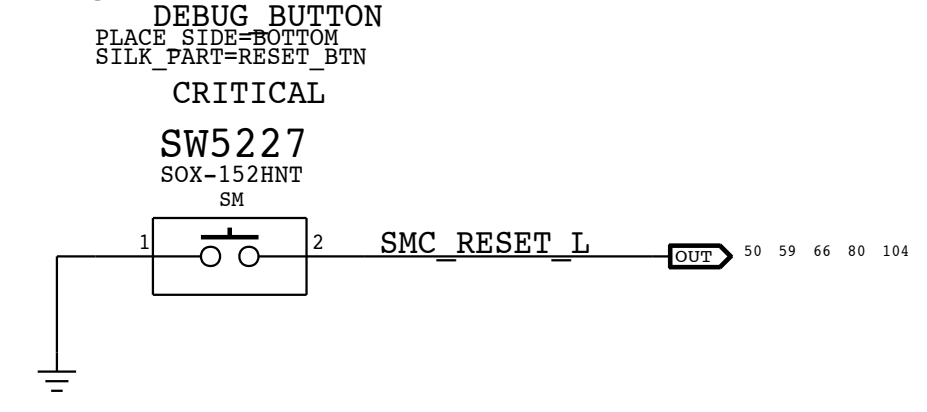
Requires EMC1412-1 or EMC1412-2 instead of EMC1412-A, new APN needs to be created.



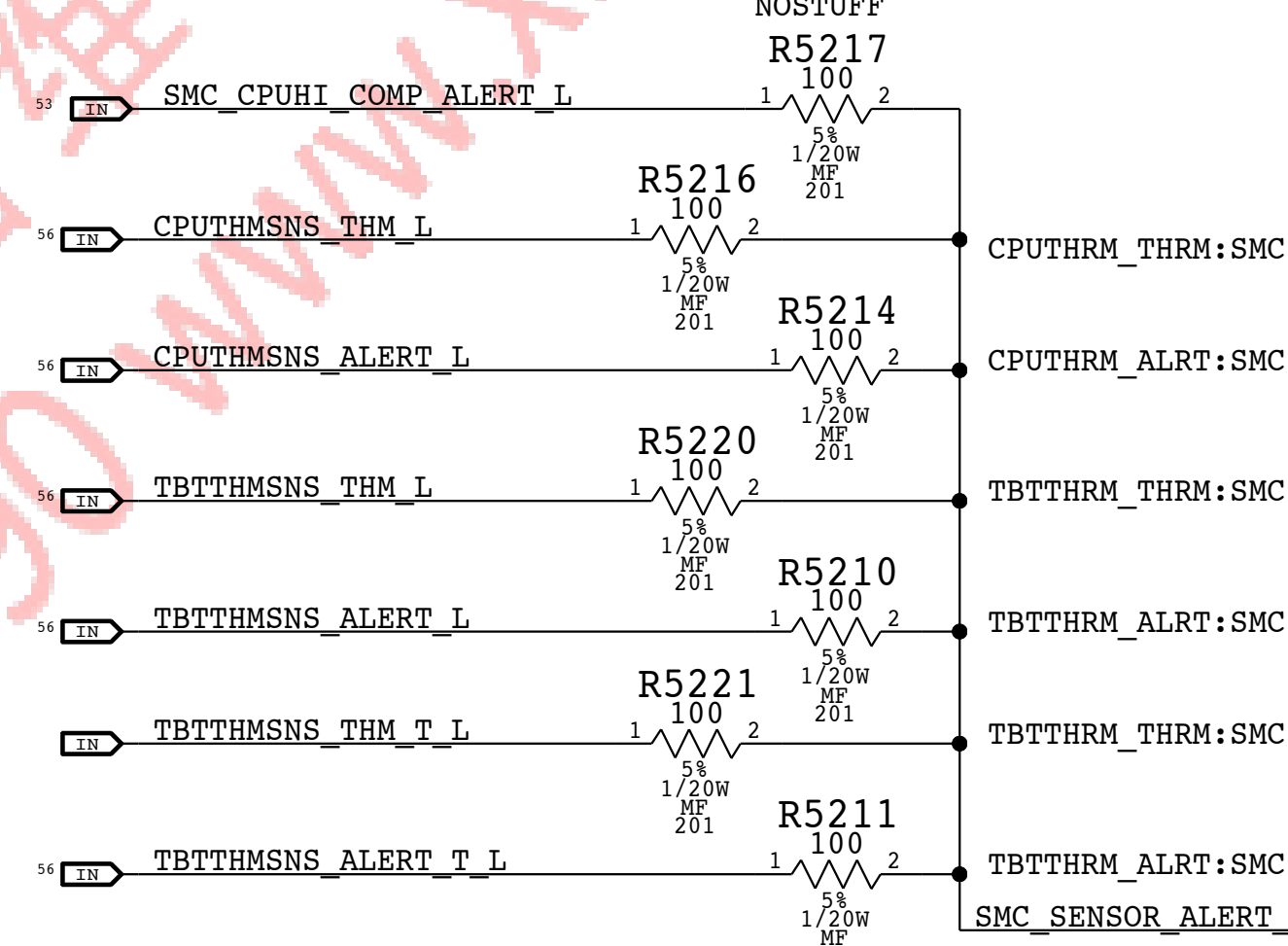
Debug Power "Buttons"



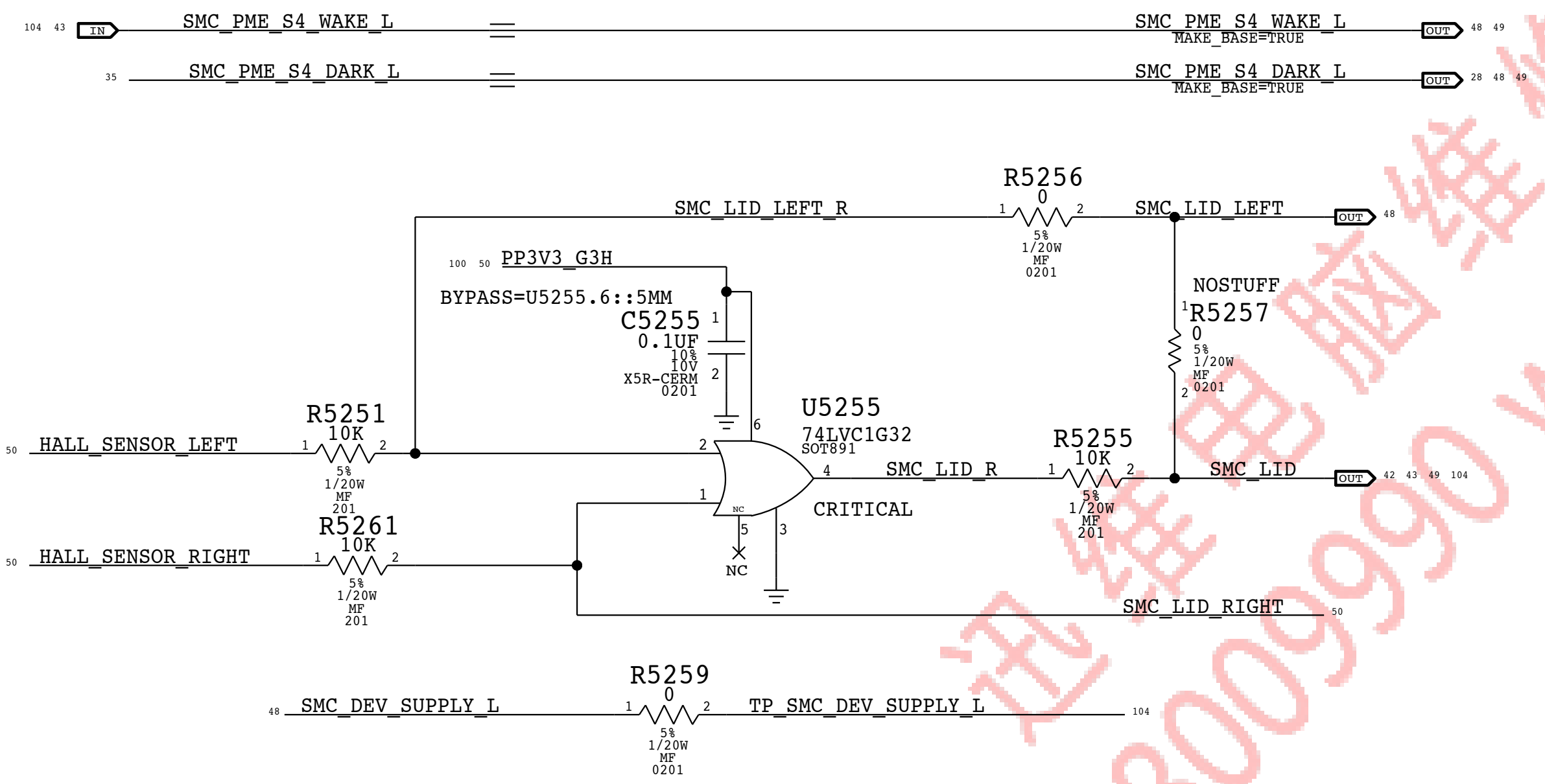
Debug RESET "Buttons"



Thermal Alerts



S4 SMC Wake Sources



SMC12 ADC Assignments

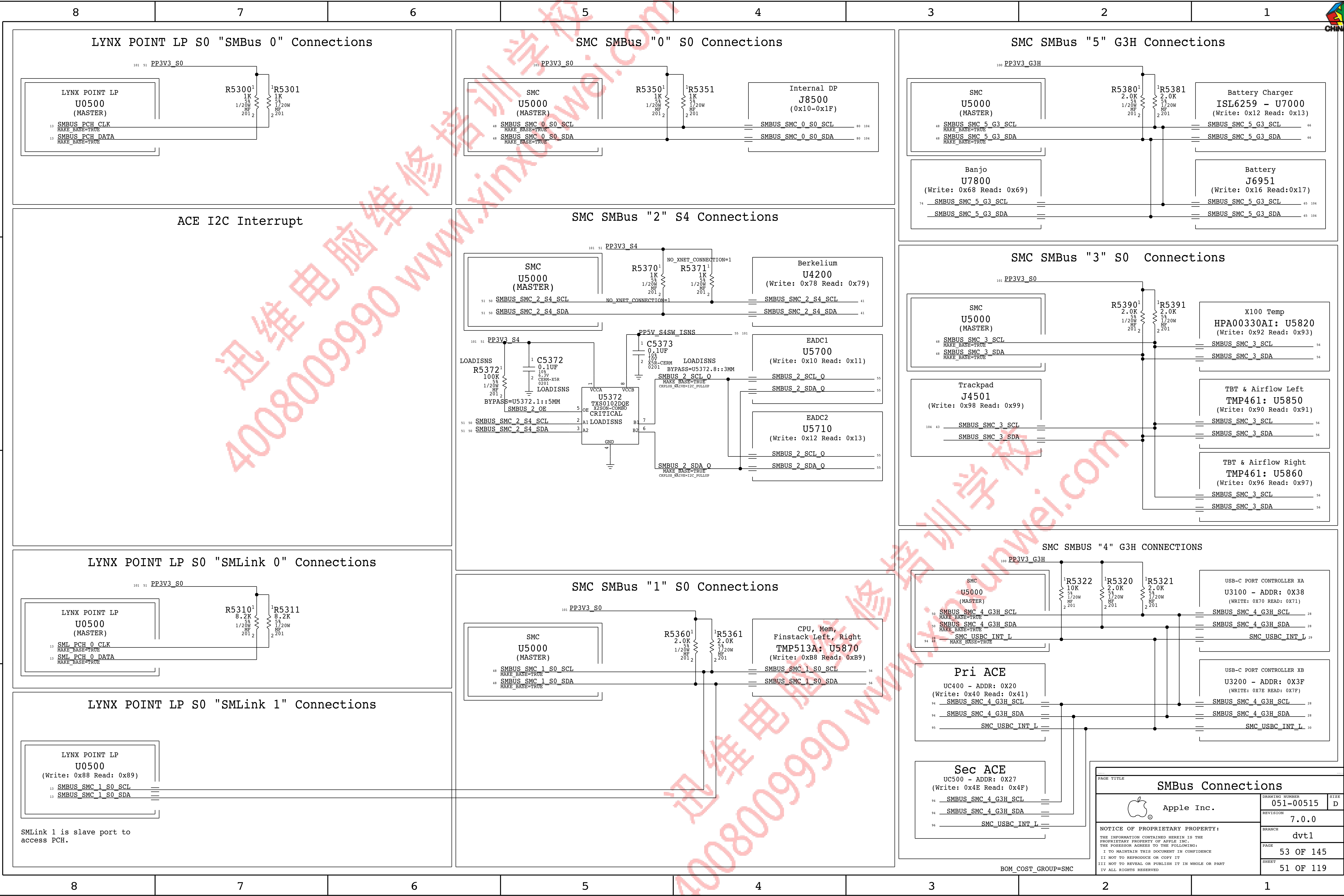
SMC_CPU_HI_ISENSE	SMC_CPU_HI_ISENSE	32
SMC_PBUS_VSENSE	SMC_PBUS_VSENSE	32
SMC_BMON_ISENSE	SMC_BMON_ISENSE	32
SMC_DCIN_ISENSE	SMC_DCIN_ISENSE	32
SMC_DCIN_VSENSE	SMC_DCIN_VSENSE	32
SMC_CPUTGT_ISENSE	SMC_CPUTGT_ISENSE	35
SMC_CPU_ISENSE	SMC_CPU_ISENSE	33
SMC_OTHER5V_HI_ISENSE	SMC_OTHER5V_HI_ISENSE	32
SMC_OTHER3V3_HI_ISENSE	SMC_OTHER3V3_HI_ISENSE	32
SMC_DDR1V2_ISENSE	SMC_DDR1V2_ISENSE	33
SMC_CPUEDRAM_ISENSE	SMC_CPUEDRAM_ISENSE	35
SMC_PCH_ISENSE	SMC_PCH_ISENSE	33
SMC_TPAD_ISENSE	SMC_TPAD_ISENSE	32
SMC_PICCOLO_ISENSE	SMC_PICCOLO_ISENSE	35
SMC_SSDNAND_ISENSE	SMC_SSDNAND_ISENSE	35
SMC_PCHPRIMCORE_ISENSE	SMC_PCHPRIMCORE_ISENSE	35
SMC_DDR1V8_ISENSE	SMC_DDR1V8_ISENSE	35
SMC_CPUSA_ISENSE	SMC_CPUSA_ISENSE	35
SMC_CPUDDR_ISENSE	SMC_CPUDDR_ISENSE	33
SMC_CPUSA_VSENSE	SMC_CPUSA_VSENSE	35
SMC_CPU_VSENSE	SMC_CPU_VSENSE	34
SMC_CPUTGT_VSENSE	SMC_CPUTGT_VSENSE	34
SMC_CPU_IMON_ISENSE	SMC_CPU_IMON_ISENSE	34
SMC_CPUTGT_IMON_ISENSE	SMC_CPUTGT_IMON_ISENSE	34

SMC12 Pin Assignments

NC_SMC_GFX_THROTTLE_L	NC_SMC_GFX_THROTTLE_L	NO_TEST=1	
NC_SMC_GFX_OVERTEMP	NC_SMC_GFX_OVERTEMP	NO_TEST=1	
NC_SMC_GFX_SELF_THROTTLE	NC_SMC_GFX_SELF_THROTTLE	NO_TEST=1	
NC_SMC_DP_HPD_L	NC_SMC_DP_HPD_L	NO_TEST=1	
NC_SYS_ONEWIRE	NC_SYS_ONEWIRE	NO_TEST=1	TRUE
NC_SMC_DEBUGPRT_EN_L	NC_SMC_DEBUGPRT_EN_L	NO_TEST=1	TRUE
SMC_LID_RIGHT	SMC_LID_RIGHT		50
NC_SMC_SPI_SMC_MISO	NC_SMC_SPI_SMC_MISO	NO_TEST=1	TRUE
NC_SMC_SPI_SMC_MOSI	NC_SMC_SPI_SMC_MOSI	NO_TEST=1	TRUE
NC_SMC_SPI_SMC_CLK	NC_SMC_SPI_SMC_CLK	NO_TEST=1	TRUE
NC_SMC_SPI_SMC_CS_L	NC_SMC_SPI_SMC_CS_L	NO_TEST=1	TRUE
SMC_VIBE_L	SMC_VIBE_L		43
SMC_PCH_SUSWARN_L	SMC_PCH_SUSWARN_L		14
SMC_PCH_SUSACK_L	SMC_PCH_SUSACK_L		14
SMC_SENSOR_PWR_EN	SMC_SENSOR_PWR_EN		78
WLAN_UART_RX	WLAN_UART_RX		35
WLAN_UART_TX	WLAN_UART_TX		35
SMC_AUX_OK	SMC_AUX_OK		49
PM_PWRBTN_L	PM_PWRBTN_L		14
SMC_AUX_OK	SMC_AUX_OK		52
SMBUS_SMC_4_G3H_SCL	SMBUS_SMC_4_G3H_SCL		48
SMBUS_SMC_4_G3H_SDA	SMBUS_SMC_4_G3H_SDA		48
SMBUS_SMC_2_S4_SCL	SMBUS_SMC_2_S4_SCL		48
SMBUS_SMC_2_S4_SDA	SMBUS_SMC_2_S4_SDA		48
SMC_OOB1_D2R_L	SMC_OOB1_D2R_L		88
SMC_OOB1_R2D_L	SMC_OOB1_R2D_L		88
SMC_ACTUATOR_DISABLE_L	SMC_ACTUATOR_DISABLE_L		43
SMC_CHGR_INT_L	SMC_CHGR_INT_L		66

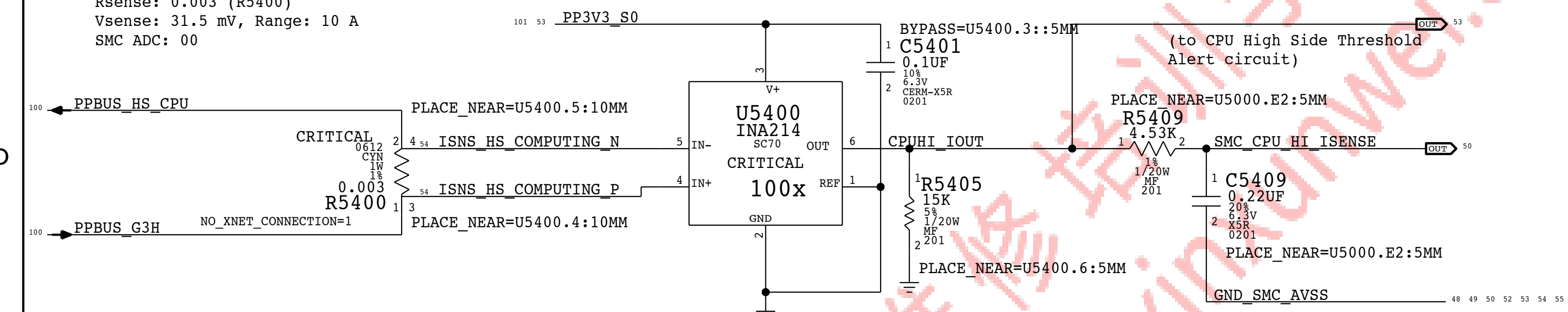
PAGE TITLE		
PAGE_TITLE=SMC Project Support		
DRAWING NUMBER		051-00515
REVISION		7.0.0
BRANCH		dvt1
PAGE		52 OF 145
SHEET		50 OF 119
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		

BOM_COST_GROUP=SMC



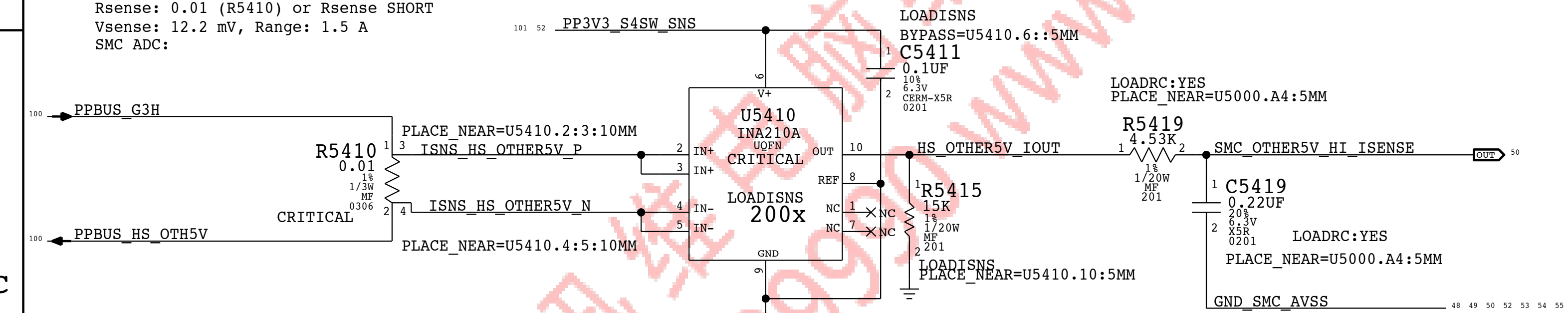
CPU High Side Current Sense (IC0R)

Gain: 100x, EDP: 10.5 A
Rsense: 0.003 (R5400)
Vsense: 31.5 mV, Range: 10 A
SMC ADC: 00



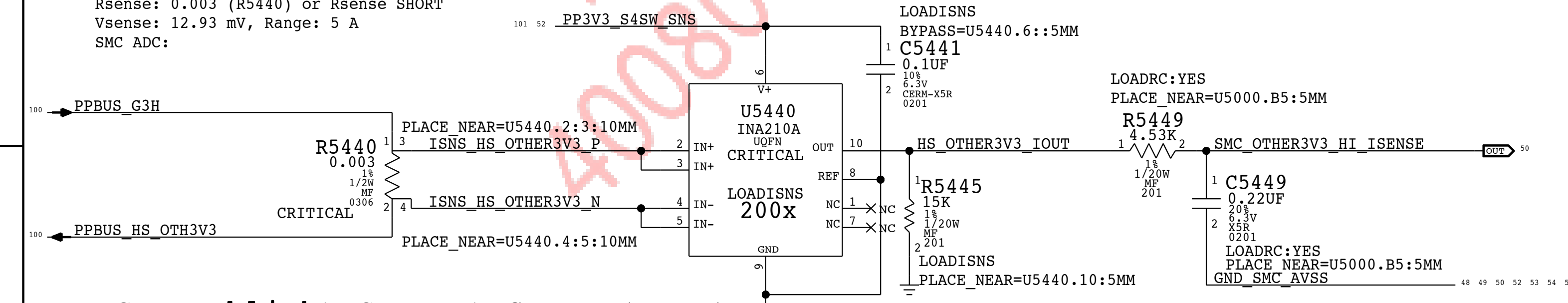
OTHER 5V High Side Current Sense (IO5R)

Gain: 200x, EDP: 1.22 A
Rsense: 0.01 (R5410) or Rsense SHORT
Vsense: 12.2 mV, Range: 1.5 A
SMC ADC:



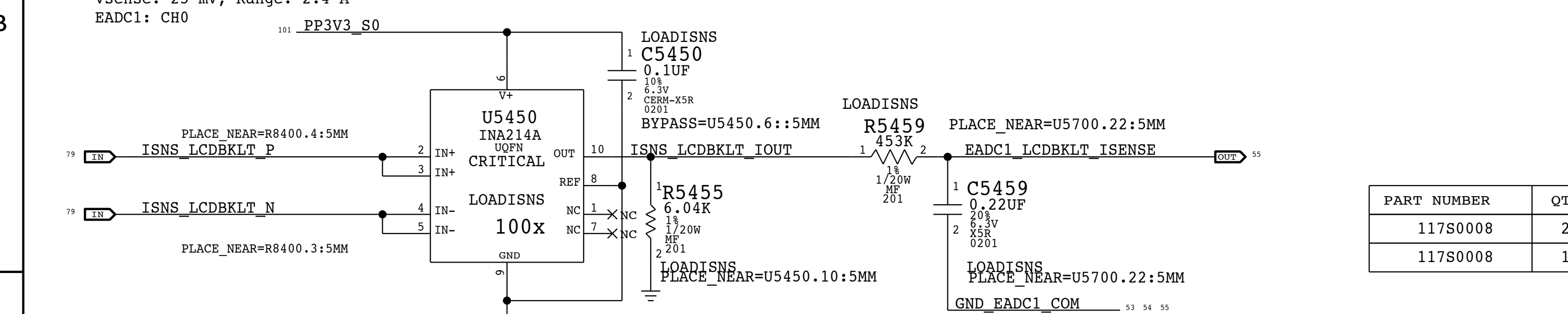
OTHER 3.3V High Side Current Sense (IO3R)

Gain: 200x, EDP: 4.31 A
Rsense: 0.003 (R5440) or Rsense SHORT
Vsense: 12.93 mV, Range: 5 A
SMC ADC:



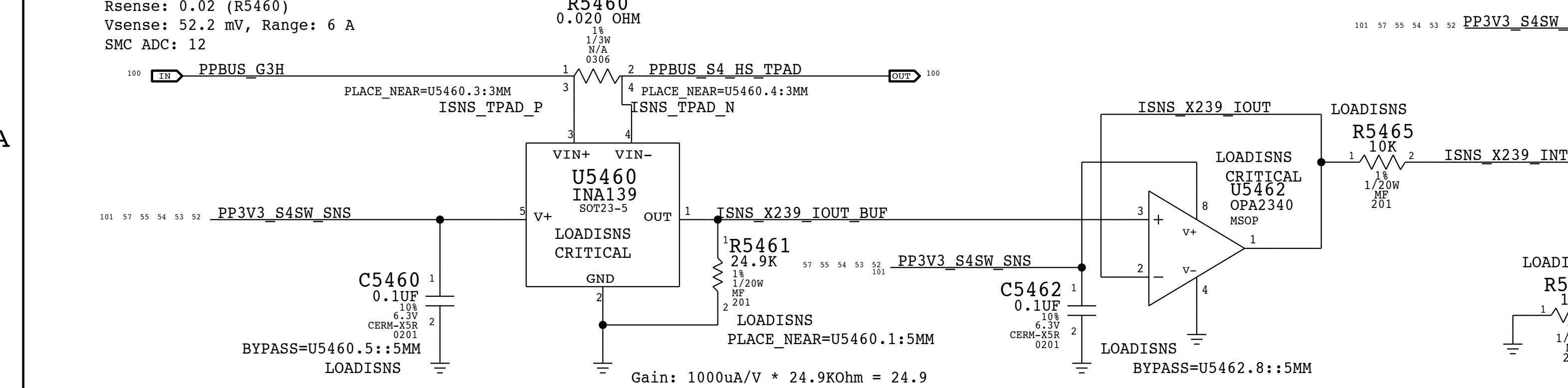
LCD Backlight Current Sense (IBLR)

Gain: 100x. EDP: 1 A
Rsense: 0.025 (R8400)
Vsense: 25 mV, Range: 2.4 A
EADC1: CH0



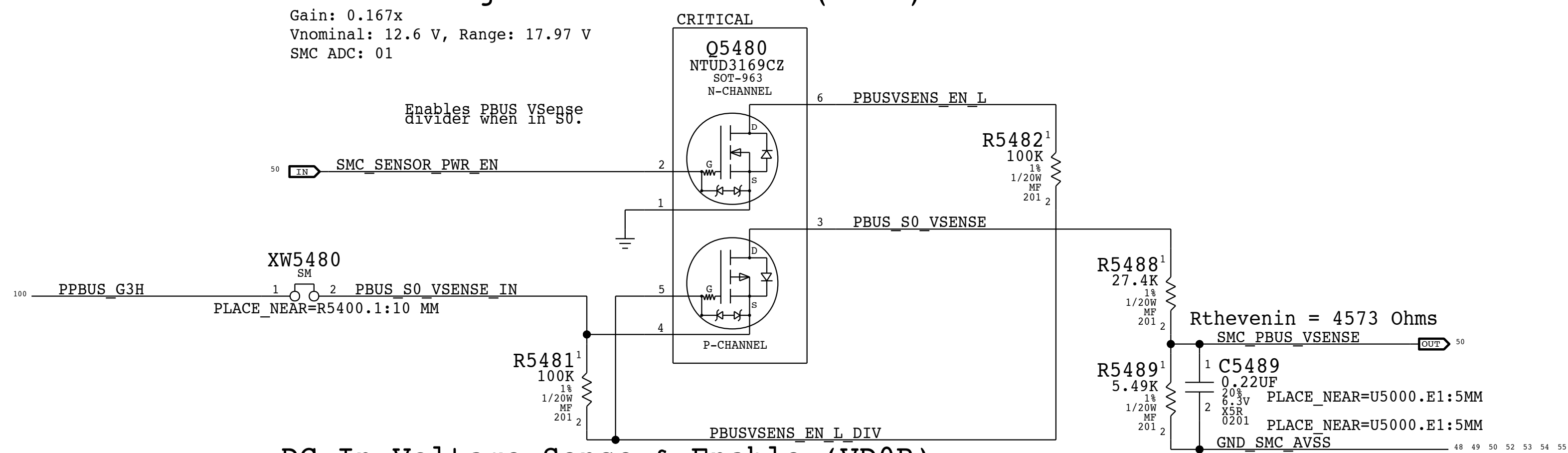
Trackpad Actuator X239 Current Sense (ITAR)

Gain: 24.9x, EDP: 2.61 A (Transient)
Rsense: 0.02 (R5460)
Vsense: 52.2 mV, Range: 6 A
SMC ADC: 12



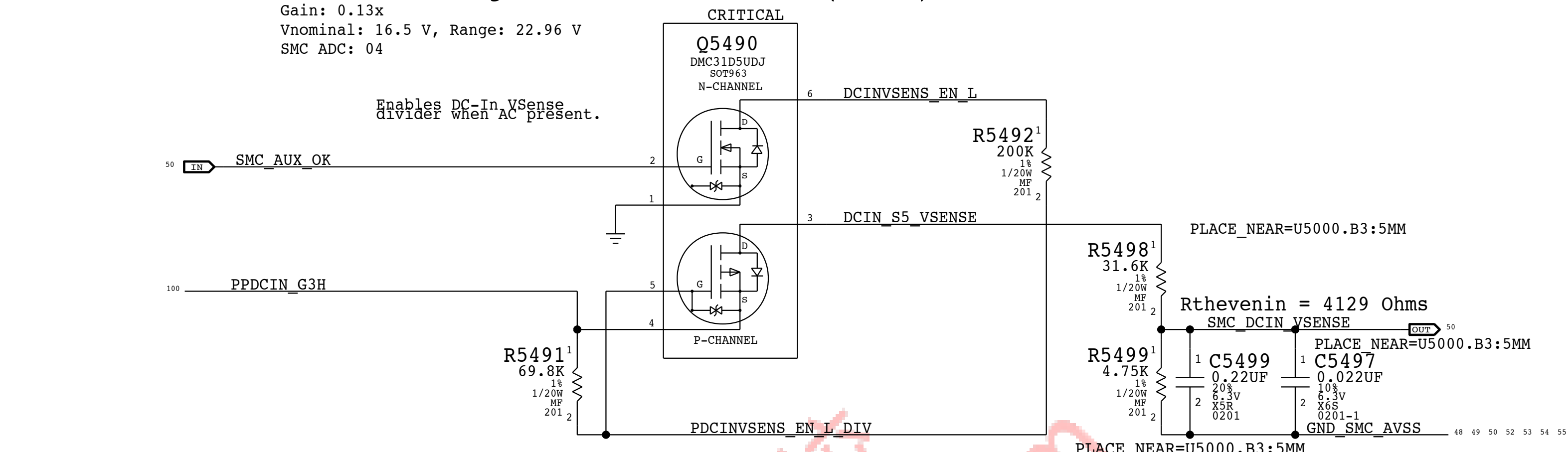
PBUS Voltage Sense & Enable (VP0R)

Gain: 0.167x
Vnominal: 12.6 V, Range: 17.97 V
SMC ADC: 01



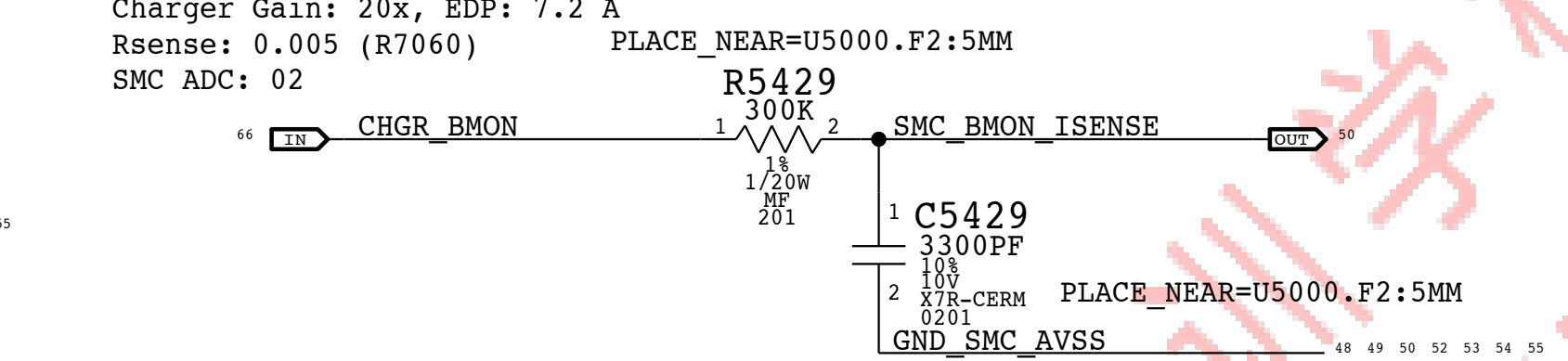
DC In Voltage Sense & Enable (VD0R)

Gain: 0.13x
Vnominal: 16.5 V, Range: 22.96 V
SMC ADC: 04



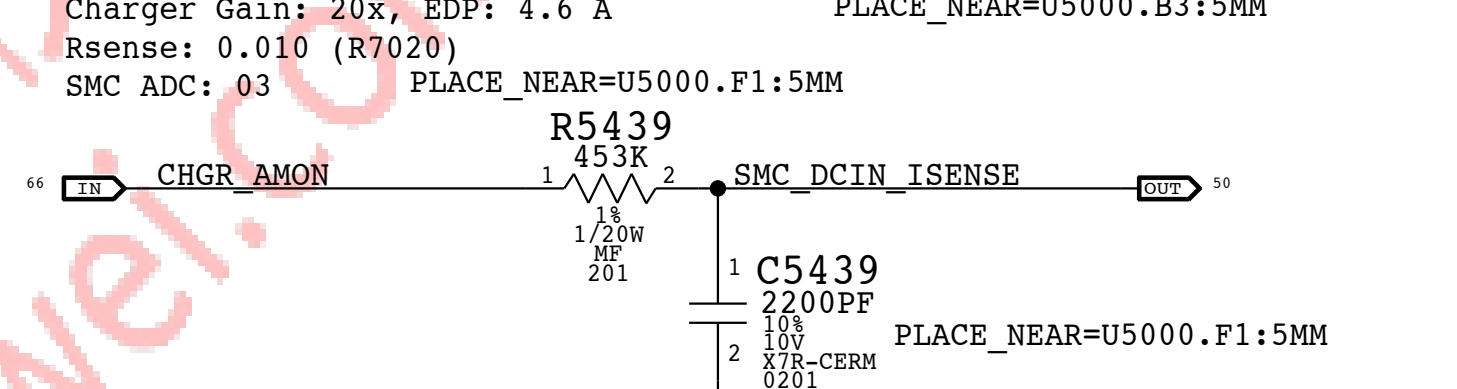
Charger (BMON) Current Sense (IPBR)

Charger Gain: 20x, EDP: 7.2 A
Rsense: 0.005 (R7060)
SMC ADC: 02



DC-IN (AMON) Current Sense (ID0R)

Charger Gain: 20x, EDP: 4.6 A
Rsense: 0.010 (R7020)
SMC ADC: 03



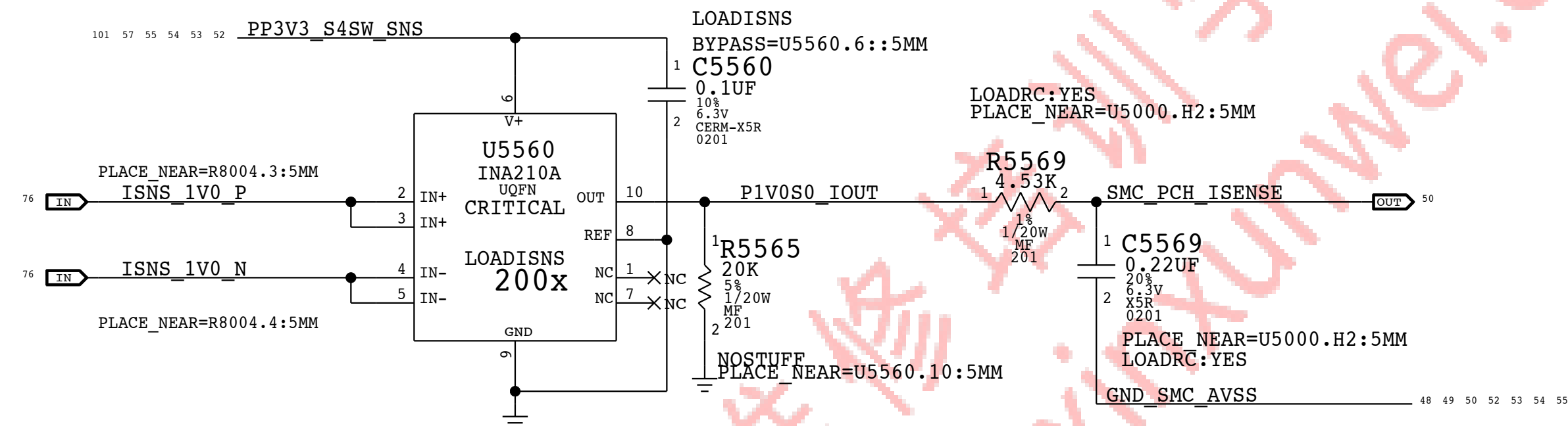
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	2	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5419,C5449		LOADRC:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5469		LOADRC:NO

Power Sensors: High Side		
	DRAWING NUMBER	051-00515
	REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I ALL RIGHTS RESERVED	BRANCH	dvt1
	PAGE	54 OF 145
BOM_COST_GROUP=SENSORS		SHEET 52 OF 119



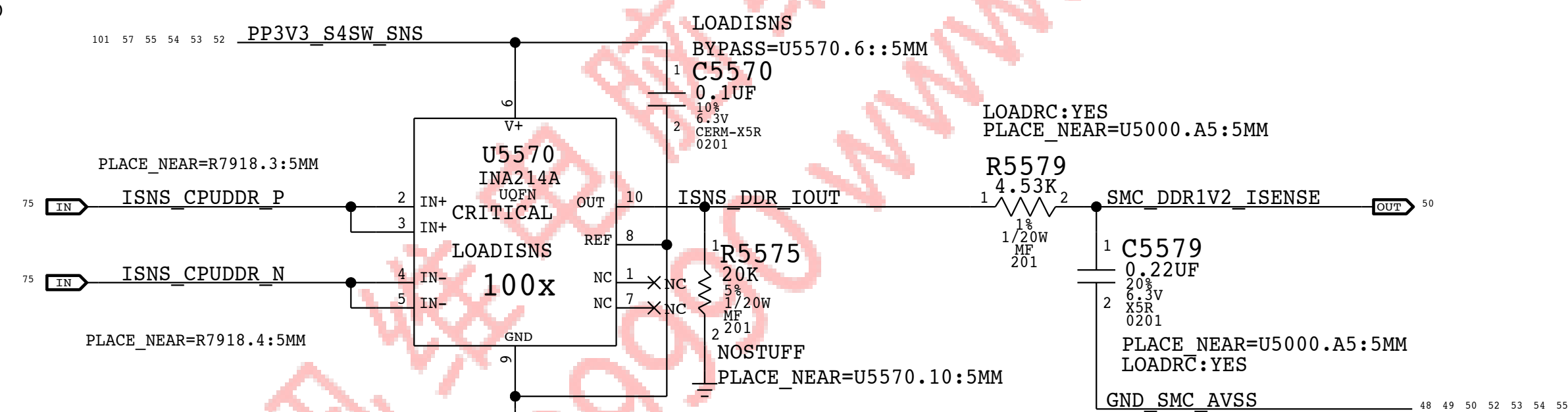
PCH 1.0V Current Sense (IS1C)

Gain: 200x, EDP: 3.29 A
Rsense: 0.003 (R8004) or Rsense SHORT
Vsense: 9.87 mV, Range: 5 A
SMC ADC: 11



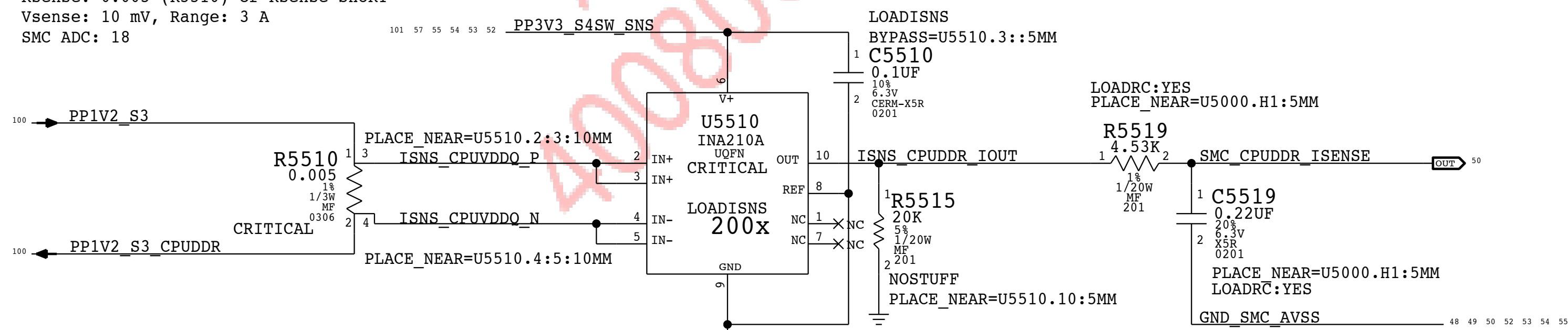
DDR 1.2V S3 (CPU & Memory) Current Sense (IM0C)

Gain: 100x, EDP: 8.21 A
Rsense: 0.003 (R7918) or XWTBD
Vsense: 24.63 mV, Range: 10 A
SMC ADC: 09



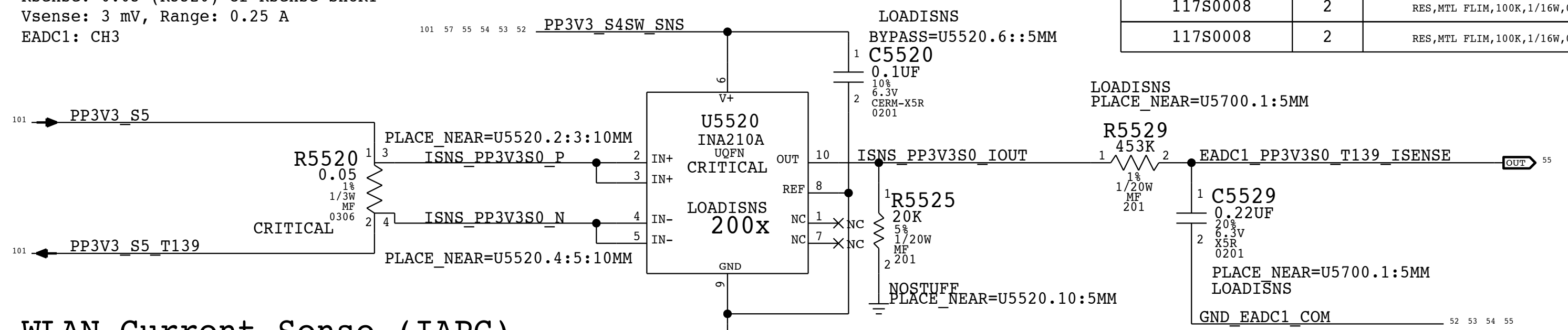
CPU DDR 1.2V S3 (CPU Only) Current Sense (IMCC)

Gain: 200x, EDP: 2 A
Rsense: 0.005 (R5510) or Rsense SHORT
Vsense: 10 mV, Range: 3 A
SMC ADC: 18



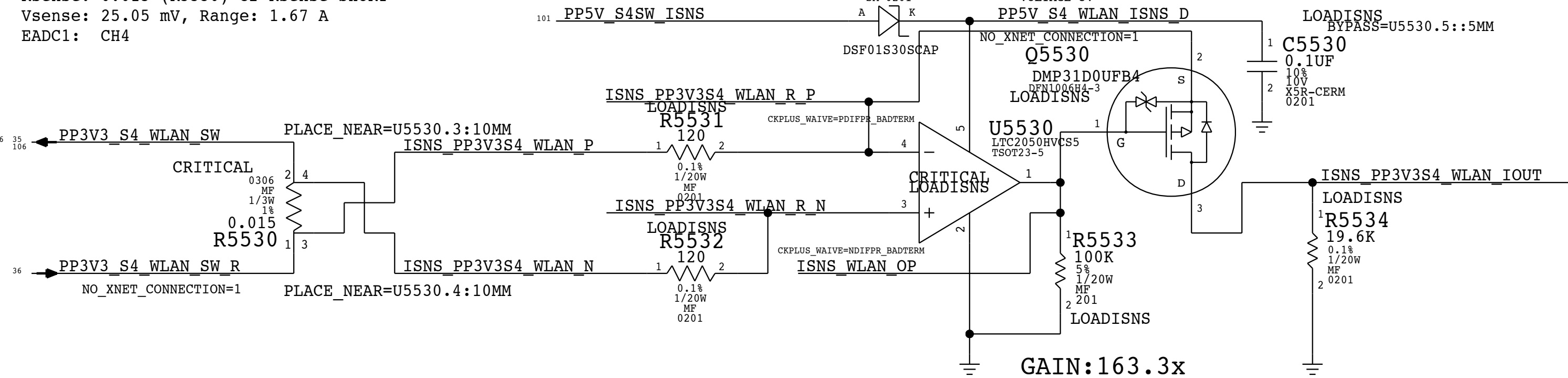
T139 Current Sense (IF3C)

Gain: 200x, EDP: 0.06 A
Rsense: 0.05 (R5520) or Rsense SHORT
Vsense: 3 mV, Range: 0.25 A
EADC1: CH3



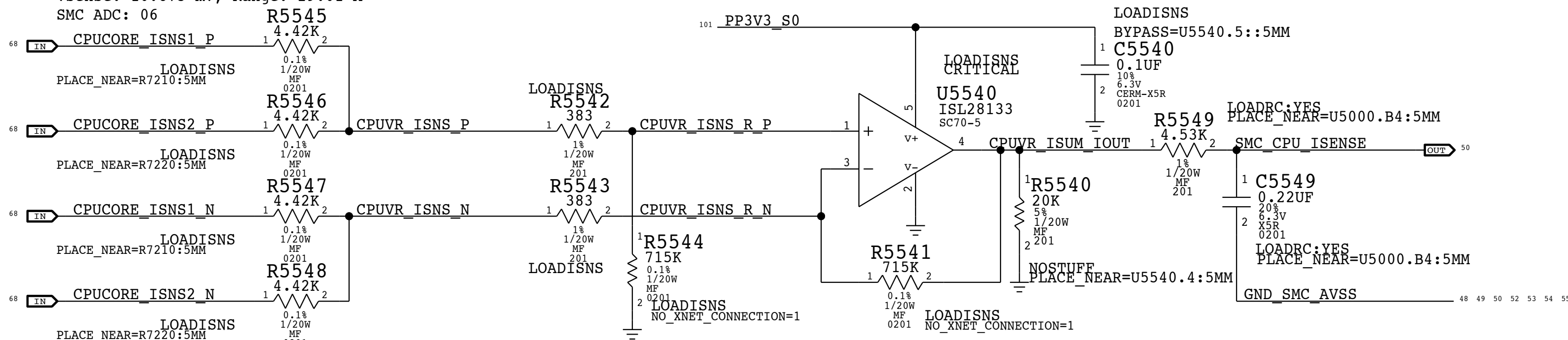
WLAN Current Sense (IAPC)

Gain: 163.3x, EDP: 1.67 A
Rsense: 0.015 (R5530) or Rsense SHORT
Vsense: 25.05 mV, Range: 1.67 A
EADC1: CH4



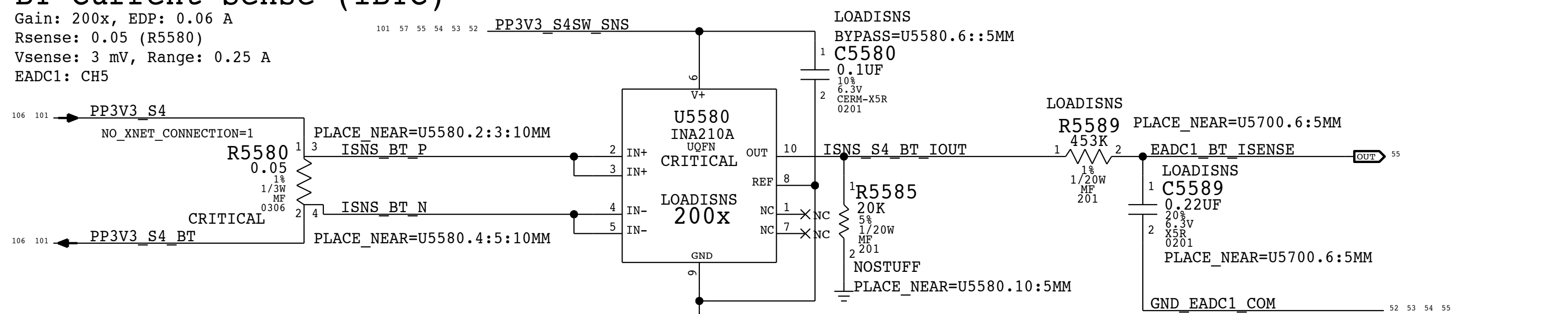
CPU Fixed Current Sense (ICAC)

Gain: 275.74x, EDP: 29 A
Rsense: 2x of 0.00075 (R7310, R7320), Rsum: 0.000375
Vsense: 10.875 mV, Range: 29.01 A
SMC ADC: 06



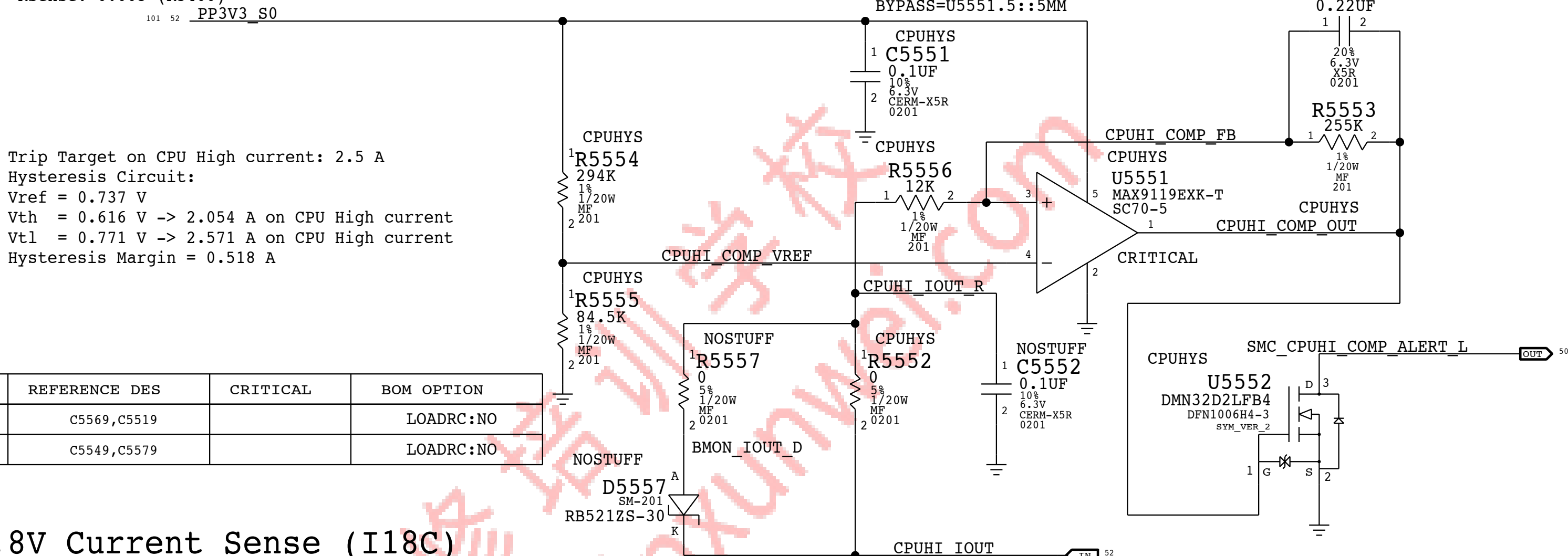
BT Current Sense (IBTC)

Gain: 200x, EDP: 0.06 A
Rsense: 0.05 (R5580)
Vsense: 3 mV, Range: 0.25 A
EADC1: CH5



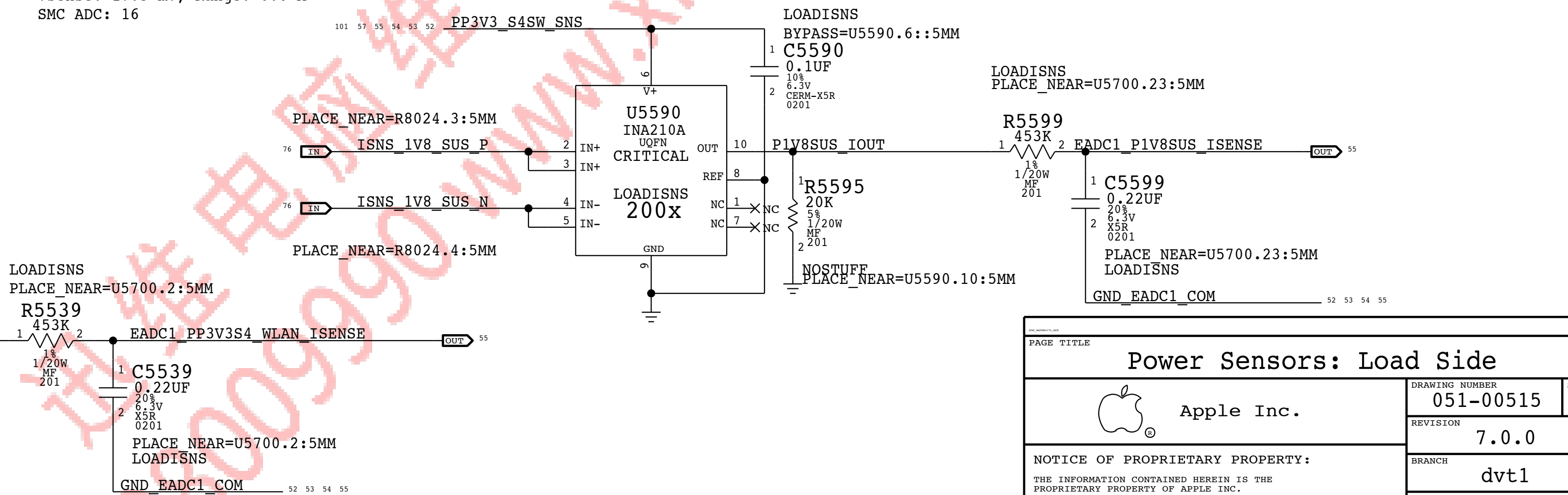
CPU High Side Current (IC0R) Threshold Alert

Gain: 100x
Rsense: 0.003 (R5400)



1.8V Current Sense (I18C)

Gain: 200x, EDP: 0.7 A
Rsense: 0.025 (R8024) or Rsense SHORT
Vsense: 17.5 mV, Range: 0.6 A
SMC ADC: 16



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	2	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5569,C5519		LOADRC:NO
117S0008	2	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5549,C5579		LOADRC:NO

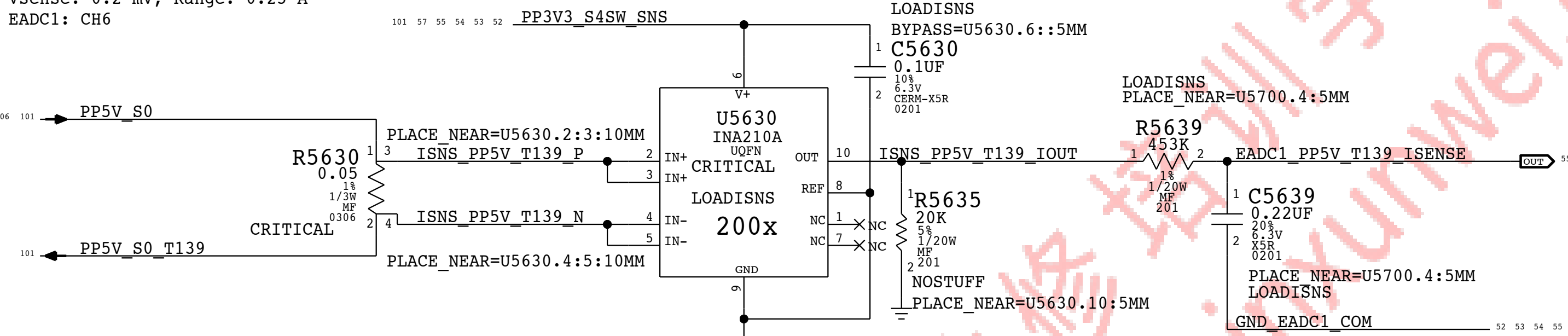
Power Sensors: Load Side

Apple Inc.	DRAWING NUMBER 051-00515	STEP D
	REVISION 7.0.0	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: 1 TO MAINTAIN THIS DOCUMENT IN CONFIDENCE 11 NOT TO REPRODUCE OR COPY IT 111 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART 1111 ALL RIGHTS RESERVED	BRANCH dvt1	PAGE 55 OF 145
	SHEET 53 OF 119	



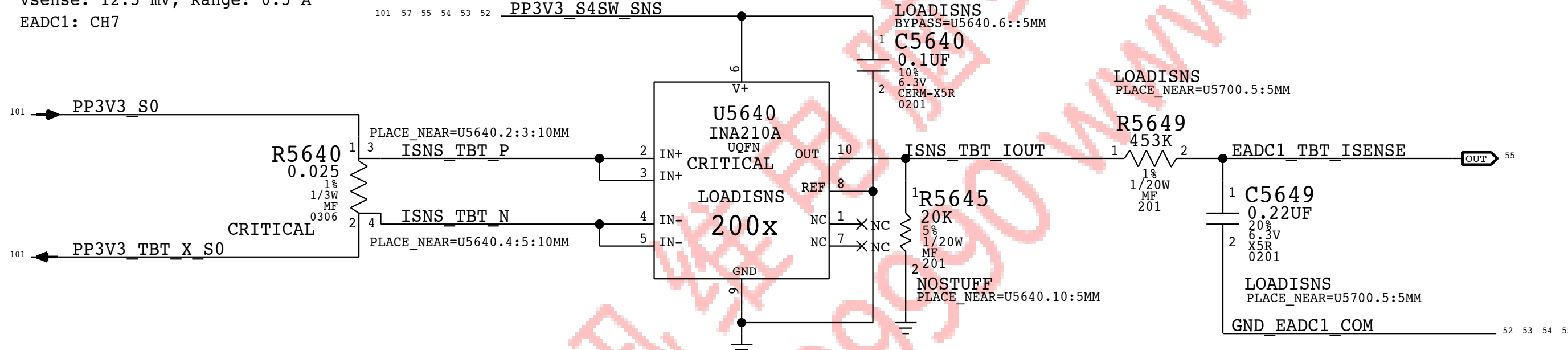
T139 5V Current Sense (IF5C)

Gain: 200x, EDP: 0.004 A
Rsense: 0.05 (R5630) or Rsense SHORT
Vsense: 0.2 mV, Range: 0.25 A
EADC1: CH6



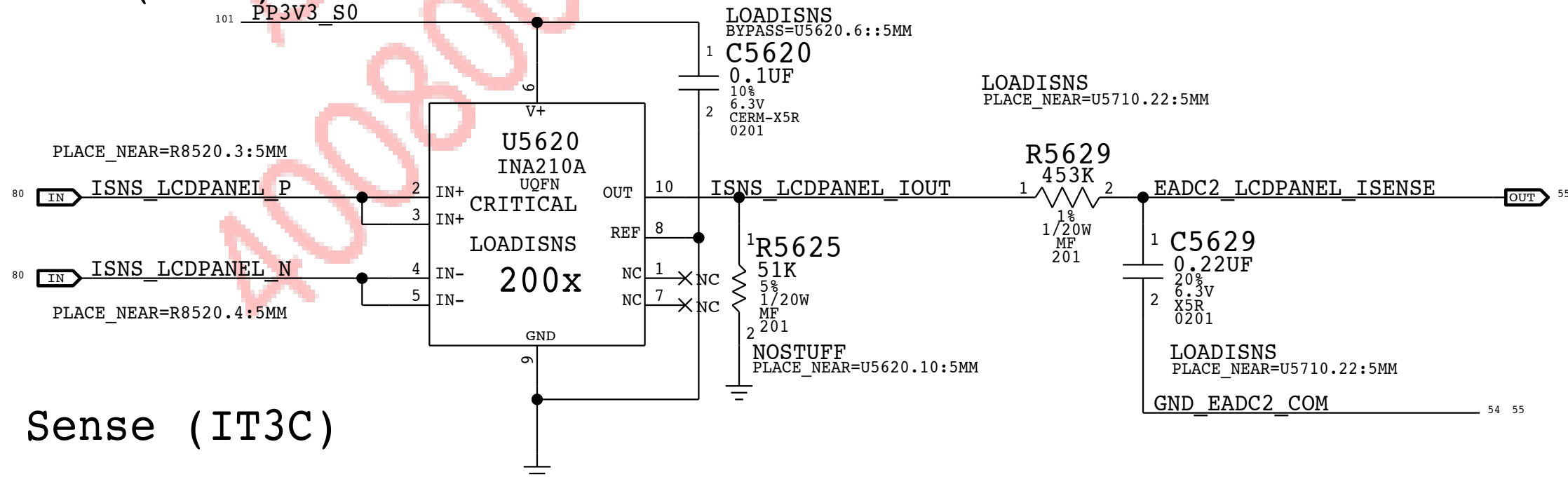
Thunderbolt TBT Current Left (IULC)

Gain: 200x, EDP: 0.5 A
Rsense: 0.025 (R5640) or Rsense SHORT
Vsense: 12.5 mV, Range: 0.5 A
EADC1: CH7



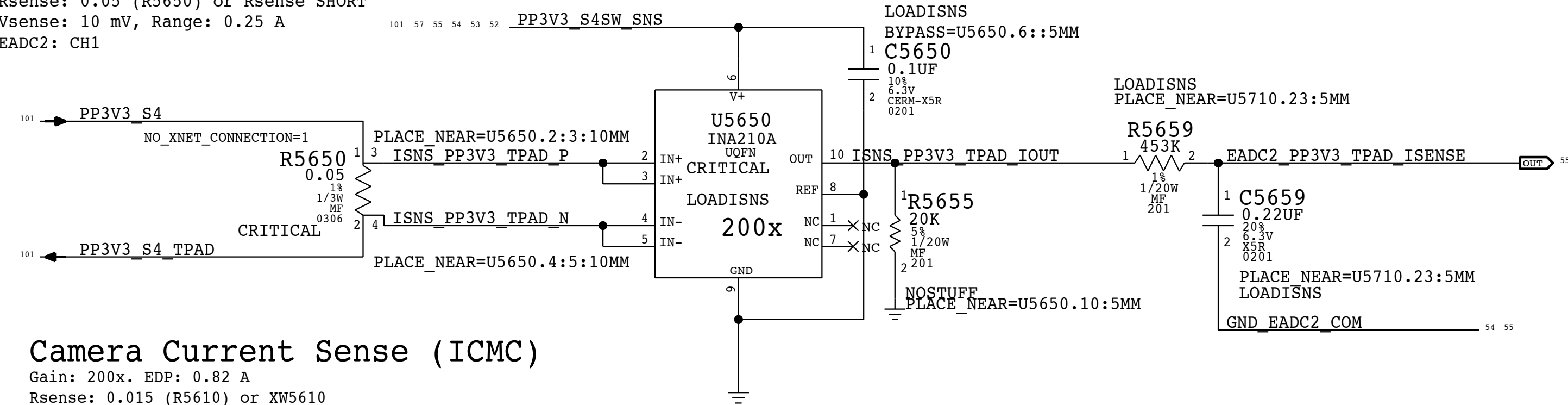
LCD Panel Current Sense (ILDC)

Gain: 200x, EDP: 1 A
RSENSE: 0.01 (R8520) or Rsense SHORT
Vsense: 5 mV, Range: 1.25 A
EADC2: CH0



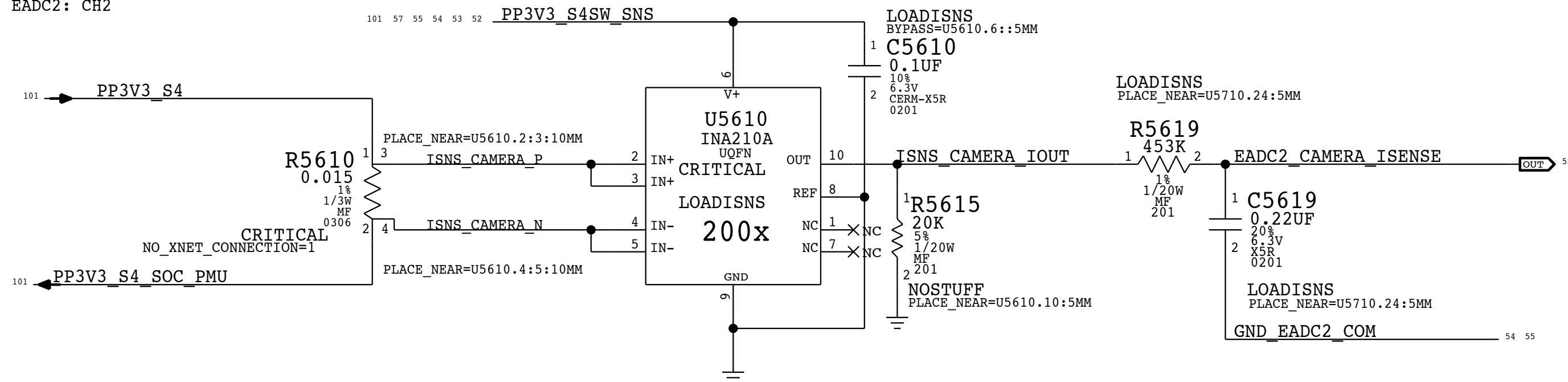
Trackpad 3V Current Sense (IT3C)

Gain: 200x, EDP: 0.2 A
Rsense: 0.05 (R5650) or Rsense SHORT
Vsense: 10 mV, Range: 0.25 A
EADC2: CH1

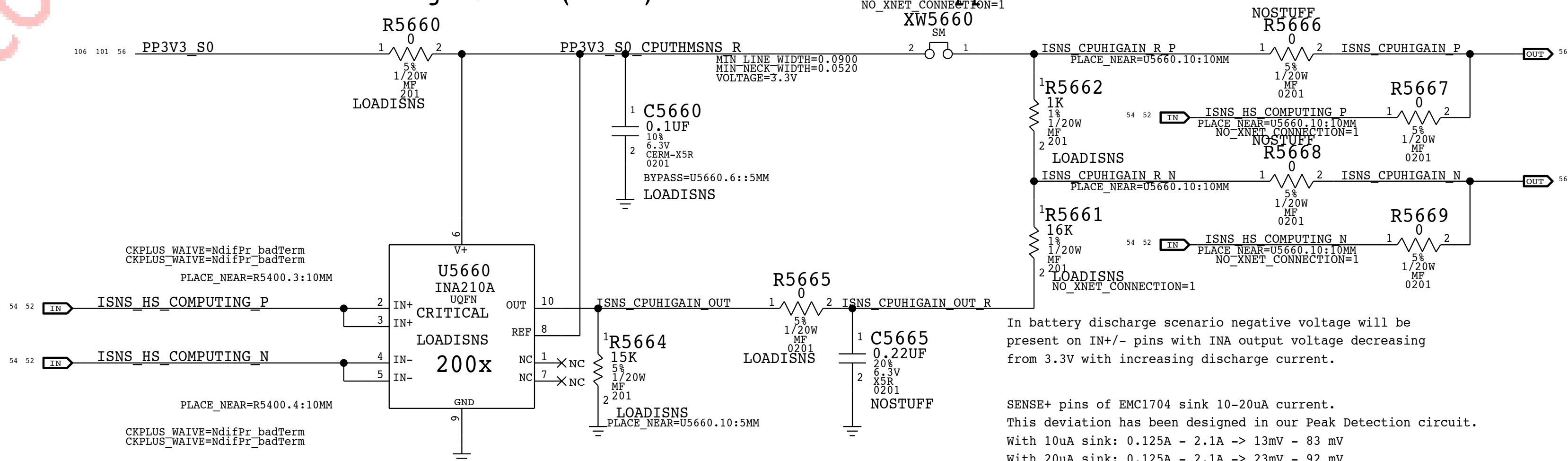


Camera Current Sense (ICMC)

Gain: 200x, EDP: 0.82 A
Rsense: 0.015 (R5610) or XW5610
Vsense: 12.3 mV, Range: 0.83 A
EADC2: CH2

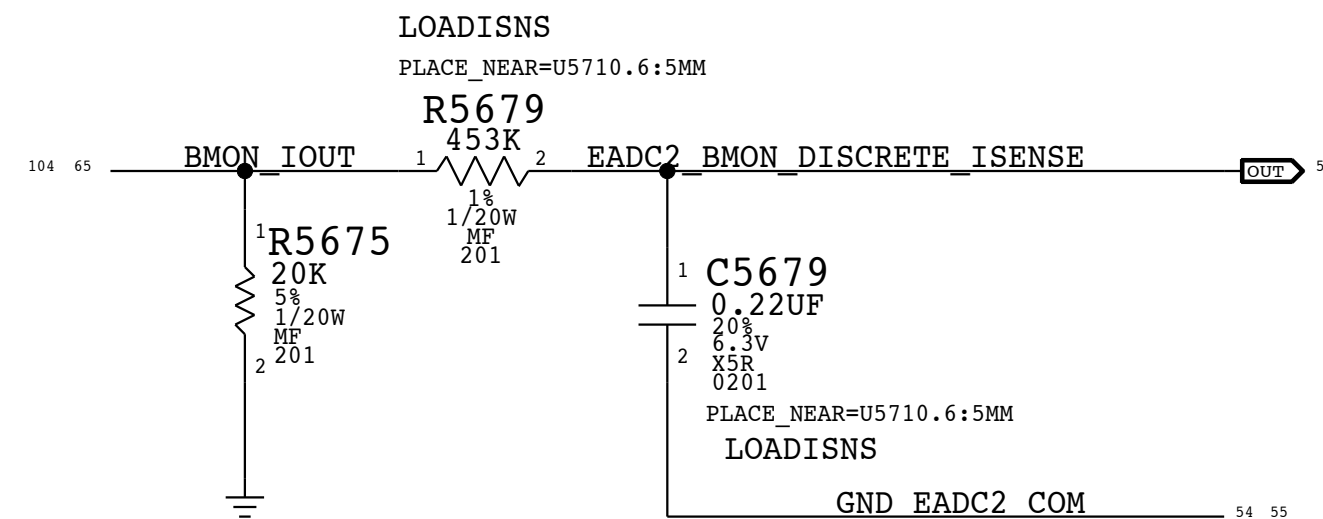


CPU High Side (IC0R) Peak Detection Support



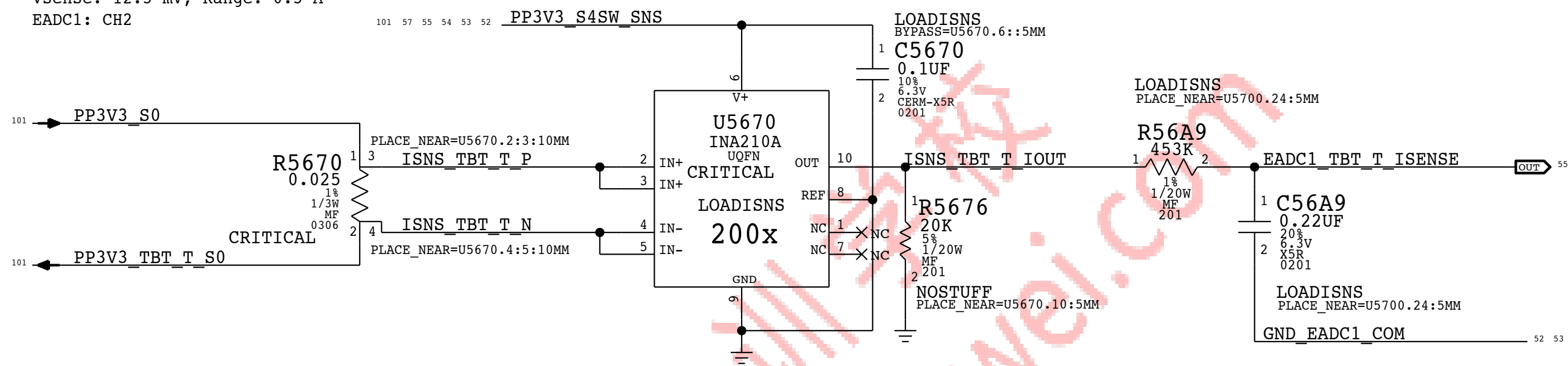
Battery Discrete Current Sense (IB0L)

Gain: 2940x, EDP: 8 A
Rsense: 0.003 (R501//R502)
Vsense: 24 mV, Range: 0.28 A
EADC2: CH5



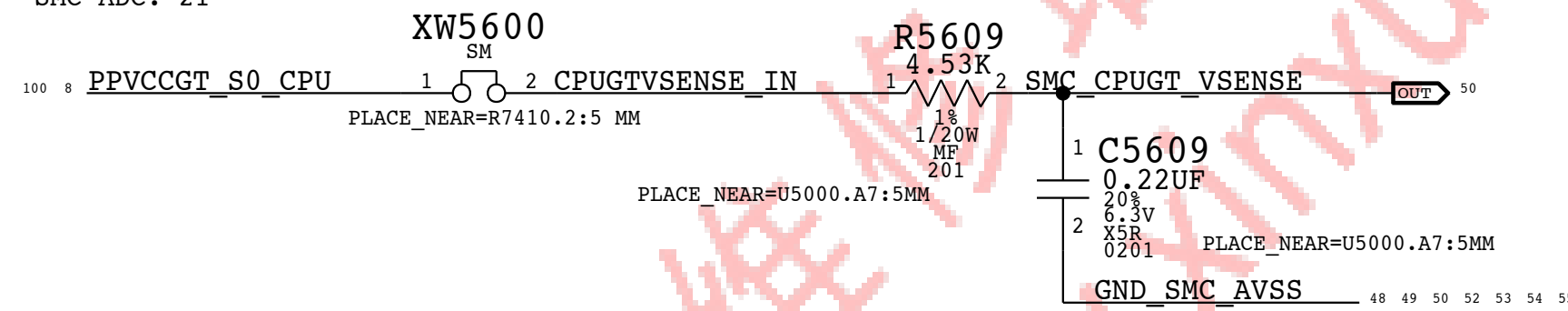
Thunderbolt TBT Current Right (IURC)

Gain: 200x, EDP: 0.5 A
Rsense: 0.025 (R5670) or Rsense SHORT
Vsense: 12.5 mV, Range: 0.5 A
EADC1: CH2



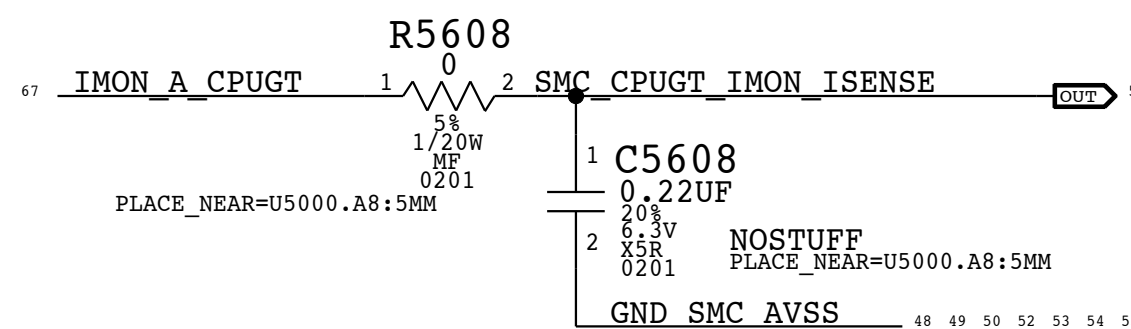
CPU GT Voltage Sense (VCGC)

SMC ADC: 21



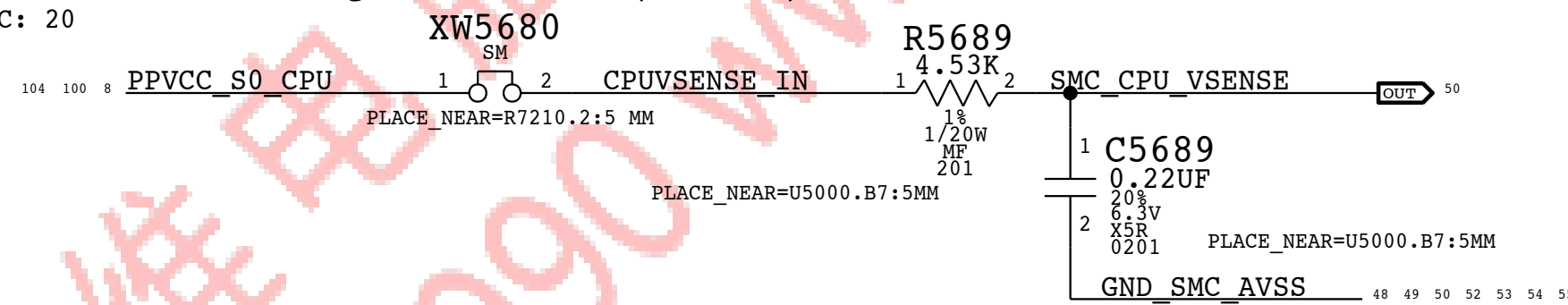
CPU GT IMON Current Sense (ICGM)

Gain: 1 A / 17.963 mV, Range: 64 A.
SMC ADC: 23



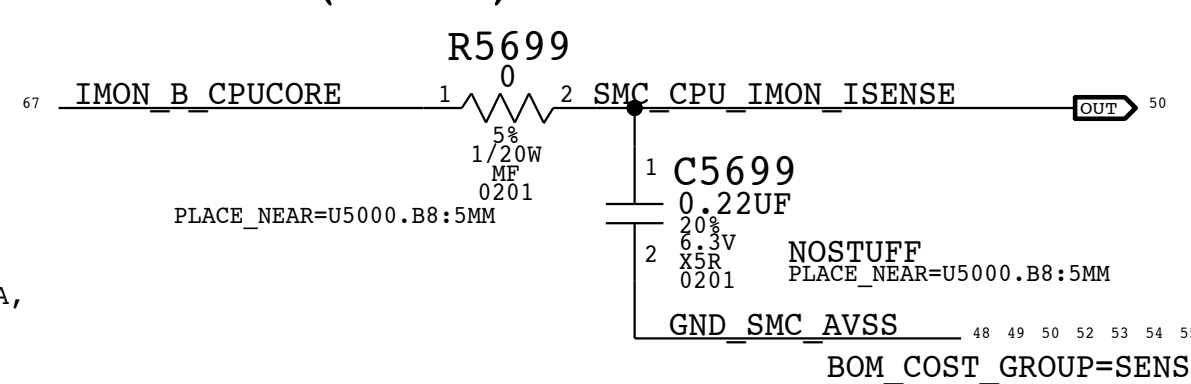
CPU Core Voltage Sense (VCAC)

SMC ADC: 20



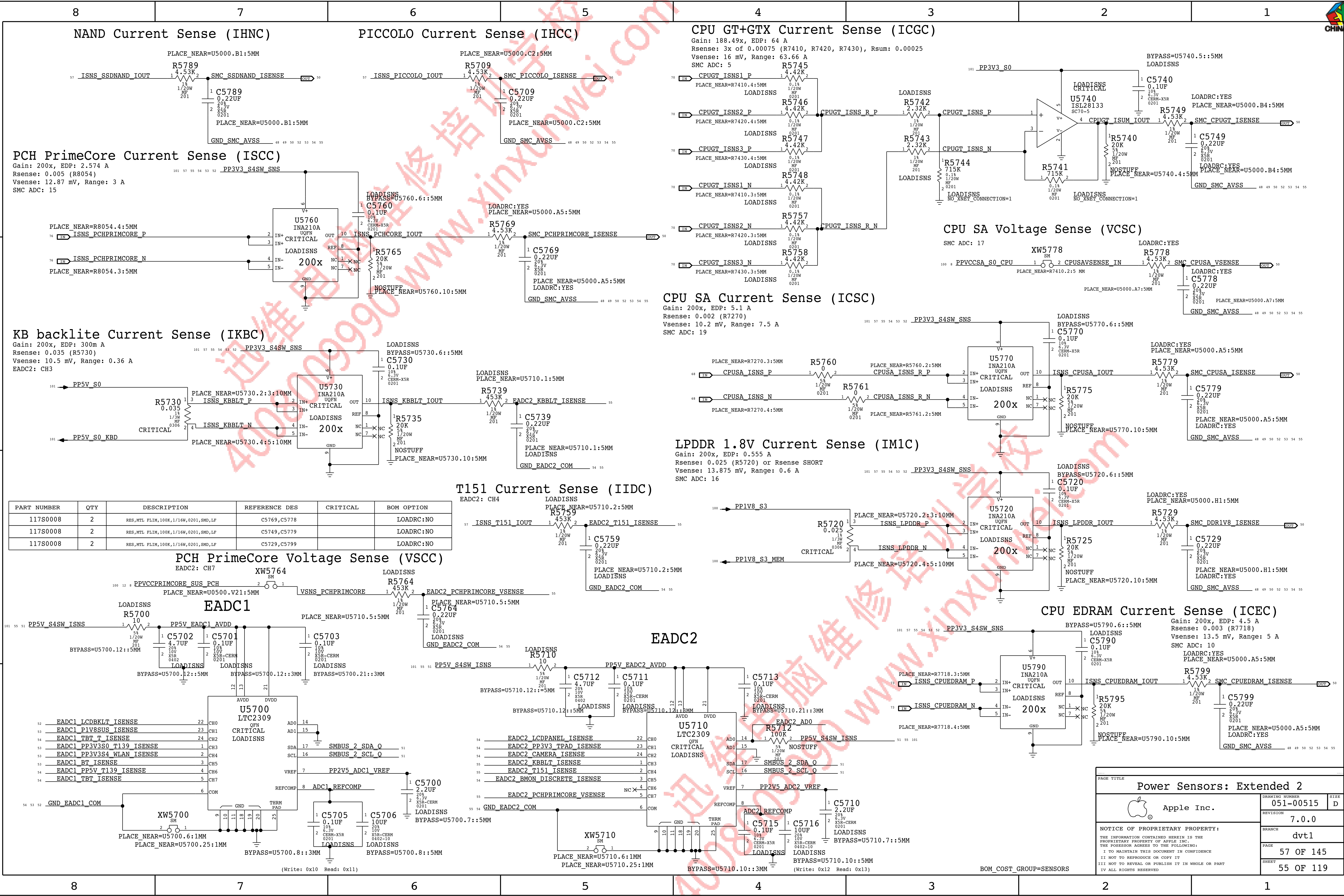
CPU Core IMON Current Sense (ICAM)

Gain: 1 A / 34.223 mV, Range: 29 A.
SMC ADC: 22



With R7150 (Ri) set to 226 Ohm,
R7210 (Rsen) set to 0.75 mOhm,
R7160 set to 82.5 kohm,
Num Phases (N) is 2, and Io (ICMax) is 29A,
then 1A of Io gives 34.223mV at the Vimon.

PAGE TITLE		
Power Sensors: Extended		
	DRAWING NUMBER	051-00515
	REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I ALL RIGHTS RESERVED	BRANCH	dvt1
	PAGE	56 OF 145
	SHEET	54 OF 119



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	2	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5769,C5778		LOADRC:NO
117S0008	2	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5749,C5779		LOADRC:NO
117S0008	2	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5729,C5799		LOADRC:NO

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
1 TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
11 NOT TO REPRODUCE OR COPY IT
111 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
1111 ALL RIGHTS RESERVED

PAGE TITLE

Power Sensors: Extended 2

DRWING NUMBER

051-00515

REVISION

7.0.0

BRANCH

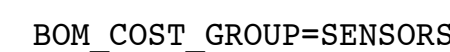
dvt1

PAGE

57 OF 145

SHEET

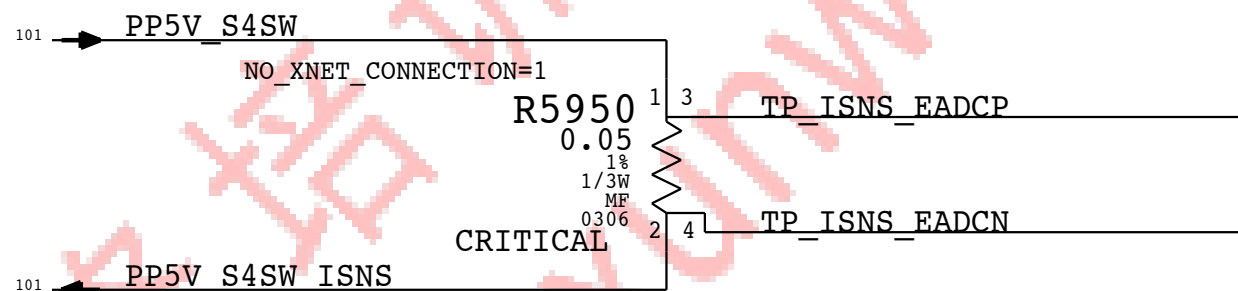
55 OF 119



 Apple Inc.		DRAWING NUMBER 051-00515		SIZE D
		REVISION 7.0.0		
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH dvt1 PAGE 58 OF 145 SHEET 56 OF 119		

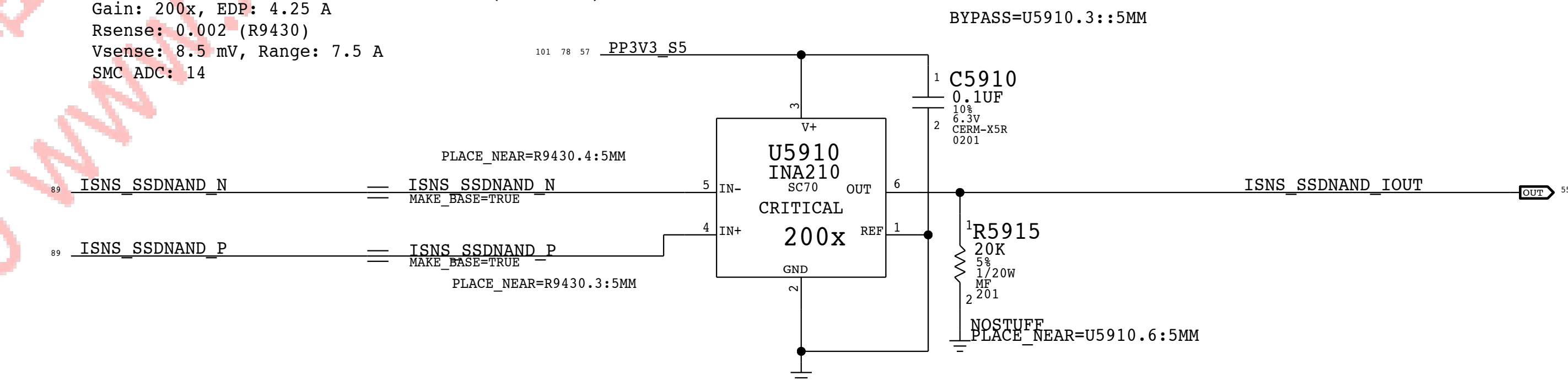
EADC Current Sense (IADC)

Gain: 200x, EDP: 200m A
Rsense: 0.050 (R5950) or Rsense SHORT
Vsense: 10 mV, Range: 0.25 A
EADC2: CH7



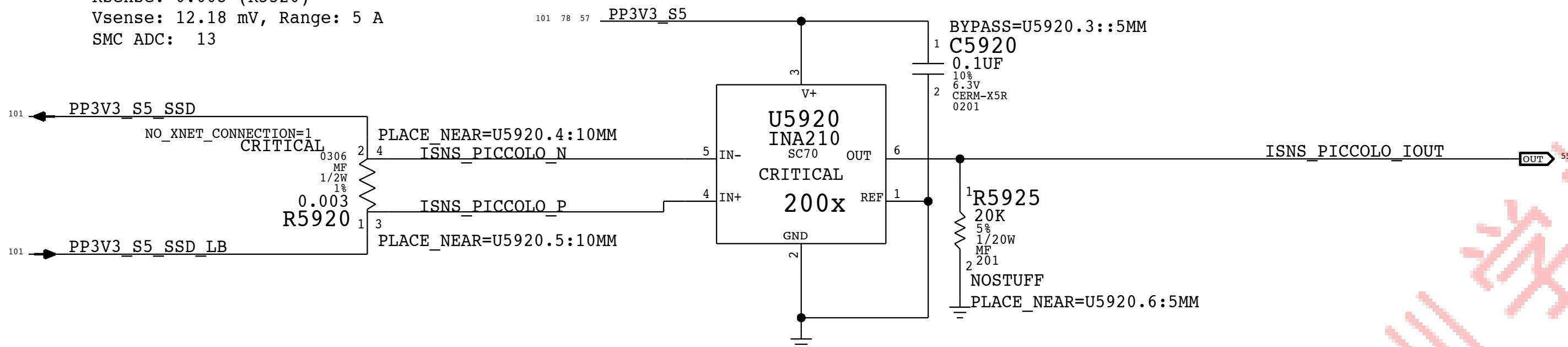
NAND Current Sense (IHNC)

Gain: 200x, EDP: 4.25 A
Rsense: 0.002 (R9430)
Vsense: 8.5 mV, Range: 7.5 A
SMC ADC: 14



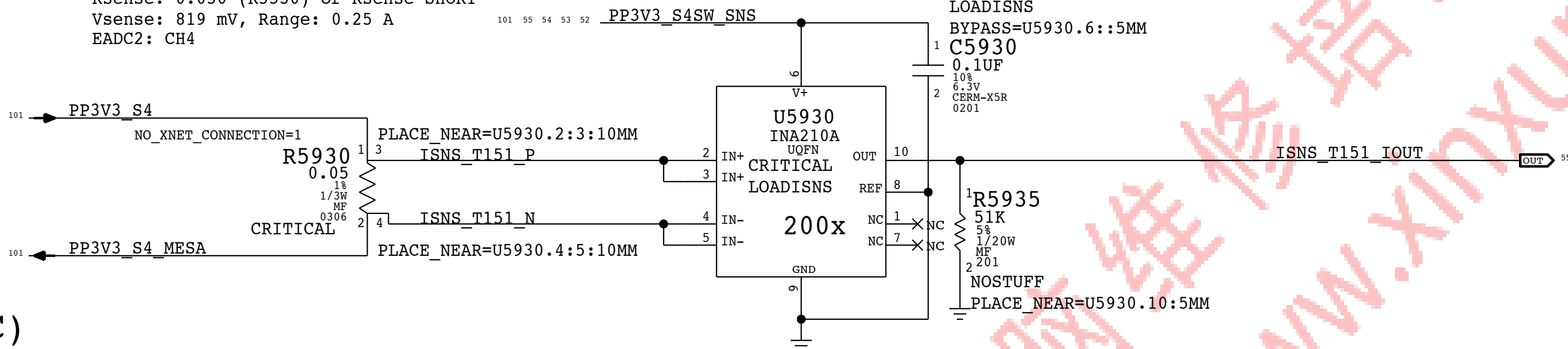
Piccolo Current Sense (IHCC)

Gain: 200x, EDP: 4.06 A
Rsense: 0.003 (R5520)
Vsense: 12.18 mV, Range: 5 A
SMC ADC: 13



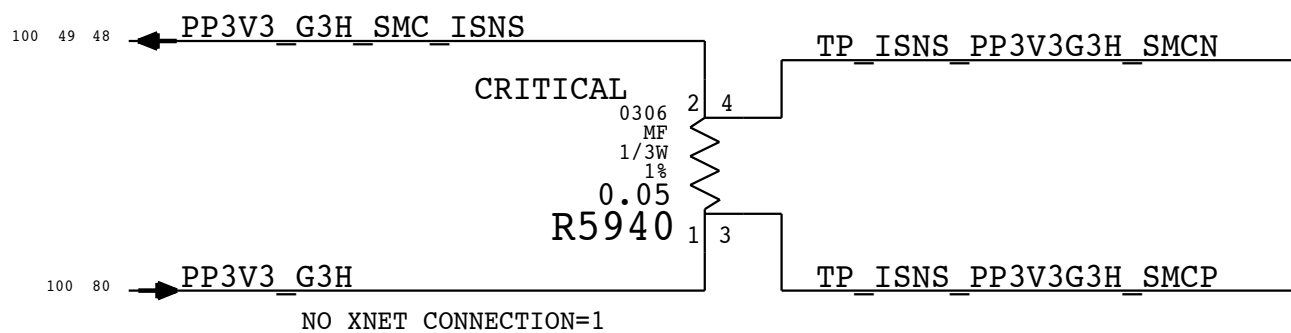
T151 Current Sense (IIDC)

Gain: 200x, EDP: 163.8m A
Rsense: 0.050 (R5530) or Rsense SHORT
Vsense: 819 mV, Range: 0.25 A
EADC2: CH4



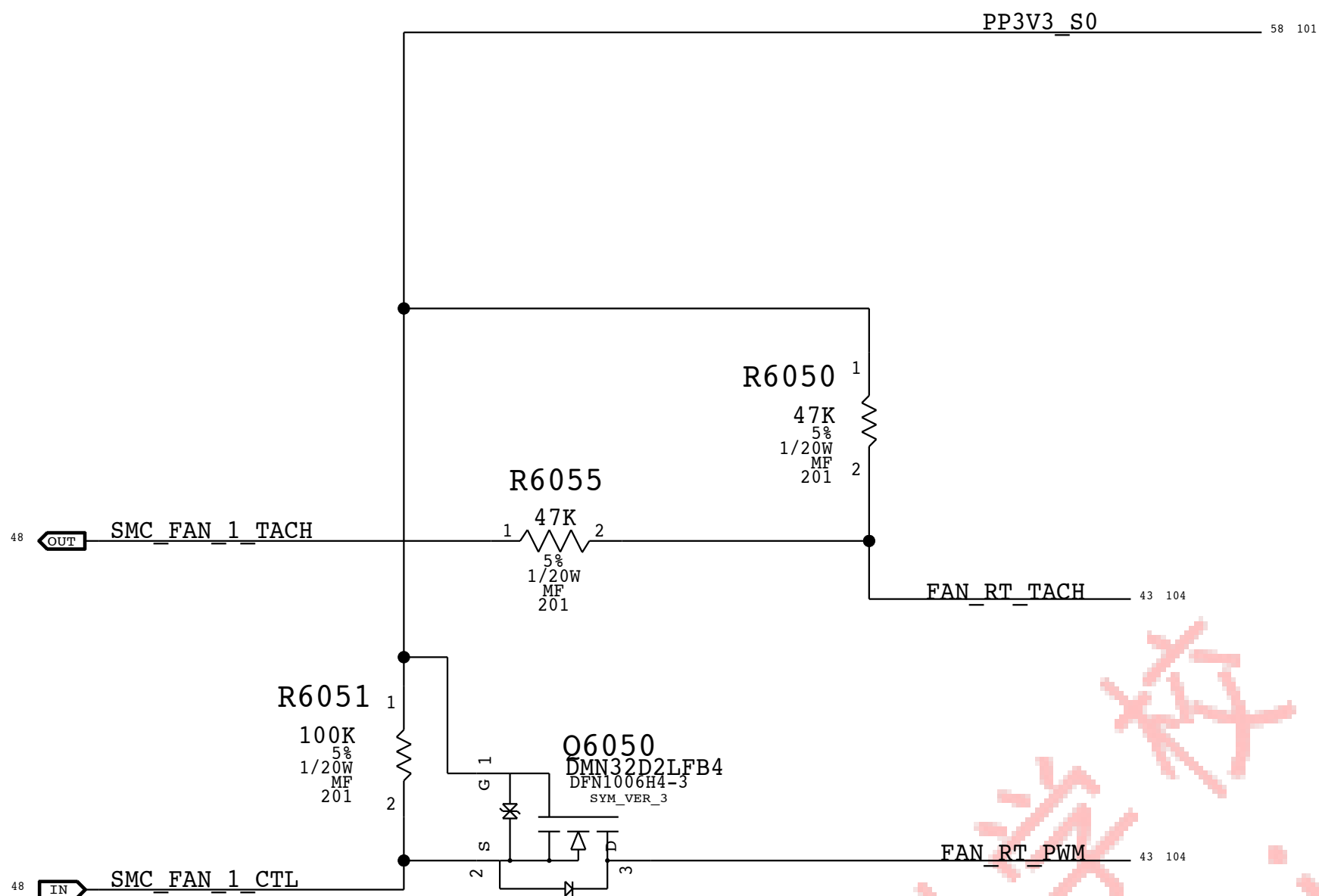
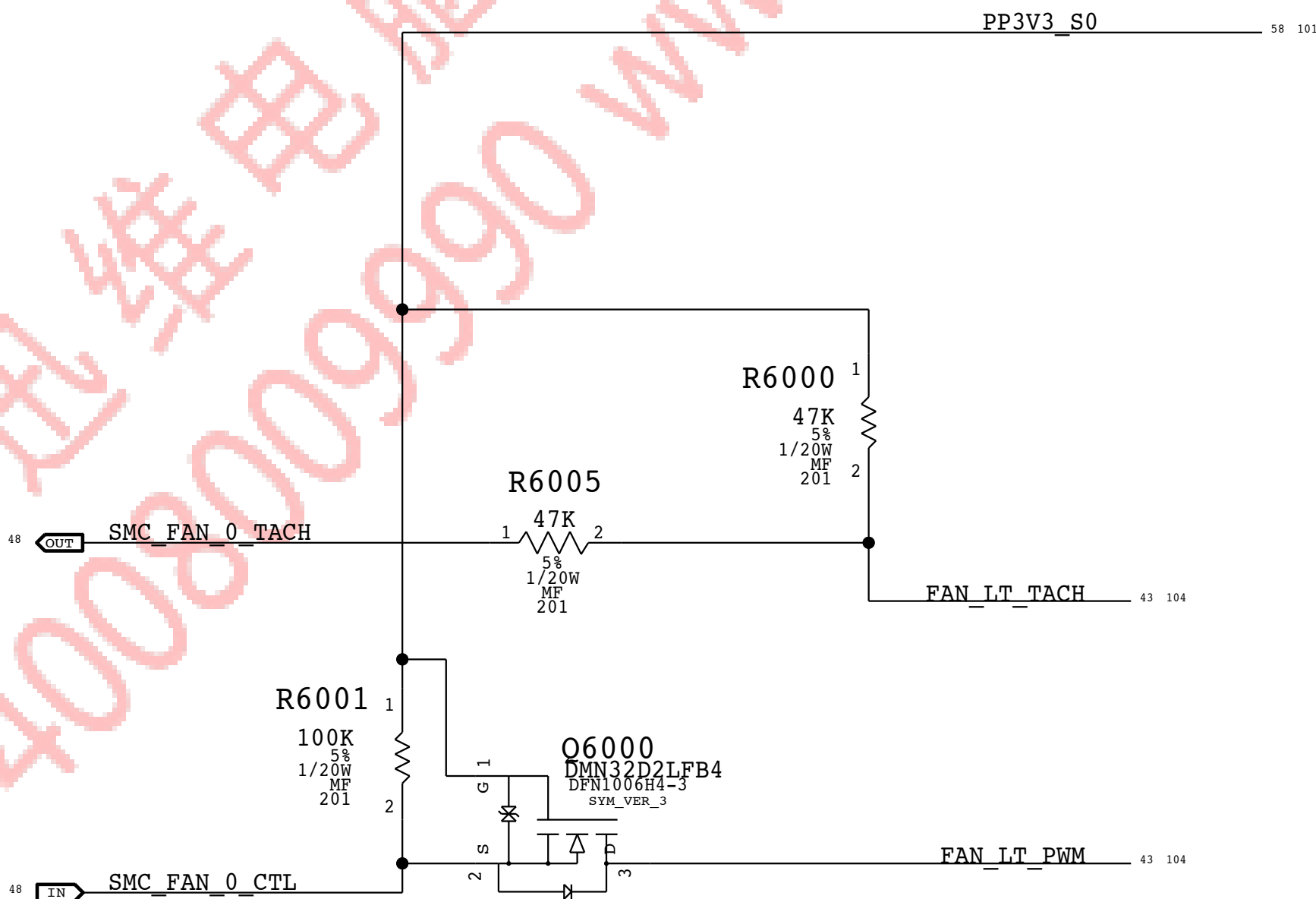
SMC Current Sense (ISMC)

Gain: 391.67x, EDP: 0.2 A
Rsense: 0.05 (R5940) or Rsense SHORT
Vsense: 10 mV, Range: 0.21A
EADC2: CH6

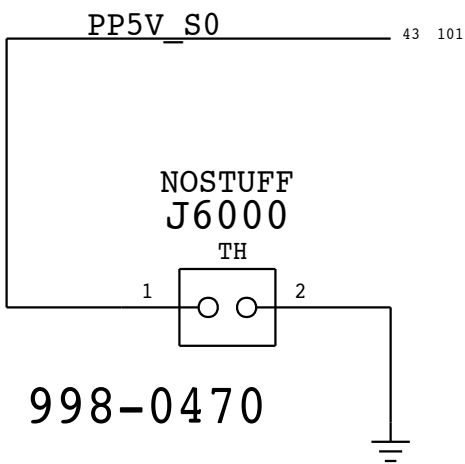


PAGE TITLE		
Power Sensors:Extended 3		
	DRAWING NUMBER	051-00515
	REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH	dvt1
	PAGE	59 OF 145
	SHEET	57 OF 119

FAN CONTROL



FOR DEBUG FAN POWER

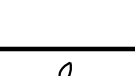


EYMC_MASTER=J79_JACK

EYMC_DATE=08/21/2015

PAGE TITLE

Fans



Apple Inc.

DRAWING NUMBER	051-00515	SIZE	D
REVISION	7.0.0		
BRANCH	dvt1		
PAGE	60 OF 145		
SHEET	58 OF 119		

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.

THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

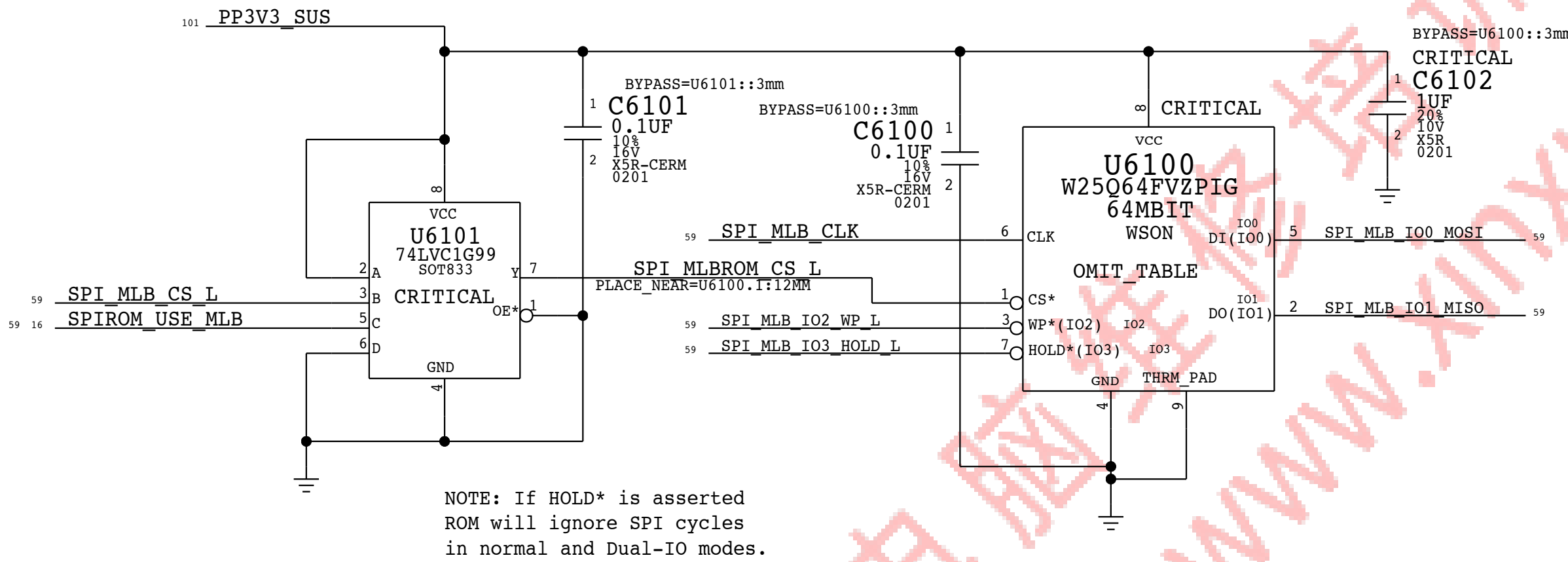
II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

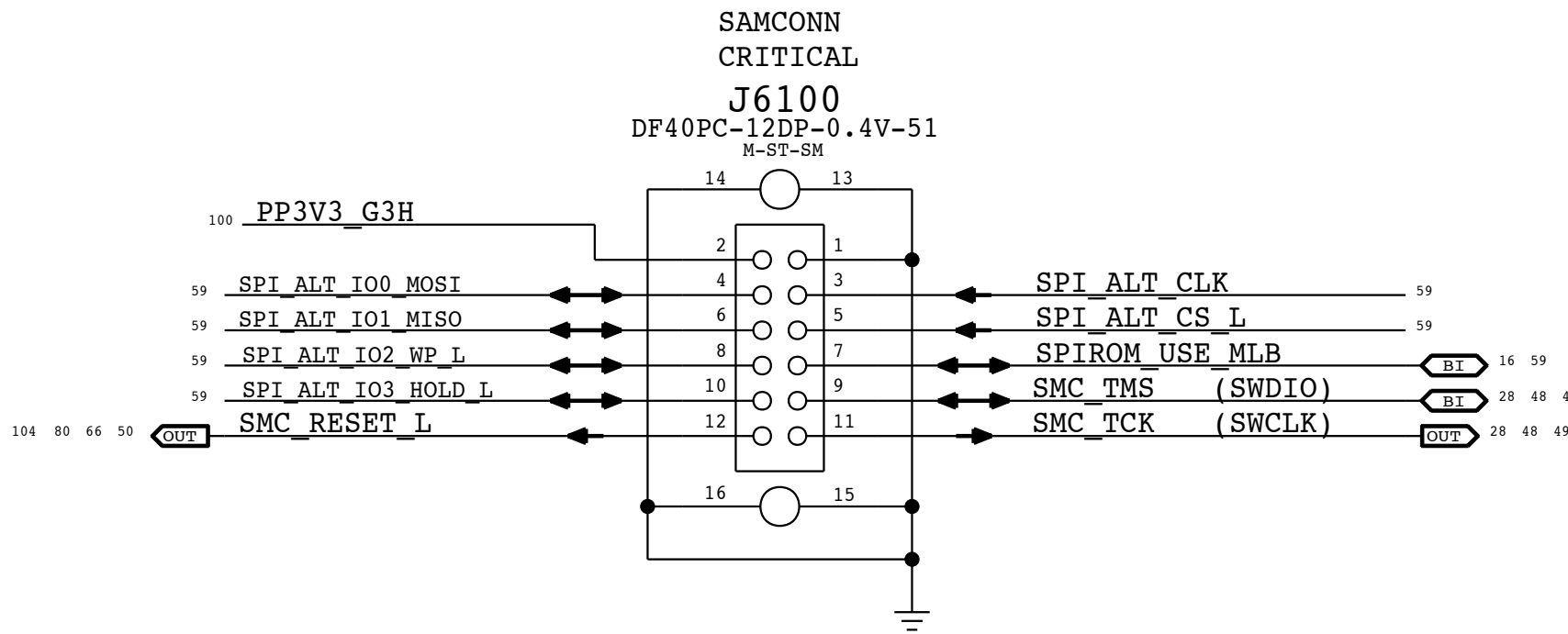
IV ALL RIGHTS RESERVED

SPI ROM

Quad-IO Mode (Mode 0 & 3) supported.
SPI Frequency: 50MHz for CPU, 20MHz for SMC.

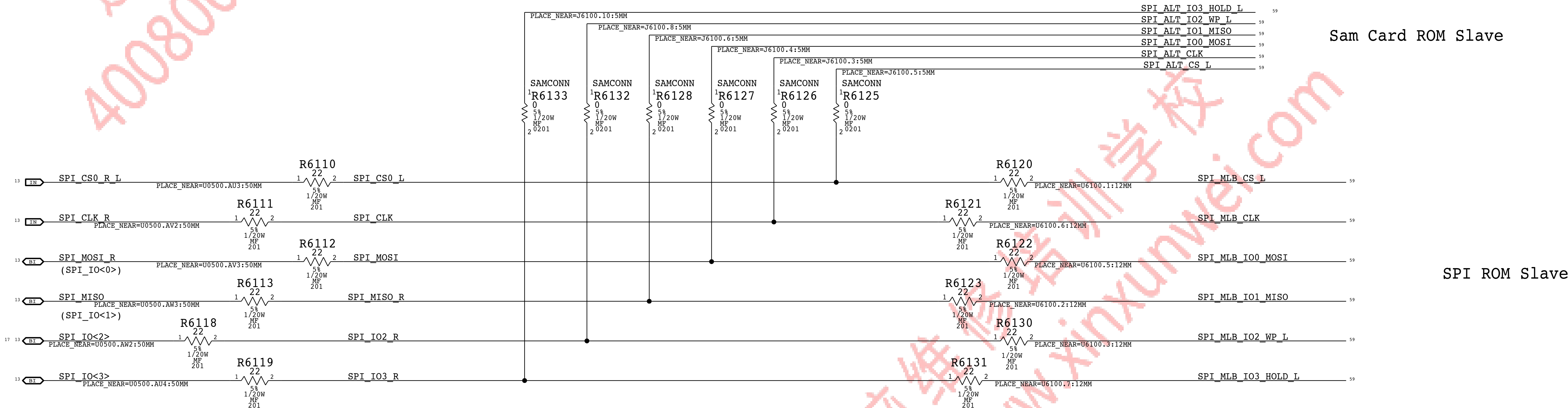


SPI+SWD SAM Connector

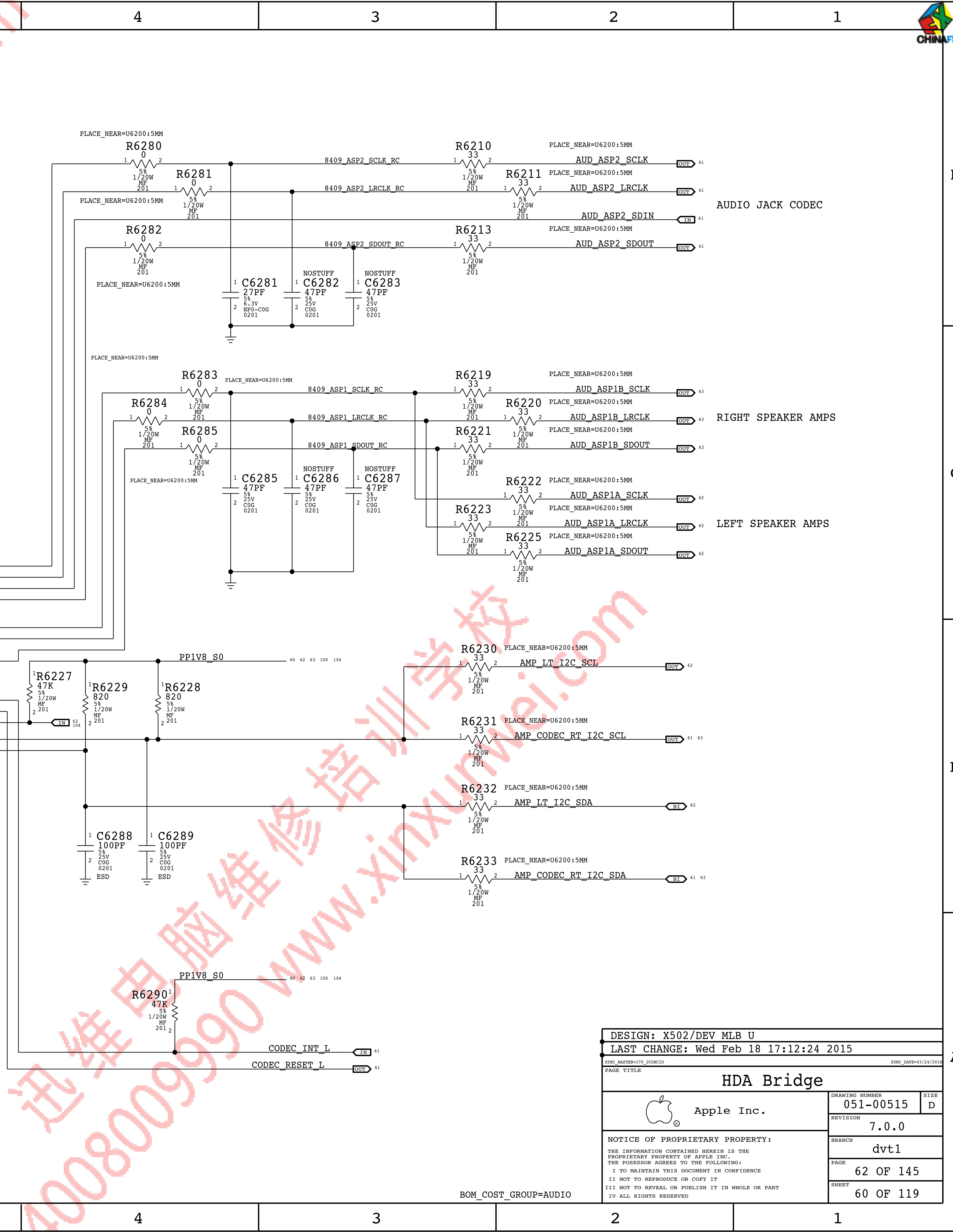


Quad SPI and QPI instructions require the non-volatile Quad Enable bit (QE) in Status Register-2 to be set. When QE=1, the /WP pin becomes IO2 and /HOLD pin becomes IO3.

SPI Bus Series Termination (Modified per PDG)



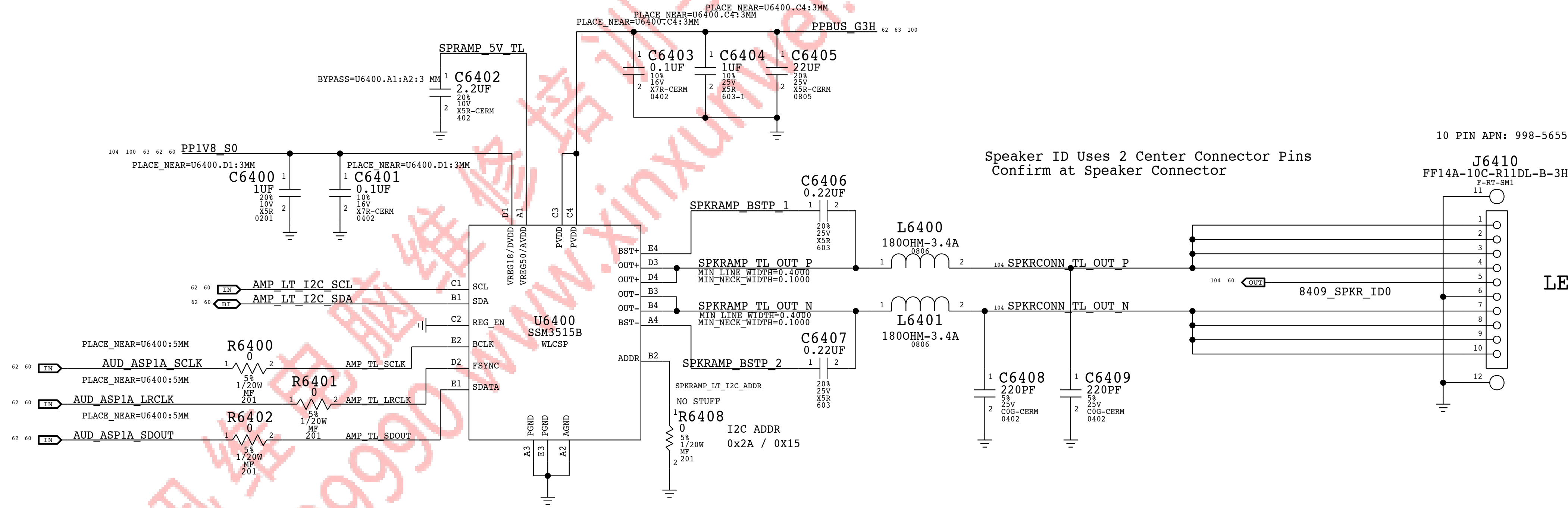
SYNC MASTER=J52_MLB		SYNC DATE=05/12/2015	
PAGE TITLE			
SPI Debug Connector			
 Apple Inc.		DRAWING NUMBER	051-00515
		REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	dvt1
		PAGE	61 OF 145
		SHEET	59 OF 119



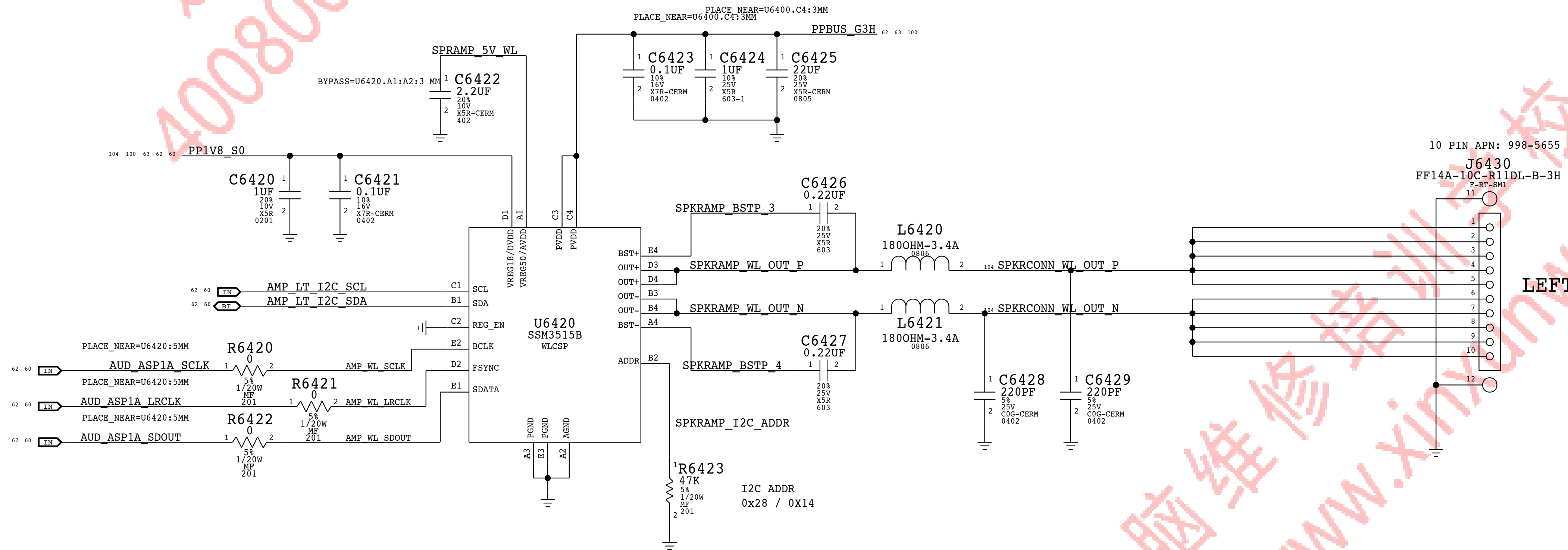
DESIGN: X502/DEV MLB U		
LAST CHANGE: Wed Feb 18 17:12:24 2015		
PAGE TITLE		
HDA Bridge		
	DRANTING NUMBER	051-00515
	REVISION	7.0.0
	BRANCH	dvt1
	PAGE	62 OF 145
NOTICE OF PROPRIETARY PROPERTY:		SHEET
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		60 OF 119
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		
I NOT TO REPRODUCE OR COPY IT		
I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		
I ALL RIGHTS RESERVED		

BOM_COST_GROUP=AUDIO

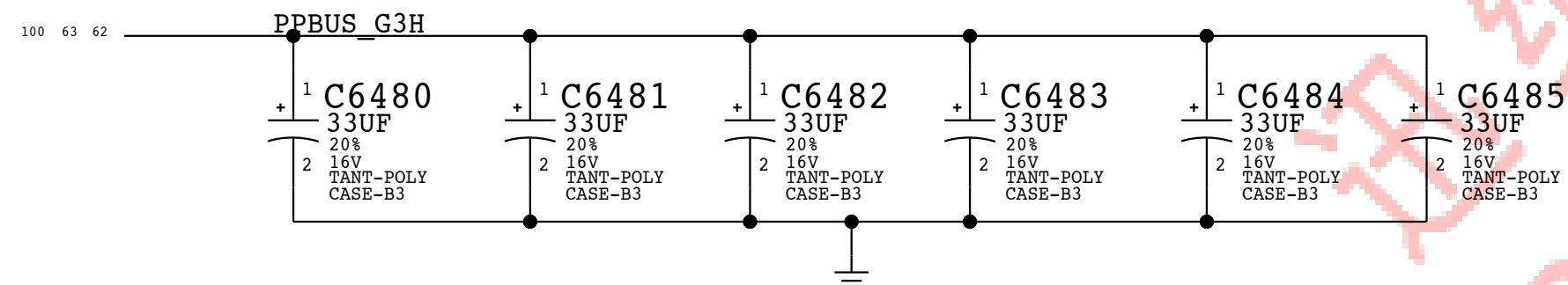
2X MONO SPEAKER AMPLIFIERS
APN: 35384074
GAIN = TBD



LEFT TWEETER SPEAKER CONNECTOR



LEFT WOOFER SPEAKER CONNECTOR



BOM_COST_GROUP=AUDIO

DESIGN: X502/DEV MLB U		
LAST CHANGE: Wed Feb 18 17:12:24 2015		
PAGE TITLE		
Left Speaker Amps & Conn		
	DRAWING NUMBER	051-00515
	REVISION	7.0.0
	BRANCH	dvt1
	PAGE	64 OF 145
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		SHEET 62 OF 119

2X MONO SPEAKER AMPLIFIERS
APN: 35384073
GAIN = TBD

8

7

6

5

4

3

2

1

D

D

C

C

B

B

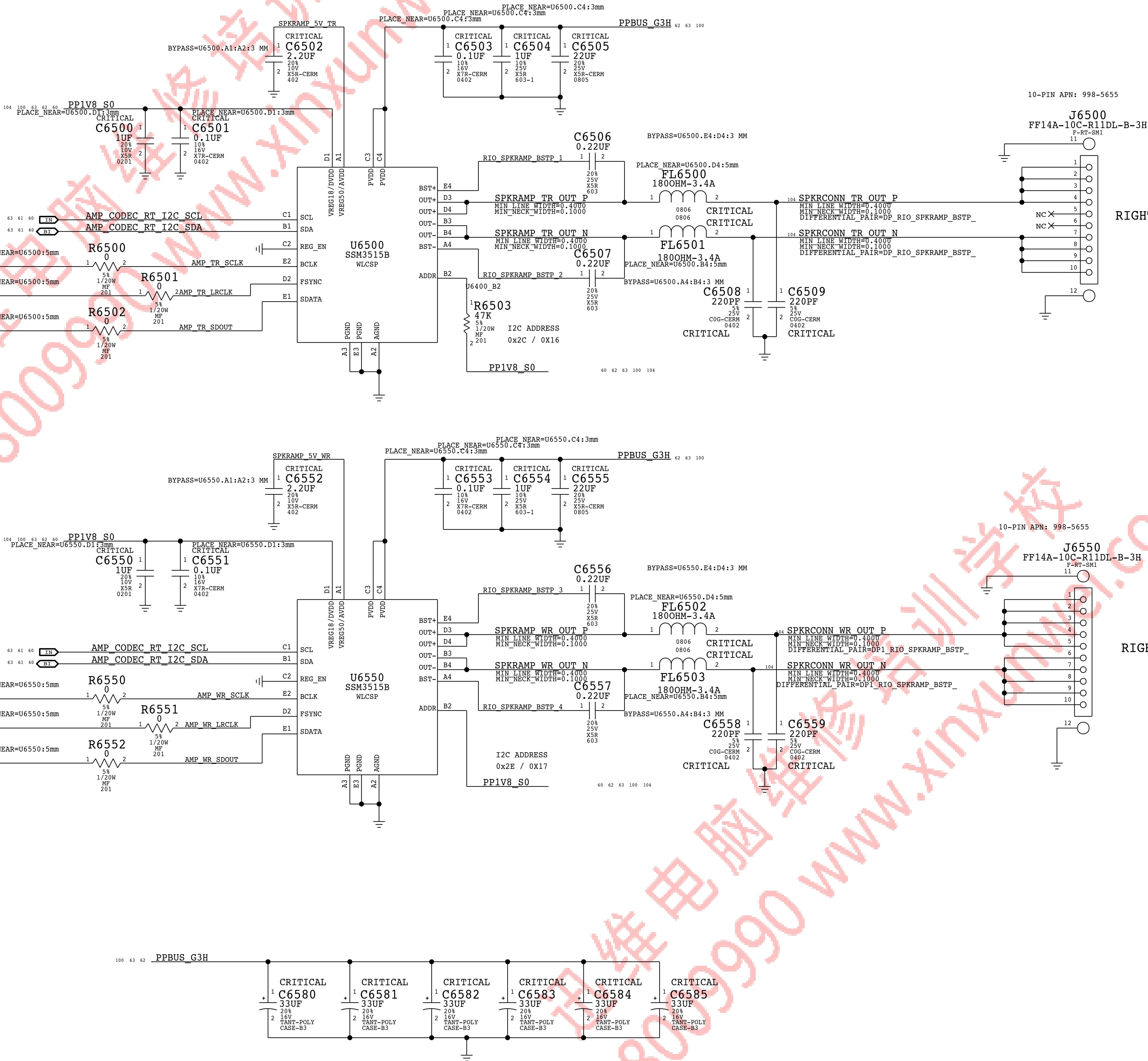
A

A

AUD_ASPIB_SCLK
AUD_ASPIB_LRCLK
AUD_ASPIB_SDOUT

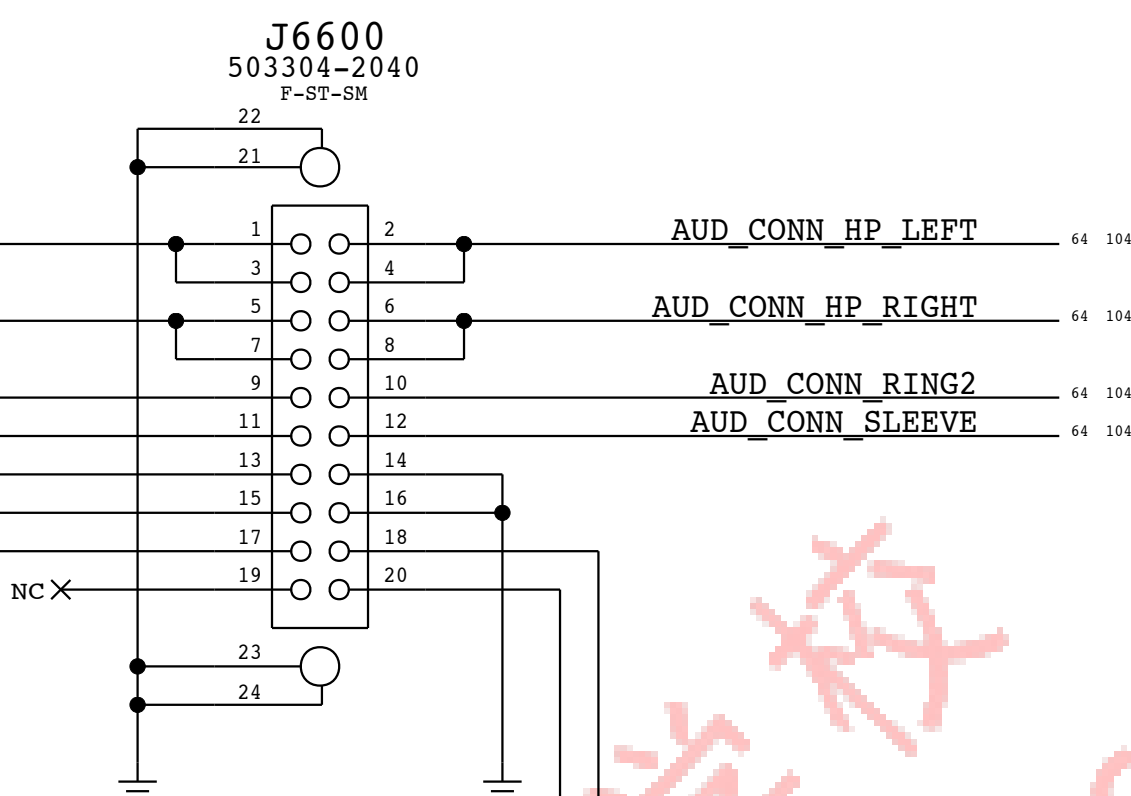
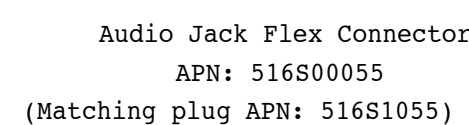
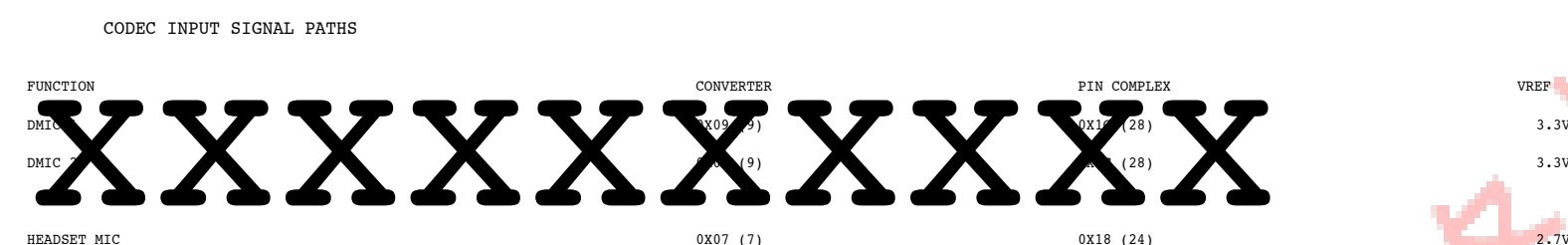
TDM SLOT 3

TDM SLOT 4

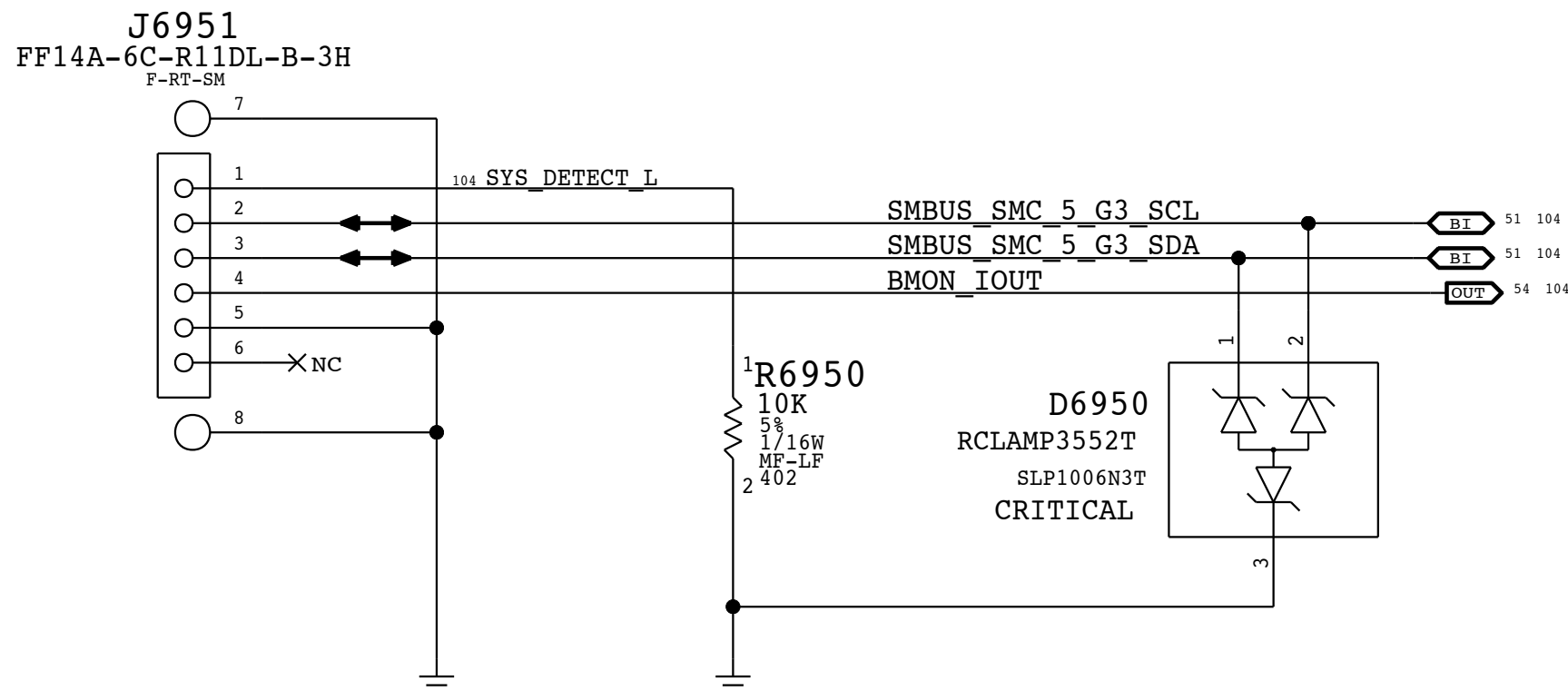
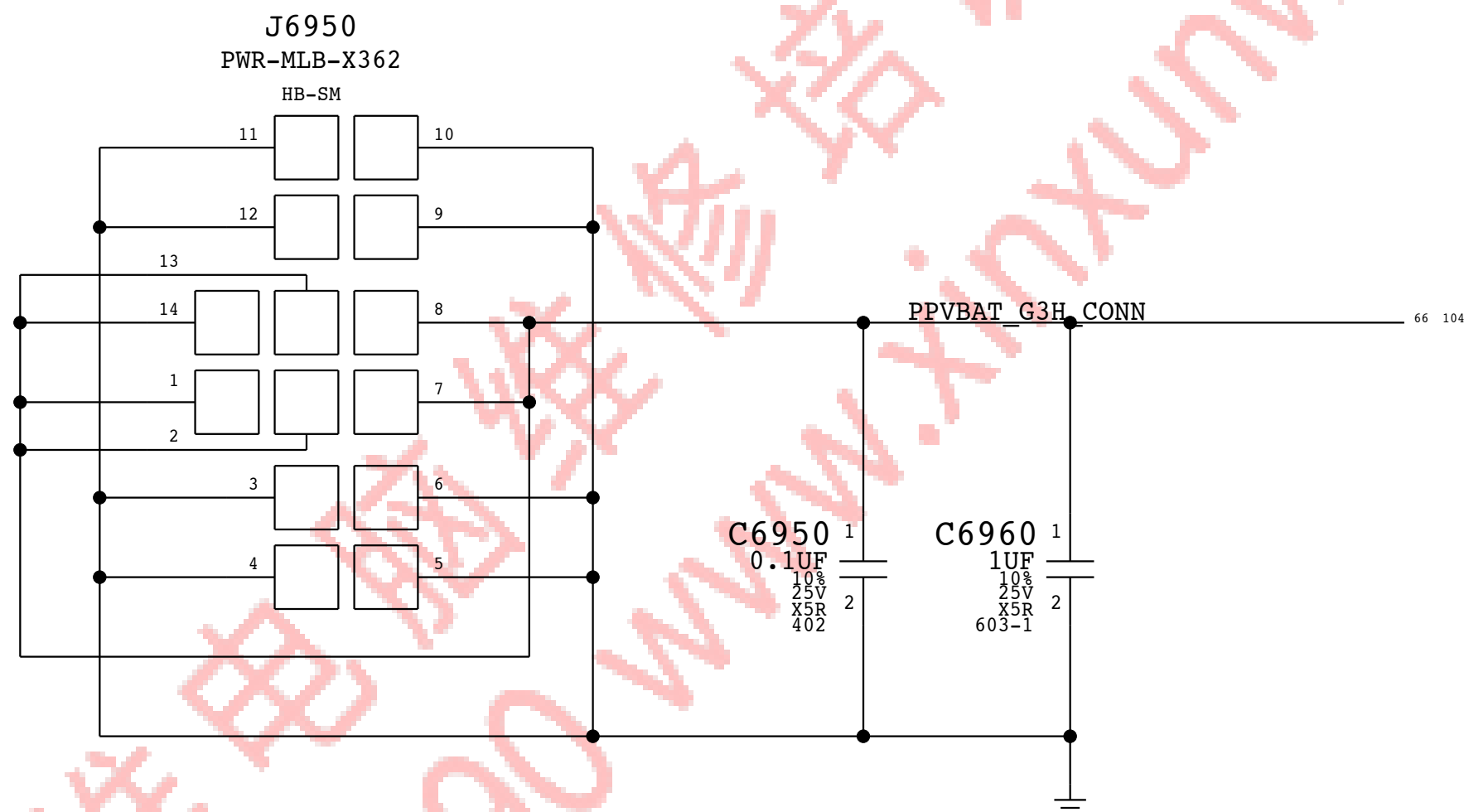


BOM_COST_GROUP=AUDIO

DESIGN: X502/DEV MLB U		
LAST CHANGE: Wed Feb 18 17:12:24 2015		
PAGE TITLE		
Right Speaker Amps & Conn		
	DRAWING NUMBER	051-00515
	REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I1 NOT TO REPRODUCE OR COPY IT I11 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I1V ALL RIGHTS RESERVED	BRANCH	dvt1
	PAGE	65 OF 145
	SHEET	63 OF 119

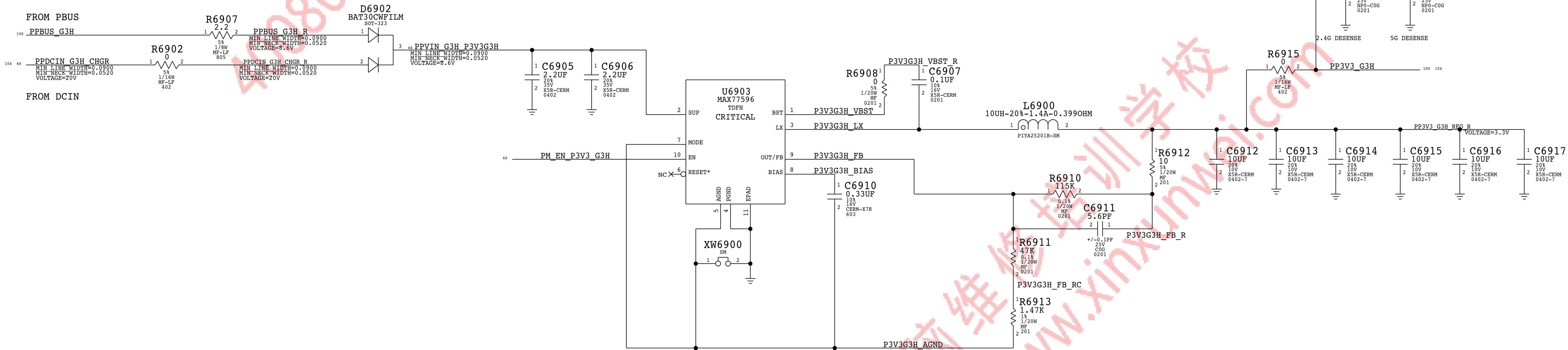


J79 Battery Hotbar Flex Pads



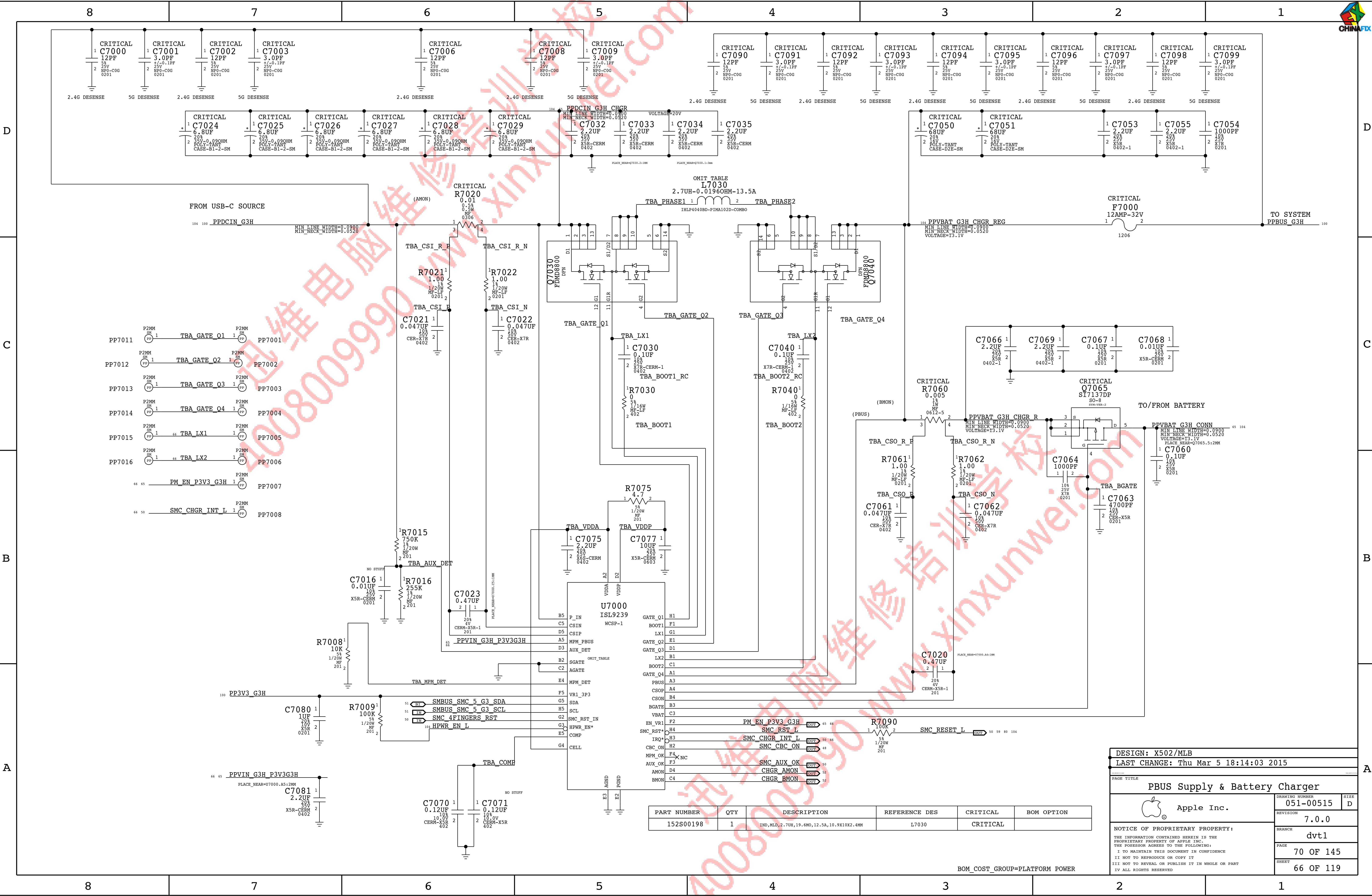
BMU POWER FLEX HOTBAR'd TO THE MLB:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
632-00566	1	PCBA,FLEX,BMU PWR,X362	J6950	CRITICAL	



PAGE TITLE		
DC-In & Battery Connectors		
 Apple Inc.	DRANTING NUMBER	051-00515
	REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH	dvt1
	PAGE	69 OF 145
	SHEET	65 OF 119

BOM_COST_GROUP=PLATFORM POWER



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
152S00198	1	IND,NLD,2.7UH,19.6mΩ,12.5A,10.9X10X2.4MM	L7030	CRITICAL	

DESIGN: X502/MLB
LAST CHANGE: Thu Mar 5 18:14:03 2015

Apple Inc.

DRATING NUMBER
051-00515

REVISION
7.0.0

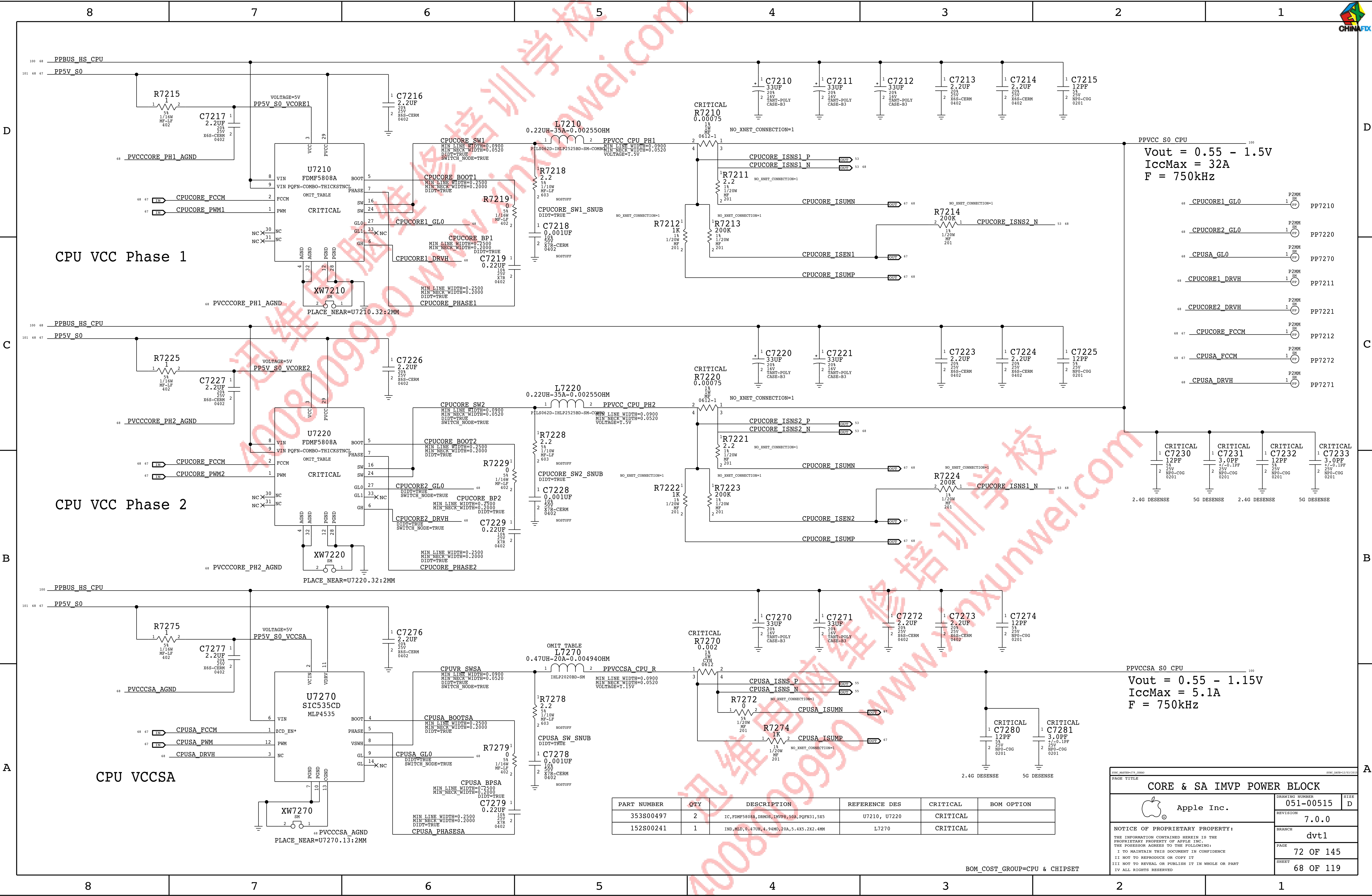
BRANCH
dvt1

PAGE
70 OF 145

SHEET
66 OF 119

NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.
THE POSSESSOR AGREES TO THE FOLLOWING:
1 TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
11 NOT TO REPRODUCE OR COPY IT
111 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
1111 ALL RIGHTS RESERVED

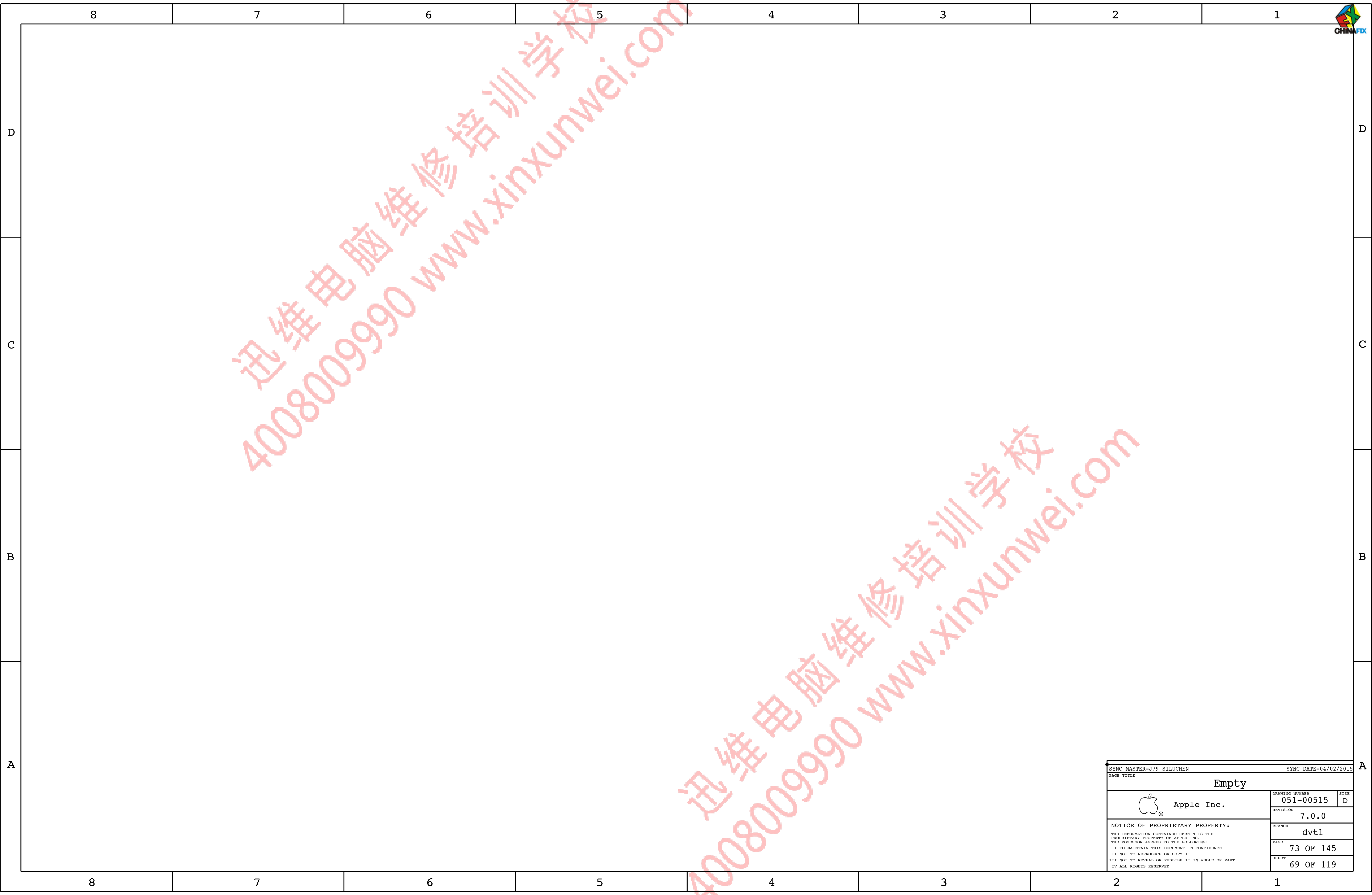
BOM_COST_GROUP=PLATFORM POWER



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00497	2	IC,FDMF5808A,DRMOS,IMVP8,50A,PQFN31,5X5	U7210, U7220	CRITICAL	
152S00241	1	IND,MLED,0.47UH,4.94NO,20A,5.4X5.2X2.4MM	L7270	CRITICAL	

CORE & SA IMVP POWER BLOCK		
	DRAWING NUMBER	051-00515
	REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: 1 TO MAINTAIN THIS DOCUMENT IN CONFIDENCE 11 NOT TO REPRODUCE OR COPY IT 111 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART 1V ALL RIGHTS RESERVED	BRANCH	dvt1
	PAGE	72 OF 145
	SHEET	68 OF 119

BOM_COST_GROUP=CPU & CHIPSET



SYNC_MASTER=J79_SILUCHEN		SYNC_DATE=04/02/2015	
PAGE TITLE			
Empty			
 Apple Inc.	DRAWING NUMBER		051-00515
	REVISION		7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH		dvt1
	PAGE		73 OF 145
	SHEET		69 OF 119

D

CPU VCCGT/GTx Phase 1

C

CPU VCCGT/GTx Phase 2

B

CPU VCCGT/GTx Phase 3

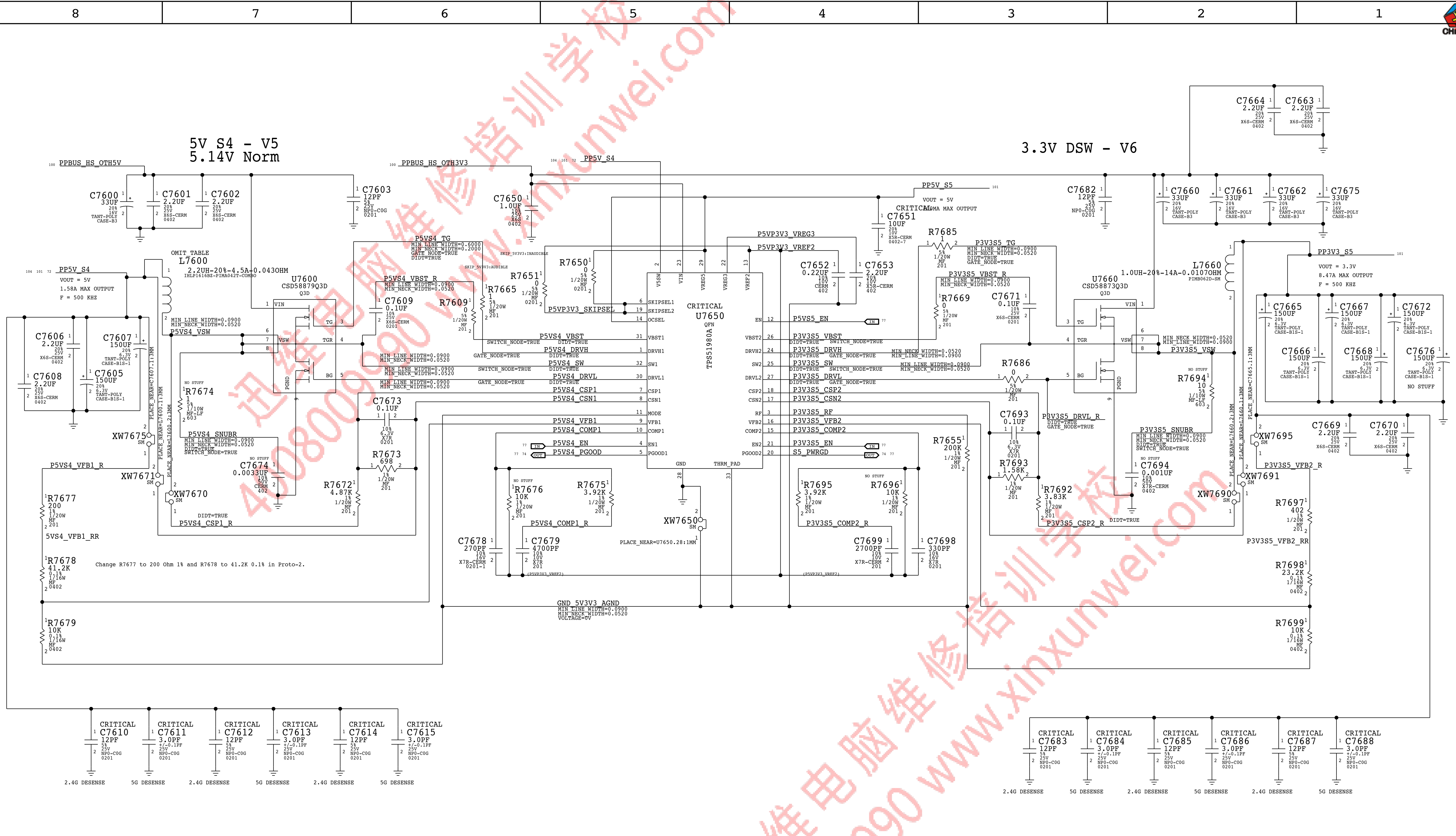
A

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00497	3	IC,FDMF5808A,DRMOS,1MVP8,50A,PQFN31,5X5	U7410, U7420, U7430	CRITICAL	

GT & GTX IMVP POWER BLOCK		DRAWING NUMBER	051-00515	STEP	D
Apple Inc.		REVISION	7.0.0		
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	dvt1		
		PAGE	74 OF 145		
		SHEET	70 OF 119		

BOM_COST_GROUP=CPU & CHIPSET

SYNC MASTER=J79 SILUCHEN		SYNC DATE=03/27/2015	
PAGE TITLE			
Empty			
 Apple Inc.	DRAWING NUMBER		STEE
	051-00515		D
	REVISION		
NOTICE OF PROPRIETARY PROPERTY:		7.0.0	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		dvt1	
		PAGE	
		75 OF 145	
		SHEET	
		71 OF 119	



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
152S00182	1	IND, PWR, 2.2UH, 20%, 4.5A, 43MOHM, 4X4MM	L7600	CRITICAL	

PAGE TITLE			Power - 5V 3.3V Supply	
		DRAWING NUMBER	051-00515	STEP D
		REVISION	7.0.0	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: 1 TO MAINTAIN THIS DOCUMENT IN CONFIDENCE 11 NOT TO REPRODUCE OR COPY IT 111 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART 1111 ALL RIGHTS RESERVED		BRANCH	dv1	
		PAGE	76 OF 145	
		SHEET	72 OF 119	

BOM_COST_GROUP=PLATFORM POWER

4

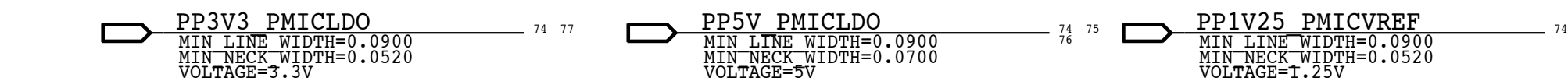


	5
--	---

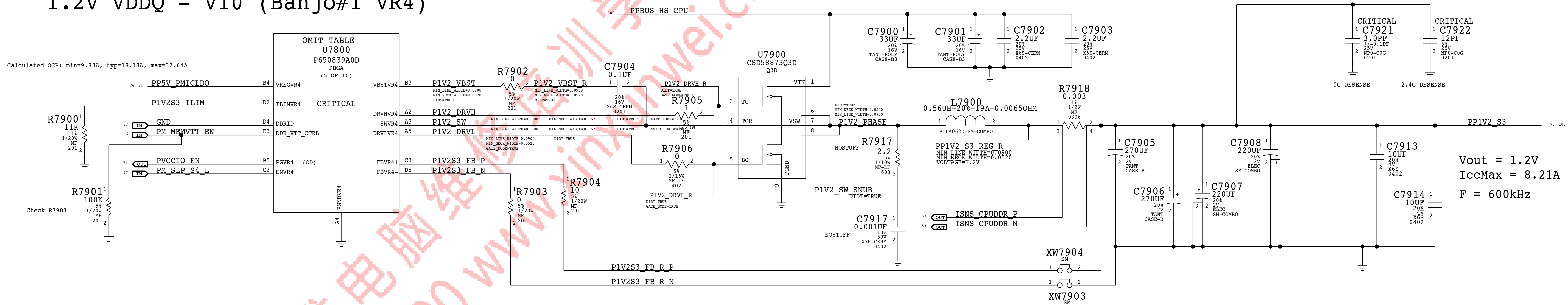
2	
---	--



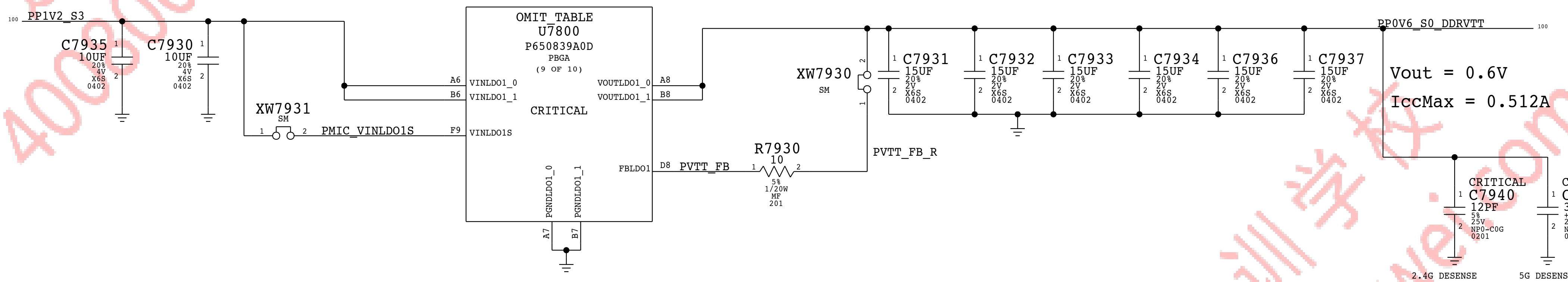
2	1
---	---

[illegible]BOM_COST_GROUP=PLATFORM POWER

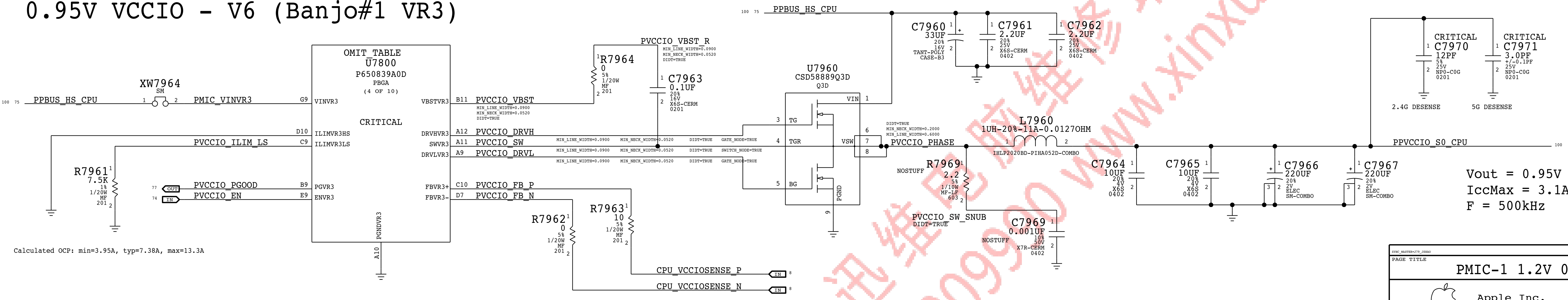
1.2V VDDQ - V10 (Banjo#1 VR4)



0.6V VTT - V13 (Banjo#1 LD01)

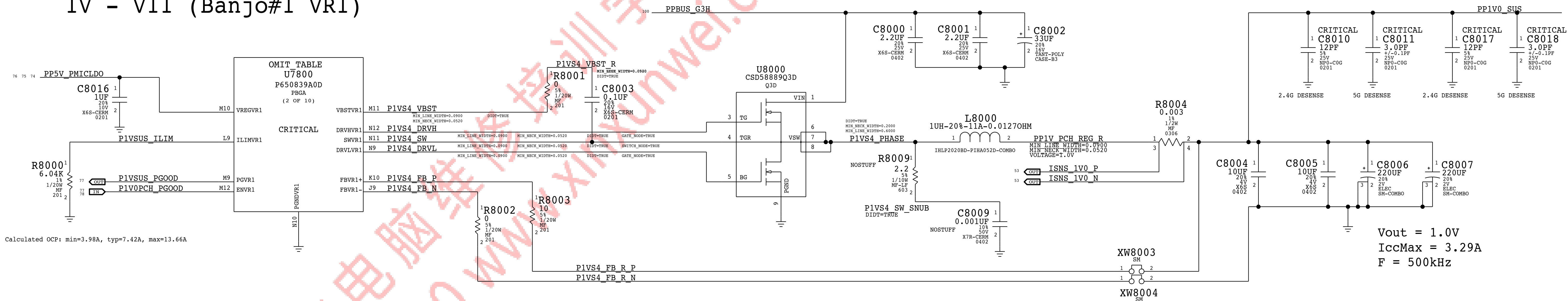


0.95V VCCIO - V6 (Banjo#1 VR3)

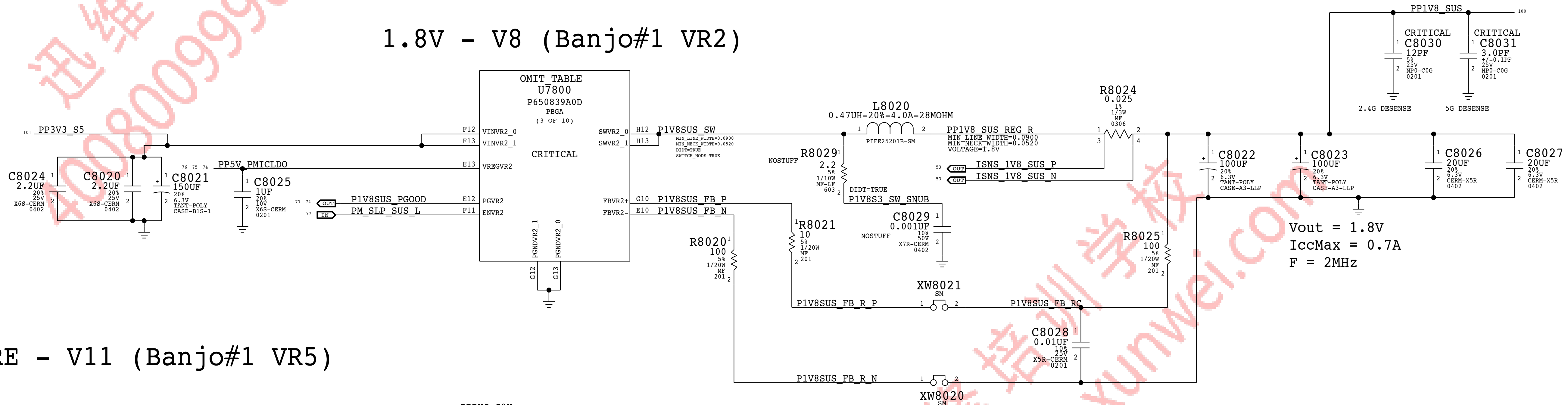
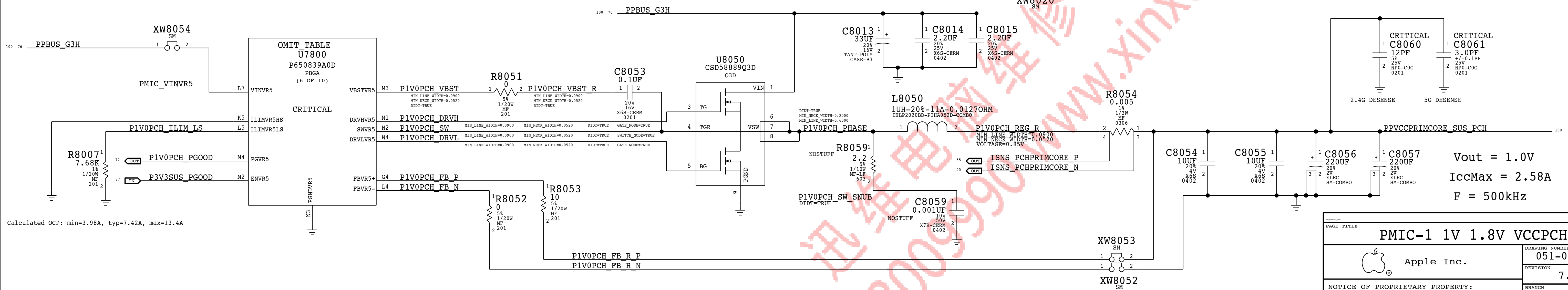


PMIC-1 1.2V 0.6V VCCIO		
Apple Inc.		
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: 1 TO MAINTAIN THIS DOCUMENT IN CONFIDENCE 11 NOT TO REPRODUCE OR COPY IT 111 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART 1111 ALL RIGHTS RESERVED		
DRAWING NUMBER 051-00515	STEP D	
REVISION 7.0.0	BRANCH dvt1	
PAGE 79 OF 145	SHEET 75 OF 119	

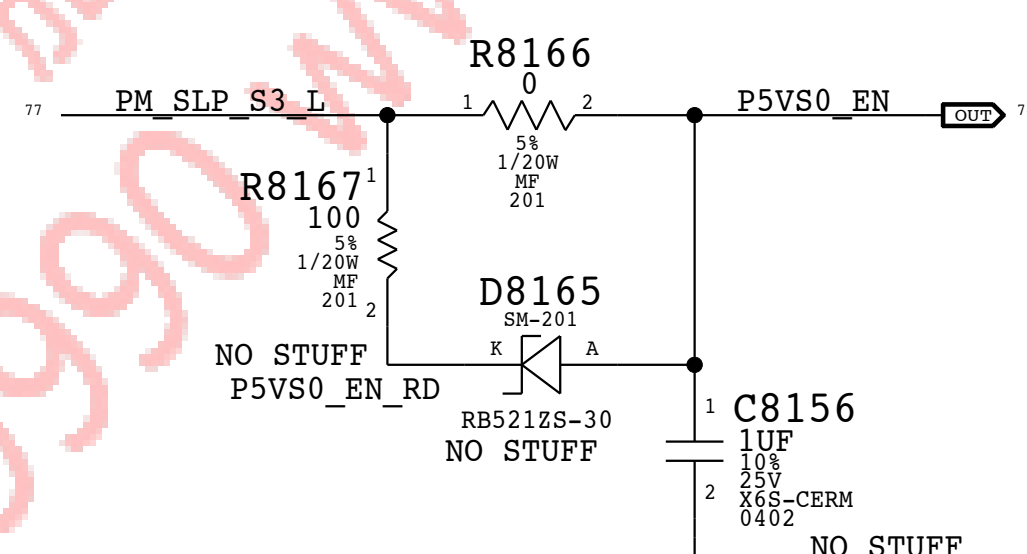
1V - V11 (Banjo#1 VR1)



1.8V - V8 (Banjo#1 VR2)

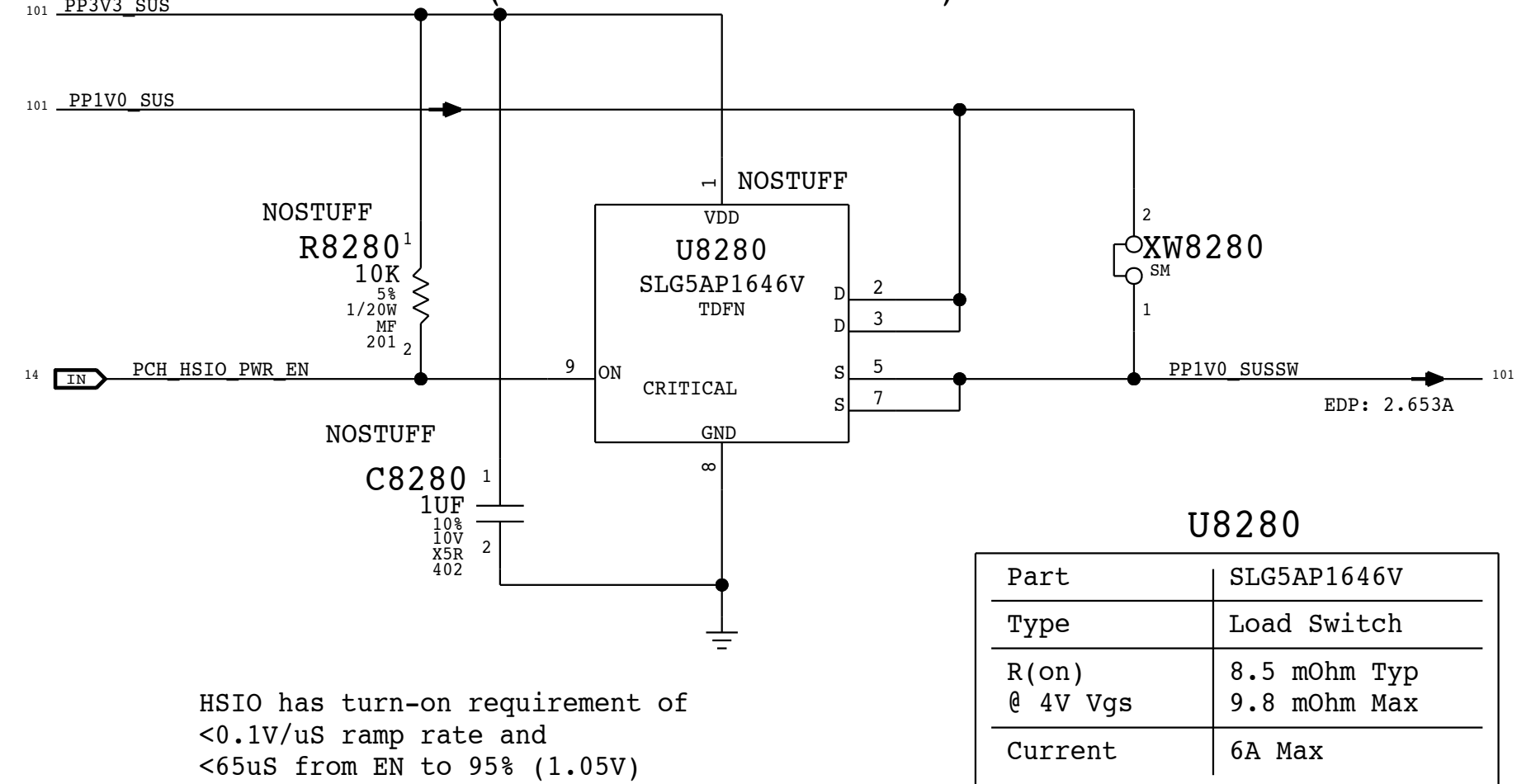
1.0V PCH CORE - V11 (Banjo#1 VR5)
0.7V LPM

PAGE TITLE		
PMIC-1 1V 1.8V VCCPCH		
	DRAWING NUMBER	051-00515
	REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: 1 TO MAINTAIN THIS DOCUMENT IN CONFIDENCE 11 NOT TO REPRODUCE OR COPY IT 111 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART 1V ALL RIGHTS RESERVED	BRANCH	dvt1
	PAGE	80 OF 145
	SHEET	76 OF 119



DRAWING NUMBER 051-00515		SHEET D
REVISION 7.0.0		
BRANCH dvt1		
PAGE 81 OF 145		
SHEET 77 OF 119		

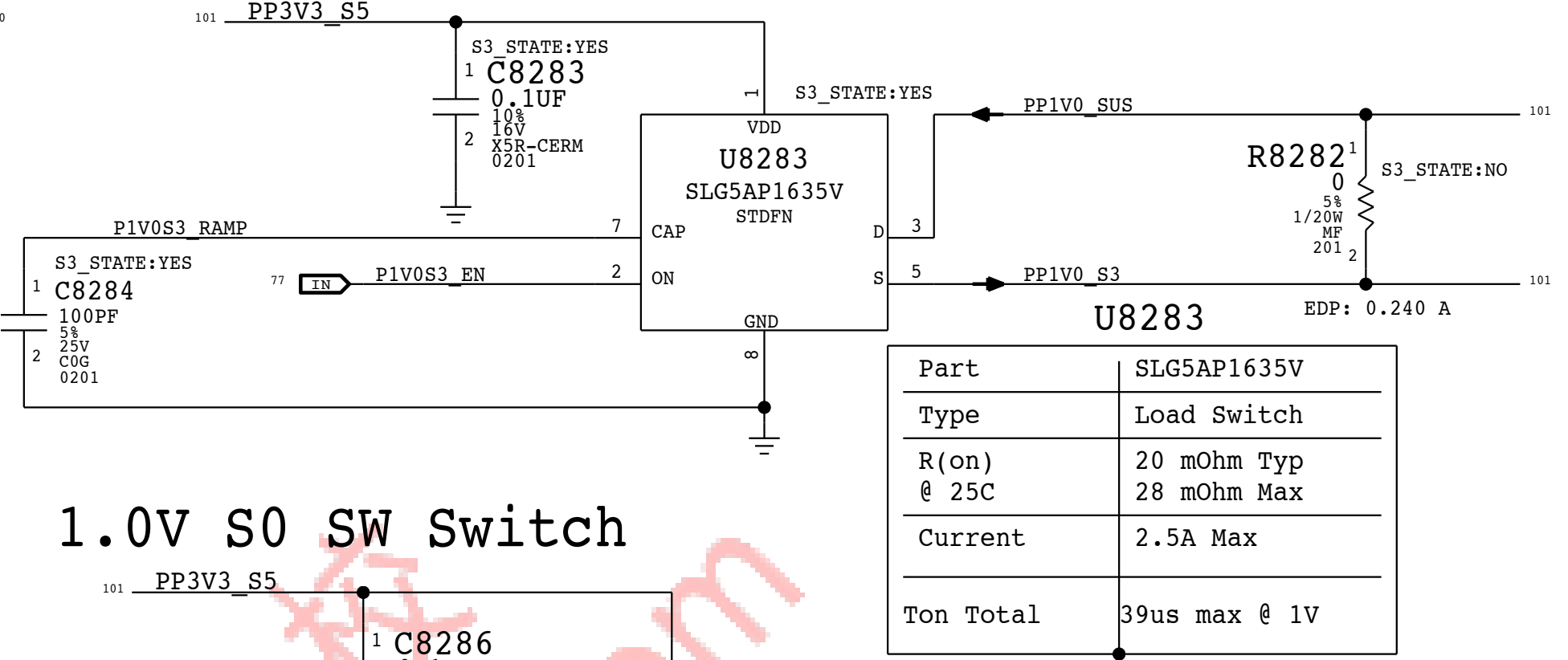
1.0V SUS SW Switch
(CURRENTLY DE-FEATURED)



Part	SLG5AP1646V
Type	Load Switch
R(on) @ 4V Vgs	8.5 mOhm Typ 9.8 mOhm Max
Current	6A Max

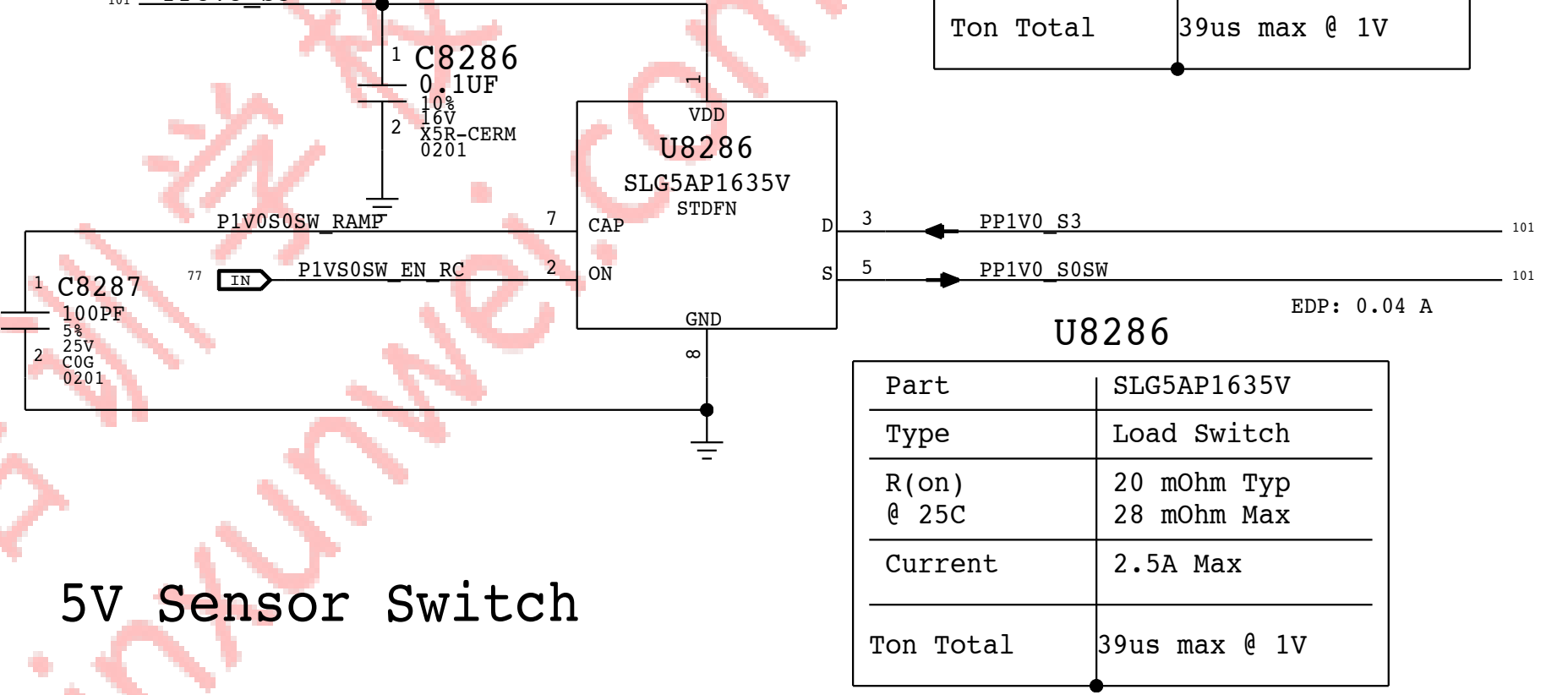
HSIO has turn-on requirement of
<0.1V/uS ramp rate and
<65uS from EN to 95% (1.05V)

1.0V S3 Switch



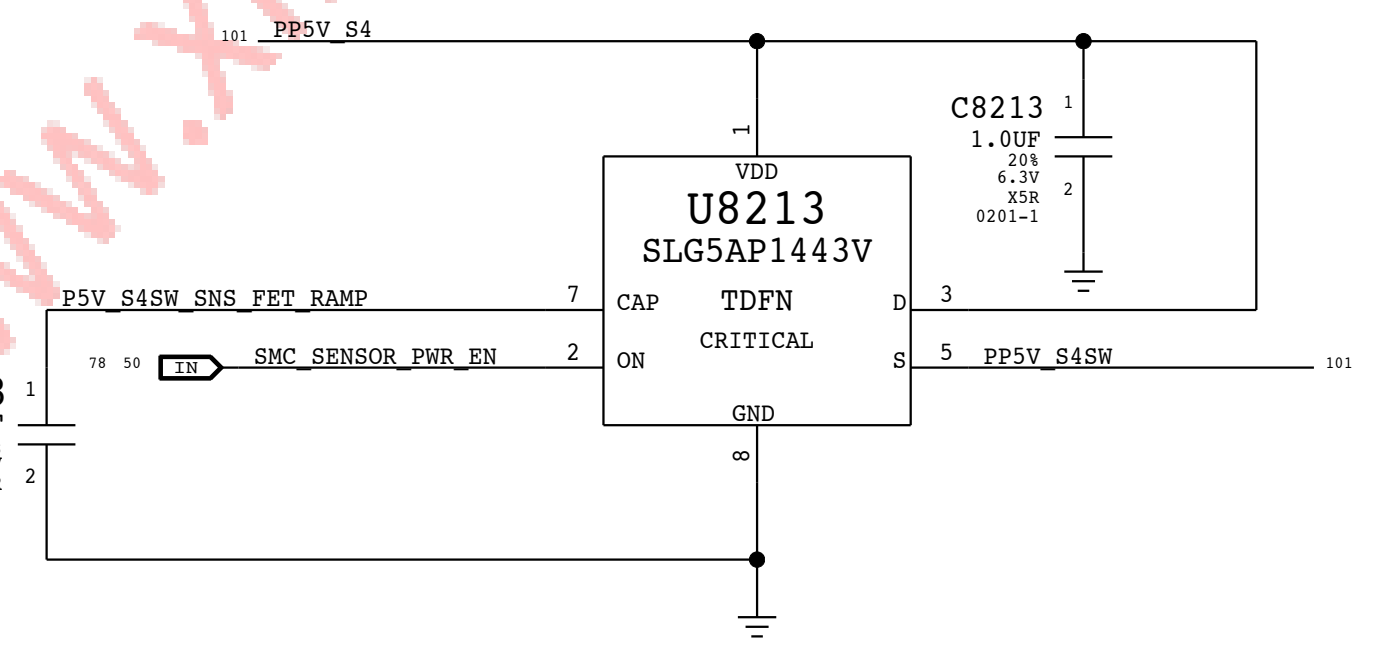
Part	SLG5AP1635V
Type	Load Switch
R(on) @ 25C	20 mOhm Typ 28 mOhm Max
Current	2.5A Max
Ton Total	39us max @ 1V

1.0V S0 SW Switch



Part	SLG5AP1635V
Type	Load Switch
R(on) @ 25C	20 mOhm Typ 28 mOhm Max
Current	2.5A Max
Ton Total	39us max @ 1V

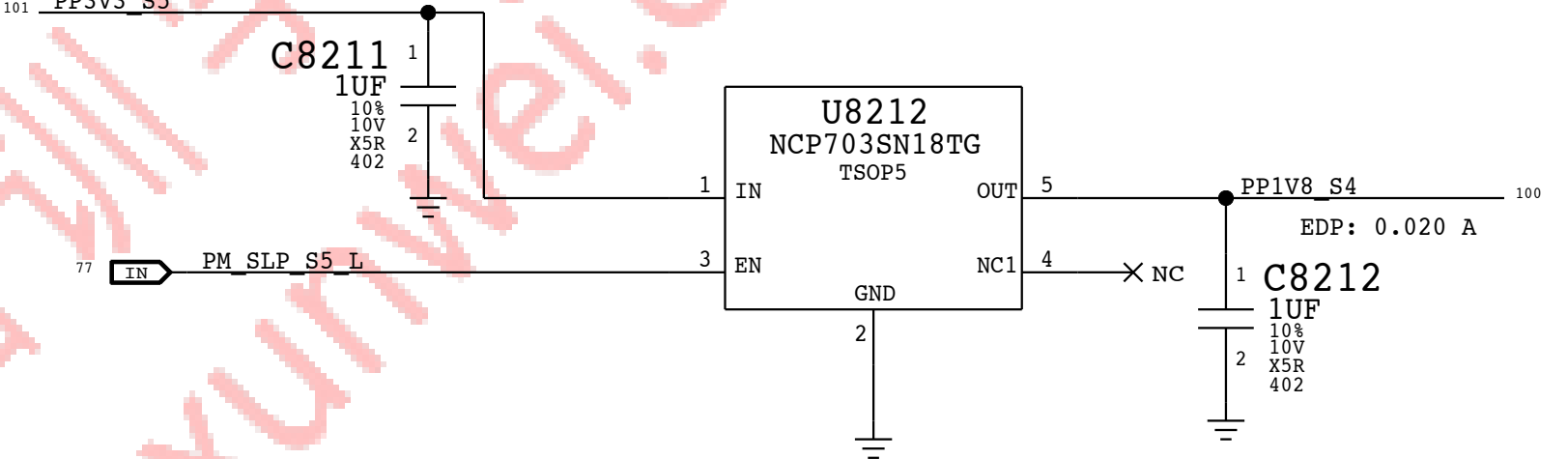
5V Sensor Switch



Part	SLG5AP1443V
Type	Load Switch
R(on) @ 25C	17 mOhm Typ 19 mOhm Max
Current	2.5A

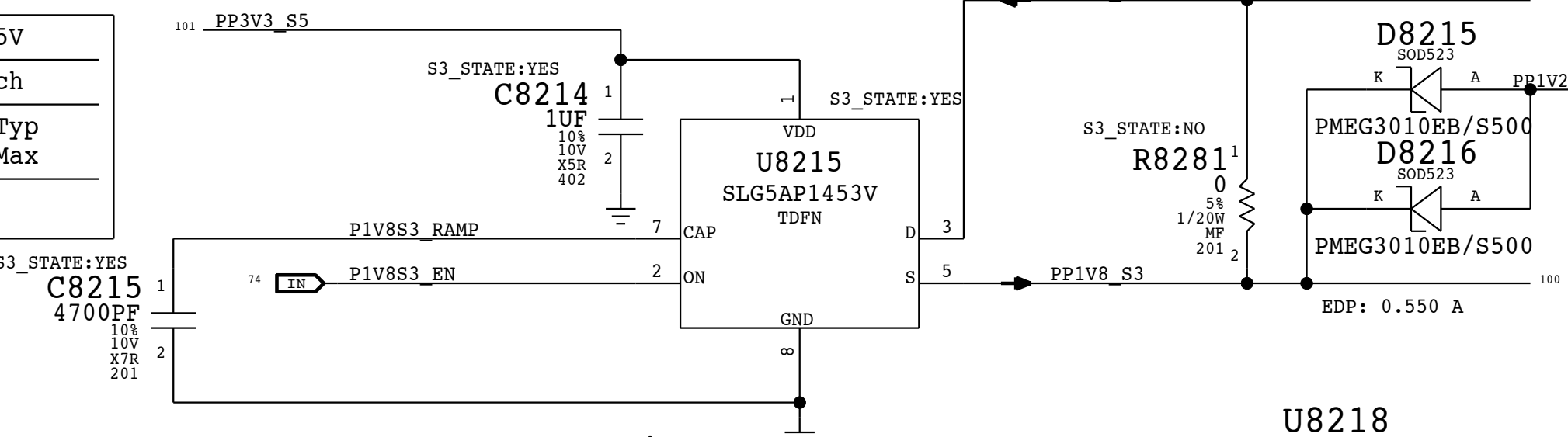
BOM_COST_GROUP=PLATFORM POWER

1.8V S4 LDO



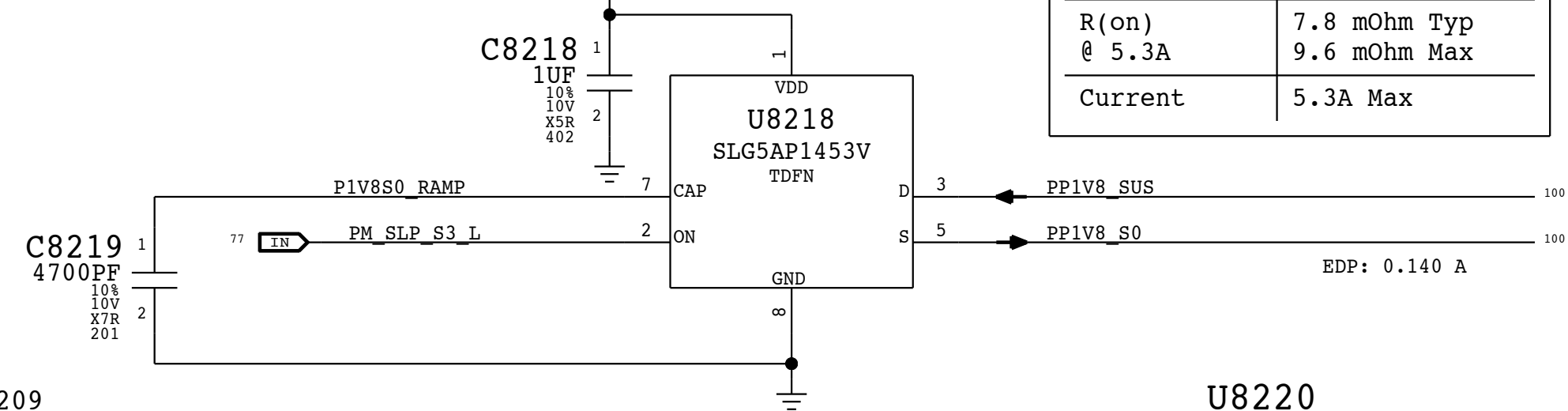
Part	SLG5AP1453V
Type	Load Switch
R(on) @ 5.3A	7.8 mOhm Typ 9.6 mOhm Max
Current	5.3A Max

1.8V S3 Switch



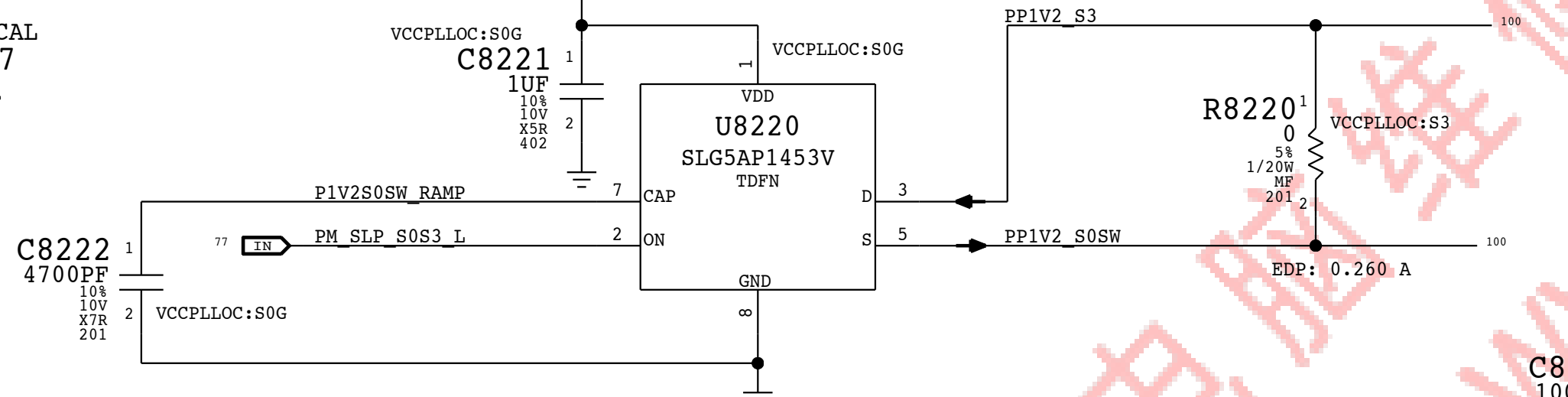
Part	SLG5AP1453V
Type	Load Switch
R(on) @ 5.3A	7.8 mOhm Typ 9.6 mOhm Max
Current	5.3A Max

1.8V S0 Switch



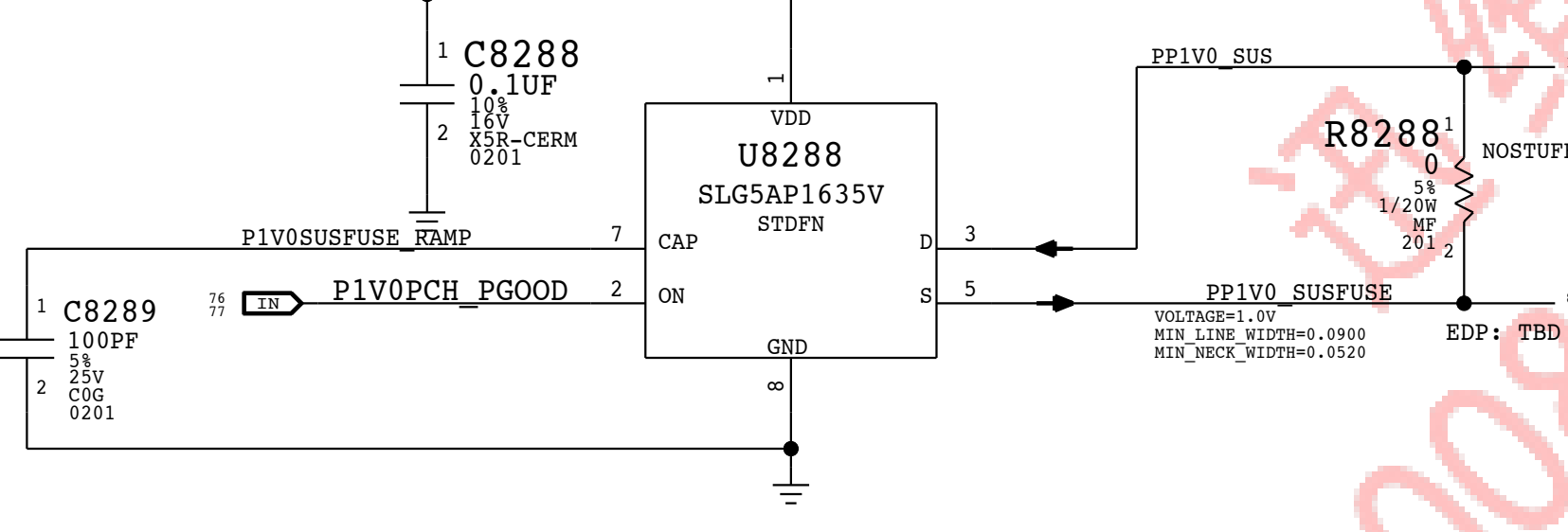
Part	SLG5AP1453V
Type	Load Switch
R(on) @ 5.3A	7.8 mOhm Typ 9.6 mOhm Max
Current	5.3A Max

1.2V S0 SW Switch

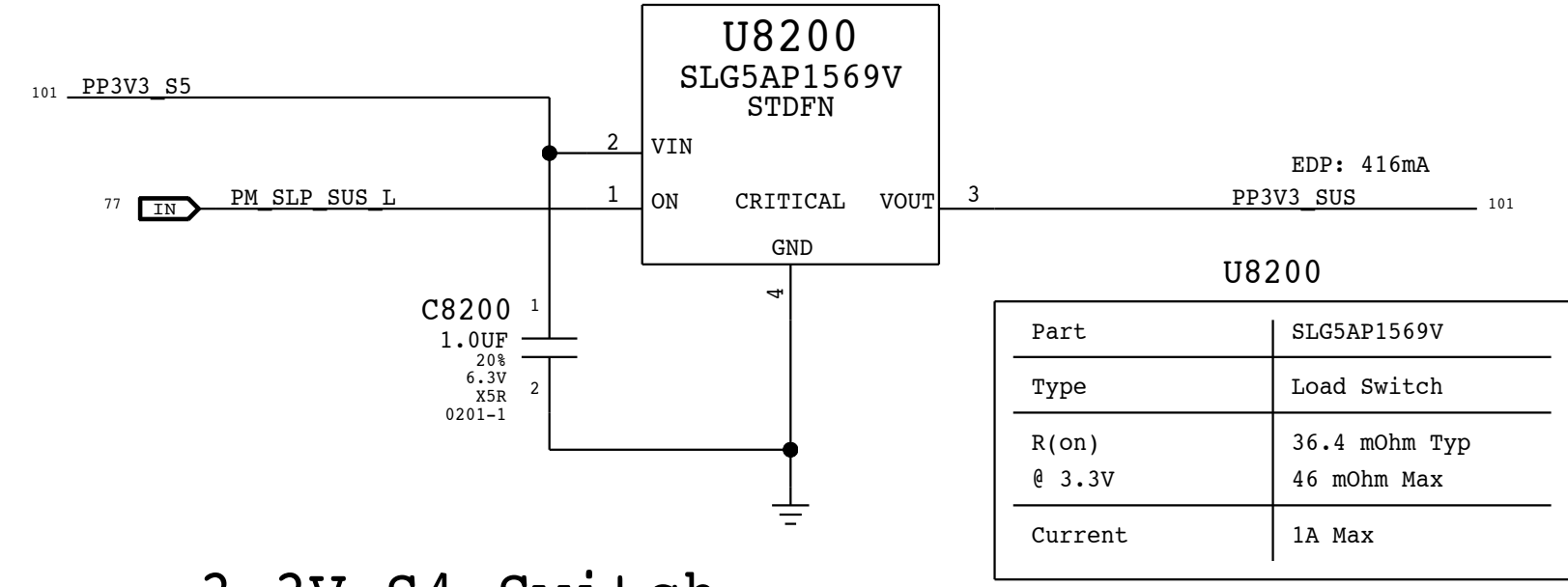


Part	SLG5AP1635V
Type	Load Switch
R(on) @ 25C	20 mOhm Typ 28 mOhm Max
Current	2.5A Max
Ton Total	39us max @ 1V
R(discharge)	300 Ohm Max

1.0V SUS FUSE Switch

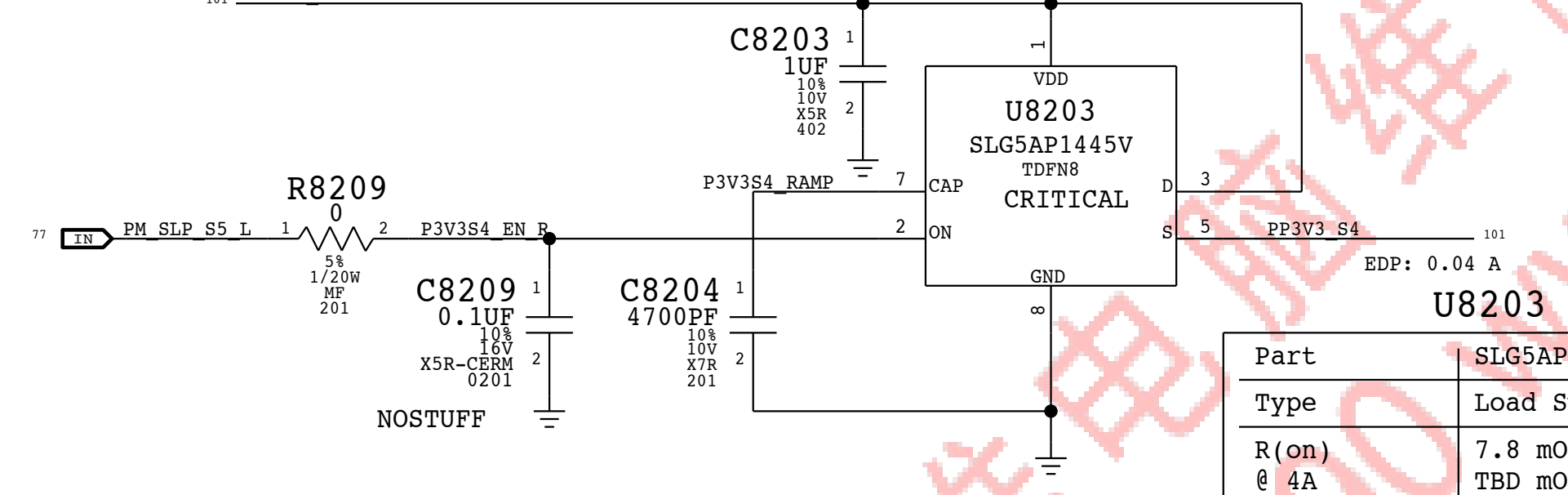


3.3V SUS Switch



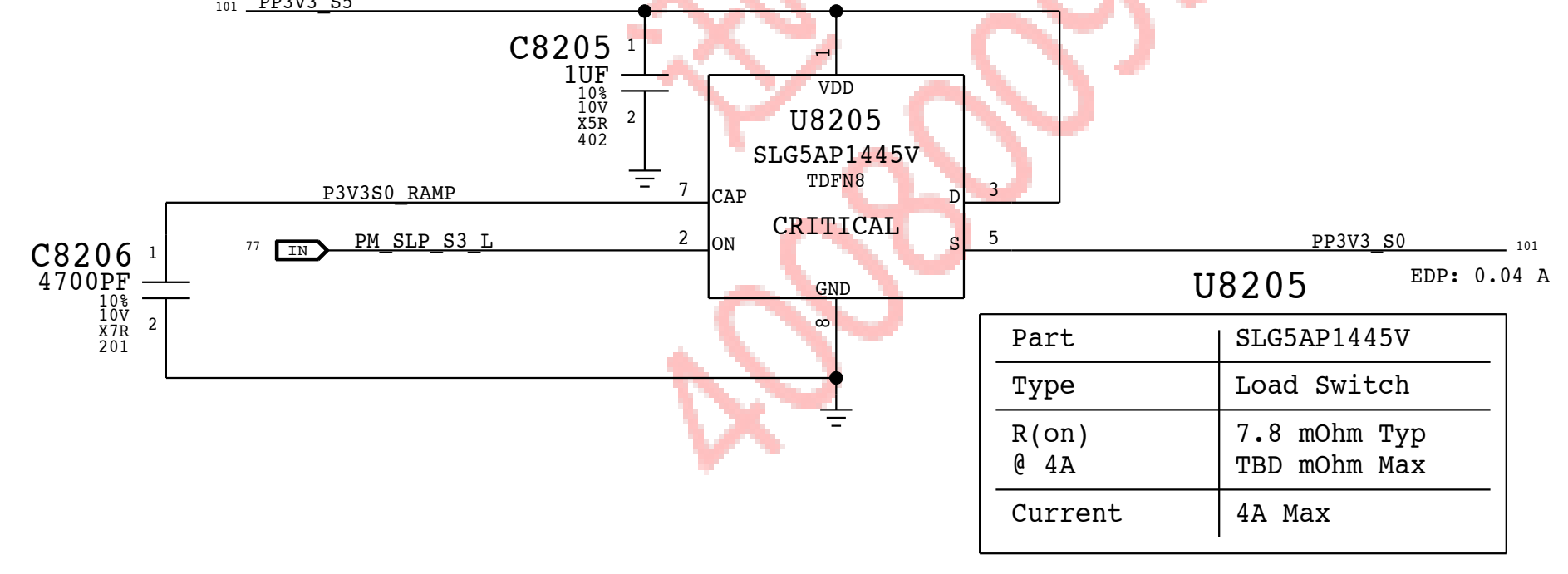
Part	SLG5AP1569V
Type	Load Switch
R(on) @ 3.3V	36.4 mOhm Typ 46 mOhm Max
Current	1A Max

3.3V S4 Switch



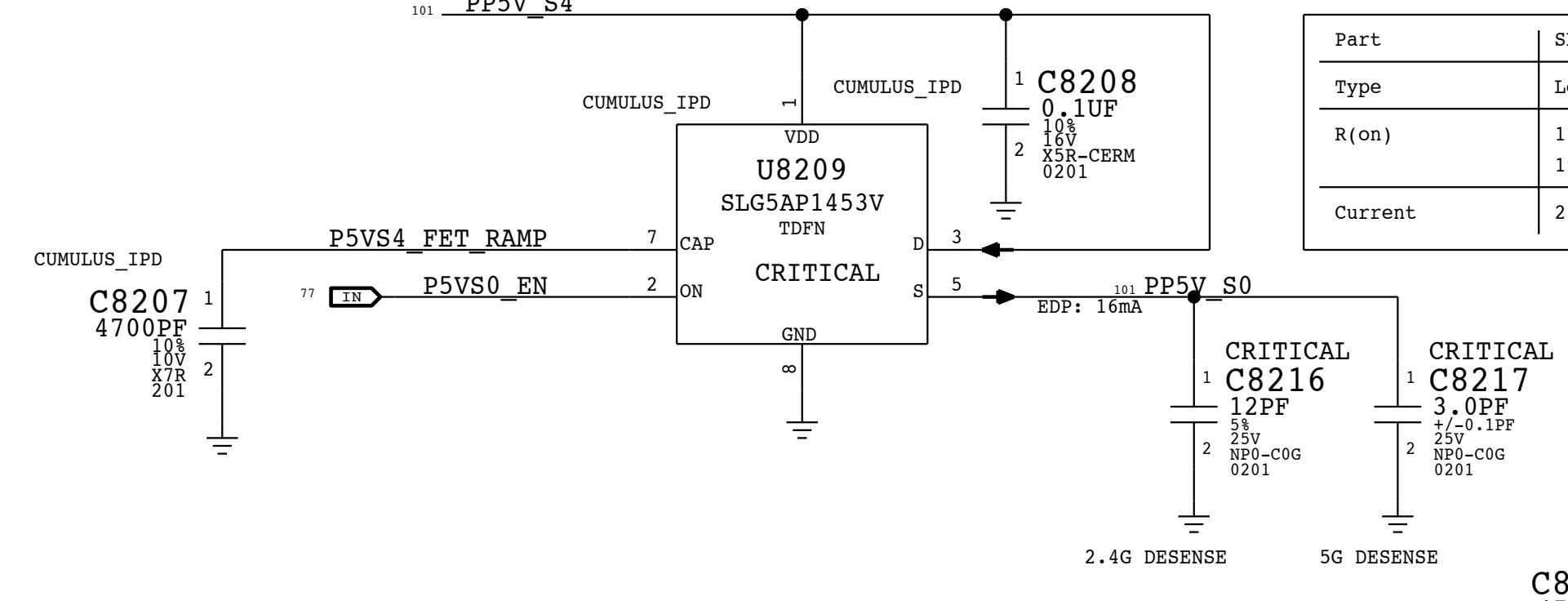
Part	SLG5AP1445V
Type	Load Switch
R(on) @ 4A	7.8 mOhm Typ TBD mOhm Max
Current	4A Max

3.3V S0 Switch



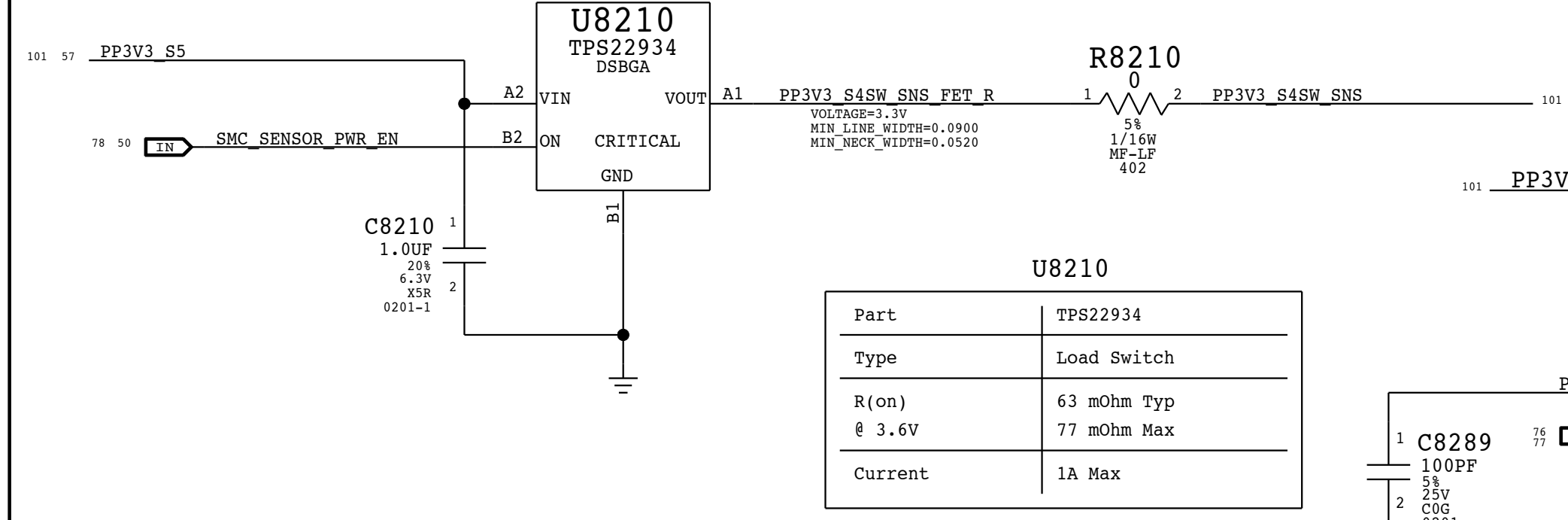
Part	SLG5AP1445V
Type	Load Switch
R(on) @ 4A	7.8 mOhm Typ TBD mOhm Max
Current	4A Max

5V S0 Switch (Cumulus vs Kona)



Part	SLG5AP1443V
Type	Load Switch
R(on)	17 mOhm Typ 19 mOhm Max
Current	2.5A

3.3V Sensor Switch



Part	TPS22934
Type	Load Switch
R(on) @ 3.6V	63 mOhm Typ 77 mOhm Max
Current	1A Max

PLATFORM RESET NO LONGER GATES THE BKLT EN AS BOTH COME FROM PCH NOW

BOM COST GROUP=DISPLAY

U8510

X604_DISP_SMC_RST_L

10

NC0 5 NC_U8510_5

NC1 11 NC_U8510_11

NO_TEST=1

NO_TEST=1

GND

7

104 90 EDP_INT_ML_N<3>

104 80 EDP_INT_ML_P<3>

25 26 I2C_BKLT_SCL

27 28 SMBUS_SMC_0_S0_SDA

29 30 SMBUS_SMC_0_S0_SCL

105 104 38 MIPI_DATA_CONN_N

105 104 38 MIPI_DATA_CONN_P

105 104 38 MIPI_CLK_CONN_N

105 104 38 MIPI_CLK_CONN_P

31 32

33 34 I2C_ALS_SDA

35 36 I2C_ALS_SCL

37 38 I2C_CAM_SCL

39 40 I2C_CAM_SDA

41 42

104 104 80 PP5V_S0SW_LCD

45 46 PP5V_S0_ALSCAM_F

47 48

49 50

51 52

53 54

55 56

57 58

59 60

61 62

63 64

65 66

67 68

PPVOUT_S0_LCDBKLT

79 80 104

C8500

1000PF

100V

X7R-CERM

0603

1

2

GND

SIGNAL_MODEL=EDP

Apple Inc.

Apple Inc.

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

BOM_COST_GROUP=DISPLAY

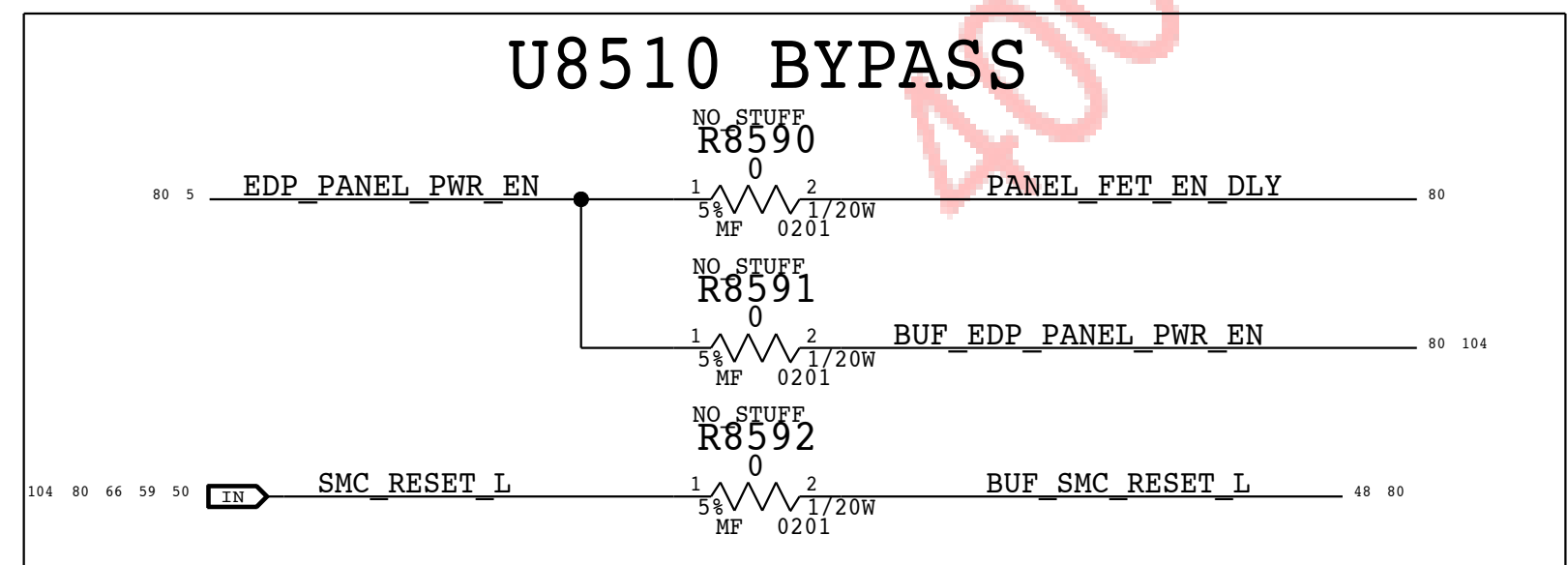
DRAWING NUMBER 051-00515

REVISION 7.0.0

BRANCH dvt1

PAGE 85 OF 145

SHEET 80 OF 119

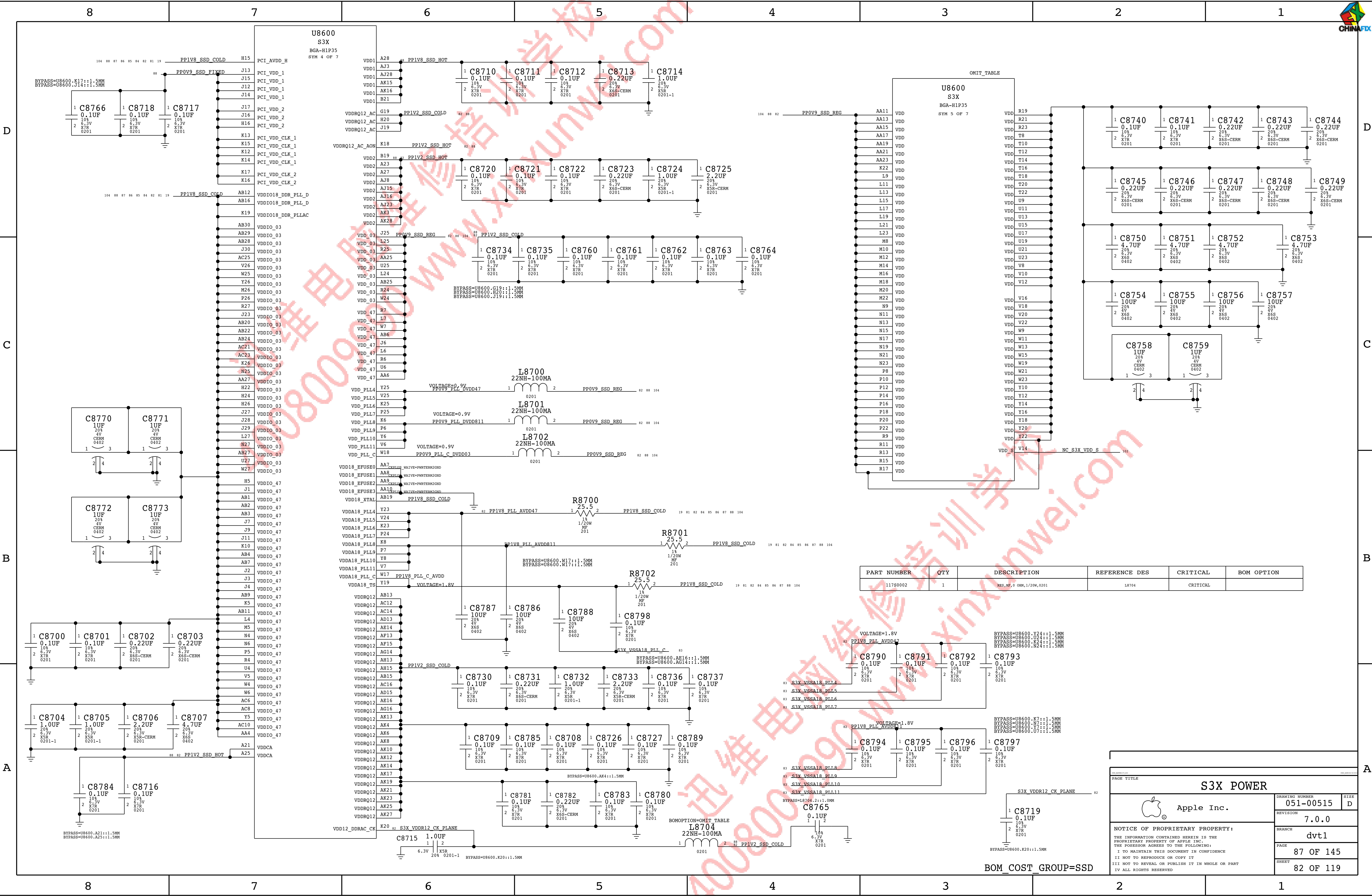


BOM_COST_GROUP=DISPLAY

PAGE TITLE		DRAWING NUMBER		SIZE	
eDP Display Connector		051-00515		D	
 Apple Inc.		REVISION		7.0.0	
NOTICE OF PROPRIETARY PROPERTY:					
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED					
BRANCH		dvt1			
PAGE		85 OF 145			
SHEET		80 OF 119			

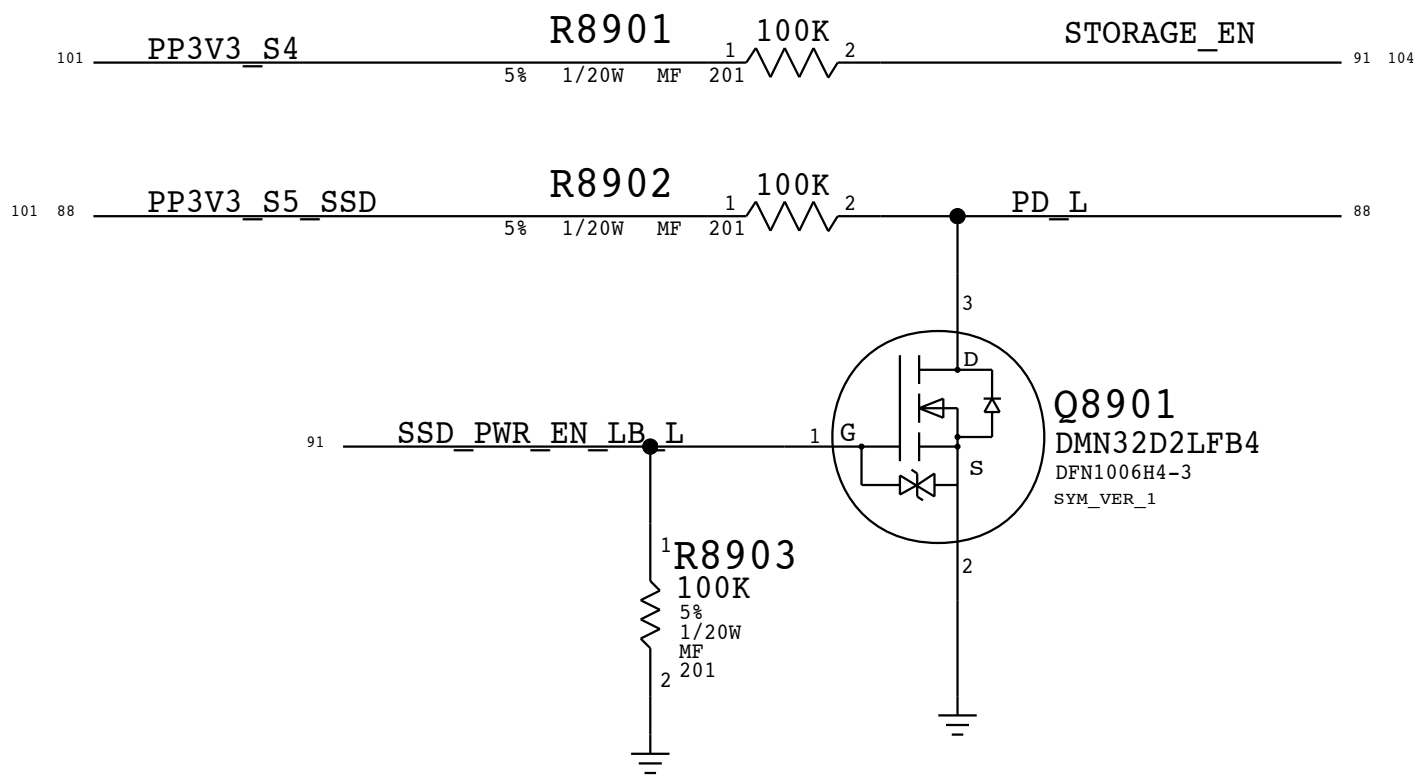
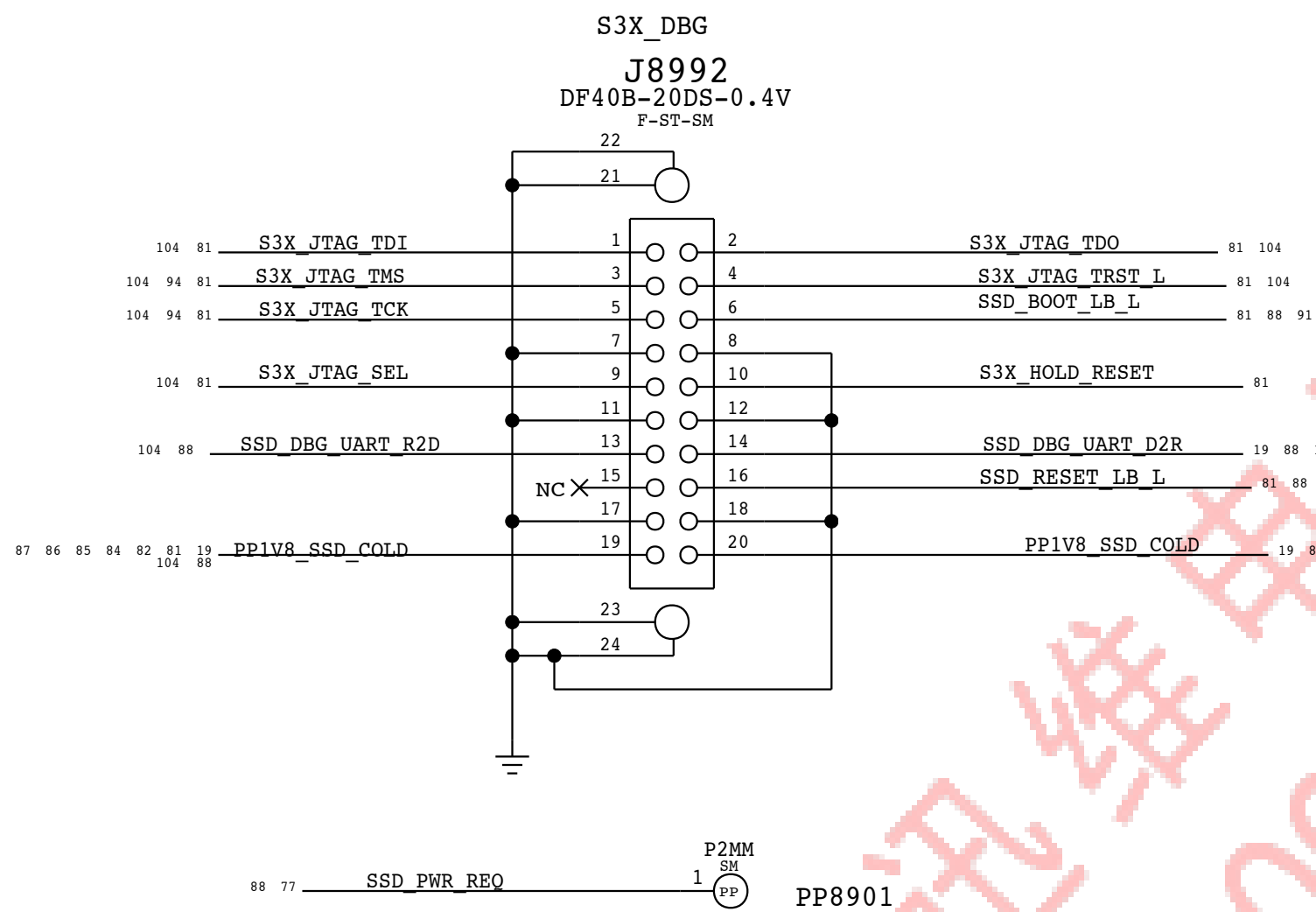
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
131S00077	1	CAP,9.5PF,50V,0201	C8601	CRITICAL	S3XCLK:INT
117S00002	1	RES,MF,0 OHM,1/20W,0201	C8601	CRITICAL	S3XCLK:EXT

BOM COST GROUP=SSD



SYNC MASTER=J79 RIO		SYNC_DATE=06/18/2015	
PAGE TITLE			
PAGE_TITLE=S3X GND			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00515		D
	REVISION		
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I II NOT TO REPRODUCE OR COPY IT I III NOT TO REVEAL OR DISCLOSE IT IN WHOLE OR PART I IV ALL RIGHTS RESERVED		7.0.0	
		BRANCH	
		dvt1	
		PAGE	88 OF 145
		SHEET	83 OF 119

JTAG (DEBUG 1)



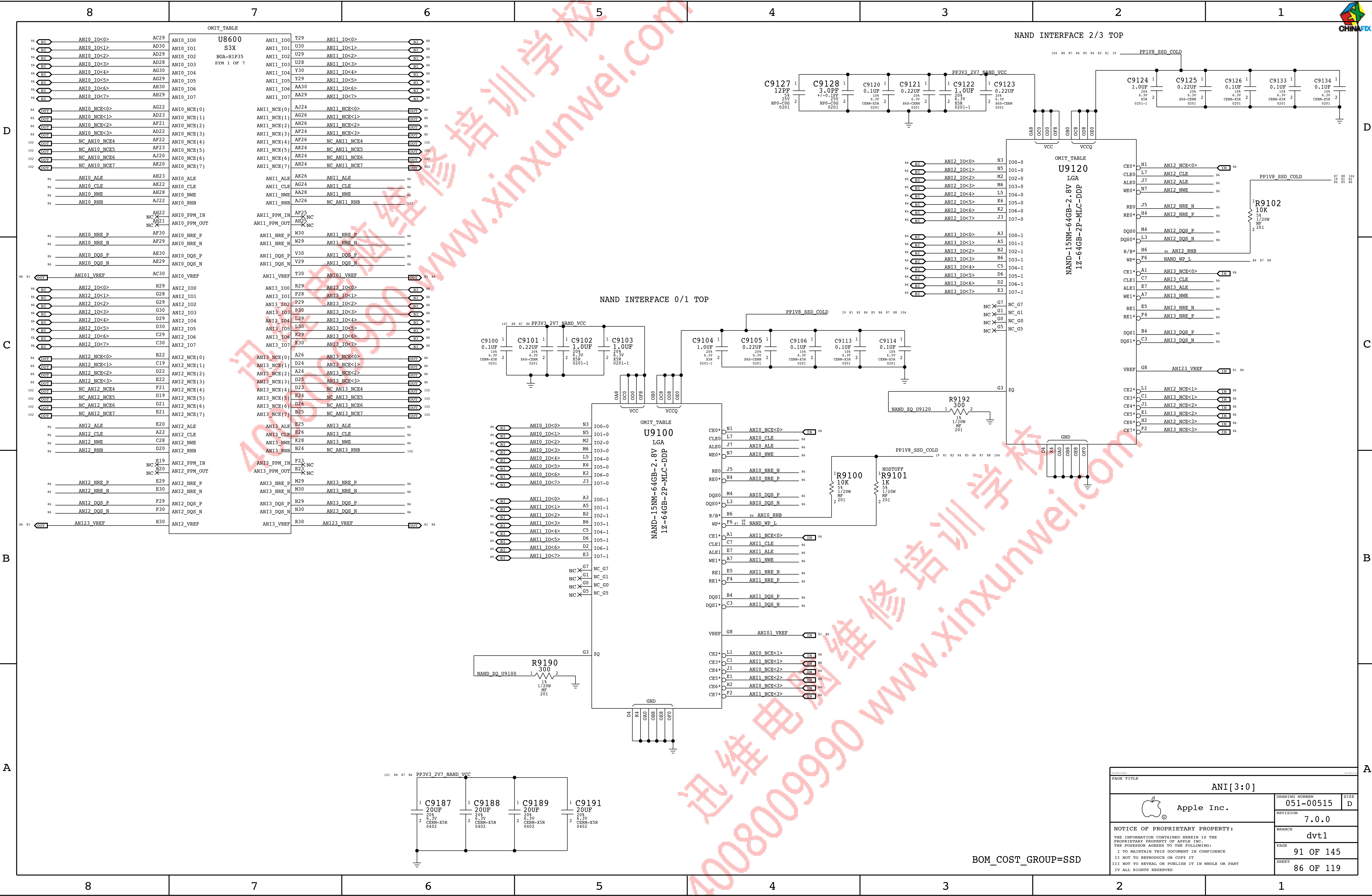
GND_VOID=TRUE		C8910	0.22UF	GND_VOID=TRUE	
91	PCIE_SSD_R2D_LB_C_P<0>	1	2	PCIE_SSD_R2D_LB_P<0>	81
GND_VOID=TRUE		C8911	0.22UF	GND_VOID=TRUE	
91	PCIE_SSD_R2D_LB_C_N<0>	1	2	PCIE_SSD_R2D_LB_N<0>	81
GND_VOID=TRUE		C8913	0.22UF	GND_VOID=TRUE	
105 15	PCIE_SSD_R2D_C_P<1>	1	2	PCIE_SSD_R2D_P<1>	81
GND_VOID=TRUE		C8912	0.22UF	GND_VOID=TRUE	
105 15	PCIE_SSD_R2D_C_N<1>	1	2	PCIE_SSD_R2D_N<1>	81
GND_VOID=TRUE		C8914	0.22UF	GND_VOID=TRUE	
105 15	PCIE_SSD_R2D_C_P<2>	1	2	PCIE_SSD_R2D_P<2>	81
GND_VOID=TRUE		C8915	0.22UF	GND_VOID=TRUE	
105 15	PCIE_SSD_R2D_C_N<2>	1	2	PCIE_SSD_R2D_N<2>	81
GND_VOID=TRUE		C8917	0.22UF	GND_VOID=TRUE	
105 15	PCIE_SSD_R2D_C_P<3>	1	2	PCIE_SSD_R2D_P<3>	81
GND_VOID=TRUE		C8916	0.22UF	GND_VOID=TRUE	
105 15	PCIE_SSD_R2D_C_N<3>	1	2	PCIE_SSD_R2D_N<3>	81

PAGE TITLE		
Connector		
 Apple Inc.	DRAWING NUMBER	051-00515
	REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH	dvt1
	PAGE	89 OF 145
	SHEET	84 OF 119

BOM_COST_GROUP=SSD

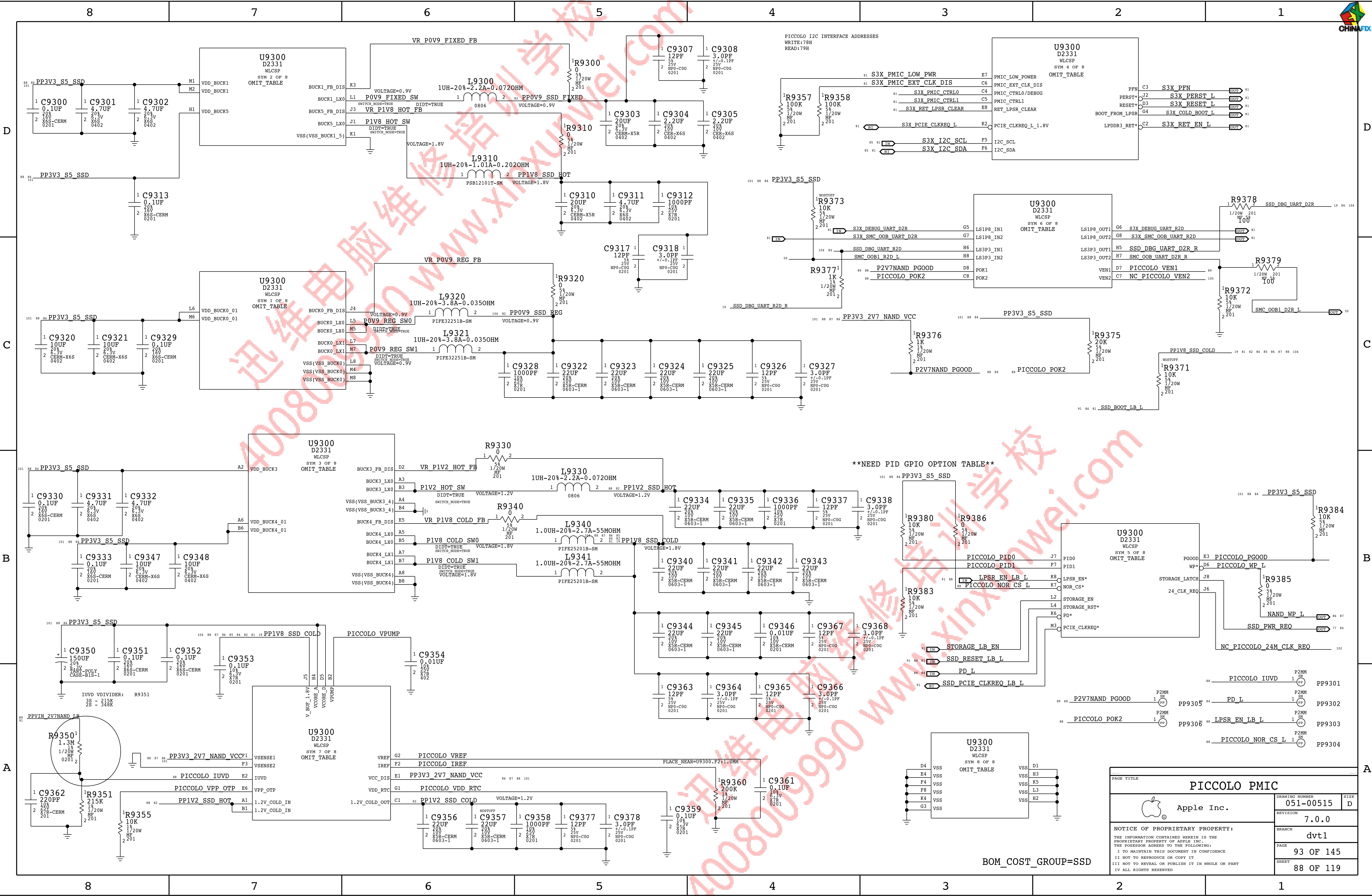
SYNC DOCUMENT'S NUMBER: PAGE TITLE NAND VR, I2C ROM, TEMP SENSORS	SYNC_DATE=09/12/2015						
Apple Inc.	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%; padding: 5px;"> DRAWING NUMBER 051-00515 </td> <td style="width: 50%; padding: 5px;"> SIZE D </td> </tr> <tr> <td colspan="2" style="padding: 5px;"> REVISION 7.0.0 </td> </tr> </table>	DRAWING NUMBER 051-00515	SIZE D	REVISION 7.0.0			
DRAWING NUMBER 051-00515	SIZE D						
REVISION 7.0.0							
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED							
<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%; padding: 5px;"> BRANCH dvt1 </td> <td style="width: 50%;"></td> </tr> <tr> <td style="padding: 5px;"> PAGE 90 OF 145 </td> <td></td> </tr> <tr> <td style="padding: 5px;"> SHEET 85 OF 119 </td> <td></td> </tr> </table>		BRANCH dvt1		PAGE 90 OF 145		SHEET 85 OF 119	
BRANCH dvt1							
PAGE 90 OF 145							
SHEET 85 OF 119							







PAGE TITLE		DRAWING NUMBER		SIZE	
ANI [7 : 4]		051-00515		D	
 Apple Inc.		REVISION		7.0.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH		dvt1	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THE DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		PAGE		92 OF 145	
		SHEET		87 OF 119	



PAGE TITLE			
PICCOLO PMIC			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00515		D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	REVISION		
	7.0.0		
	BRANCH		
	dvt1		
	PAGE		
	93 OF 145		
	SHEET		
	88 OF 119		

BOM_COST_GROUP=SSD



D

C

B

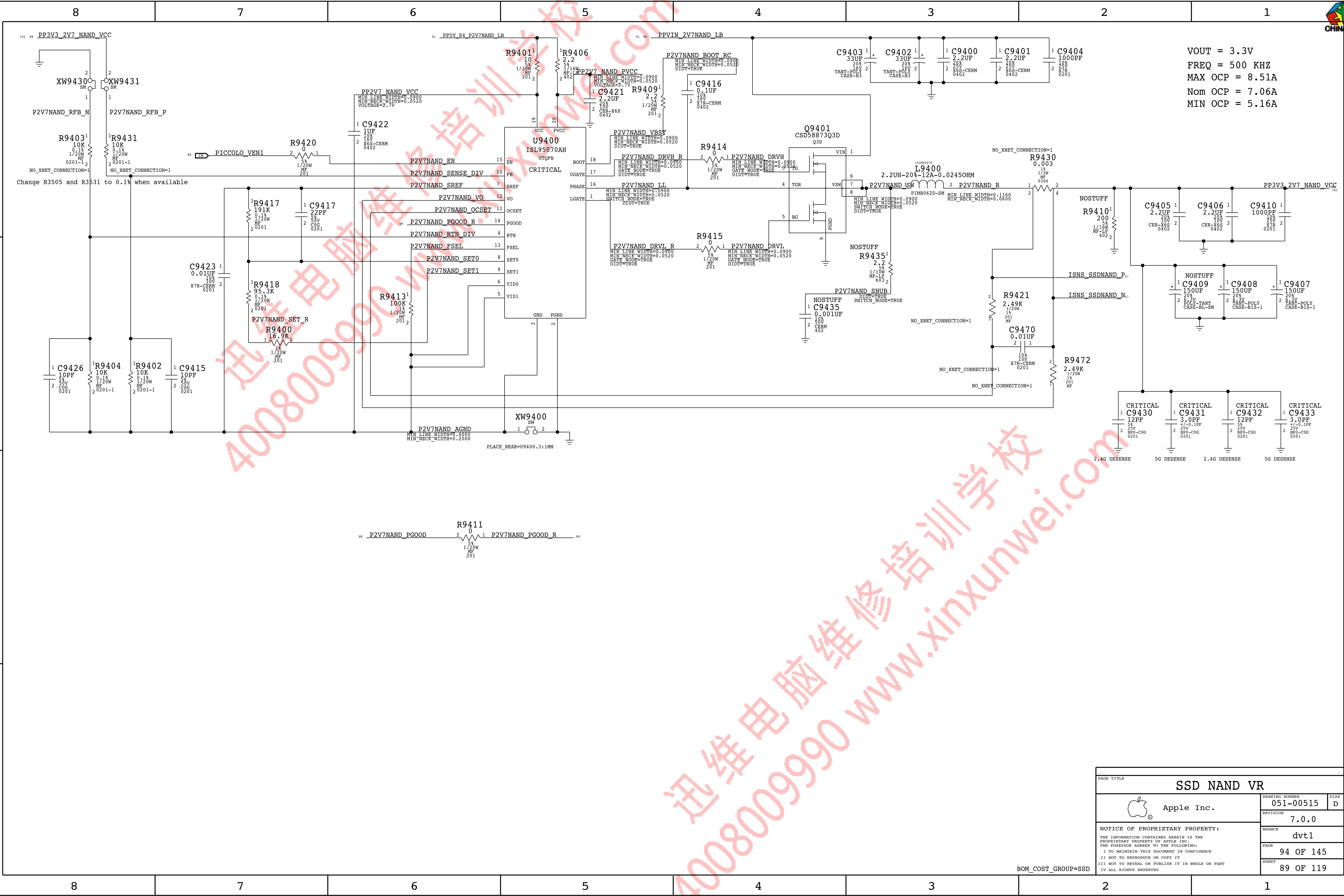
A

D

C

B

A

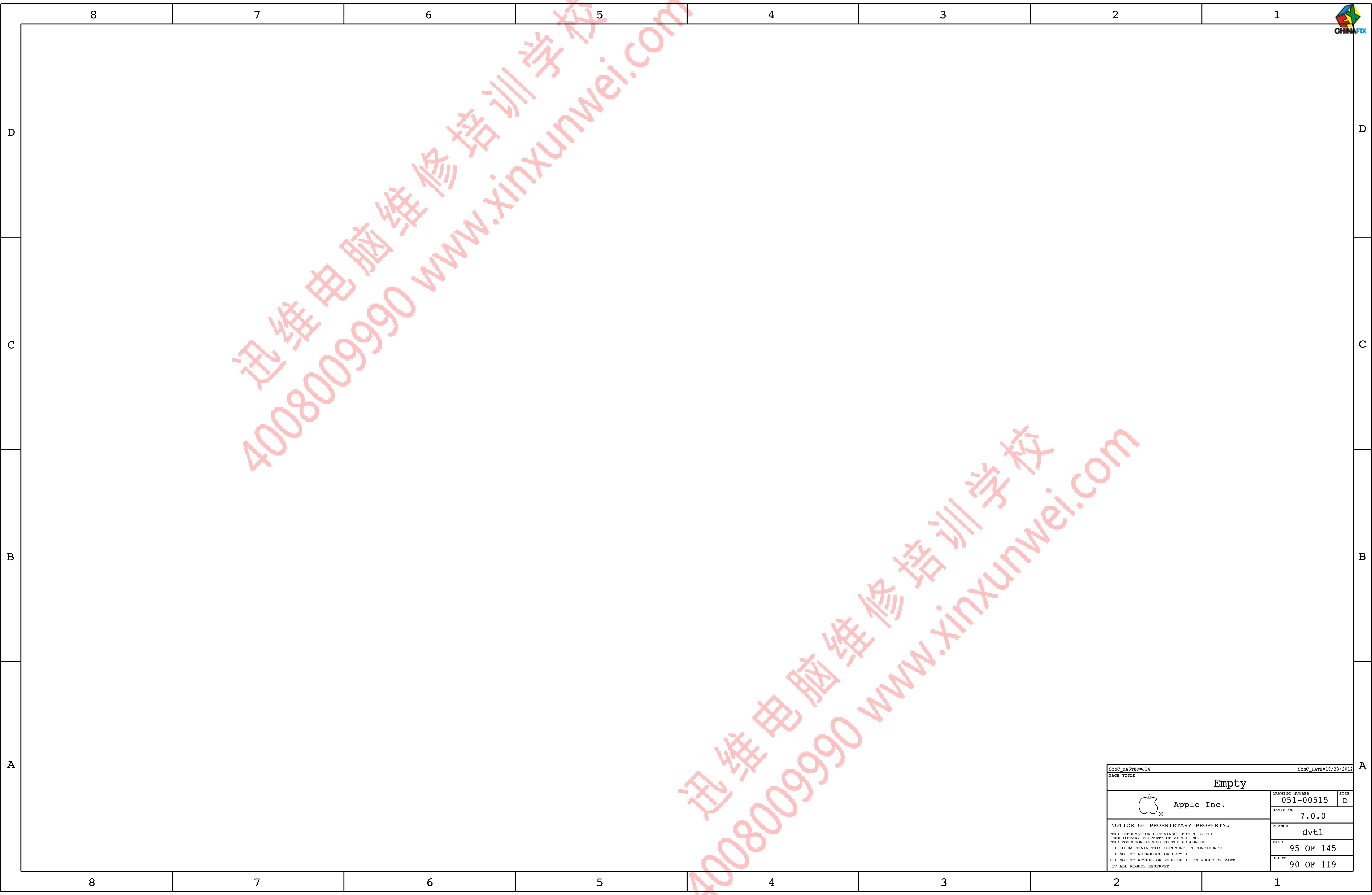


VOUT = 3.3V
FREQ = 500 KHZ
MAX OCP = 8.51A
Nom OCP = 7.06A
MIN OCP = 5.16A

R9411
P2V7NAND_PGOOD
P2V7NAND_PGOOD_R

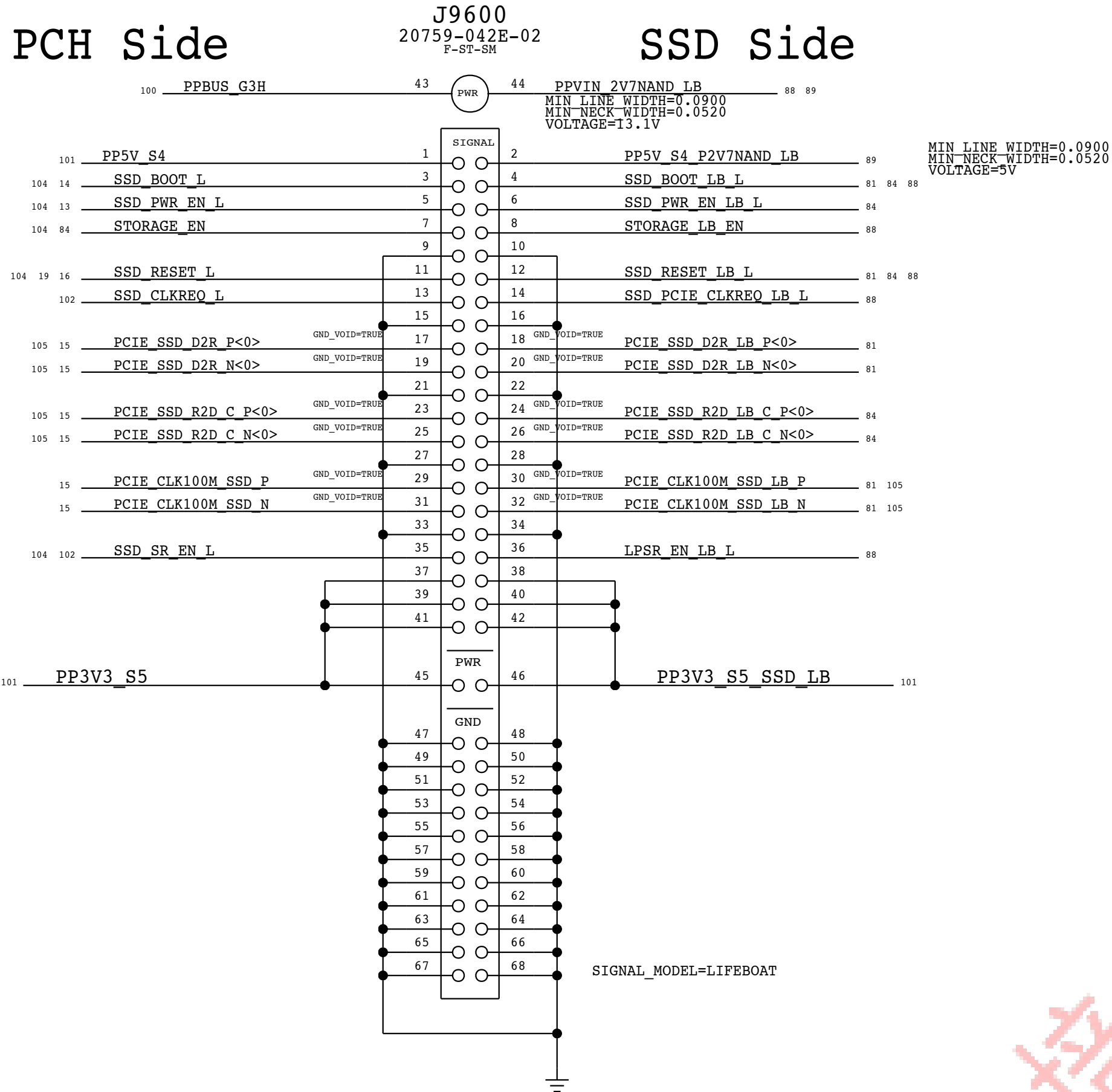
PAGE TITLE		
SSD NAND VR		
 Apple Inc.	DRANTING NUMBER	051-00515
	REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH	dvt1
	PAGE	94 OF 145
	SHEET	89 OF 119

BOM_COST_GROUP=SSD



SYNC_MASTER=J14		SYNC_DATE=10/23/2012	
PAGE TITLE			
Empty			
 Apple Inc.	DRAWING NUMBER	051-00515	SIZE D
	REVISION	7.0.0	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH	dvt1	
	PAGE	95 OF 145	
	SHEET	90 OF 119	

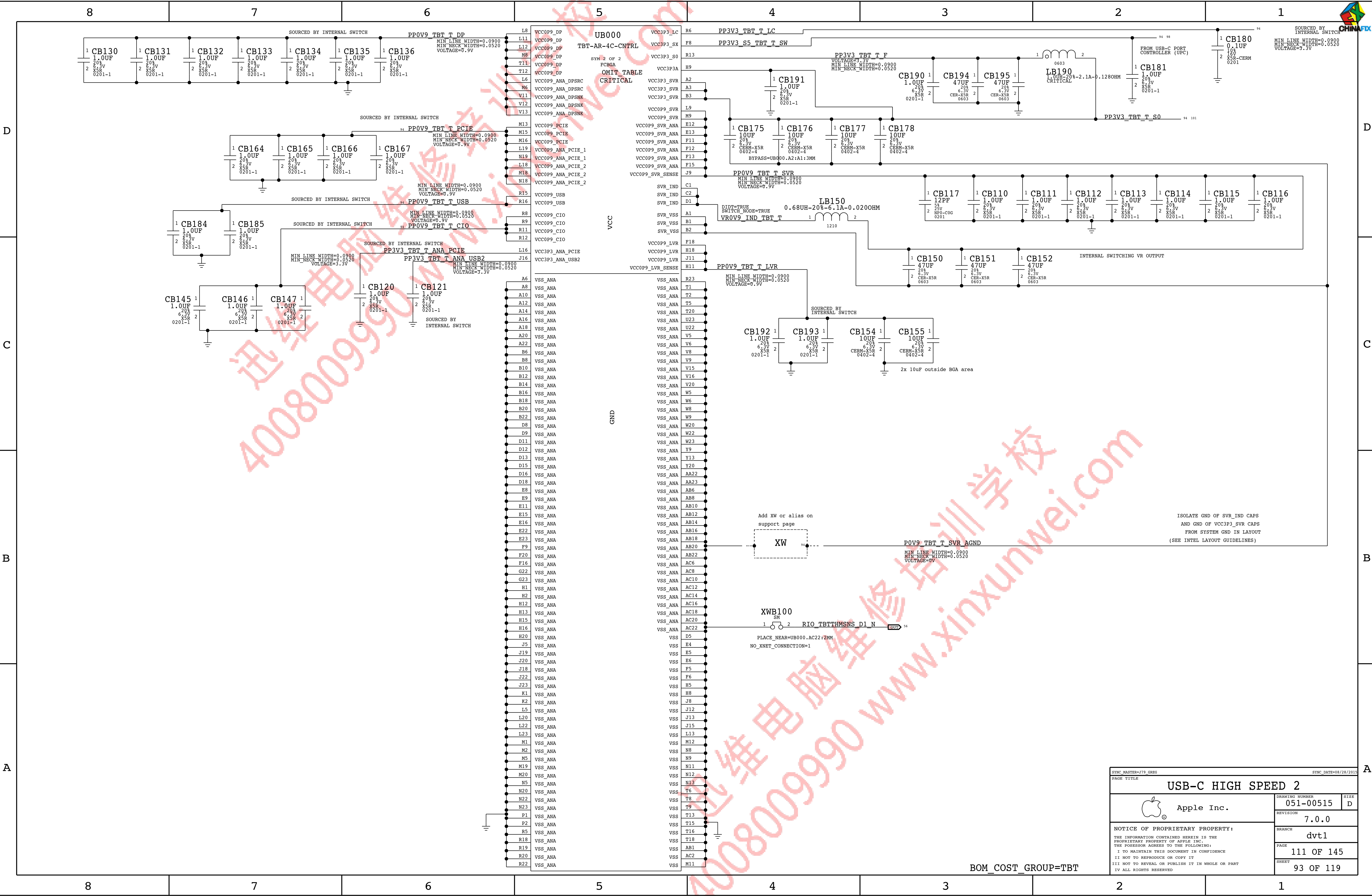
LIFEBOAT



BOM_COST_GROUP=SSD

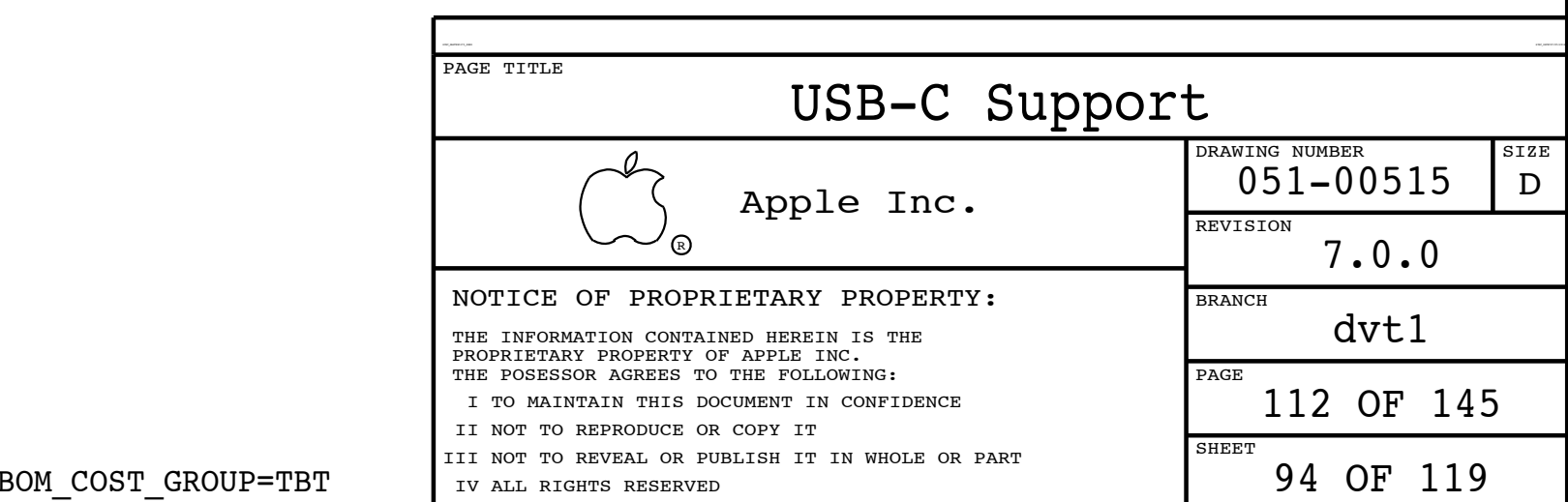
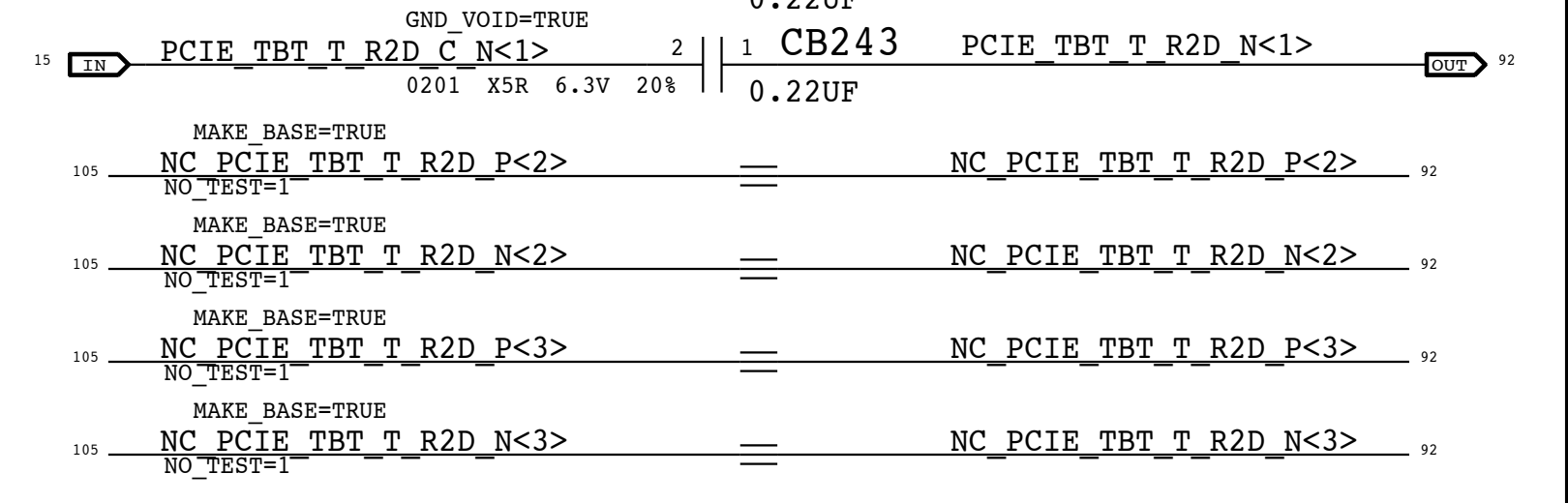
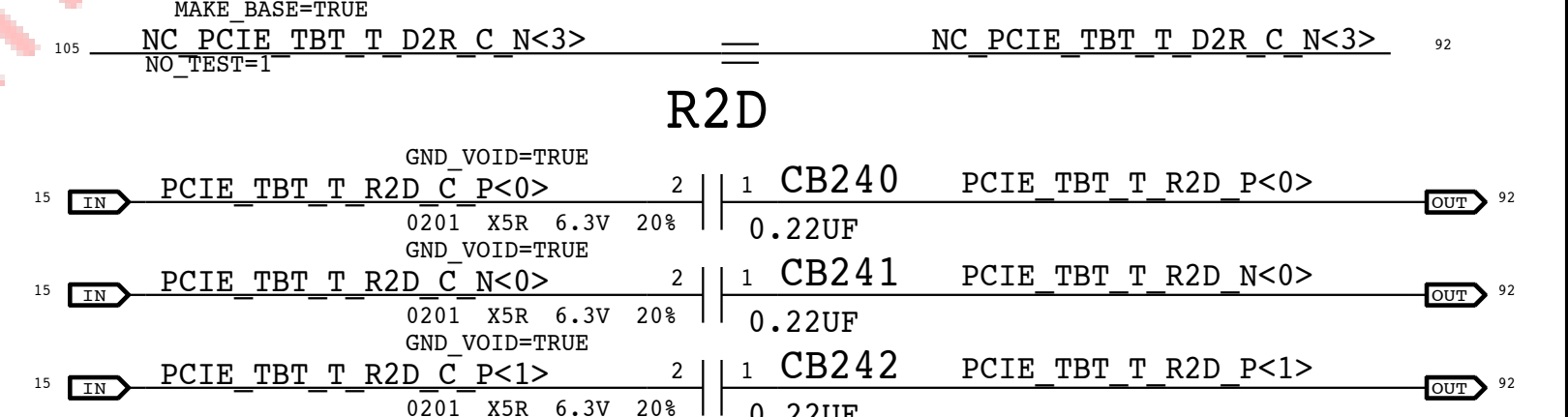
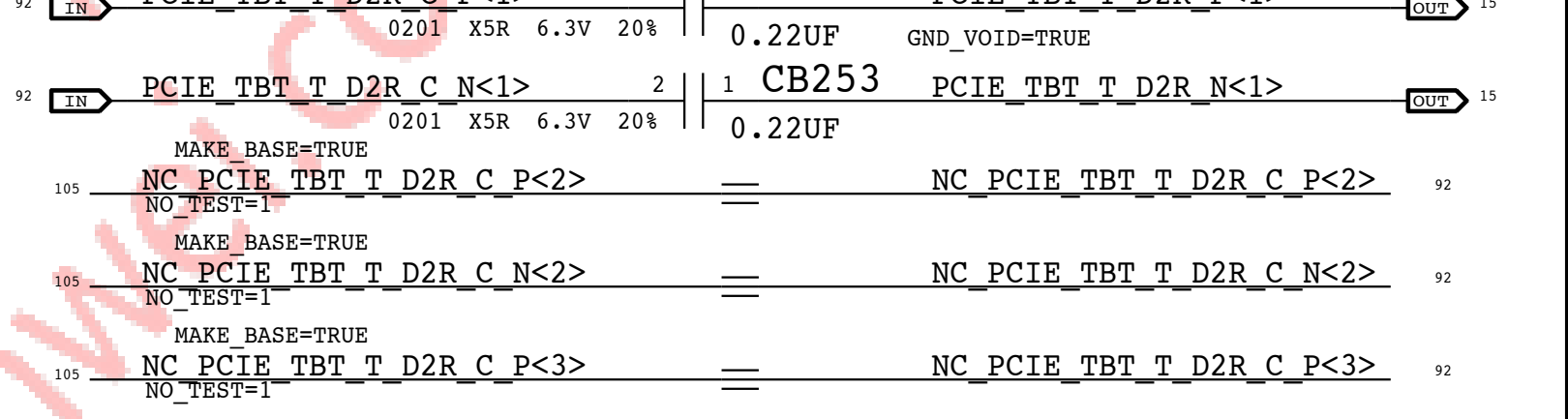
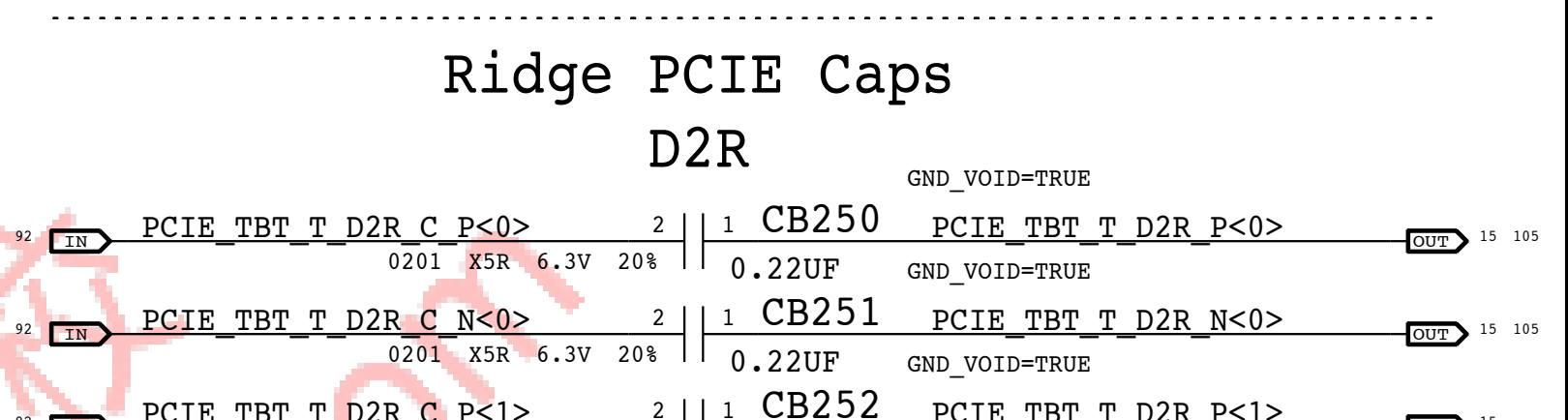
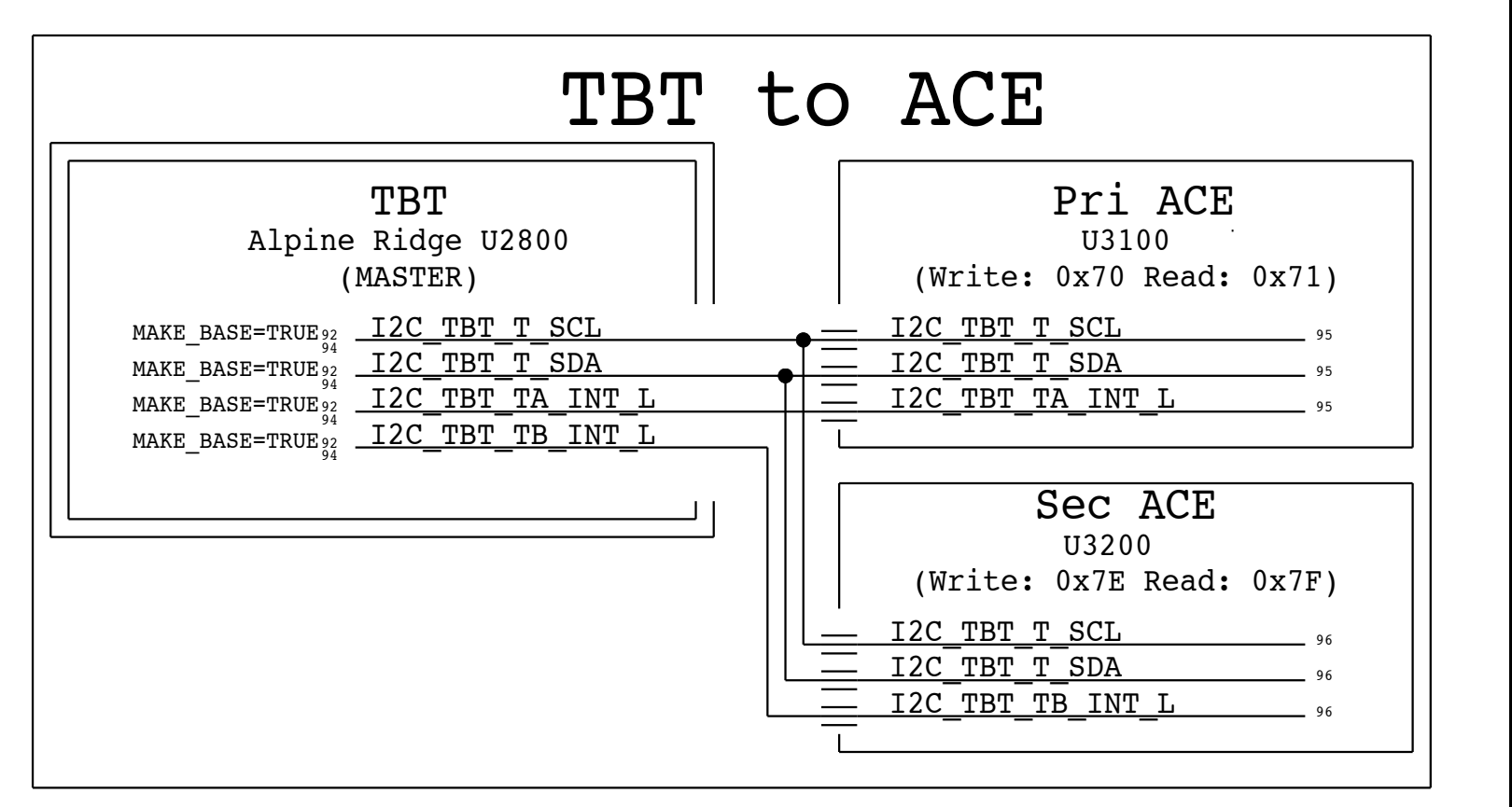
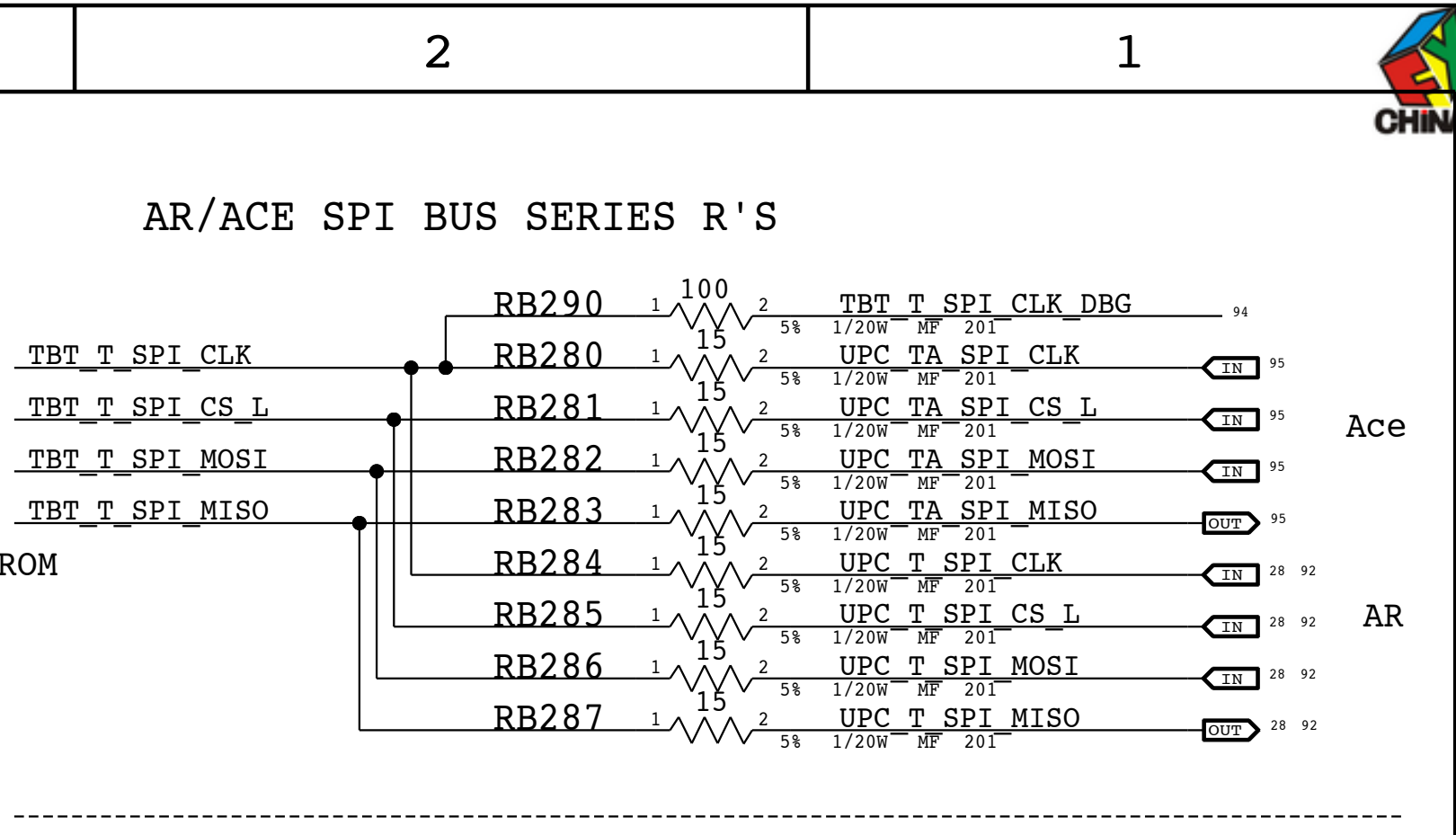
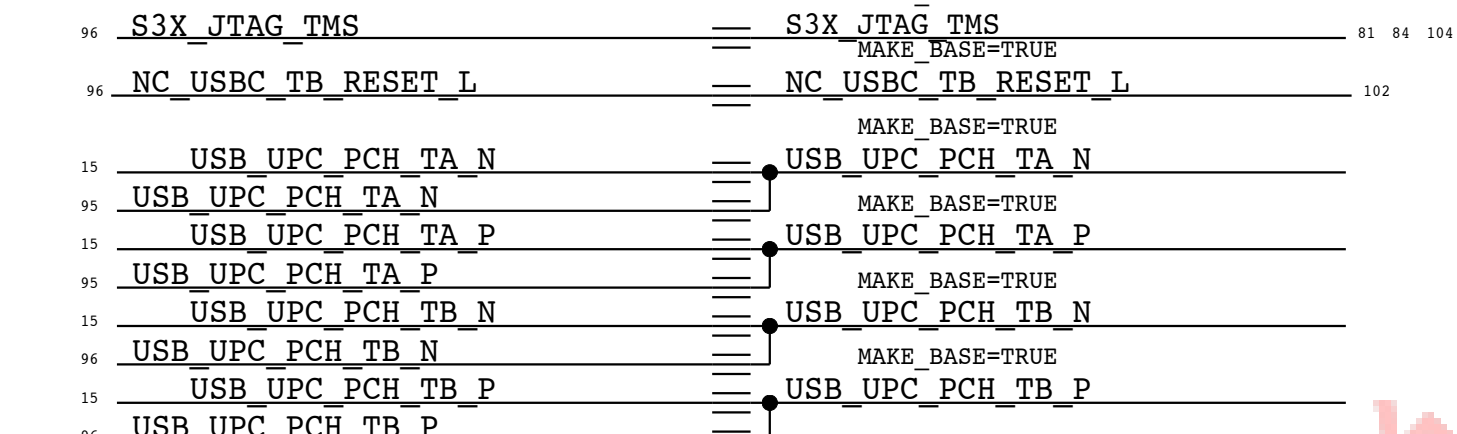
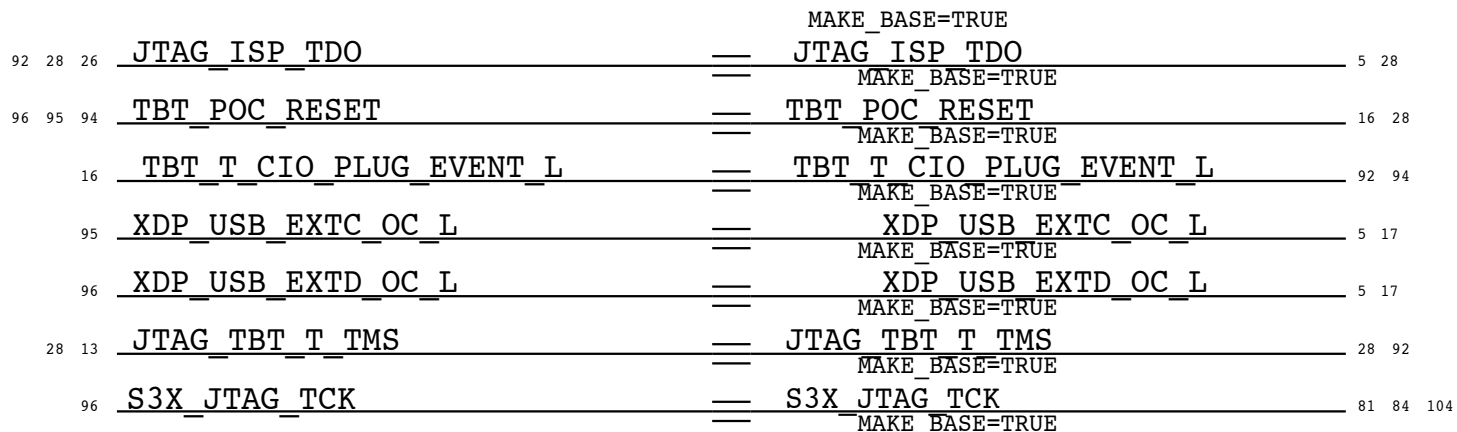
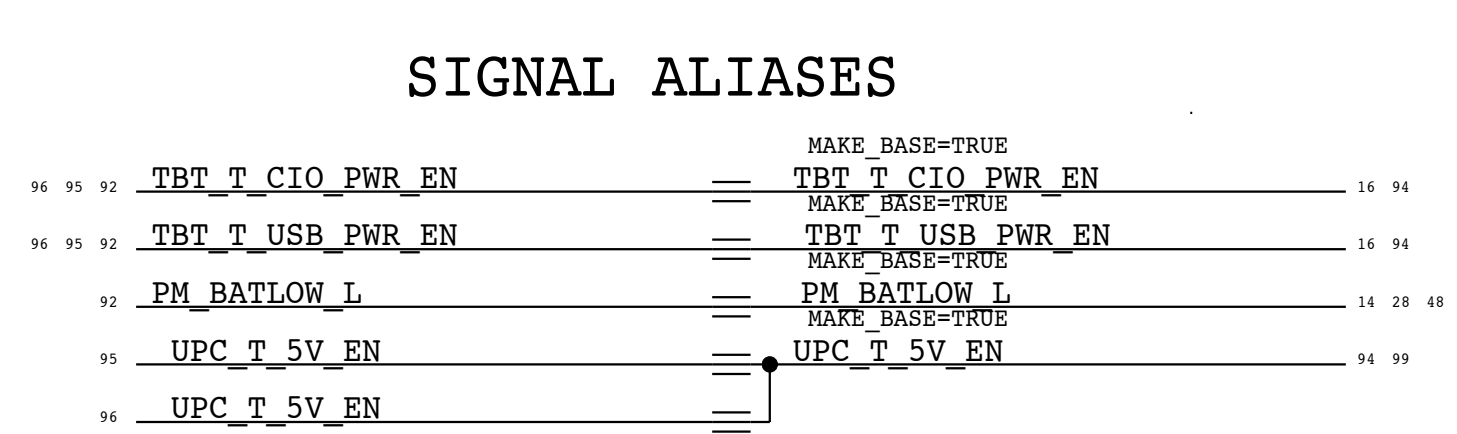
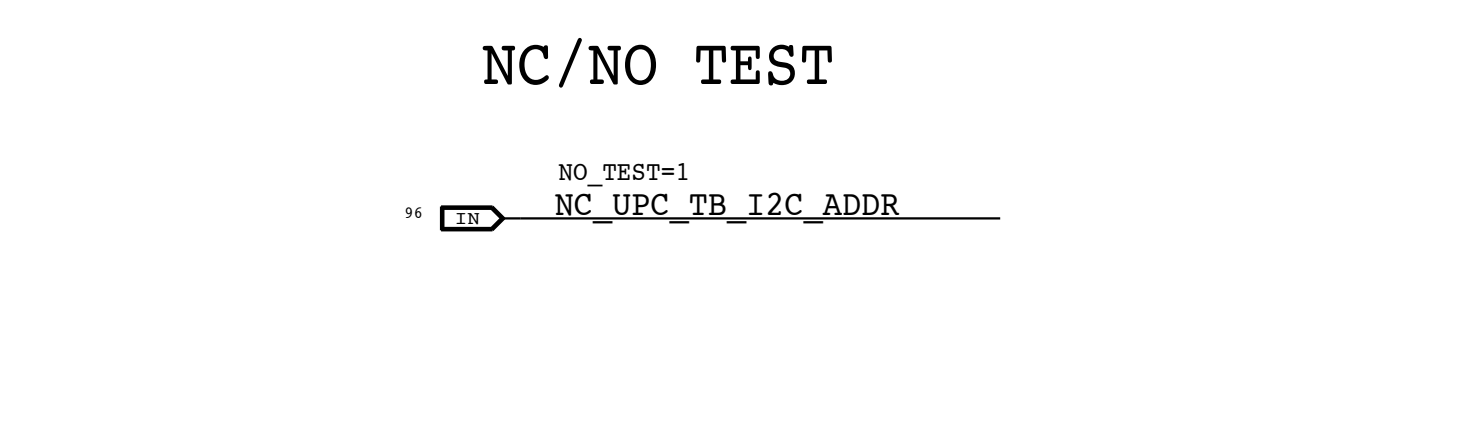
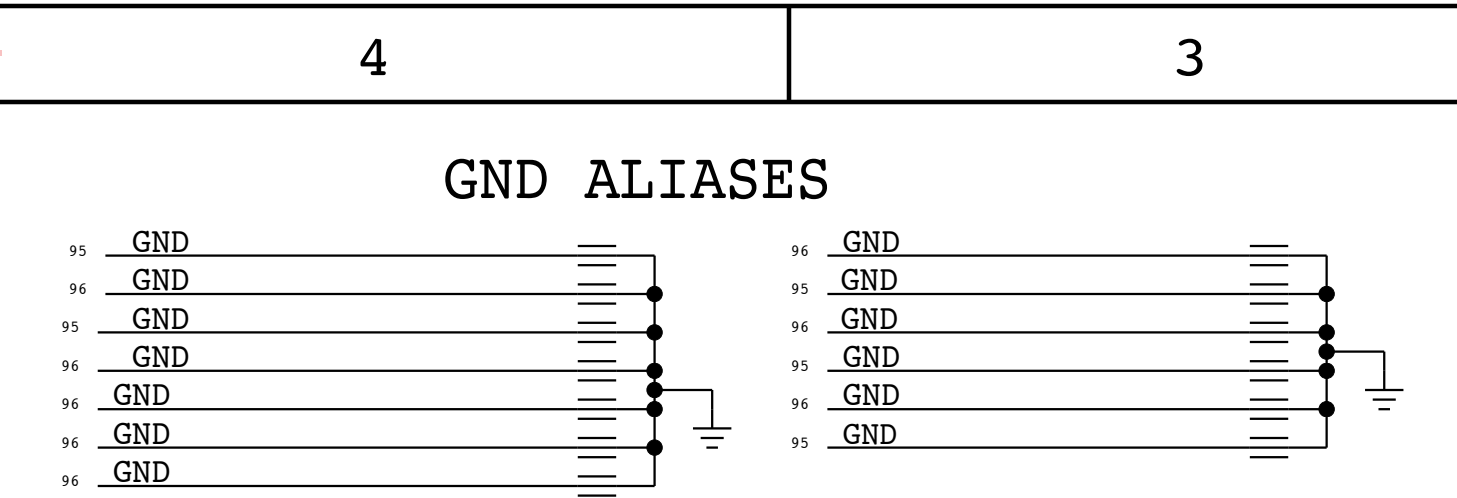
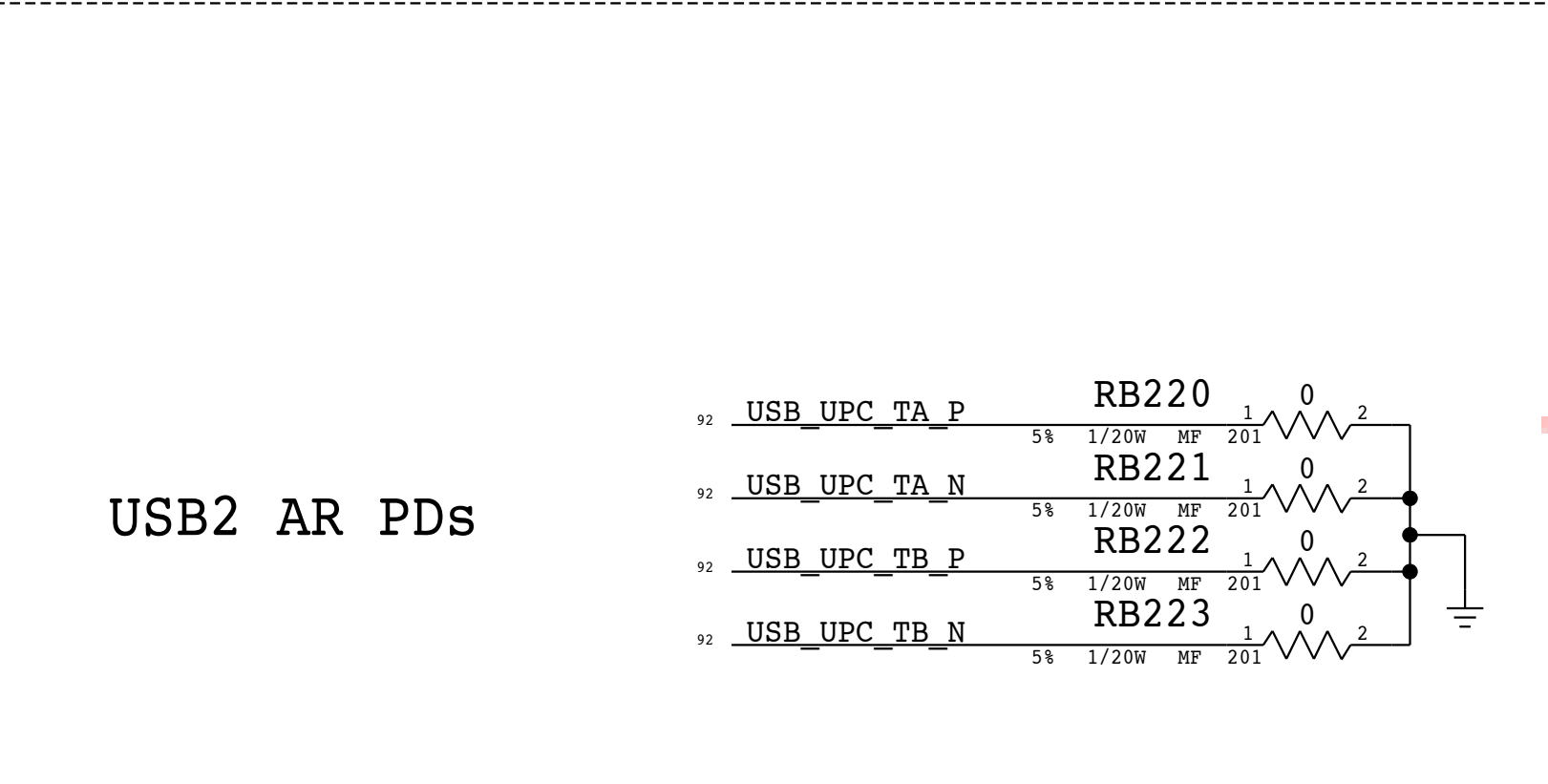
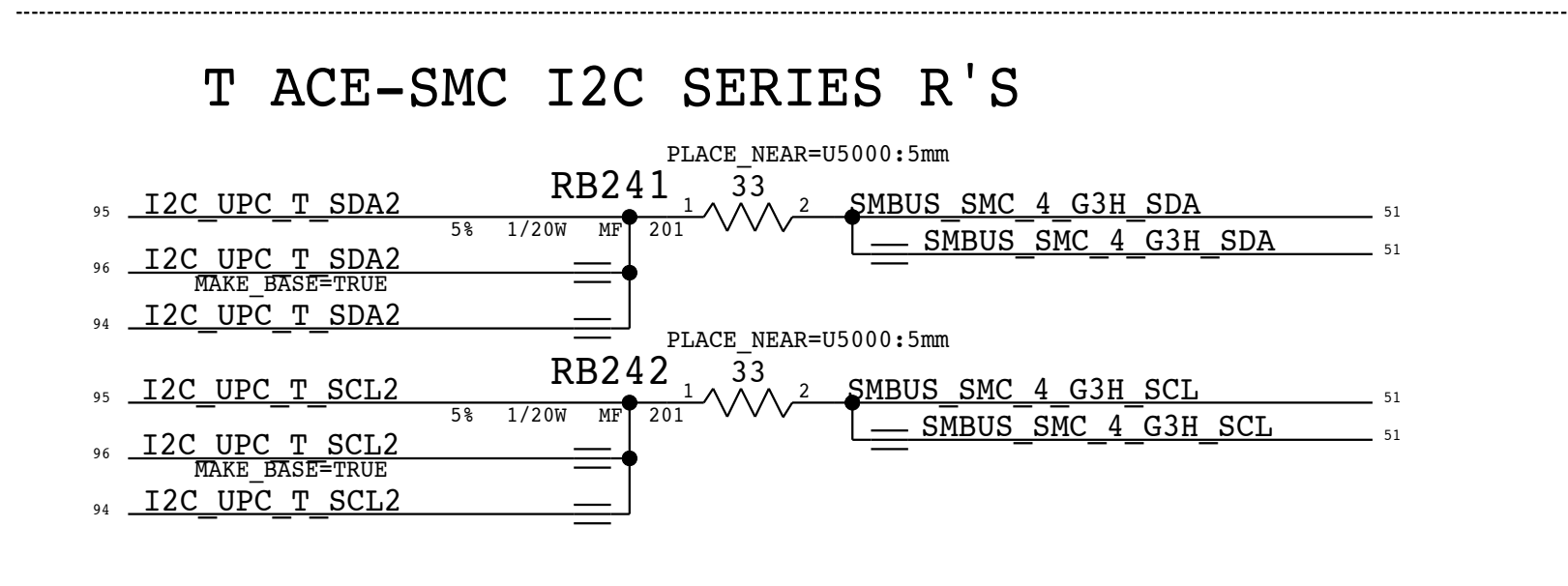
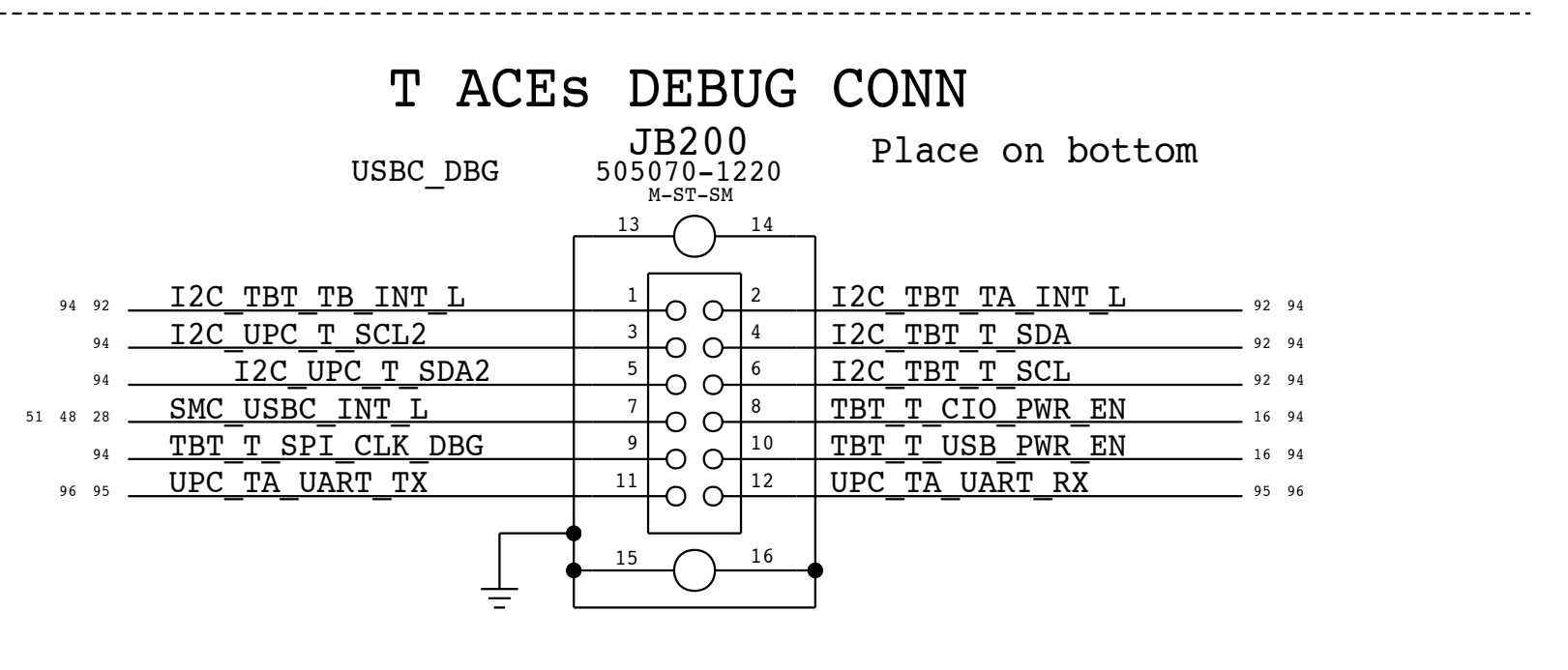
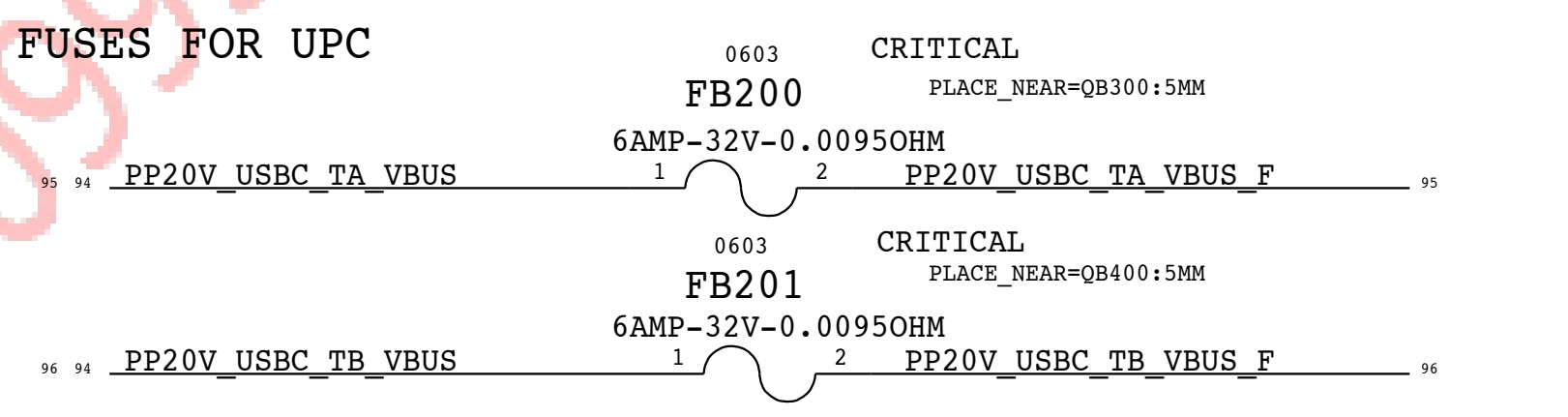
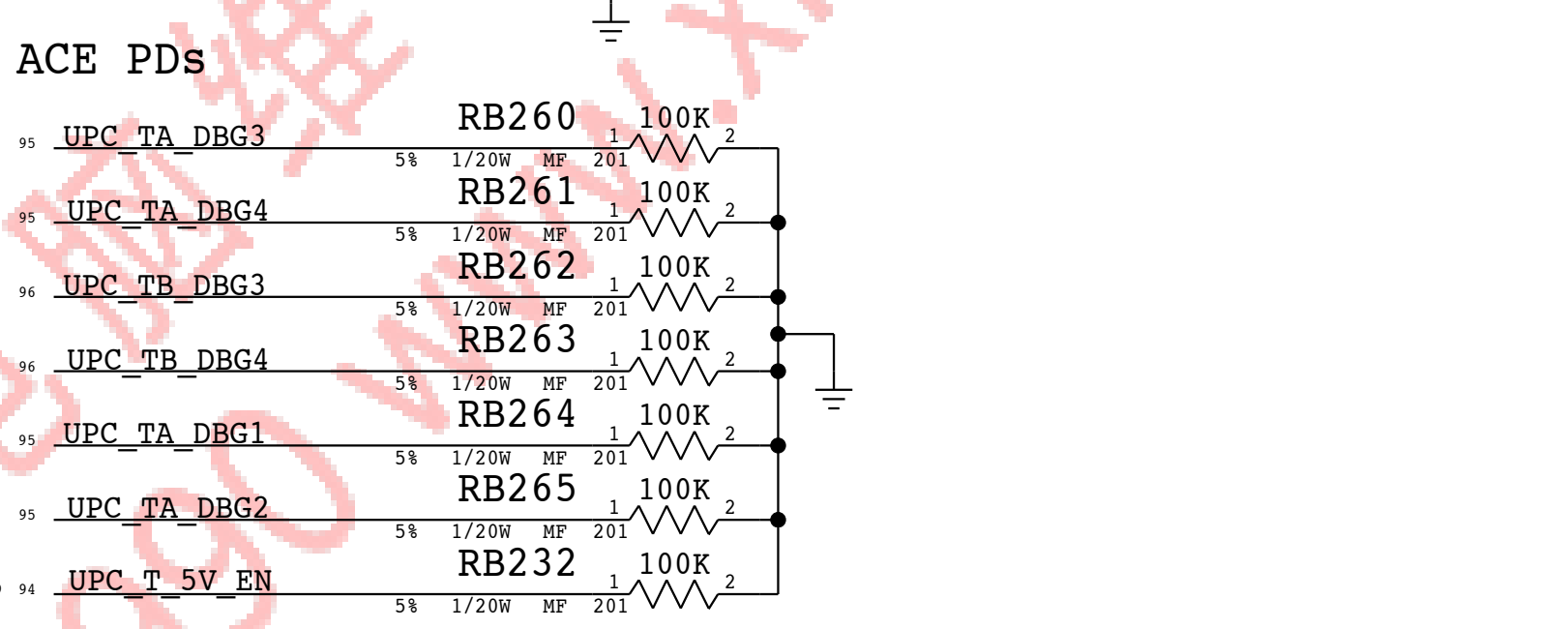
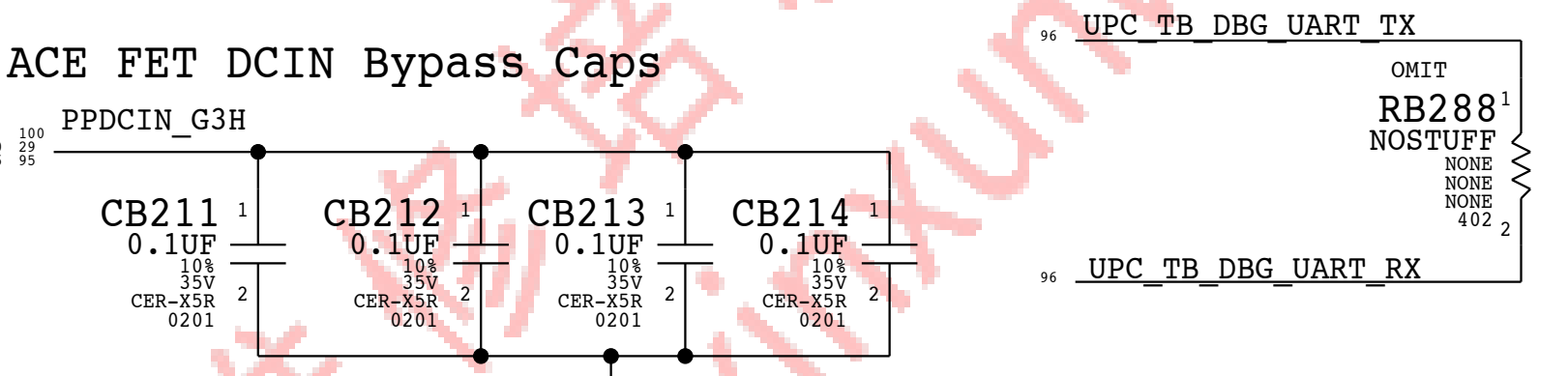
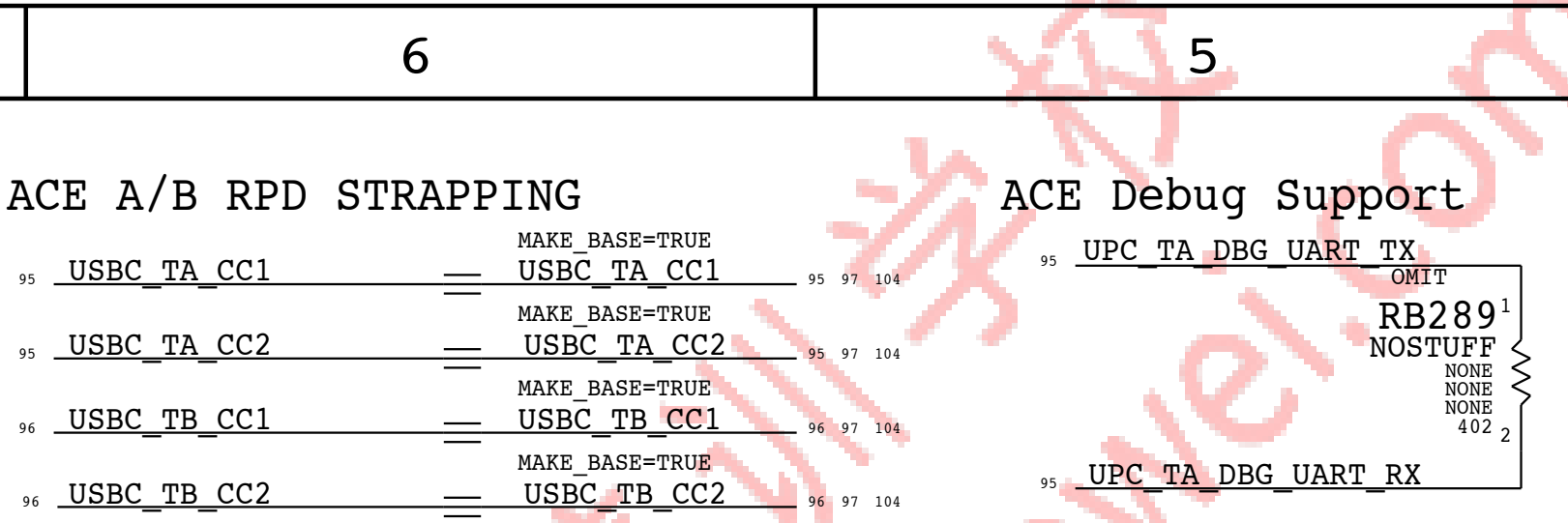
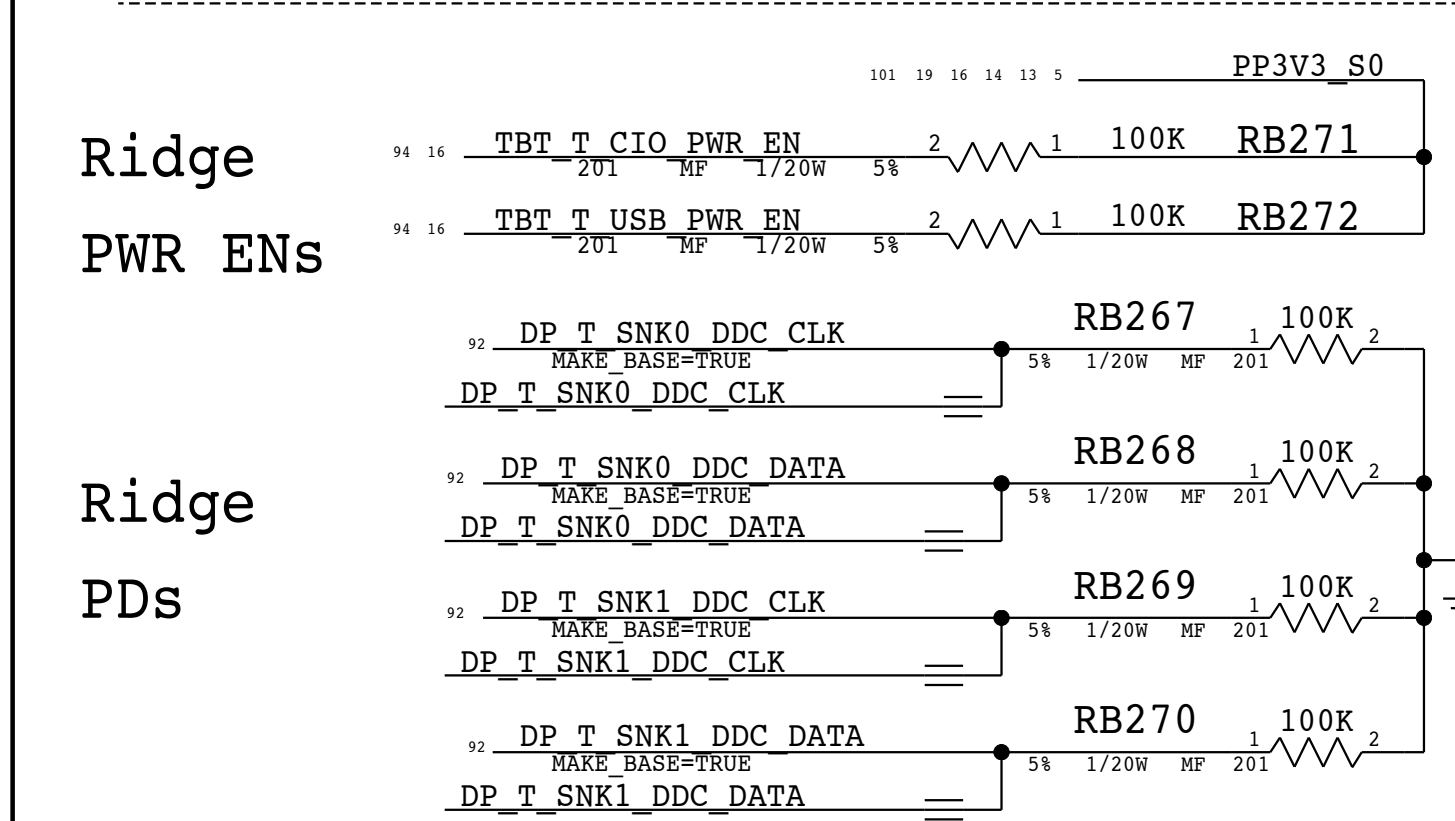
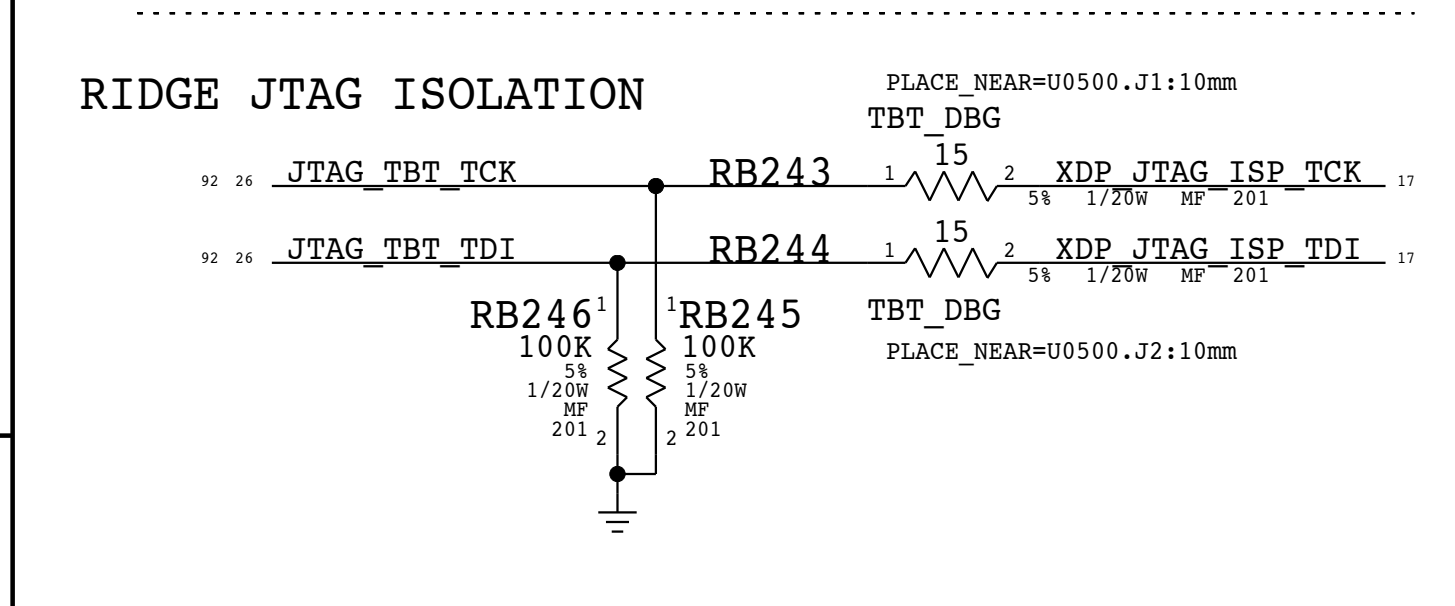
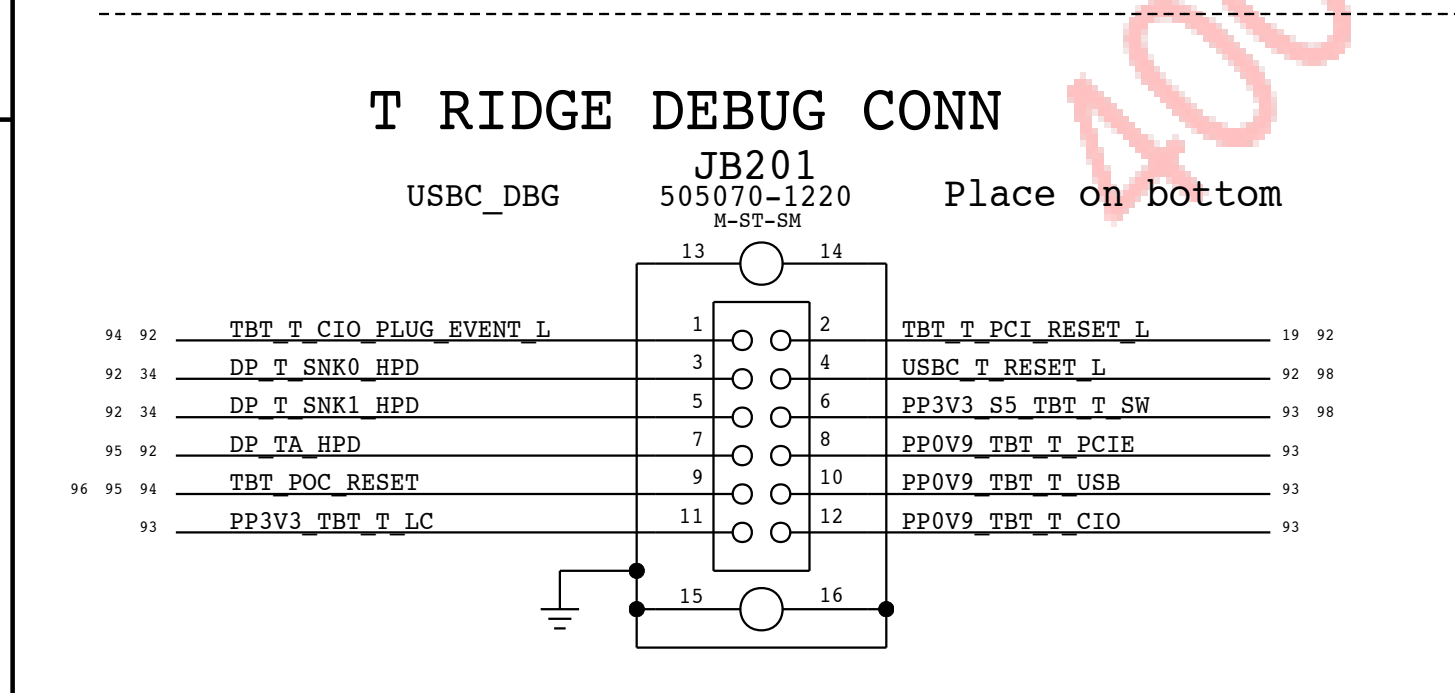
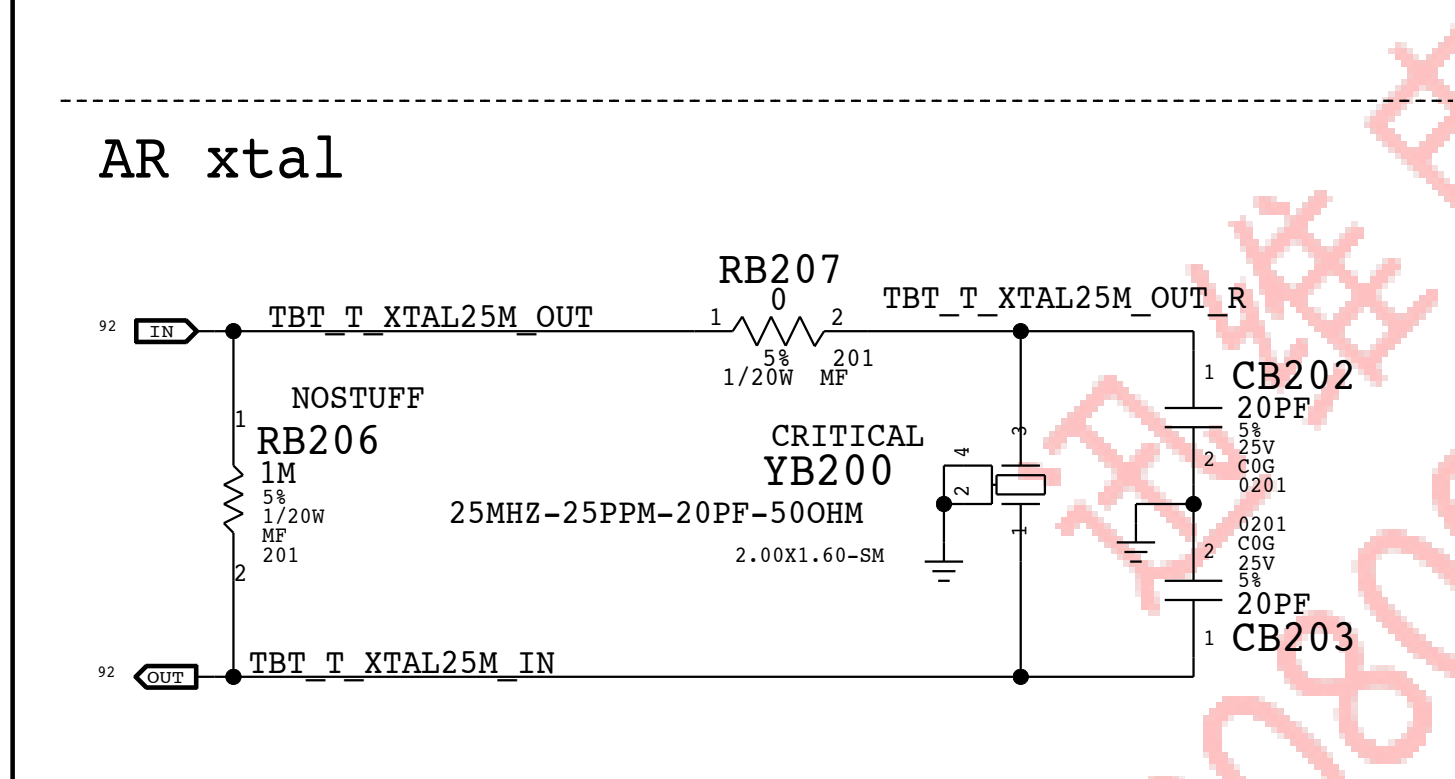
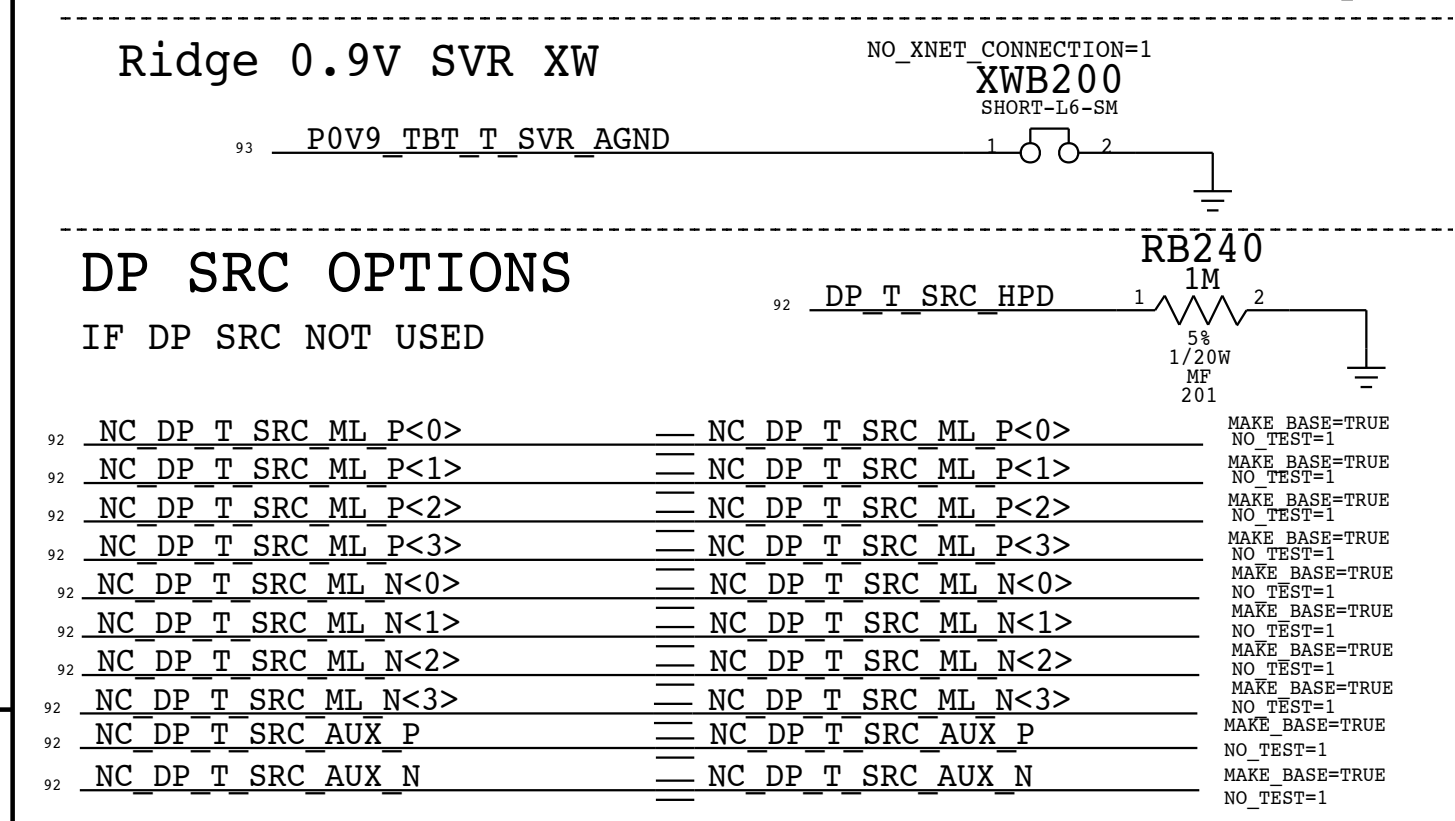
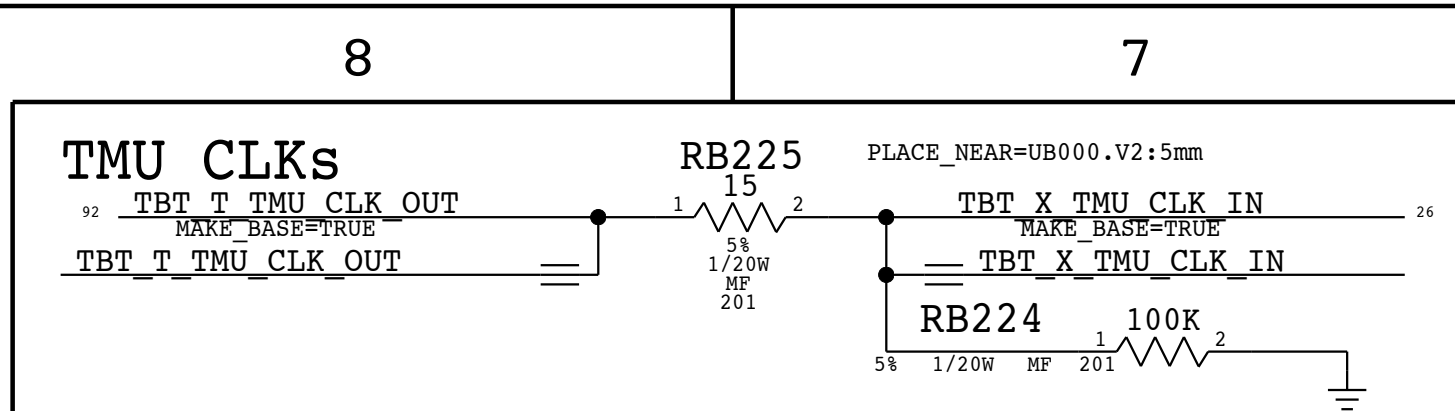
SYNC MASTER=J79_RUENJOU			SYNC DATE=09/09/2015			
PAGE TITLE						
LIFEBOAT						
 Apple Inc.			DRAWING NUMBER		SHEET	
			051-00515		D	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			REVISION		7.0.0	
			BRANCH		dvt1	
			PAGE		96 OF 145	
			SHEET		91 OF 119	



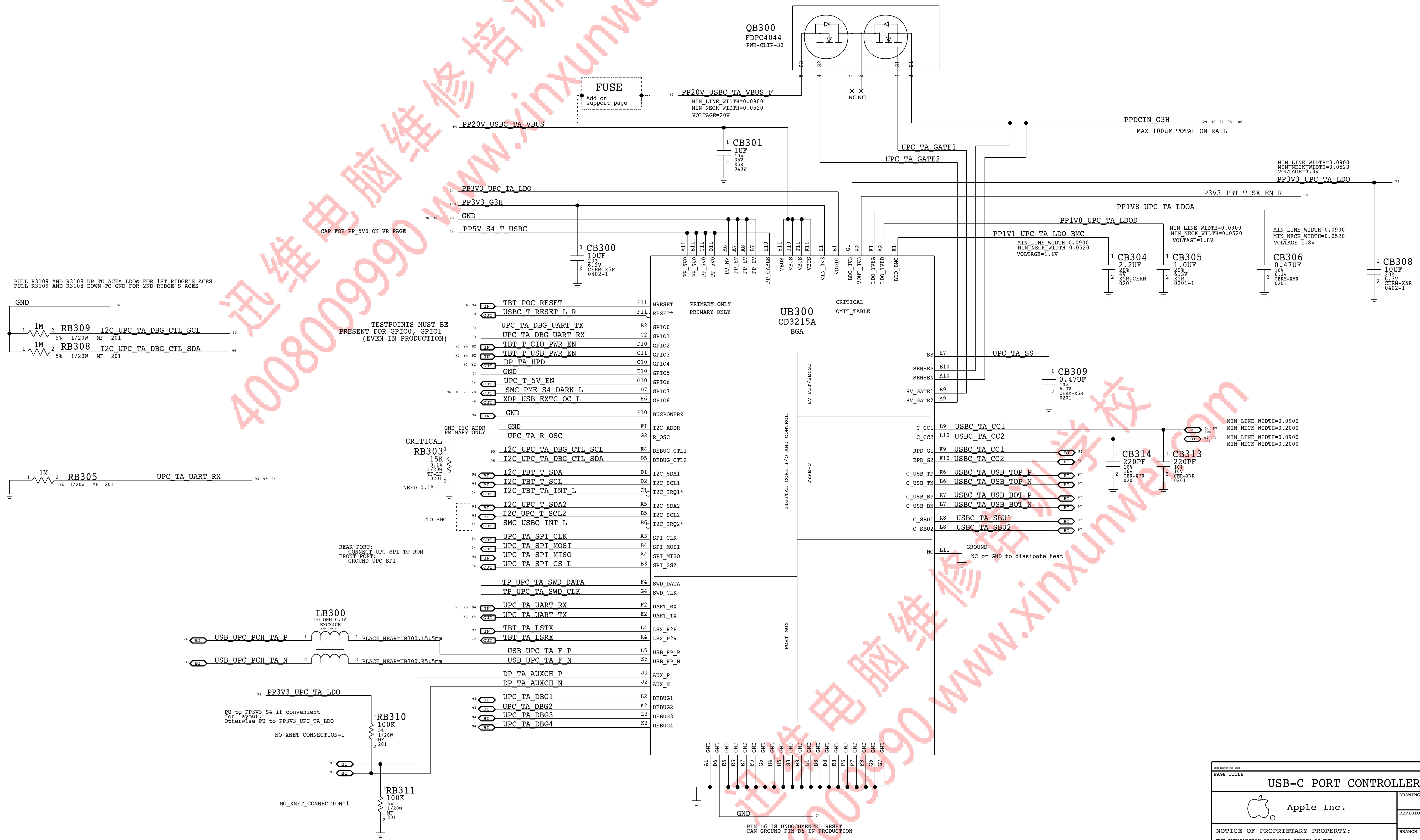


SYNC_MASTER=J79_GREG		SYNC_DATE=08/28/2015	
PAGE TITLE			
USB-C HIGH SPEED 2			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00515		D
	REVISION		
		7.0.0	
NOTICE OF PROPRIETARY PROPERTY:			
BRANCH			
dvt1			
PAGE			
111 OF 145			
SHEET			
93 OF 119			

BOM_COST_GROUP=TBT



PRIMARY ACE USB-C PORT CONTROLLER (UPC)



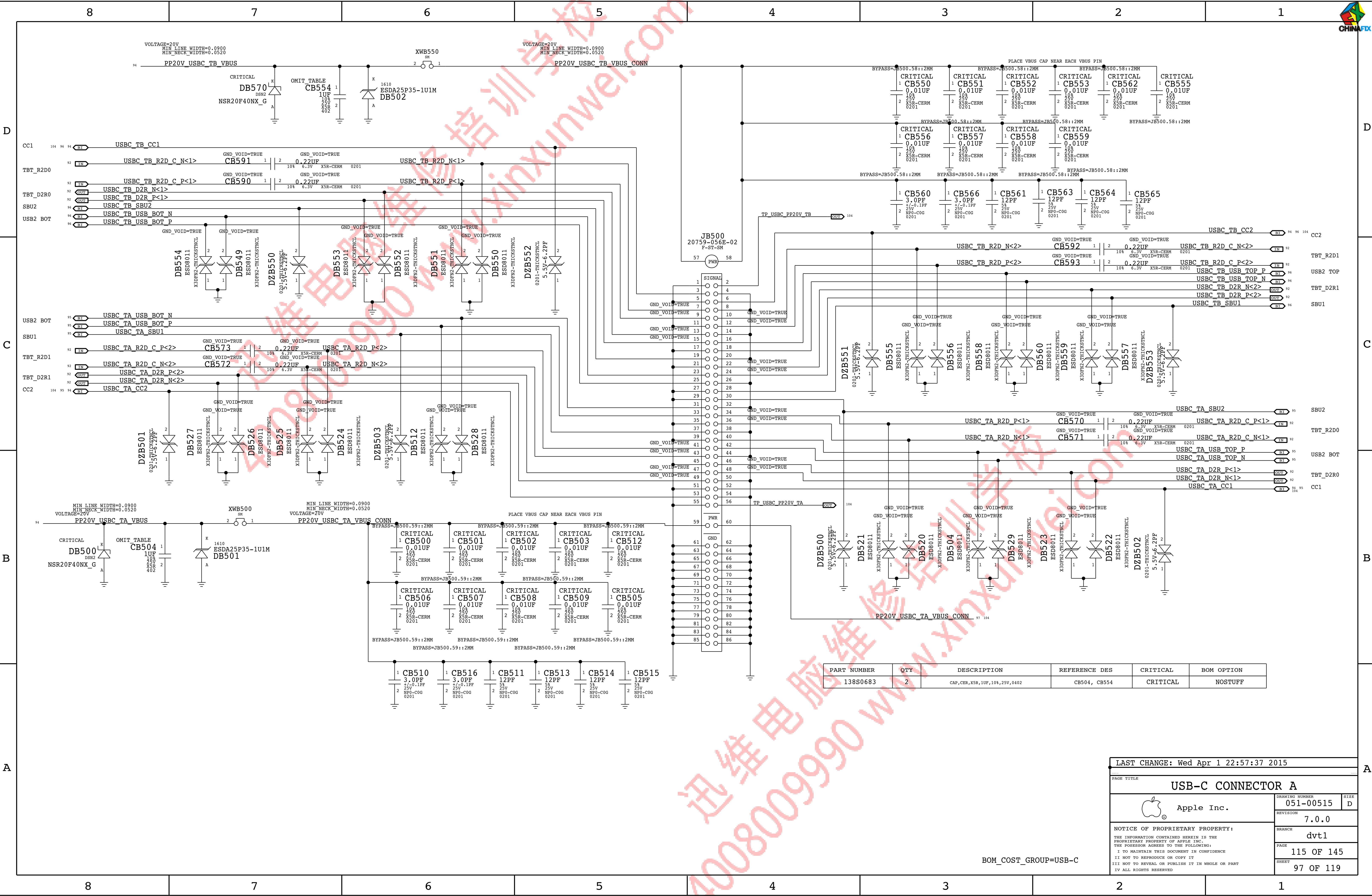
BOM_COST_GROUP=USB-C

PAGE TITLE		
USB-C PORT CONTROLLER A		
	DRAWING NUMBER	051-00515
	REVISION	7.0.0
	BRANCH	dvt1
	PAGE	113 OF 145
NOTICE OF PROPRIETARY PROPERTY:		SHEET
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		95 OF 119
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		
I NOT TO REPRODUCE OR COPY IT		
I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		
I ALL RIGHTS RESERVED		

BOM COST GROUP=USB-C



FORM NO:2025-01/17-01		FORM NO:2025-01/17-01	
PAGE TITLE		PAGE TITLE	
USB-C PORT CONTROLLER B			
	Apple Inc.	DRAWING NUMBER	051-00515
		REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	dvt1
		PAGE	114 OF 145
		SHEET	96 OF 119

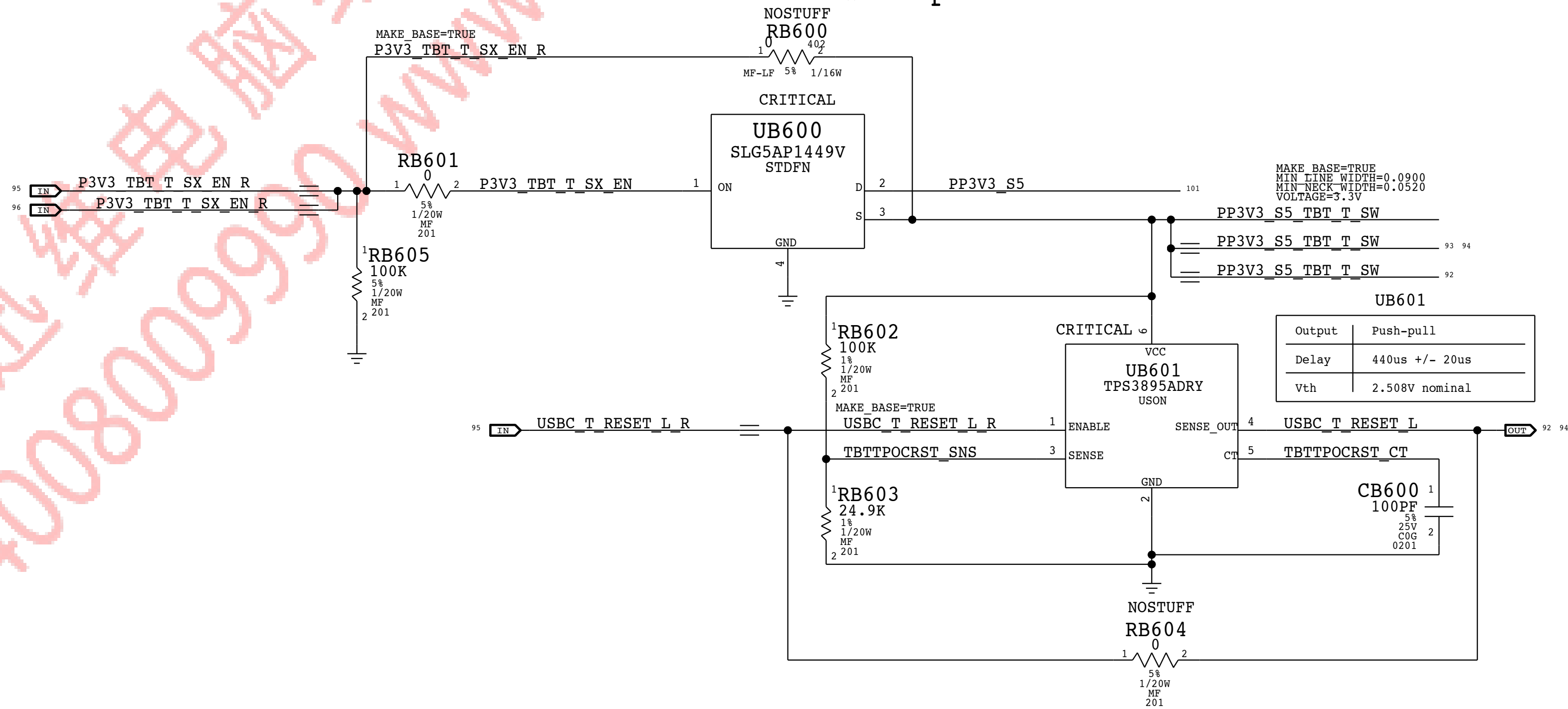


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0683	2	CAP,CER,X5R,1UF,10%,25V,0402	CB504, CB554	CRITICAL	NOSTUFF

LAST CHANGE: Wed Apr 1 22:57:37 2015			
PAGE TITLE			
USB-C CONNECTOR A			
	DRAWING NUMBER	051-00515	STAR D
	REVISION	7.0.0	
	BRANCH	dvt1	
	PAGE	115 OF 145	
NOTICE OF PROPRIETARY PROPERTY:		SHEET	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:		97 OF 119	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
I NOT TO REPRODUCE OR COPY IT			
I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			
I ALL RIGHTS RESERVED			

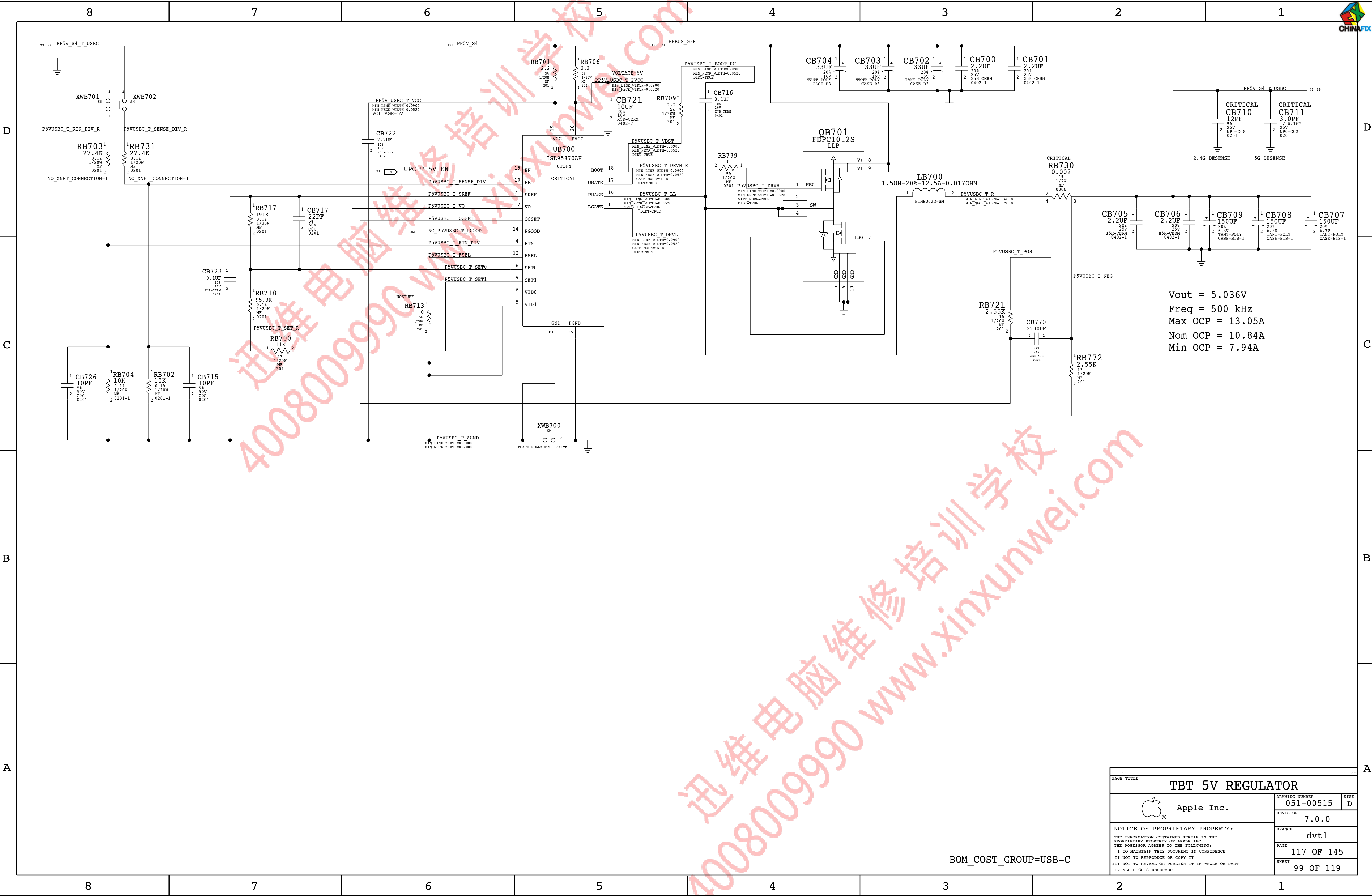
BOM_COST_GROUP=USB-C

TBT T "POC" Power-up Reset



BOM_COST_GROUP=USB-C

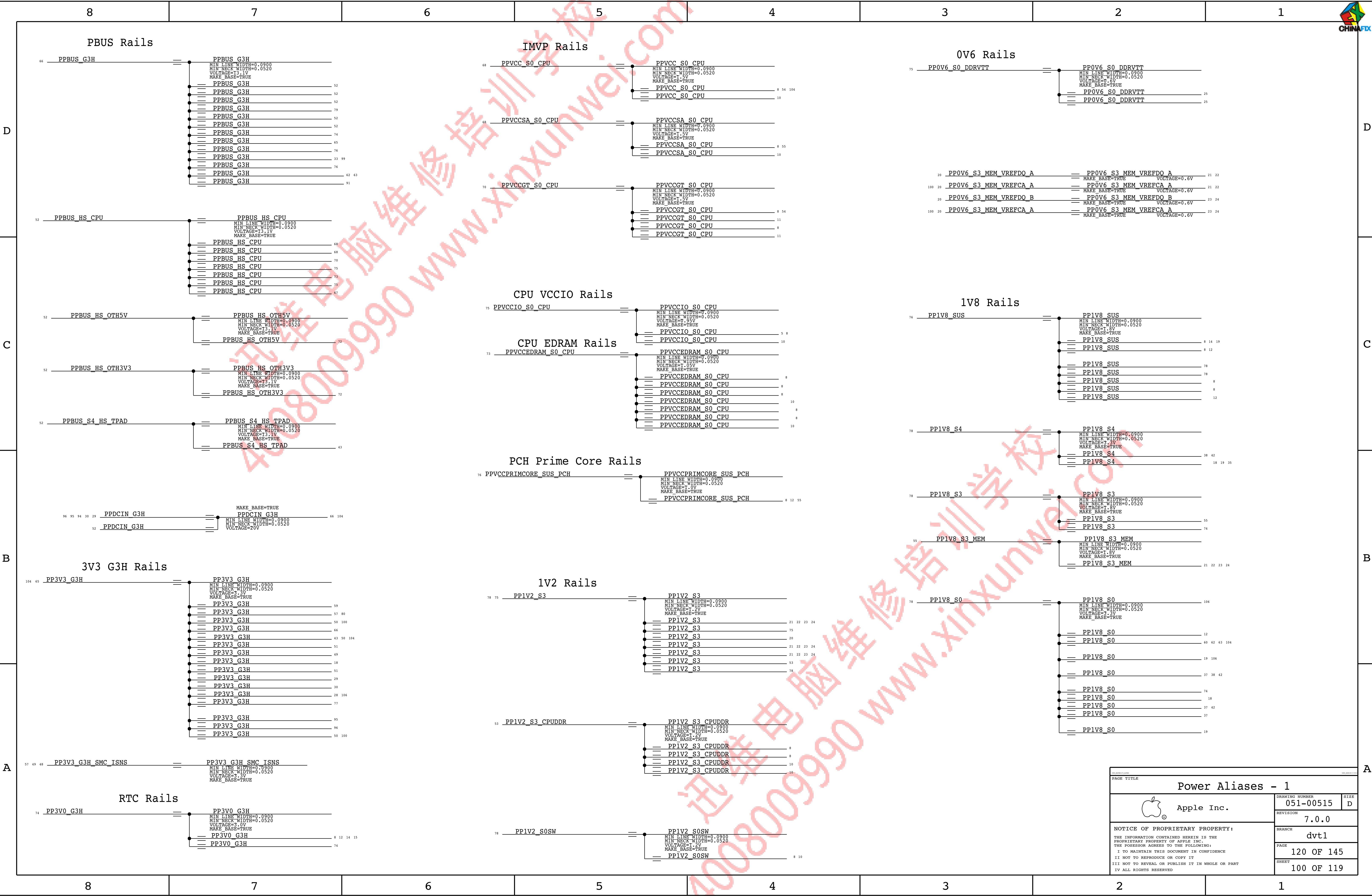
DESIGN: j130/dev m1b u		
LAST CHANGE: Wed Apr 1 22:57:37 2015		
PAGE TITLE		
USB-C CONNECTOR B		
 Apple Inc.	DRAWING NUMBER	051-00515
	REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH	dvt1
	PAGE	116 OF 145
	SHEET	98 OF 119



Vout = 5.036V
Freq = 500 kHz
Max OCP = 13.05A
Nom OCP = 10.84A
Min OCP = 7.94A

PAGE TITLE		
TBT 5V REGULATOR		
	DRANTING NUMBER	051-00515
	REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH	dvt1
	PAGE	117 OF 145
	SHEET	99 OF 119

BOM_COST_GROUP=USB-C



PAGE TITLE		
Power Aliases - 1		
 Apple Inc.	DRAWING NUMBER	051-00515
	REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I ALL RIGHTS RESERVED	BRANCH	dvt1
	PAGE	120 OF 145
	SHEET	100 OF 119



D

C

B

A

1

2

3

4

5

6

7

8

1

2

3

4

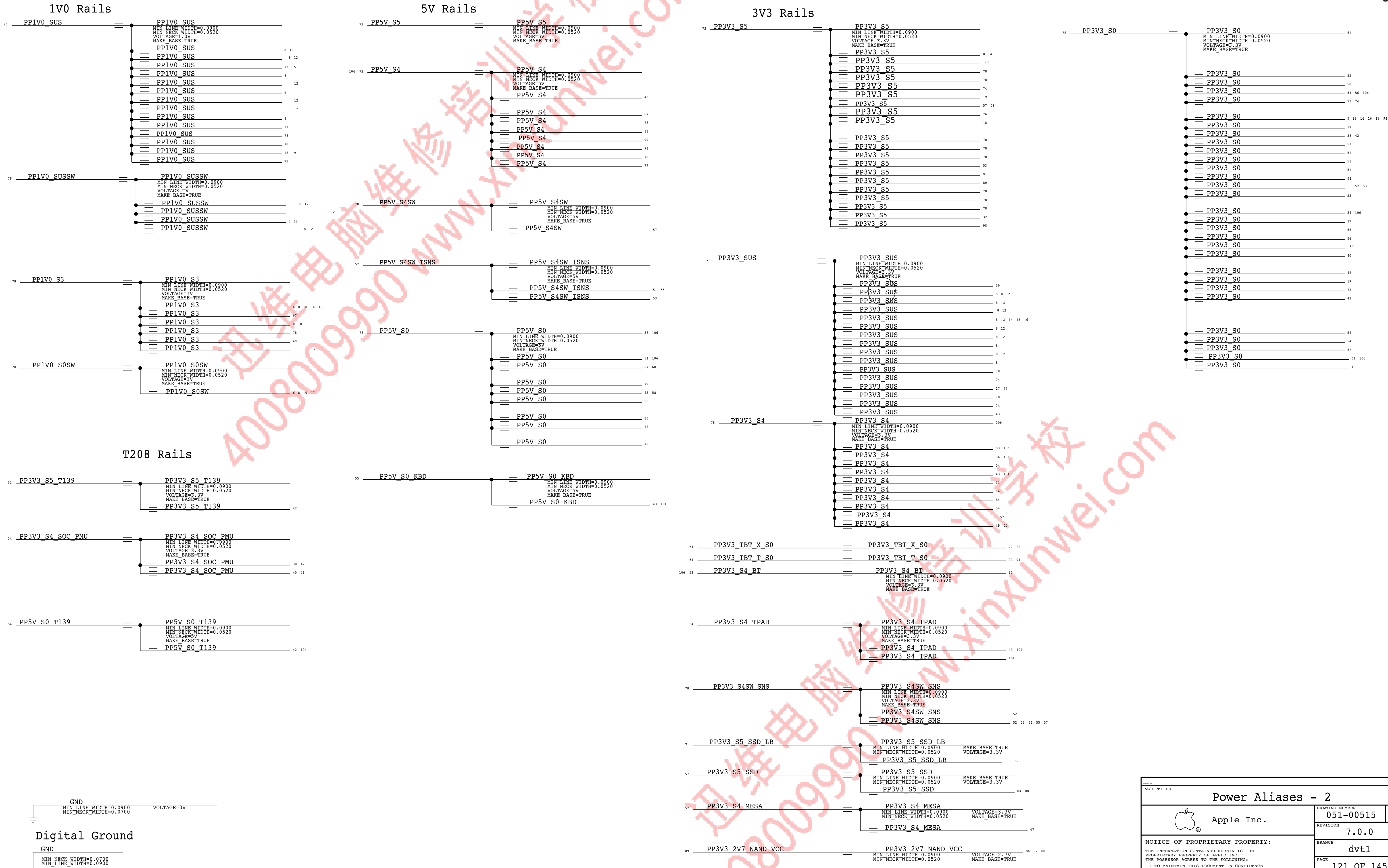
5

6

7

8

PAGE TITLE			Power Aliases - 2	
 Apple Inc.		DRAWING NUMBER	051-00515	STAR
		REVISION	7.0.0	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: 1 TO MAINTAIN THIS DOCUMENT IN CONFIDENCE 11 NOT TO REPRODUCE OR COPY IT 111 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART 111 ALL RIGHTS RESERVED		BRANCH	dvt1	
		PAGE	121 OF 145	
		SHEET	101 OF 119	



Memory Bit & Byte Swizzle

MAKE_BASE=TRUE

MEM_A_DQ<0>	MEM_A_DQ<0>
MEM_A_DQ<1>	MEM_A_DQ<1>
MEM_A_DQ<2>	MEM_A_DQ<2>
MEM_A_DQ<3>	MEM_A_DQ<3>
MEM_A_DQ<4>	MEM_A_DQ<4>
MEM_A_DQ<5>	MEM_A_DQ<5>
MEM_A_DQ<6>	MEM_A_DQ<6>
MEM_A_DQ<7>	MEM_A_DQ<7>
MEM_A_DQ<8>	MEM_A_DQ<8>
MEM_A_DQ<9>	MEM_A_DQ<9>
MEM_A_DQ<10>	MEM_A_DQ<10>
MEM_A_DQ<11>	MEM_A_DQ<11>
MEM_A_DQ<12>	MEM_A_DQ<12>
MEM_A_DQ<13>	MEM_A_DQ<13>
MEM_A_DQ<14>	MEM_A_DQ<14>
MEM_A_DQ<15>	MEM_A_DQ<15>
MEM_A_DQ<16>	MEM_A_DQ<16>
MEM_A_DQ<17>	MEM_A_DQ<17>
MEM_A_DQ<18>	MEM_A_DQ<18>
MEM_A_DQ<19>	MEM_A_DQ<19>
MEM_A_DQ<20>	MEM_A_DQ<20>
MEM_A_DQ<21>	MEM_A_DQ<21>
MEM_A_DQ<22>	MEM_A_DQ<22>
MEM_A_DQ<23>	MEM_A_DQ<23>
MEM_A_DQ<24>	MEM_A_DQ<24>
MEM_A_DQ<25>	MEM_A_DQ<25>
MEM_A_DQ<26>	MEM_A_DQ<26>
MEM_A_DQ<27>	MEM_A_DQ<27>
MEM_A_DQ<28>	MEM_A_DQ<28>
MEM_A_DQ<29>	MEM_A_DQ<29>
MEM_A_DQ<30>	MEM_A_DQ<30>
MEM_A_DQ<31>	MEM_A_DQ<31>
MEM_A_DQ<32>	MEM_A_DQ<32>
MEM_A_DQ<33>	MEM_A_DQ<33>
MEM_A_DQ<34>	MEM_A_DQ<34>
MEM_A_DQ<35>	MEM_A_DQ<35>
MEM_A_DQ<36>	MEM_A_DQ<36>
MEM_A_DQ<37>	MEM_A_DQ<37>
MEM_A_DQ<38>	MEM_A_DQ<38>
MEM_A_DQ<39>	MEM_A_DQ<39>
MEM_A_DQ<40>	MEM_A_DQ<40>
MEM_A_DQ<41>	MEM_A_DQ<41>
MEM_A_DQ<42>	MEM_A_DQ<42>
MEM_A_DQ<43>	MEM_A_DQ<43>
MEM_A_DQ<44>	MEM_A_DQ<44>
MEM_A_DQ<45>	MEM_A_DQ<45>
MEM_A_DQ<46>	MEM_A_DQ<46>
MEM_A_DQ<47>	MEM_A_DQ<47>
MEM_A_DQ<48>	MEM_A_DQ<48>
MEM_A_DQ<49>	MEM_A_DQ<49>
MEM_A_DQ<50>	MEM_A_DQ<50>
MEM_A_DQ<51>	MEM_A_DQ<51>
MEM_A_DQ<52>	MEM_A_DQ<52>
MEM_A_DQ<53>	MEM_A_DQ<53>
MEM_A_DQ<54>	MEM_A_DQ<54>
MEM_A_DQ<55>	MEM_A_DQ<55>
MEM_A_DQ<56>	MEM_A_DQ<56>
MEM_A_DQ<57>	MEM_A_DQ<57>
MEM_A_DQ<58>	MEM_A_DQ<58>
MEM_A_DQ<59>	MEM_A_DQ<59>
MEM_A_DQ<60>	MEM_A_DQ<60>
MEM_A_DQ<61>	MEM_A_DQ<61>
MEM_A_DQ<62>	MEM_A_DQ<62>
MEM_A_DQ<63>	MEM_A_DQ<63>

MAKE_BASE=TRUE

MEM_A_DQS_P<0>	MEM_A_DQS_P<0>
MEM_A_DQS_N<0>	MEM_A_DQS_N<0>
MEM_A_DQS_P<1>	MEM_A_DQS_P<1>
MEM_A_DQS_N<1>	MEM_A_DQS_N<1>
MEM_A_DQS_P<2>	MEM_A_DQS_P<2>
MEM_A_DQS_N<2>	MEM_A_DQS_N<2>
MEM_A_DQS_P<3>	MEM_A_DQS_P<3>
MEM_A_DQS_N<3>	MEM_A_DQS_N<3>
MEM_A_DQS_P<4>	MEM_A_DQS_P<4>
MEM_A_DQS_N<4>	MEM_A_DQS_N<4>
MEM_A_DQS_P<5>	MEM_A_DQS_P<5>
MEM_A_DQS_N<5>	MEM_A_DQS_N<5>
MEM_A_DQS_P<6>	MEM_A_DQS_P<6>
MEM_A_DQS_N<6>	MEM_A_DQS_N<6>
MEM_A_DQS_P<7>	MEM_A_DQS_P<7>
MEM_A_DQS_N<7>	MEM_A_DQS_N<7>

MAKE_BASE=TRUE

MEM_B_DQ<0>	MEM_B_DQ<0>
MEM_B_DQ<1>	MEM_B_DQ<1>
MEM_B_DQ<2>	MEM_B_DQ<2>
MEM_B_DQ<3>	MEM_B_DQ<3>
MEM_B_DQ<4>	MEM_B_DQ<4>
MEM_B_DQ<5>	MEM_B_DQ<5>
MEM_B_DQ<6>	MEM_B_DQ<6>
MEM_B_DQ<7>	MEM_B_DQ<7>
MEM_B_DQ<8>	MEM_B_DQ<8>
MEM_B_DQ<9>	MEM_B_DQ<9>
MEM_B_DQ<10>	MEM_B_DQ<10>
MEM_B_DQ<11>	MEM_B_DQ<11>
MEM_B_DQ<12>	MEM_B_DQ<12>
MEM_B_DQ<13>	MEM_B_DQ<13>
MEM_B_DQ<14>	MEM_B_DQ<14>
MEM_B_DQ<15>	MEM_B_DQ<15>
MEM_B_DQ<16>	MEM_B_DQ<16>
MEM_B_DQ<17>	MEM_B_DQ<17>
MEM_B_DQ<18>	MEM_B_DQ<18>
MEM_B_DQ<19>	MEM_B_DQ<19>
MEM_B_DQ<20>	MEM_B_DQ<20>
MEM_B_DQ<21>	MEM_B_DQ<21>
MEM_B_DQ<22>	MEM_B_DQ<22>
MEM_B_DQ<23>	MEM_B_DQ<23>
MEM_B_DQ<24>	MEM_B_DQ<24>
MEM_B_DQ<25>	MEM_B_DQ<25>
MEM_B_DQ<26>	MEM_B_DQ<26>
MEM_B_DQ<27>	MEM_B_DQ<27>
MEM_B_DQ<28>	MEM_B_DQ<28>
MEM_B_DQ<29>	MEM_B_DQ<29>
MEM_B_DQ<30>	MEM_B_DQ<30>
MEM_B_DQ<31>	MEM_B_DQ<31>
MEM_B_DQ<32>	MEM_B_DQ<32>
MEM_B_DQ<33>	MEM_B_DQ<33>
MEM_B_DQ<34>	MEM_B_DQ<34>
MEM_B_DQ<35>	MEM_B_DQ<35>
MEM_B_DQ<36>	MEM_B_DQ<36>
MEM_B_DQ<37>	MEM_B_DQ<37>
MEM_B_DQ<38>	MEM_B_DQ<38>
MEM_B_DQ<39>	MEM_B_DQ<39>
MEM_B_DQ<40>	MEM_B_DQ<40>
MEM_B_DQ<41>	MEM_B_DQ<41>
MEM_B_DQ<42>	MEM_B_DQ<42>
MEM_B_DQ<43>	MEM_B_DQ<43>
MEM_B_DQ<44>	MEM_B_DQ<44>
MEM_B_DQ<45>	MEM_B_DQ<45>
MEM_B_DQ<46>	MEM_B_DQ<46>
MEM_B_DQ<47>	MEM_B_DQ<47>
MEM_B_DQ<48>	MEM_B_DQ<48>
MEM_B_DQ<49>	MEM_B_DQ<49>
MEM_B_DQ<50>	MEM_B_DQ<50>
MEM_B_DQ<51>	MEM_B_DQ<51>
MEM_B_DQ<52>	MEM_B_DQ<52>
MEM_B_DQ<53>	MEM_B_DQ<53>
MEM_B_DQ<54>	MEM_B_DQ<54>
MEM_B_DQ<55>	MEM_B_DQ<55>
MEM_B_DQ<56>	MEM_B_DQ<56>
MEM_B_DQ<57>	MEM_B_DQ<57>
MEM_B_DQ<58>	MEM_B_DQ<58>
MEM_B_DQ<59>	MEM_B_DQ<59>
MEM_B_DQ<60>	MEM_B_DQ<60>
MEM_B_DQ<61>	MEM_B_DQ<61>
MEM_B_DQ<62>	MEM_B_DQ<62>
MEM_B_DQ<63>	MEM_B_DQ<63>

MAKE_BASE=TRUE

MEM_B_DQS_P<0>	MEM_B_DQS_P<0>
MEM_B_DQS_N<0>	MEM_B_DQS_N<0>
MEM_B_DQS_P<1>	MEM_B_DQS_P<1>
MEM_B_DQS_N<1>	MEM_B_DQS_N<1>
MEM_B_DQS_P<2>	MEM_B_DQS_P<2>
MEM_B_DQS_N<2>	MEM_B_DQS_N<2>
MEM_B_DQS_P<3>	MEM_B_DQS_P<3>
MEM_B_DQS_N<3>	MEM_B_DQS_N<3>
MEM_B_DQS_P<4>	MEM_B_DQS_P<4>
MEM_B_DQS_N<4>	MEM_B_DQS_N<4>
MEM_B_DQS_P<5>	MEM_B_DQS_P<5>
MEM_B_DQS_N<5>	MEM_B_DQS_N<5>
MEM_B_DQS_P<6>	MEM_B_DQS_P<6>
MEM_B_DQS_N<6>	MEM_B_DQS_N<6>
MEM_B_DQS_P<7>	MEM_B_DQS_P<7>
MEM_B_DQS_N<7>	MEM_B_DQS_N<7>





D

C

B

A

8	7	6	5	4	3	2	1
PP9901 PP9902 PP9903 PP9904 PP9918 PP9910 PP9911	DP_DDI1_ML_C_N<0> DP_DDI1_ML_C_P<0> DP_DDI2_ML_C_N<0> DP_DDI2_ML_C_P<0> SMC_PECI_L PCIE_SSD_D2R_N<0> PCIE_SSD_D2R_P<0>	MIPIC_DATA_P MIPIC_DATA_N MIPIC_CLK_P MIPIC_CLK_N	PP9949 PP9950 PP9951 PP9952	NO_TESTS			
PP9919 PP9920	PCIE_TBT_X_D2R_N<0> PCIE_TBT_X_D2R_P<0>	MIPID_DATA_CONN_P MIPID_DATA_CONN_N MIPID_CLK_CONN_P MIPID_CLK_CONN_N					
PP9927 PP9928	PCIE_TBT_T_D2R_N<0> PCIE_TBT_T_D2R_P<0>	MIPI_DATA_CONN_P MIPI_DATA_CONN_N MIPI_CLK_CONN_P MIPI_CLK_CONN_N					
PP9933 PP9934	USB_CAMERA_DFR_N USB_CAMERA_DFR_P						
PP9941 PP9942 PP9943 PP9944 PP9945 PP9946 PP9947 PP9948	PCIE_CLK100M_SSD_LB_N PCIE_CLK100M_SSD_LB_P PCIE_CLK100M_TBT_X_N PCIE_CLK100M_TBT_X_P PCIE_CLK100M_TBT_T_N PCIE_CLK100M_TBT_T_P PCIE_CLK100M_AP_N PCIE_CLK100M_AP_P						
PP9801 PP9802 PP9803 PP9804 PP9808 PP9819	XDP_USB_EXT_A_OC_L XDP_USB_EXT_B_OC_L XDP_USB_EXT_C_OC_L XDP_USB_EXT_D_OC_L HDA_SDINO LPC_CLK24M_SMC						
TP9820	XDP_BPM_L<0>						
8	7	6	5	4	3	2	1

SYNC_MASTER=YHARTANTO_J44		SYNC_DATE=12/18/2012			
PAGE TITLE					
ICT FCT 2					
 Apple Inc.		DRAWING NUMBER	051-00515	SIZE	D
		REVISION			7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH			
		dvt1			
		PAGE			
		125 OF 145			
		SHEET			
		105 OF 119			



D

C

B

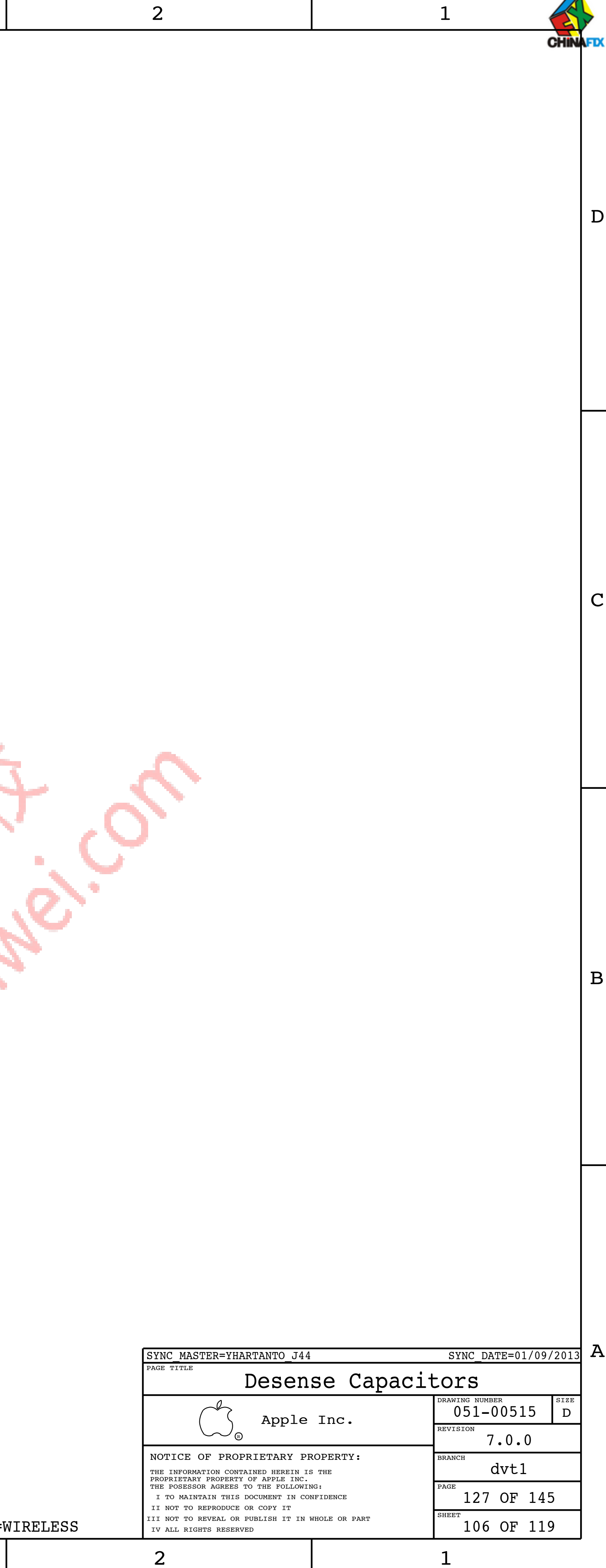
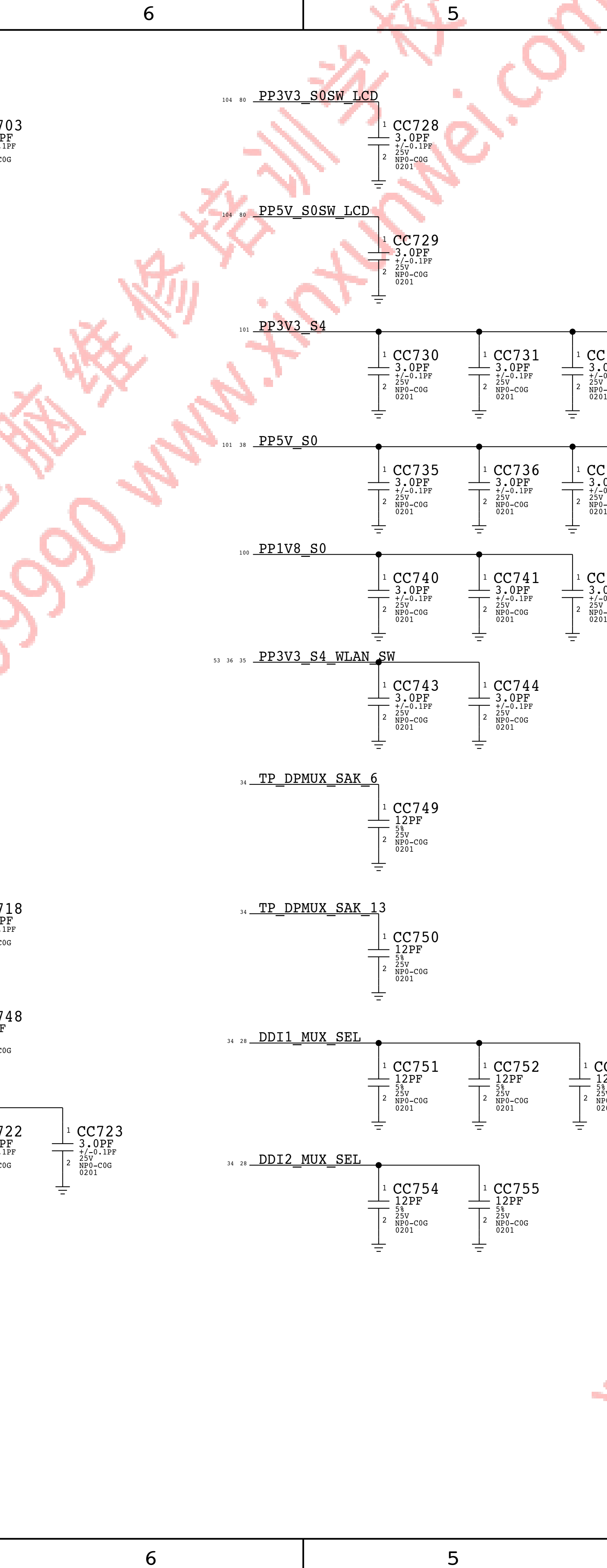
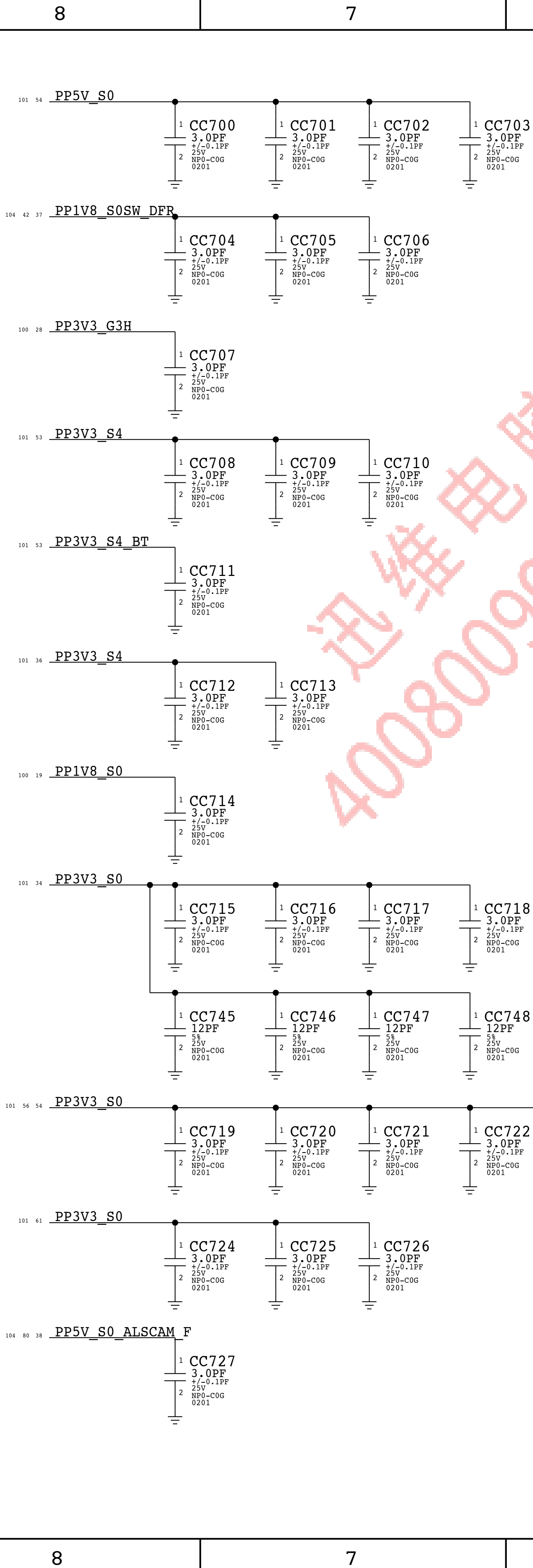
A

D

C

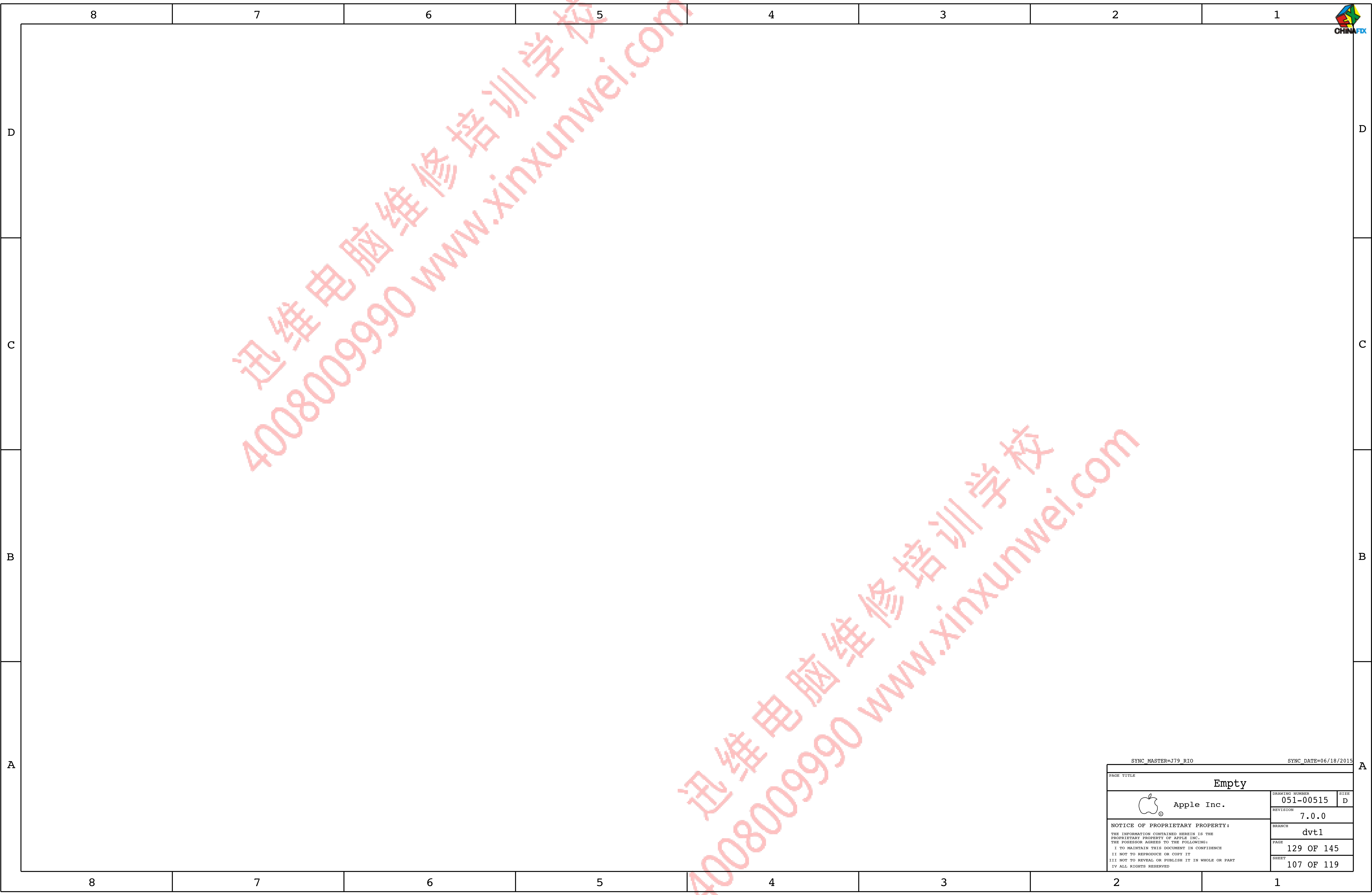
B

A



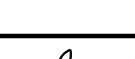
BOM_COST_GROUP=WIRELESS

SYNC_MASTER=YHARTANTO J44		SYNC DATE=01/09/2013	
PAGE TITLE			
Desense Capacitors			
 Apple Inc.	DRAWING NUMBER	051-00515	SIZE
	REVISION	7.0.0	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	dvt1
		PAGE	127 OF 145
		SHEET	106 OF 119



SYNC MASTER=J79 RIO		SYNC DATE=06/18/2015		
PAGE TITLE				
Empty				
 Apple Inc.	DRAWING NUMBER		051-00515	SIZE
	REVISION		7.0.0	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH		dvt1	
	PAGE		129 OF 145	
	SHEET		107 OF 119	

	8	7	6	5	4	3	2	1	
D	J79 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS								
C									
B									
A									
	8	7	6	5	4	3	2	1	

SYNC_MASTER=YHARTANTO_J44		SYNC_DATE=12/14/2012	
PAGE TITLE			
PCB Rule Definitions			
 Apple Inc.	DRAWING NUMBER	051-00515	SIZE
	REVISION	7.0.0	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH	dvt1	
	PAGE	130 OF 145	
	SHEET	108 OF 119	

[illegible]

Thunderbolt SPI Signal Constraints

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_SPI	*	=2x_DIELECTRIC	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBTD_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
TBTDP_*	*	*	TBTDP_2OTHER
TBTDP_*	=SAME	*	TBTDP_2SAME
TBTDP_TX	*_RX	*	TBTDP_TXRX
TBTDP_RX	*_TX	*	TBTDP_TXRX

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
HDMI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2SAME	*	=3x_DIELECTRIC	?	DP_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
DP_2OTHER	*	=4x_DIELECTRIC	?	DP_2OTHER	TOP,BOTTOM	=6x_DIELECTRIC	?
HDMICKL_2OTHER	*	=7x_DIELECTRIC	?	HDMICKL_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?
HDMICKL_2DPHDMI	*	=4x_DIELECTRIC	?	HDMICKL_2DPHDMI	TOP,BOTTOM	=6x_DIELECTRIC	?
HDMIDATA_2SAME	*	=3x_DIELECTRIC	?	HDMIDATA_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
HDMIDATA_2OTHER	*	=4x_DIELECTRIC	?	HDMIDATA_2OTHER	TOP,BOTTOM	=6x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
HDMI_DATA	*	*	HDMIDATA_20THER
HDMI_DATA	=SAME	*	HDMIDATA_2SAME
HDMI_DATA	TBTDP_TX	*	HDMIDATA_2SAME
HDMI_DATA	TBTDP_RX	*	TBTDP_TXRX
HDMI_CLK	*	*	HDMICLK_20THER
HDMI_CLK	HDMI_DATA	*	HDMICLK_2DPHDMI
HDMI_CLK	DISPLAYPORT	*	HDMICLK_2DPHDMI
HDMI_CLK	TBTDP_TX	*	HDMICLK_2DPHDMI

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DISPLAYPORT	*	*	DP_20THER
DISPLAYPORT	=SAME	*	DP_2SAME
DISPLAYPORT	HDMI_DATA	*	DP_2SAME
DISPLAYPORT	TBTDP_TX	*	DP_2SAME
DISPLAYPORT	TBTDP_RX	*	TBTDP_TXRX

ELECTRICAL CONST SET	NET TYPE	
	PHYSICAL	SPACING

SYNC_MASTER=J75 JACK		SYNC_DATE=05/19/2015	
PAGE TITLE			
TBT DP HDMI Constraints			
	DRAWING NUMBER		STEP
	051-00515		D
Apple Inc.	REVISION		
	7.0.0		
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH		
	dvt1		
	PAGE		
	134 OF 145		
	SHEET		
	112 OF 119		

SYNC MASTER=THARTANTO J44		SYNC DATE=01/02/2013	
PAGE TITLE			
SMC Constraints			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00515		D
	REVISION		
			7.0.0
NOTICE OF PROPRIETARY PROPERTY:			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			
BRANCH		dvt1	
PAGE		137 OF 145	
SHEET		115 OF 119	

87654321

CHINAFOX

Camera Net Properties

ELECTRICAL CONST SET

NET TYPE

PHYSICALSPACING

	8	7	6	5	4	3	2	1																																																																																																																																																																																																																																																																		
D	Alternate Parts								D																																																																																																																																																																																																																																																																	
	<table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>107S00033</td><td>107S00034</td><td></td><td>ALL</td><td></td></tr><tr><td>138S0738</td><td>138S1101</td><td></td><td>ALL</td><td>Samsung alt to Murata</td></tr><tr><td>138S0846</td><td>138S0811</td><td></td><td>ALL</td><td>Samsung alt to Murata</td></tr><tr><td>152S00359</td><td>152S00253</td><td></td><td>ALL</td><td>Chillisin alt to Cyttec</td></tr><tr><td>371S0704</td><td>371S00077</td><td></td><td>ALL</td><td>NXP alt to Diodes</td></tr><tr><td>376S1053</td><td>376S0604</td><td></td><td>ALL</td><td>Diodes alt to Fairchild</td></tr><tr><td>376S1106</td><td>376S0678</td><td></td><td>ALL</td><td>Fairchild alt to Vishay</td></tr><tr><td>740S00027</td><td>740S0159</td><td></td><td>ALL</td><td>Bourns alt to Little Fuse</td></tr><tr><td>107S0249</td><td>107S0251</td><td></td><td>ALL</td><td></td></tr></table>				PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES		COMMENTS:	107S00033	107S00034		ALL		138S0738	138S1101		ALL	Samsung alt to Murata	138S0846	138S0811		ALL	Samsung alt to Murata	152S00359	152S00253		ALL	Chillisin alt to Cyttec	371S0704	371S00077		ALL	NXP alt to Diodes	376S1053	376S0604		ALL	Diodes alt to Fairchild	376S1106	376S0678		ALL	Fairchild alt to Vishay	740S00027	740S0159		ALL	Bourns alt to Little Fuse	107S0249	107S0251		ALL		<table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>197S00046</td><td>197S00036</td><td></td><td>ALL</td><td>Epson w/ TXC</td></tr><tr><td>197S00047</td><td>197S00036</td><td></td><td>ALL</td><td>Kyocera w/ TXC</td></tr><tr><td>197S00048</td><td>197S00036</td><td></td><td>ALL</td><td>Murata w/ TXC</td></tr><tr><td>197S00053</td><td>197S00050</td><td></td><td>ALL</td><td>Kyocera w/ TXC</td></tr><tr><td>197S00054</td><td>197S00050</td><td></td><td>ALL</td><td>NDK w/ TXC</td></tr><tr><td>197S00055</td><td>197S00050</td><td></td><td>ALL</td><td>Murata w/ TXC</td></tr><tr><td>311S0596</td><td>311S0593</td><td></td><td>ALL</td><td>NXP w/ Diodes</td></tr><tr><td>107S0276</td><td>107S00020</td><td></td><td>ALL</td><td>Cyntec w/ TFT</td></tr><tr><td>107S00021</td><td>107S0284</td><td></td><td>ALL</td><td>TFT w/ Yageo</td></tr><tr><td>152S00343</td><td>152S1682</td><td></td><td>ALL</td><td>NXP w/ Diodes</td></tr><tr><td>107S00087</td><td>107S00029</td><td></td><td>ALL</td><td>TFT w/ Yageo</td></tr><tr><td>128S00058</td><td>128S00018</td><td></td><td>ALL</td><td>NEC w/ Rohm</td></tr><tr><td>138S0706</td><td>138S0739</td><td></td><td>ALL</td><td>NEC w/ Vishay</td></tr><tr><td>138S0945</td><td>138S0739</td><td></td><td>ALL</td><td>NEC w/ Rohm</td></tr><tr><td>152S00358</td><td>152S00208</td><td></td><td>ALL</td><td>Murata w/ Chillisin</td></tr><tr><td>152S00400</td><td>152S00361</td><td></td><td>ALL</td><td>Murata w/ Cyntec</td></tr><tr><td>152S1872</td><td>152S00361</td><td></td><td>ALL</td><td>Murata w/ Cyntec</td></tr><tr><td>155S00034</td><td>155S0706</td><td></td><td>ALL</td><td>Taiyo w/ Murata</td></tr></table>				PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	197S00046	197S00036		ALL	Epson w/ TXC	197S00047	197S00036		ALL	Kyocera w/ TXC	197S00048	197S00036		ALL	Murata w/ TXC	197S00053	197S00050		ALL	Kyocera w/ TXC	197S00054	197S00050		ALL	NDK w/ TXC	197S00055	197S00050		ALL	Murata w/ TXC	311S0596	311S0593		ALL	NXP w/ Diodes	107S0276	107S00020		ALL	Cyntec w/ TFT	107S00021	107S0284		ALL	TFT w/ Yageo	152S00343	152S1682		ALL	NXP w/ Diodes	107S00087	107S00029		ALL	TFT w/ Yageo	128S00058	128S00018		ALL	NEC w/ Rohm	138S0706	138S0739		ALL	NEC w/ Vishay	138S0945	138S0739		ALL	NEC w/ Rohm	152S00358	152S00208		ALL	Murata w/ Chillisin	152S00400	152S00361		ALL	Murata w/ Cyntec	152S1872	152S00361		ALL	Murata w/ Cyntec	155S00034	155S0706		ALL	Taiyo w/ Murata	<table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>138S00104</td><td>138S0978</td><td></td><td>ALL</td><td>Murata w/ Taiyo</td></tr><tr><td>311S00072</td><td>311S0657</td><td></td><td>ALL</td><td>NXP w/ On Semi</td></tr><tr><td>311S00090</td><td>311S00028</td><td></td><td>ALL</td><td>On Semi w/ TI</td></tr><tr><td>353S00854</td><td>353S4342</td><td></td><td>ALL</td><td>TI w/ ST</td></tr><tr><td>377S0077</td><td>377S0183</td><td></td><td>ALL</td><td>Infineon w/ ST</td></tr></table>				PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	138S00104	138S0978		ALL	Murata w/ Taiyo	311S00072	311S0657		ALL	NXP w/ On Semi	311S00090	311S00028		ALL	On Semi w/ TI	353S00854	353S4342		ALL	TI w/ ST	377S0077	377S0183		ALL	Infineon w/ ST																																																																														
	PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:																																																																																																																																																																																																																																																																					
	107S00033	107S00034		ALL																																																																																																																																																																																																																																																																						
	138S0738	138S1101		ALL	Samsung alt to Murata																																																																																																																																																																																																																																																																					
	138S0846	138S0811		ALL	Samsung alt to Murata																																																																																																																																																																																																																																																																					
	152S00359	152S00253		ALL	Chillisin alt to Cyttec																																																																																																																																																																																																																																																																					
	371S0704	371S00077		ALL	NXP alt to Diodes																																																																																																																																																																																																																																																																					
	376S1053	376S0604		ALL	Diodes alt to Fairchild																																																																																																																																																																																																																																																																					
	376S1106	376S0678		ALL	Fairchild alt to Vishay																																																																																																																																																																																																																																																																					
740S00027	740S0159		ALL	Bourns alt to Little Fuse																																																																																																																																																																																																																																																																						
107S0249	107S0251		ALL																																																																																																																																																																																																																																																																							
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:																																																																																																																																																																																																																																																																						
197S00046	197S00036		ALL	Epson w/ TXC																																																																																																																																																																																																																																																																						
197S00047	197S00036		ALL	Kyocera w/ TXC																																																																																																																																																																																																																																																																						
197S00048	197S00036		ALL	Murata w/ TXC																																																																																																																																																																																																																																																																						
197S00053	197S00050		ALL	Kyocera w/ TXC																																																																																																																																																																																																																																																																						
197S00054	197S00050		ALL	NDK w/ TXC																																																																																																																																																																																																																																																																						
197S00055	197S00050		ALL	Murata w/ TXC																																																																																																																																																																																																																																																																						
311S0596	311S0593		ALL	NXP w/ Diodes																																																																																																																																																																																																																																																																						
107S0276	107S00020		ALL	Cyntec w/ TFT																																																																																																																																																																																																																																																																						
107S00021	107S0284		ALL	TFT w/ Yageo																																																																																																																																																																																																																																																																						
152S00343	152S1682		ALL	NXP w/ Diodes																																																																																																																																																																																																																																																																						
107S00087	107S00029		ALL	TFT w/ Yageo																																																																																																																																																																																																																																																																						
128S00058	128S00018		ALL	NEC w/ Rohm																																																																																																																																																																																																																																																																						
138S0706	138S0739		ALL	NEC w/ Vishay																																																																																																																																																																																																																																																																						
138S0945	138S0739		ALL	NEC w/ Rohm																																																																																																																																																																																																																																																																						
152S00358	152S00208		ALL	Murata w/ Chillisin																																																																																																																																																																																																																																																																						
152S00400	152S00361		ALL	Murata w/ Cyntec																																																																																																																																																																																																																																																																						
152S1872	152S00361		ALL	Murata w/ Cyntec																																																																																																																																																																																																																																																																						
155S00034	155S0706		ALL	Taiyo w/ Murata																																																																																																																																																																																																																																																																						
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:																																																																																																																																																																																																																																																																						
138S00104	138S0978		ALL	Murata w/ Taiyo																																																																																																																																																																																																																																																																						
311S00072	311S0657		ALL	NXP w/ On Semi																																																																																																																																																																																																																																																																						
311S00090	311S00028		ALL	On Semi w/ TI																																																																																																																																																																																																																																																																						
353S00854	353S4342		ALL	TI w/ ST																																																																																																																																																																																																																																																																						
377S0077	377S0183		ALL	Infineon w/ ST																																																																																																																																																																																																																																																																						
C	<table><tr><td>128S0364</td><td>128S0264</td><td></td><td>ALL</td><td>Kemet w/ Panasonic</td></tr><tr><td>128S0325</td><td>128S0397</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00009</td><td>128S00007</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00029</td><td>128S00007</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00070</td><td>128S00007</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00010</td><td>128S00011</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00031</td><td>128S00011</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00026</td><td>128S00011</td><td></td><td>ALL</td><td></td></tr><tr><td>132S00064</td><td>132S0409</td><td></td><td>ALL</td><td></td></tr><tr><td>138S0614</td><td>138S0578</td><td></td><td>ALL</td><td></td></tr></table>				128S0364	128S0264		ALL	Kemet w/ Panasonic	128S0325	128S0397		ALL		128S00009	128S00007		ALL		128S00029	128S00007		ALL		128S00070	128S00007		ALL		128S00010	128S00011		ALL		128S00031	128S00011		ALL		128S00026	128S00011		ALL		132S00064	132S0409		ALL		138S0614	138S0578		ALL		<table><tr><td>353S00711</td><td>353S2073</td><td></td><td>ALL</td><td>On Semi w/ TI</td></tr></table>				353S00711	353S2073		ALL	On Semi w/ TI	<table><tr><td>740S00019</td><td>740S00007</td><td></td><td>ALL</td><td>Bourns w/ Polytronics</td></tr><tr><td>155S00189</td><td>155S0342</td><td></td><td>ALL</td><td>Murata w/ Taiyo</td></tr><tr><td>138S0714</td><td>138S0713</td><td></td><td>ALL</td><td>Murata w/ Samsung</td></tr><tr><td>138S0715</td><td>138S0732</td><td></td><td>ALL</td><td>Murata w/ Samsung</td></tr><tr><td>107S00086</td><td>107S00056</td><td></td><td>ALL</td><td>TFT w/ Cyntec</td></tr><tr><td>138S0875</td><td>138S0678</td><td></td><td>ALL</td><td>Taiyo w/ Mur&SS</td></tr><tr><td>138S0786</td><td>138S0705</td><td></td><td>ALL</td><td>Murata w/ Samsung</td></tr><tr><td>152S2052</td><td>152S1954</td><td></td><td>ALL</td><td>Taiyo w/ Cyntec</td></tr><tr><td>152S2015</td><td>152S1958</td><td></td><td>ALL</td><td>Taiyo w/ Cyntec</td></tr><tr><td>138S0789</td><td>138S0941</td><td></td><td>ALL</td><td>Murata w/ SS</td></tr><tr><td>107S00101</td><td>107S00005</td><td></td><td>ALL</td><td>Cyntec w/ Yageo</td></tr><tr><td>107S00102</td><td>107S00017</td><td></td><td>ALL</td><td>Cyntec w/ Yageo</td></tr><tr><td>107S00100</td><td>107S00057</td><td></td><td>ALL</td><td>Cyntec w/ TFT</td></tr><tr><td>107S00103</td><td>107S00058</td><td></td><td>ALL</td><td>Cyntec w/ Yageo</td></tr><tr><td>107S00104</td><td>107S00061</td><td></td><td>ALL</td><td>Cyntec w/ Yageo</td></tr><tr><td>107S00105</td><td>107S00062</td><td></td><td>ALL</td><td>Cyntec w/ Yageo</td></tr><tr><td>152S00403</td><td>152S00322</td><td></td><td>ALL</td><td>Murata w/ Chillisin</td></tr></table>				740S00019	740S00007		ALL	Bourns w/ Polytronics	155S00189	155S0342		ALL	Murata w/ Taiyo	138S0714	138S0713		ALL	Murata w/ Samsung	138S0715	138S0732		ALL	Murata w/ Samsung	107S00086	107S00056		ALL	TFT w/ Cyntec	138S0875	138S0678		ALL	Taiyo w/ Mur&SS	138S0786	138S0705		ALL	Murata w/ Samsung	152S2052	152S1954		ALL	Taiyo w/ Cyntec	152S2015	152S1958		ALL	Taiyo w/ Cyntec	138S0789	138S0941		ALL	Murata w/ SS	107S00101	107S00005		ALL	Cyntec w/ Yageo	107S00102	107S00017		ALL	Cyntec w/ Yageo	107S00100	107S00057		ALL	Cyntec w/ TFT	107S00103	107S00058		ALL	Cyntec w/ Yageo	107S00104	107S00061		ALL	Cyntec w/ Yageo	107S00105	107S00062		ALL	Cyntec w/ Yageo	152S00403	152S00322		ALL	Murata w/ Chillisin	<table><tr><td>128S00062</td><td>128S00067</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00069</td><td>128S00067</td><td></td><td>ALL</td><td></td></tr><tr><td>132S00012</td><td>132S0401</td><td></td><td>ALL</td><td></td></tr><tr><td>138S0660</td><td>138S0684</td><td></td><td>ALL</td><td></td></tr><tr><td>138S1103</td><td>138S0719</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00097</td><td>138S0750</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00111</td><td>138S00036</td><td></td><td>ALL</td><td></td></tr><tr><td>155S00203</td><td>155S0894</td><td></td><td>ALL</td><td></td></tr><tr><td>197S00082</td><td>197S00081</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00105</td><td>311S0233</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00060</td><td>311S0273</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00118</td><td>311S0489</td><td></td><td>ALL</td><td></td></tr><tr><td>311S0271</td><td>311S00008</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00104</td><td>311S00091</td><td></td><td>ALL</td><td></td></tr><tr><td>311S0437</td><td>311S00112</td><td></td><td>ALL</td><td></td></tr><tr><td>335S00213</td><td>335S0888</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00880</td><td>353S3452</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00878</td><td>353S00599</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00879</td><td>353S00754</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00750</td><td>353S00877</td><td></td><td>ALL</td><td></td></tr><tr><td>371S00089</td><td>371S00085</td><td></td><td>ALL</td><td></td></tr><tr><td>377S0178</td><td>377S00031</td><td></td><td>ALL</td><td></td></tr></table>				128S00062	128S00067		ALL		128S00069	128S00067		ALL		132S00012	132S0401		ALL		138S0660	138S0684		ALL		138S1103	138S0719		ALL		138S00097	138S0750		ALL		138S00111	138S00036		ALL		155S00203	155S0894		ALL		197S00082	197S00081		ALL		311S00105	311S0233		ALL		311S00060	311S0273		ALL		311S00118	311S0489		ALL		311S0271	311S00008		ALL		311S00104	311S00091		ALL		311S0437	311S00112		ALL		335S00213	335S0888		ALL		353S00880	353S3452		ALL		353S00878	353S00599		ALL		353S00879	353S00754		ALL		353S00750	353S00877		ALL		371S00089	371S00085		ALL		377S0178	377S00031		ALL	
	128S0364	128S0264		ALL	Kemet w/ Panasonic																																																																																																																																																																																																																																																																					
	128S0325	128S0397		ALL																																																																																																																																																																																																																																																																						
	128S00009	128S00007		ALL																																																																																																																																																																																																																																																																						
	128S00029	128S00007		ALL																																																																																																																																																																																																																																																																						
	128S00070	128S00007		ALL																																																																																																																																																																																																																																																																						
	128S00010	128S00011		ALL																																																																																																																																																																																																																																																																						
	128S00031	128S00011		ALL																																																																																																																																																																																																																																																																						
	128S00026	128S00011		ALL																																																																																																																																																																																																																																																																						
	132S00064	132S0409		ALL																																																																																																																																																																																																																																																																						
138S0614	138S0578		ALL																																																																																																																																																																																																																																																																							
353S00711	353S2073		ALL	On Semi w/ TI																																																																																																																																																																																																																																																																						
740S00019	740S00007		ALL	Bourns w/ Polytronics																																																																																																																																																																																																																																																																						
155S00189	155S0342		ALL	Murata w/ Taiyo																																																																																																																																																																																																																																																																						
138S0714	138S0713		ALL	Murata w/ Samsung																																																																																																																																																																																																																																																																						
138S0715	138S0732		ALL	Murata w/ Samsung																																																																																																																																																																																																																																																																						
107S00086	107S00056		ALL	TFT w/ Cyntec																																																																																																																																																																																																																																																																						
138S0875	138S0678		ALL	Taiyo w/ Mur&SS																																																																																																																																																																																																																																																																						
138S0786	138S0705		ALL	Murata w/ Samsung																																																																																																																																																																																																																																																																						
152S2052	152S1954		ALL	Taiyo w/ Cyntec																																																																																																																																																																																																																																																																						
152S2015	152S1958		ALL	Taiyo w/ Cyntec																																																																																																																																																																																																																																																																						
138S0789	138S0941		ALL	Murata w/ SS																																																																																																																																																																																																																																																																						
107S00101	107S00005		ALL	Cyntec w/ Yageo																																																																																																																																																																																																																																																																						
107S00102	107S00017		ALL	Cyntec w/ Yageo																																																																																																																																																																																																																																																																						
107S00100	107S00057		ALL	Cyntec w/ TFT																																																																																																																																																																																																																																																																						
107S00103	107S00058		ALL	Cyntec w/ Yageo																																																																																																																																																																																																																																																																						
107S00104	107S00061		ALL	Cyntec w/ Yageo																																																																																																																																																																																																																																																																						
107S00105	107S00062		ALL	Cyntec w/ Yageo																																																																																																																																																																																																																																																																						
152S00403	152S00322		ALL	Murata w/ Chillisin																																																																																																																																																																																																																																																																						
128S00062	128S00067		ALL																																																																																																																																																																																																																																																																							
128S00069	128S00067		ALL																																																																																																																																																																																																																																																																							
132S00012	132S0401		ALL																																																																																																																																																																																																																																																																							
138S0660	138S0684		ALL																																																																																																																																																																																																																																																																							
138S1103	138S0719		ALL																																																																																																																																																																																																																																																																							
138S00097	138S0750		ALL																																																																																																																																																																																																																																																																							
138S00111	138S00036		ALL																																																																																																																																																																																																																																																																							
155S00203	155S0894		ALL																																																																																																																																																																																																																																																																							
197S00082	197S00081		ALL																																																																																																																																																																																																																																																																							
311S00105	311S0233		ALL																																																																																																																																																																																																																																																																							
311S00060	311S0273		ALL																																																																																																																																																																																																																																																																							
311S00118	311S0489		ALL																																																																																																																																																																																																																																																																							
311S0271	311S00008		ALL																																																																																																																																																																																																																																																																							
311S00104	311S00091		ALL																																																																																																																																																																																																																																																																							
311S0437	311S00112		ALL																																																																																																																																																																																																																																																																							
335S00213	335S0888		ALL																																																																																																																																																																																																																																																																							
353S00880	353S3452		ALL																																																																																																																																																																																																																																																																							
353S00878	353S00599		ALL																																																																																																																																																																																																																																																																							
353S00879	353S00754		ALL																																																																																																																																																																																																																																																																							
353S00750	353S00877		ALL																																																																																																																																																																																																																																																																							
371S00089	371S00085		ALL																																																																																																																																																																																																																																																																							
377S0178	377S00031		ALL																																																																																																																																																																																																																																																																							
B	<table><tr><td>155S0914</td><td>155S0897</td><td></td><td>ALL</td><td></td></tr><tr><td>155S00190</td><td>155S0897</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00004</td><td>311S0370</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00013</td><td>311S0508</td><td></td><td>ALL</td><td>NXP w/ Diodes</td></tr><tr><td>353S00107</td><td>353S3239</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00525</td><td>353S4471</td><td></td><td>ALL</td><td></td></tr><tr><td>372S0186</td><td>372S0185</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00014</td><td>376S0761</td><td></td><td>ALL</td><td></td></tr><tr><td>376S00086</td><td>376S0761</td><td></td><td>ALL</td><td></td></tr><tr><td>376S1080</td><td>376S0820</td><td></td><td>ALL</td><td></td></tr></table>				155S0914	155S0897		ALL		155S00190	155S0897		ALL		311S00004	311S0370		ALL		311S00013	311S0508		ALL	NXP w/ Diodes	353S00107	353S3239		ALL		353S00525	353S4471		ALL		372S0186	372S0185		ALL		376S00014	376S0761		ALL		376S00086	376S0761		ALL		376S1080	376S0820		ALL		<table><tr><td>376S00074</td><td>376S0855</td><td></td><td>ALL</td><td></td></tr><tr><td>376S1089</td><td>376S1128</td><td></td><td>ALL</td><td></td></tr><tr><td>740S0144</td><td>740S0118</td><td></td><td>ALL</td><td></td></tr><tr><td>740S00028</td><td>740S0118</td><td></td><td>ALL</td><td></td></tr><tr><td>740S00003</td><td>740S0135</td><td></td><td>ALL</td><td></td></tr><tr><td>998-04070</td><td>998-04071</td><td></td><td>ALL</td><td>Hynix alt to SS</td></tr></table>				376S00074	376S0855		ALL		376S1089	376S1128		ALL		740S0144	740S0118		ALL		740S00028	740S0118		ALL		740S00003	740S0135		ALL		998-04070	998-04071		ALL	Hynix alt to SS																																																																																																																																																																																		
	155S0914	155S0897		ALL																																																																																																																																																																																																																																																																						
	155S00190	155S0897		ALL																																																																																																																																																																																																																																																																						
	311S00004	311S0370		ALL																																																																																																																																																																																																																																																																						
	311S00013	311S0508		ALL	NXP w/ Diodes																																																																																																																																																																																																																																																																					
	353S00107	353S3239		ALL																																																																																																																																																																																																																																																																						
	353S00525	353S4471		ALL																																																																																																																																																																																																																																																																						
	372S0186	372S0185		ALL																																																																																																																																																																																																																																																																						
	376S00014	376S0761		ALL																																																																																																																																																																																																																																																																						
	376S00086	376S0761		ALL																																																																																																																																																																																																																																																																						
376S1080	376S0820		ALL																																																																																																																																																																																																																																																																							
376S00074	376S0855		ALL																																																																																																																																																																																																																																																																							
376S1089	376S1128		ALL																																																																																																																																																																																																																																																																							
740S0144	740S0118		ALL																																																																																																																																																																																																																																																																							
740S00028	740S0118		ALL																																																																																																																																																																																																																																																																							
740S00003	740S0135		ALL																																																																																																																																																																																																																																																																							
998-04070	998-04071		ALL	Hynix alt to SS																																																																																																																																																																																																																																																																						
A									A																																																																																																																																																																																																																																																																	
FPIC PARTS/01/10/04 FPIC PARTS/01/10/04 Alternate Parts Alternate Parts DRANTING NUMBER 051-00515 STEP D REVISION 7.0.0 BRANCH dvt1 PAGE 145 OF 145 SHEET 119 OF 119 NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED																																																																																																																																																																																																																																																																										
8	7	6	5	4	3	2	1																																																																																																																																																																																																																																																																			